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Performance Analysis of Linear Codes under Maximum-Likelihood Decoding: A Tutorial

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Chapter 1

A Short Overview

Chapter Overview: Upper and lower bounds on the error probability of linear codes under maximum-likelihood (ML) decoding are shortly surveyed and applied to ensembles of codes on graphs. For upper bounds, focus is put on Gallager bounding techniques and their relation to a variety of other reported bounds. Within the class of lower bound, we address de Caen's based bounds and their improvements, sphere-packing bounds and recent research work related to these bounds, and information-theoretic bounds on the bit error probability of codes defined on graphs. This serves as an introductory chapter, and a comprehensive overview is provided in the continuation of this monograph.

1.1 Introduction

Consider the classical coded communication model of transmitting one of equally likely signals over a communication channel. Since the error performance of coded communication systems rarely admits exact expressions, tight analytical upper and lower bounds emerge as a useful theoretical and engineering tool for assessing performance and for gaining insight into the effect of the main system parameters. As specific good codes are hard to identify, the performance of ensembles of codes is usually considered. The Fano [107] and Gallager [123] bounds were introduced as efficient tools to determine the error exponents of the ensemble of random codes, providing informative results up to the ultimate capacity limit. Since the advent of information theory, the search for efficient coding systems has motivated the introduction of efficient bounding techniques tailored to specific codes or some carefully chosen ensembles of codes. A classical example is the adaptation of the Fano upper bounding technique [107] to specific codes, as reported in the seminal dissertation by Gallager [122] (to be referred to as the 1961 Gallager-Fano bound). The incentive for introducing and applying such bounds has strengthened with the introduction of various families of codes defined on graphs which closely approach the channel capacity with feasible complexity (e.g., turbo codes, repeat-accumulate codes, and low-density parity-check (LDPC) codes). Clearly, the desired bounds must not be subject to the union bound limitation, since for long blocks these ensembles of turbo-like codes perform reliably at rates which are considerably above the cutoff rate (R_0) of the channel (recalling that union bounds for long codes are not informative at the portion of the rate region above R_0, where the performance of these capacity-approaching codes is most appealing). Although maximum-likelihood (ML) decoding is in general prohibitively complex for long codes, the derivation of upper and lower bounds on the ML decoding error probability is of interest, providing an ultimate indication of the system performance. Further, the structure of efficient codes is usually not available, necessitating efficient bounds on performance to rely only on basic features, such as the distance spectrum and the input-output weight enumeration function (IOWEF) of the examined code (for the evaluation of the block and bit error probabilities, respectively, of a specific code or ensemble). These latter features can be found by analytical methods (see e.g., [214]).

In Chapter 2–4, we present various reported upper bounds on the ML decoding error probability. These include the bounds of Berlekamp [38], Divsalar [86], Duman-Salehi [94], Engdahl-Zigangirov [103], Gallager-Fano 1961 bound [122], Hughes [152], Poltyrev [259], Sason-Shamai [283], Shulman-Feder [303], Viterbi ([347] and [348]), Yousefi-Khandani ([371] and [372]) and others. We demonstrate in Chapters 3 and 4 the underlying connections that exist between them; the bounds are based on the distance spectrum or the IOWEFs of the codes. The focus of this presentation is directed towards the application of efficient bounding techniques on ML decoding performance, which are not subjected to the deficiencies of the union bounds and therefore provide useful results at rates reasonably higher than the cut-off rate. In Chapters 2–4 and references therein, improved upper bounds are applied to block codes and turbo-like codes. In addressing the Gallager bounds and their variations, we focus in [296] (and more extensively in Chapter 4) on the Duman and Salehi variation which originates from the standard Gallager bound. A large class of efficient recent bounds (or their Chernoff versions) is demonstrated to be a special case of the generalized second version of the Duman and Salehi bounds. Implications and applications of these observations are addressed in Chapter 4.

We also address here lower bounds on the ML decoding error probability (see Chapters 6–7), and exemplify these bounds on linear block codes with a special emphasis on ensembles of codes defined on graphs. Here we overview a class of bounds which are based on de Caen's bound and an improved version of it, sphere-packing bounds, and other information-theoretic lower bounds which are suitable for linear codes defined on bipartite Tanner graphs (e.g., LDPC codes).

1.2 General Approach for the Derivation of Improved Upper Bounds

In Chapters 3–4, we present many improved upper bounds on the ML decoding error probability which are tighter than the union bound. Let $\mathbf{r}$ be the received signal vector, and $\mathcal{R}$ be an arbitrary region around the transmitted signal point. The basic concept which is common to the derivation of the upper bounds within the class discussed in Chapter 3 and 4 is the following:

$$\begin{aligned} \Pr(\text{error}) &= \Pr(\text{error}, \mathbf{r} \in \mathcal{R}) + \Pr(\text{error}, \mathbf{r} \notin \mathcal{R}) \\ &\leq \Pr(\text{error}, \mathbf{r} \in \mathcal{R}) + \Pr(\mathbf{r} \notin \mathcal{R}). \end{aligned} \tag{1.1}$$

The region $\mathcal{R}$ in (1.1) is interpreted as the "good region". The idea is to use the union bound only for the joint event where the decoder fails to decode correctly, and in addition, the received signal vector falls inside the region $\mathcal{R}$ (i.e., the union bound is used for upper bounding the first term in the right-hand side (RHS) of (1.1)). On the other hand, the second term in the RHS of (1.1) represents the probability of the event where the received signal vector falls outside the region $\mathcal{R}$, and which is typically the dominant term for very low SNR, is calculated only one time (and it is not part of the event where the union bound is used). We note that in the case where the region $\mathcal{R}$ is the whole observation space, the basic approach which is suggested above provides the union bound. However, since the upper bound in (1.1) is valid for an arbitrary region $\mathcal{R}$ in the observation space, many improved upper bounds can be derived by an appropriate selection of this region. These bounds could be therefore interpreted as geometric bounds (see [86] and [296]). As we will see, the choice of the region $\mathcal{R}$ is very significant in this bounding technique; different choices of this region have resulted in various different improved upper bounds which are considered extensively in Chapters 3 and 4. For instance, the tangential bound of Berlekamp [38] used the basic inequality in (1.1) to provide a considerably tighter bound than the union bound at low SNR values. This was achieved by separating the radial and tangential components of the Gaussian noise with a half-space as the underlying region $\mathcal{R}$. For the derivation of the sphere bound [140], Herzberg and Poltyrev have chosen the region $\mathcal{R}$ in (3.1) to be a sphere around the transmitted signal vector, and optimized the

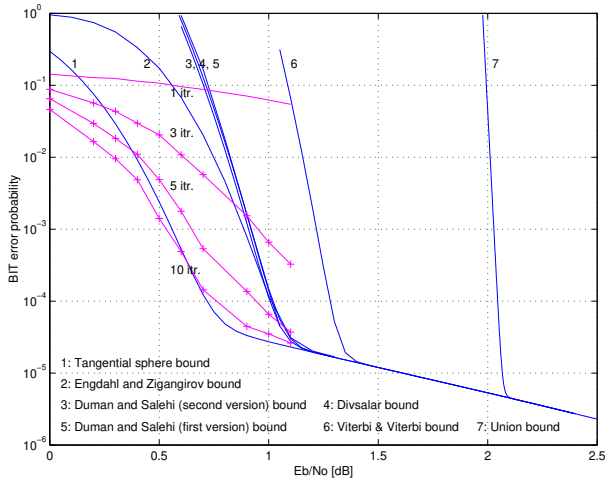


Figure 1.1: Various bounds for the ensemble of rate$-\frac{1}{3}$ turbo codes whose components are recursive systematic convolutional codes with generators $G_1(D) = G_2(D) = \left[1, \frac{1+D^4}{1+D+D^2+D^3+D^4}\right]$. There is no puncturing of the parity bits, and the uniform interleaver between the two parallel concatenated (component) codes is of length $N = 1000$. It is assumed that the transmission of the codes takes place over a binary-input AWGN channel. The upper bounds on the bit error probability under optimal ML decoding are compared with computer simulations of the iterative sum-product algorithm with up to 10 iterations.

radius of the sphere in order to get the tightest upper bound within this form. Divsalar bound [86] is another simple and tight bound which relies on the basic inequality (1.1). The geometrical region $\mathcal{R}$ in Divsalar bound was chosen to be a sphere; in addition to the optimization of the radius of this sphere, the center of the sphere which does not necessarily coincide with the transmitted signal vector was optimized too. Finally, the tangential-sphere bound (TSB) which was proposed for binary linear block codes by Poltyrev [259] and for M-ary PSK block coded-modulation schemes by Herzberg and Poltyrev [141] selected $\mathcal{R}$ as a circular cone of half-angle θ, whose central line passes through the origin and the transmitted signal. It is one of the tightest upper bounds known to-date for linear codes whose transmission takes place over a binary-input AWGN channel (see Fig. 1.1 and [282, 283, 371]).

We note that the bounds mentioned above are only a sample of various bounds reported in Chapter 3; all of these bounds rely on the inequality (1.1) where the geometric region $\mathcal{R}$ characterizes the resulting upper bounds on the decoding error probability. After providing the general approach, we outline some connections between these bounds and demonstrate a few possible applications.

1.3 On Gallager Bounds: Variations and Applications

In addressing the Gallager bounding techniques and their variations, we focus in Chapter 4 on a generalization of the second version of the Duman and Salehi (DS2) bound [95]. The Duman and Salehi bounding technique (see [95] and [296]) originates from the 1965 Gallager bound [123]. Let $\psi_N^m(\underline{y})$ be an arbitrary probability measure (which may also depend on the transmitted codeword

$\underline{x}^m$). The 1965 Gallager bound then yields that for arbitrary $\lambda, \rho > 0$:

$$P_{\mathrm{e}|m} \leq \sum_{\underline{y}} \psi_N^m(\underline{y})\ \psi_N^m(\underline{y})^{-1}\ p_N(\underline{y}|\underline{x}^m)\ \left(\sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)}\right)^{\lambda}\right)^{\rho}$$

$$= \sum_{\underline{y}} \psi_N^m(\underline{y})\ \left(\psi_N^m(\underline{y})^{-\frac{1}{\rho}}\ p_N(\underline{y}|\underline{x}^m)^{\frac{1}{\rho}} \sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)}\right)^{\lambda}\right)^{\rho}.$$

By invoking Jensen's inequality to the RHS of the last inequality, the DS2 bound follows:

$$P_{\mathrm{e}|m} \leq \left(\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)^{\frac{1}{\rho}}\ \psi_N^m(\underline{y})^{1-\frac{1}{\rho}}\ \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)}\right)^{\lambda}\right)^{\rho}$$

for arbitrary parameters $0 \leq \rho \leq 1$ and $\lambda \geq 0$.

Let $G_N^m(\underline{y})$ be an arbitrary non-negative function of $\underline{y}$, and let the probability density function $\psi_N^m(\underline{y})$ be

$$\psi_N^m(\underline{y}) = \frac{G_N^m(\underline{y})\ p_N(\underline{y}|\underline{x}^m)}{\sum\limits_{\underline{y}} G_N^m(\underline{y})\ p_N(\underline{y}|\underline{x}^m)}\ . \tag{1.2}$$

The functions $G_N^m(\underline{y})$ and $\psi_N^m(\underline{y})$ are referred to as the *un-normalized and normalized tilting measures*, respectively. The substitution of (1.2) into the RHS of the last inequality gives

$$P_{\mathrm{e}|m} \leq \left(\sum_{\underline{y}} G_N^m(\underline{y})\ p_N(\underline{y}|\underline{x}^m)\right)^{1-\rho}$$
$$\cdot \left\{\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ G_N^m(\underline{y})^{1-\frac{1}{\rho}}\ \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)}\right)^{\lambda}\right\}^{\rho}$$

where $0 \leq \rho \leq 1$ and $\lambda \geq 0$. This upper bound on the decoding error probability was also derived in [86, Eq. (62)]. For the class of memoryless channels, and for the choice of $\psi_N^m(\underline{y})$ as $\psi_N^m(\underline{y}) = \prod\limits_{i=1}^{N} \psi^m(y_i)$ (recalling that the function ψ^m may also depend on the transmitted codeword $\underline{x}^m$), this upper bound is relatively easily evaluated (similarly to the standard union bounds) for particular block codes. In that case, the latter bound is calculable in terms of the distance spectrum, not requiring the fine details of the code structure. A similar upper bound on the bit error probability is expressible in terms of the IOWEFs of the codes (or the average IOWEFs of code ensembles).

By generalizing the framework of the DS2 bound, a large class of efficient bounds (or their Chernoff versions) is demonstrated to follow from this bound. Implications and applications of these observations are pointed out in [296], including the fully interleaved fading channel, resorting to either matched or mismatched decoding. The proposed approach can be generalized to geometrically uniform non-binary codes, finite state channels, bit-interleaved coded-modulation systems, parallel channels [199], and it can be also used for the derivation of upper bounds on the conditional decoding error probability. In Chapter 4, we present the suitability of variations on the Gallager bounds as bounding techniques for random and deterministic codes, which partially rely on insightful observations made by Divsalar [86]. Focus is put in [296] on geometric interpretations of the 1961 Gallager-Fano bound (see [107] and [122]). The interconnections between many

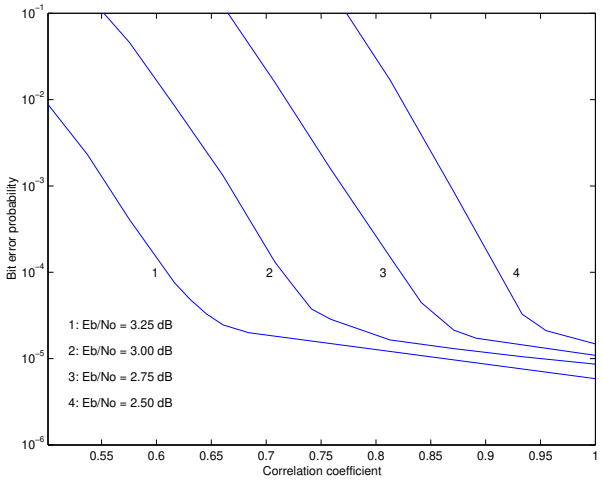


Figure 1.2: A comparison between upper bounds on the bit error probability for the ensemble of turbo codes considered in Example 1.1 where the transmission of these codes takes place over a fully interleaved Rayleigh fading channel with mismatched decoding. The bounds are based on the combination of the generalized DS2 bound and the tight form of the union bound applied to every constant Hamming-weight subcode. These bounds are plotted for $\frac{E_b}{N_0} = 2.50, 2.75, 3.00$ and 3.25 dB, as a function of the correlation coefficient between the actual i.i.d Rayleigh fading samples and their Rayleigh distributed estimations.

reported upper bounds are illustrated in Chapter 4, where it is shown that the generalized DS2 bound particularizes to these upper bounds by proper selections of the tilting measure. Further details, extensions and examples are provided in Chapter 4.

The TSB [259] happens often to be the tightest reported upper bound for block codes which are transmitted over the binary-input AWGN channel and ML decoded (see e.g., [282] and [283]). However, in the random coding setting, it fails to reproduce the random coding error exponent (see [259]), while the DS2 bound does. In fact, also the Shulman-Feder bound [303] which is a special case of the latter bound achieves capacity for the ensemble of fully random block codes. This substantiates the claim that there is no uniformly best bound. However, we note that the loosened version of the TSB [86] (which involves the Chernoff inequality) maintains the asymptotic (i.e, for infinite block length) exponential tightness of the TSB of Poltyrev [259], and it is a special case of the DS2 bound.

In the following, we exemplify the use of the DS2 bounding technique for fully interleaved fading channels with faulty measurements of the fading samples.

Example 1.1. The Generalized DS2 bound for the Mismatched Regime. In [296], we apply the generalized DS2 bound to study the robustness of a mismatched decoding that is based on ML decoding with respect to the faulty channel measurements. We examine here the robustness of the decoder in case that a BPSK modulated signal is transmitted through a fully interleaved Rayleigh fading channel. For simplicity, the bounds are applied to the case of a perfect phase estimation of the i.i.d fading samples (in essence reducing the problem to a real channel). We also assume here that the estimated and real magnitudes of the Rayleigh fading samples have a joint distribution of two correlated bivariate Rayleigh variables with an average power of unity.

The bounds in Fig. 1.2 refer to the ensemble of uniformly interleaved rate $-\frac{1}{3}$ turbo codes whose components are recursive systematic convolutional codes: $G_1(D) = G_2(D) = \left[1, \frac{1+D^4}{1+D+D^2+D^3+D^4}\right]$

without puncturing of parity bits, and an interleaver length of $N = 1000$. Since for a fully interleaved Rayleigh fading channel with *perfect* side information on the fading samples, the channel matched cutoff rate corresponds to $\frac{E_b}{N_0} = 3.23$ dB then, according to the upper bounds depicted in Fig. 1.2, the ensemble performance of these turbo codes (associated with the ML decoding) is sufficiently robust in case of mismatched decoding, even in a portion of the rate region exceeding the channel matched cutoff rate. The proposed upper bounds depicted here were efficiently implemented in software, indicating their feasible computational complexity.

1.4 Lower Bounds on the Decoding Error Probability

1.4.1 De Caen Inequality and Variations

D. de Caen [59] suggested a lower bound on the probability of a finite union of events. While an elementary result (essentially, Cauchy-Schwartz inequality), this bound was used to compute lower bounds on the decoding error probability of linear block codes via their distance distribution (see [168] for the binary symmetric channel (BSC), and [294] for the Gaussian channel). In [71], Cohen and Merhav improved de Caen's inequality by introducing a weighting function which is subject to optimization. Their improved bound is presented in the following statement and, like de Caen's inequality, it follows from the Cauchy-Schwartz inequality.

Theorem 1.1. [71, Theorem 2.1] Let $\{A_i\}_{i\in\mathcal{I}}$ be an arbitrary set of events in a probability space $(\Omega, \mathcal{F}, P)$, then the probability of the union of these events is lower bounded by

$$P\left(\bigcup_{i\in\mathcal{I}} A_i\right) \geq \sum_{i\in\mathcal{I}}\left\{\frac{\left(\sum\limits_{x\in A_i} p(x)m_i(x)\right)^2}{\sum\limits_{j\in\mathcal{I}}\sum\limits_{x\in A_i\cap A_j} p(x)m_i(x)^2}\right\} \tag{1.3}$$

where $m_i(x) \geq 0$ is any real function on Ω such that the sums on the RHS of (1.3) converge. Further, equality in (1.3) is achieved when

$$m_i(x) = m^*(x) = \frac{1}{\deg(x)}\ , \quad \forall\, i \in \mathcal{I} \tag{1.4}$$

where for each $x \in \Omega$

$$\deg(x) \triangleq |\{i \in \mathcal{I} \mid x \in A_i\}|. \tag{1.5}$$

The lower bound on the union of events in Theorem 1.1 particularizes to de Caen's inequality by the particular choice of the weighting functions $m_i(x) = 1$ for all $i \in \mathcal{I}$, which then gives:

$$P\left(\bigcup_{i\in\mathcal{I}} A_i\right) \geq \sum_{i\in\mathcal{I}} \frac{P(A_i)^2}{\sum\limits_{j\in\mathcal{I}} P(A_i\cap A_j)^2}\ .$$

Cohen and Merhav relied on (1.3) for the derivation of improved lower bounds on the decoding error probability of linear codes under optimal ML decoding. They exemplified their bounds for BPSK modulated signals which are equally likely to be transmitted among M signals, and the examined communication channels were a BSC and an AWGN channel. In this context, the element x in

the RHS of (1.3) is replaced by the received vector $\mathbf{r}$ at the output of the communication channel (whose length K is equal to the dimension of the code), and A_i (where $i = 1, 2, \ldots, M-1$) consists of all the vectors of length K which are closer in the Euclidean sense to the signal $\mathbf{s}_i$ rather than the transmitted signal $\mathbf{s}_0$. Following [294], the bounds in [71] get (after some loosening in their tightness) final forms which solely depend on the distance spectrum of the code. Recently, two lower bounds on the ML decoding error probability of linear binary block codes were derived by Behnamfar et al. [30] for BPSK-modulated AWGN channels. These bounds are easier for numerical calculation, but are looser than Cohen-Merhav bounds for low to moderate SNRs.

Note that de Caen's based lower bounds on the decoding error probability (see [30], [71], [168] and [294]) are applicable for *specific* codes but not for ensembles; this restriction is due to the fact that Jensen's inequality does not allow to replace the distance spectrum of a linear code in these bounds by the average distance spectrum of ensembles.

1.4.2 Sphere-Packing Bounds Revisited for Moderate Block Lengths

In the asymptotic case where the block length of a code tends to infinity, the best known lower bound on the decoding error probability for high levels of noise is the 1967 sphere-packing (SP67) bound [300]. Like the random coding bound of Gallager [123], the sphere-packing bound decreases exponentially with the block length. Further, the error exponent of the SP67 bound is a convex function of the rate which is known to be tight at the portion of the rate region between the critical rate (R_{c}) and the channel capacity; for this important rate region, the error exponent of the SP67 bound coincides with the error exponent of the random coding bound [300, Part 1]. For the AWGN channel, the 1959 sphere-packing (SP59) bound was derived by Shannon [299] by first showing that the error probability of any code whose codewords lie on a sphere must be greater than the error probability of a code of the same length and rate whose codewords are uniformly distributed over that sphere.

The reason that the SP67 bound fails to provide useful results for codes of small to moderate block length is due to the original focus in [300] on asymptotic analysis. In their paper [334], Valembois and Fossorier have recently revisited the SP67 bound in order to make it applicable for codes of moderate block lengths, and also to extend its field of application to continuous output channels (e.g, the AWGN channel which is the communication channel model of the SP59 bound of Shannon [299]). The motivation for the study in [334] was strengthened due to the outstanding performance of codes defined on graphs even of moderate block length. The remarkable improvement in the tightness of the SP67 was exemplified in [334] for the case of the AWGN channel with BPSK signaling, and it was shown that the improved version of the SP67 is an interesting alternative to the SP59 bound [299].

1.4.3 Lower Bounds for Codes on Graphs

In [290, Theorem 2.5], Sason and Urbanke derived an information-theoretic lower bound on the bit error probability for binary linear block codes under ML decoding. The bound is expressed in terms of the *density* of an arbitrary parity-check matrix which represents a binary linear block code, and it is valid for any memoryless binary-input output-symmetric channel. The motivation for the derivation of this bound was due to the fact that the decoding complexity per iteration and the parity-check density are strongly linked when such a binary linear code is represented by a bipartite graph and an iterative message-passing decoding algorithm is applied to decode this code. The general idea used for the derivation of the lower bound on the bit error probability in [290,

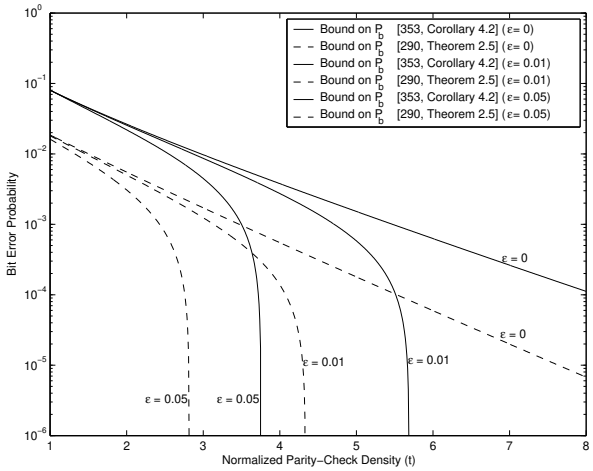


Figure 1.3: Lower bounds on the bit error probability for any binary linear block code transmitted over a binary-input AWGN channel whose capacity is $\frac{1}{2}$ bits per channel use. The bounds are depicted in terms of the normalized density of an arbitrary parity-check matrix which represents the code, and the curves correspond to code rates which are a fraction $1-\varepsilon$ of the channel capacity (for different values of ε). The bounds depicted in dashed lines are based on [290, Theorem 2.5], and the bounds in solid lines are given in [353, Corollary 4.2].

Theorem 2.5] was the derivation of a lower bound on the normalized conditional entropy $\frac{H(\mathbf{X}|\mathbf{Y})}{n}$, on one hand, where $\mathbf{X}$ and $\mathbf{Y}$ are the transmitted codeword and the received vector, respectively, and n designates the block length of the code; this lower bound on the conditional entropy was combined with the Fano inequality, in order to finally obtain a lower bound on the bit error probability. The bound in [290, Theorem 2.5] is expressed in terms of the *normalized density* of an arbitrary parity-check matrix which represents the binary linear code; this quantity is defined to be equal to $t = \left(\frac{R}{2-R}\right)\Delta$ where R is the code rate (in bits per channel use) and Δ designates the number of ones per information bit in the parity-check matrix. The reason for the scaling factor of $\frac{R}{2-R}$ is due to the fact that by doing so, the case where $t = 1$ corresponds to a bipartite graph without cycles, and otherwise $t > 1$ (the fundamental cycles in the bipartite graph grows linearly with t [290]). The tightness of the lower bounds was demonstrated in [290] for the binary erasure channel (BEC). Except for the BEC and BSC, the tightness of these bounds was recently improved by Wiechman and Sason [353]. The improved bound in [353] is based on improving the tightness of the lower bound on the conditional entropy; to this end, the derivation of the latter bound relies on the soft values of the log-likelihood ratio (LLR) at the output of the channel, instead of a two-level quantization of the LLR which was used in [290] (thus degrading the original memoryless binary-input output-symmetric channel into a BSC). Another refinement in the tightness of the resulting lower bound on the bit error probability was obtained by using the Fano inequality to get the inequality $\frac{H(\mathbf{X}|\mathbf{Y})}{n} \leq R h_2(P_b)$ where R is the code rate and $h_2(x) = -x\log_2(x) - (1-x)\log_2(1-x)$ designates the binary entropy function to the base 2 (the factor R in this upper bound was previously replaced by 1 in the derivation of [290, Theorem 2.5]).

A lower bound on the BER is shown in Fig. 1.3 for the binary-input AWGN channel. These bounds are plotted as a function of the normalized density of an arbitrary parity-check matrix. In this example, the capacity of the channel is equal to $\frac{1}{2}$ bit per channel use, and the bounds are depicted for binary linear block codes whose rate is a fraction $1-\varepsilon$ of the channel capacity. The plots in Fig. 1.3 show the superiority of the lower bound in [353, Eq. (78)] over the bound in [290, Theorem 2.5]. In order to exemplify the superiority of the former bound over the latter bound,

assume for example that one wishes to design a binary LDPC code which achieves a bit-error probability of 10^{-6} at a rate which is 99% of the channel capacity. The curve of the lower bound from [290] for $\varepsilon = 0.01$ implies that the normalized density of an arbitrary parity-check matrix which represents the code should be at least 4.33, while the curve depicting the bound from [353, Eq. (78)] strengthens this requirement to a normalized density (of each parity-check matrix) of at least 5.68. Translating this into terms of parity-check density (which is also the complexity per iteration for iterative message-passing decoding) yields minimal parity-check densities of 13.16 and 17.27, respectively (the minimal parity-check density is given by $\Delta_{\min} = \frac{(2-R)t_{\min}}{R}$). It is reflected from Fig. 1.3 that as the gap to capacity ε tends to zero, the lower bound on the normalized density (t) of an arbitrary parity-check matrix grows significantly; it actually behaves like $\log \frac{1}{\varepsilon}$, and becomes unbounded as the gap to capacity vanishes.

Chapter 2

Union Bounds: How Tight Can They Be ?

Chapter overview: Union bounds are reviewed shortly in this chapter, and their limited tightness for ensembles of turbo-like codes is addressed.

2.1 Union Bounds

Union bounds are based on the trivial inequality which states that the probability of a union of events is upper bounded by the sum of the probabilities of the individual events. Let $\{A_i\}_{i=1}^{M}$ designate a set of M events, then we get the inequality

$$\text{Prob}\left(\bigcup_{i=1}^{M} A_i\right) \leq \sum_{i=1}^{M} \text{Prob}\,(A_i). \tag{2.1}$$

It is evident that (2.1) turns to be an equality if these events are disjoint. Otherwise, it could be a very loose bound on the probability of a union of events (for instance, the right-hand side of (2.1) may exceed unity which makes the union bound useless).

Let us consider a binary linear code $\mathcal{C}$ whose transmission takes place over a memoryless, binary-input and output-symmetric (MBIOS) channel. Since the linearity of the code and the symmetry of the channel imply that the error probability under ML decoding does not depend on the transmitted codeword, then the average decoding error probability is equal to the conditional error probability given that the all-zero codeword was transmitted. Let $\{\mathbf{c}_0, \mathbf{c}_1, \ldots, \mathbf{c}_{M-1}\}$ be the set of codewords of the code $\mathcal{C}$ where M stands for the number of codewords, and let $\mathbf{c}_0$ stand for the all-zero codeword which, for simplicity of the analysis, is assumed to be the transmitted codeword. Let $\text{Prob}\,(\mathbf{c}_0 \to \mathbf{c}_i)$ be the pairwise error probability, i.e., the probability for deciding that another codeword $\mathbf{c}_i$ of the code $\mathcal{C}$ (where $1 \leq i \leq M-1$) is more likely to be the transmitted codeword than the codeword $\mathbf{c}_0$ which was actually transmitted (the decision is based on the received signal vector at the output of the channel demodulator). Then it follows immediately from the union bound in (2.1) that

$$\text{Prob}\,(\text{error}) \leq \sum_{i=1}^{M-1} \text{Prob}\,(\mathbf{c}_0 \to \mathbf{c}_i) \tag{2.2}$$

For simplicity we will consider here the case where the codewords are BPSK modulated before their transmission through the channel. The *input-output weight distribution* of an (N, K) binary linear block code $\mathcal{C}$, call it $A_{w,h}^{\mathcal{C}}$, is the number of codewords which are encoded by K information

bits whose (input) Hamming weight is equal to w, and the (output) Hamming weight of the N-bit codeword is h (where $0 \leq w \leq K$ and $0 \leq h \leq N$). The *weight distribution* of the code $\mathcal{C}$, call it $A_h^{\mathcal{C}}$, denotes the number of codewords of the code $\mathcal{C}$ which have a Hamming weight h (the weight distribution is also called the *distance spectrum* of the code $\mathcal{C}$, and clearly $A_h^{\mathcal{C}} = \sum_{w=0}^{K} A_{w,h}^{\mathcal{C}}$). Based on the Bhattacharyya bound, the pairwise error probability between any two codewords differing in h positions is upper bounded by z^h where z stands for the *Bhattacharyya constant*, i.e.,

$$z \triangleq -\ln \int_{-\infty}^{\infty} \sqrt{p(y|X=1)\, p(y|X=0)}\, dy \tag{2.3}$$

and $p(y|x)$ is the *pdf* of the MBIOS channel. From (2.2) and the Bhattacharyya bound, union bounds on the block error probability (P_{e}) and the bit error probability (P_{b}) of the code $\mathcal{C}$ get the form

$$P_{\mathrm{e}} \leq \sum_{h=1}^{N} \sum_{w=1}^{K} A_{w,h}^{\mathcal{C}} z^h \ , \tag{2.4}$$

$$P_{\mathrm{b}} \leq \sum_{h=1}^{N} \sum_{w=1}^{K} \frac{w}{K} A_{w,h}^{\mathcal{C}} z^h \ . \tag{2.5}$$

In order to write the union bound in a compact form, let us define the two-variables polynomial

$$A^{\mathcal{C}}(W,Z) \triangleq \sum_{h=0}^{N} \sum_{w=0}^{K} A_{w,h}^{\mathcal{C}} W^w Z^h \tag{2.6}$$

which designates the *input-output weight enumerator function* (IOWEF) of $\mathcal{C}$.[1] Then, the union bounds on the block error probability (2.4) and the bit error probability (2.5) are written as

$$P_{\mathrm{e}} \leq A^{\mathcal{C}}(1,z) - 1 \ , \qquad P_{\mathrm{b}} \leq \frac{1}{K} \left. \frac{\partial A^{\mathcal{C}}(W,Z)}{\partial W} \right|_{W=1,\, Z=z} \tag{2.7}$$

where z designates the Bhattacharyya constant in (2.3). The substraction of 1 in the upper bound on the block error probability follows from the fact that the all-zero codeword contributes to the IOWEF of the linear code $\mathcal{C}$ an addition by 1.

For the binary-input AWGN channel, the Bhattacharyya constant is equal to $z = e^{-\frac{E_{\mathrm{s}}}{N_0}}$ where $\frac{E_{\mathrm{s}}}{N_0}$ stands for the energy per symbol to the one-sided spectral noise density. Clearly, $\frac{E_{\mathrm{s}}}{N_0} = \frac{RE_{\mathrm{b}}}{N_0}$ where $R = \frac{K}{N}$ is the rate of the code $\mathcal{C}$, and $\frac{E_{\mathrm{b}}}{N_0}$ stands for the energy per information bit to the one-sided spectral noise density. The pairwise error probability between two BPSK modulated codewords differing in h positions and coherently detected over the AWGN channel is equal to $Q\left(\sqrt{\frac{2hRE_{\mathrm{b}}}{N_0}}\right)$ where

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-\frac{t^2}{2}}\, dt \tag{2.8}$$

is the probability that a random Gaussian variable with zero mean and unit variance exceeds the value x. Since $Q(x) \leq \frac{1}{2}\, e^{-\frac{x^2}{2}}$ for $x \geq 0$, then it follows that for the AWGN channel, multiplying the Bhattacharyya bound in (2.7) by a factor of one-half still gives a valid exponential upper bound. A tighter upper bound on the bit error probability is derived by using Craig's identity [72, 307]:

$$Q(x) = \frac{1}{\pi} \int_{0}^{\frac{\pi}{2}} e^{-\frac{x^2}{2\sin^2\theta}}\, d\theta \ , \quad x \geq 0 \ . \tag{2.9}$$

[1] A general technique for the calculation of IOWEFs of convolutional codes was proposed by McEliece [214]. For analytical methods to calculate the weight distribution of binary linear block codes, we refer the reader to [67, 75, 76, 173, 315, 358] and references therein.

With the aid of (2.9), we obtain the following upper bounds on the block and bit error probabilities of a binary linear block code $\mathcal{C}$

$$P_{\mathrm{e}} \leq \frac{1}{\pi} \int_0^{\frac{\pi}{2}} A^{\mathcal{C}}(W,Z) - 1 \Big|_{W=1,\, Z=e^{-\frac{RE_{\mathrm{b}}}{N_0 \sin^2 \theta}}} d\theta \,, \tag{2.10}$$

$$P_{\mathrm{b}} \leq \frac{1}{\pi K} \int_0^{\frac{\pi}{2}} \frac{\partial A^{\mathcal{C}}(W,Z)}{\partial W} \Big|_{W=1,\, Z=e^{-\frac{RE_{\mathrm{b}}}{N_0 \sin^2 \theta}}} d\theta \tag{2.11}$$

replacing the upper bounds

$$P_{\mathrm{e}} \leq \frac{1}{2} \left[A^{\mathcal{C}} \left(1, e^{-\frac{RE_{\mathrm{b}}}{N_0}} \right) - 1 \right], \quad P_{\mathrm{b}} \leq \frac{1}{2K} \frac{\partial A^{\mathcal{C}}(W,Z)}{\partial W} \Big|_{W=1,\, Z=e^{-\frac{RE_{\mathrm{b}}}{N_0}}}$$

which are obviously looser.

The weakness of the ubiquitous union bounding technique stems from the overlap between the decision regions in the calculation of the individual pairwise error probabilities. It is therefore not surprising that for long enough block codes which include lots of codewords, the union bounds become useless at low SNR values (the union bounds even exceed unity at low SNR, as will be later exemplified in Section 3.2.11); in fact, the union bound becomes useless at the rate portion between the cutoff rate and the channel capacity, where the performance of capacity-approaching codes (e.g., turbo codes [40, 41] and LDPC codes [208, 271]) is most appealing (see also the tutorial paper [114] which considers coding techniques for linear Gaussian channels, and the drawback of union bounds). This clearly motivates the need for improved upper bounds on the decoding error probability which are considered in the continuation of this chapter.

2.2 Union Bounds for Turbo-Like Codes

For long enough linear block codes, union bounds are not informative at rates exceeding the cut-off rate of the channel. Hence, union bounds which depend on the distance spectrum of the examined codes appear in general to be useless at the rate region where the performance of turbo-like codes is most appealing. The reader is referred to various reported works which explore the ensemble performance of turbo-like codes via union bounds. Union bounds on the ensemble performance of parallel and serially concatenated (turbo) codes with component codes which are time-varying recursive systematic convolutional codes are studied in [285, 324], union bounds on the ensemble performance of uniformly interleaved repeat-accumulate codes are studied in [85], and coding theorems for turbo-like ensembles which rely on union bounds are introduced in [160]. In general, union bounds are not informative at low signal to noise ratios, and only provide some insight about the interleaver gain which is obtained by various ensembles of turbo-like codes. In the continuation to this monograph, we will present numerical results for union bounds as a benchmark in order to exemplify the improvement in the tightness of various upper bounds over ubiquitous union bounds.

Chapter 3

Improved Upper Bounds For Gaussian and Fading Channels

Chapter overview: In recent years there has been renewed interest in deriving tight analytical bounds on the error performance of specific codes and ensembles. In this chapter, we present many reported upper bounds on the maximum-likelihood (ML) decoding error probability (including Berlekamp, Divsalar, Duman-Salehi, Engdahl-Zigangirov, Gallager, Hughes, Poltyrev, Sason-Shamai, Shulman-Feder, Viterbi, Yousefi-Khandani and others), and demonstrate the underlying connections that exist between them; the bounds are based on the distance spectrum or the input-output weight enumerators of the codes. The focus of this chapter is directed towards the application of efficient bounding techniques on ML decoding performance, which are not subjected to the deficiencies of the union bounds and therefore provide useful results at rates reasonably higher than the cut-off rate, where union bounds are usually useless. We apply improved upper bounds to block codes and turbo-like codes.

3.1 The Methodology of the Bounding Technique

In this chapter, we present many improved upper bounds which are tighter than the union bound. Let $\mathbf{r}$ be the received signal vector and let $\mathcal{R}$ be an arbitrary region around the transmitted signal point. Then the basic concept which is common to all of the improved upper bounds in this chapter is the following inequality which forms an upper bound on the decoding error probability:

$$\text{Prob}\,(\text{error}) \leq \text{Prob}\,(\text{error}, \mathbf{r} \in \mathcal{R}) + \text{Prob}\,(\mathbf{r} \notin \mathcal{R}). \tag{3.1}$$

The region $\mathcal{R}$ in (3.1) is interpreted as the "good region". The idea is to use the union bound only for the joint event where the decoder fails to decode correctly, and in addition, the received signal vector falls inside the region $\mathcal{R}$ (i.e., the union bound is used for upper bounding the first term in the right-hand side of (3.1)). On the other hand, the second term in the right-hand side in (3.1) represents the probability of the event where the received signal vector falls outside the region $\mathcal{R}$, and which is typically the dominant term for very low SNR, is calculated only one time (and it is not part of the event where the union bound is used). We note that in the case where the region $\mathcal{R}$ is the whole observation space, the basic approach which is suggested above provides the union bound. However, since the upper bound in (3.1) is valid for an arbitrary region $\mathcal{R}$ in the observation space, many improved upper bounds can be derived by an appropriate selection of this region. As we will see, the choice of the region $\mathcal{R}$ is very significant in this bounding technique; different choices of this region have resulted in various different improved upper bounds which will be considered in the continuation of this chapter. For instance, the tangential bound of Berlekamp [38] (which will be presented in Section 3.2.3) used the basic inequality in (3.1) to provide a considerably tighter bound than the union bound at low SNR values. This was achieved by separating the radial and tangential components of the Gaussian noise with a half-space as the underlying region $\mathcal{R}$. For the

derivation of the sphere bound [140], Herzberg and Poltyrev have chosen the region $\mathcal{R}$ in (3.1) to be a sphere around the transmitted signal vector, and optimized the radius of the sphere in order to get the tightest upper bound within this form. Divsalar bound [86] is another simple and tight bound which relies on the basic inequality (3.1) (the bound is presented in Section 3.2.4). The geometrical region $\mathcal{R}$ in Divsalar bound was chosen to be a sphere; in addition to the optimization of the radius of this sphere, the center of the sphere which does not necessarily coincide with the transmitted signal vector was optimized too. Finally, the tangential-sphere bound which was proposed for binary linear block codes by Poltyrev [259] and for M-ary PSK block coded-modulation schemes by Herzberg and Poltyrev [141] selected $\mathcal{R}$ as a conical region. It is one of the tightest upper bounds known to-date, and we will therefore choose to present it in detail in Section 3.2.1.

We note that the bounds mentioned above are only a sample of the so many bounds reported in this chapter, and all of them rely on the basic inequality (3.1). It was therefore important to provide the general approach at this stage, and then go into the details, show connections between these bounds and demonstrate their possible applications.

3.2 Improved Upper Bounds for the Gaussian Channel

3.2.1 Tangential-Sphere Bound (TSB)

The TSB was originally derived by Poltyrev [259], and then Herzberg and Poltyrev re-derived the TSB and applied it to the performance evaluation of PSK block coded modulation schemes [141]. In the following, we present the TSB and also discuss its geometrical interpretation (in this occasion, we wish to correct few printing typos which appear in the derivation of this bound in [141, 259], and to improve the lucidity of its derivation).

Consider a binary linear block code $\mathcal{C}$, and assume that its codewords are mapped to signals with constant energy. Let n and R be the block length and the rate of the code $\mathcal{C}$, respectively, and let E_{b} and E_{s} designate the energy per information bit and per coded symbol, respectively. By assumption, all the transmitted signals can be interpreted as points on an n-dimensional sphere with center at the origin and radius $r_{\mathrm{c}} = \sqrt{nE_{\mathrm{s}}}$ where $E_{\mathrm{s}} = RE_{\mathrm{b}}$. Since the channel is binary-input, output-symmetric and memoryless, and the code $\mathcal{C}$ is binary and linear, then without any loss of generality, one can assume that the all-zero codeword ($\mathbf{c}_0 = \underline{0}$) is transmitted (i.e., the conditional error probability does not depend on the transmitted codeword). We therefore assume in our analysis that $\mathbf{s}_0$ is the transmitted signal over the AWGN channel.

Referring to Fig. 3.1, let $C_n(\theta)$ designate an n-dimensional circular cone with a half-angle θ whose central line passes through the origin and the transmitted signal ($\mathbf{s}_0$). Let $\mathbf{z} = (z_1, z_2, \ldots, z_n)$ designate an n-dimensional noise vector which corresponds to n orthogonal projections of the AWGN. Let z_1 be the radial component of $\mathbf{z}$ (see Fig. 3.1), so the other $n-1$ components of $\mathbf{z}$ are orthogonal to its radial component. Since $\mathbf{z}$ is a Gaussian vector and its components are uncorrelated, then the n components of $\mathbf{z}$ are i.i.d., and each component has a zero mean and variance $\sigma^2 = \frac{N_0}{2}$. From Fig. 3.1, we obtain that

$$\begin{aligned} r &= \sqrt{nE_{\mathrm{s}}}\tan\theta \\ r_{z_1} &= \left(\sqrt{nE_{\mathrm{s}}} - z_1\right)\tan\theta \\ \beta_k(z_1) &= \left(\sqrt{nE_{\mathrm{s}}} - z_1\right)\tan\zeta = \frac{\sqrt{nE_{\mathrm{s}}} - z_1}{\sqrt{nE_{\mathrm{s}} - \frac{\delta_k^2}{4}}}\,\frac{\delta_k}{2}\,. \end{aligned} \tag{3.2}$$

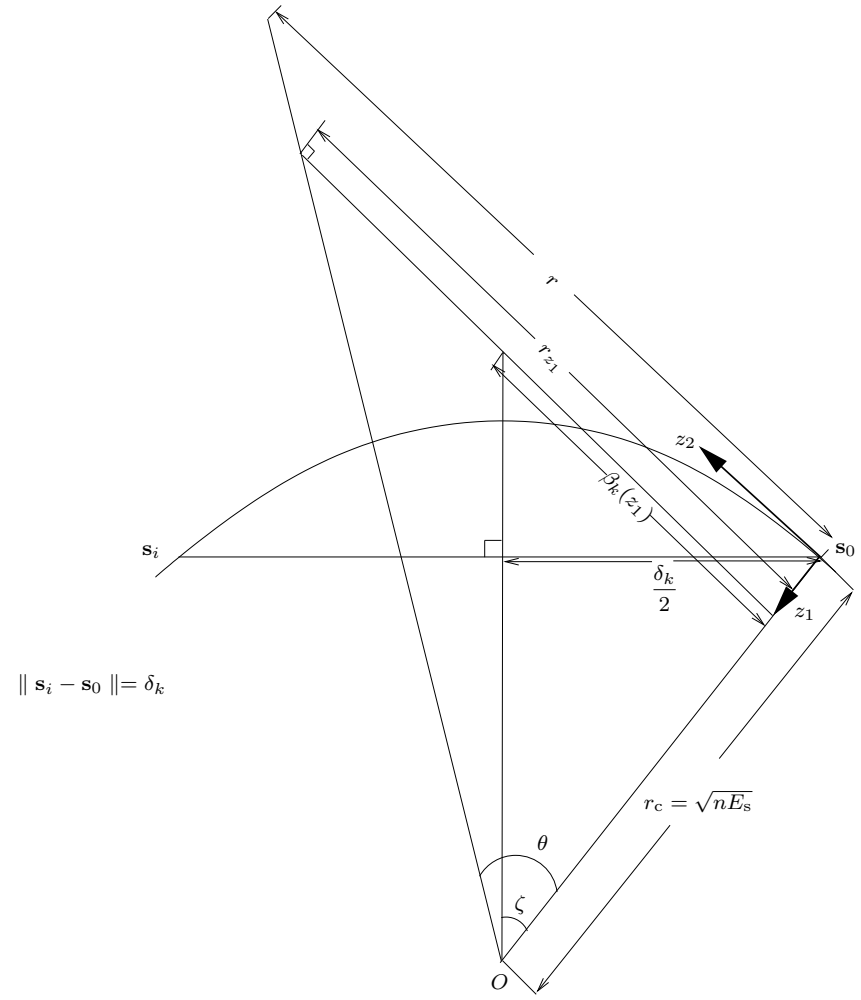


Figure 3.1: The geometric interpretation of the TSB [141, 259].

The random variable $Y \triangleq \sum_{i=2}^{n} z_i^2$ is χ^2 distributed with $n-1$ degrees of freedom, so its *pdf* is

$$f_Y(y) = \frac{y^{\frac{n-3}{2}}\ e^{-\frac{y}{2\sigma^2}}\ U(y)}{2^{\frac{n-1}{2}} \sigma^{n-1}\ \Gamma\left(\frac{n-1}{2}\right)}\ , \quad y \geq 0$$

where $U(\cdot)$ designates the unit step function, i.e., it is equal to 1 for non-negative arguments, and is zero otherwise. The function $\Gamma(\cdot)$ designates the complete Gamma function

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\ dt, \quad \text{Real}(x) > 0. \tag{3.3}$$

Conditioned on the value of the radial component of the noise, z_1, let $E(z_1)$ be the event of deciding erroneously (under ML decoding) on a codeword which is different from the transmitted codeword. Given the radial component of the noise, z_1, let $E_k(z_1)$ designate the event of deciding under ML decoding in favor of any other signal ($\mathbf{s}_i$) whose Euclidean distance from the transmitted signal ($\mathbf{s}_0$) is equal to δ_k.

Let $\mathbf{r} = \mathbf{s}_0 + \mathbf{z}$ be the received vector at the output of the binary-input AWGN channel. Given the value of the radial component of the noise vector, z_1, the conditional error probability satisfies

$$\text{Prob}\left(E(z_1)|\ z_1\right) \leq \text{Prob}\left(E(z_1),\ \mathbf{r} \in C_n(\theta)\ |\ z_1\right) + \text{Prob}\left(\mathbf{r} \notin C_n(\theta)\ |\ z_1\right) \tag{3.4}$$

and from the union bound

$$\text{Prob}\left(E(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) \leq \sum_k S_k \text{ Prob}\left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) \tag{3.5}$$

where S_k designates the number of the constant-energy signals ($\mathbf{s}_i$) in the considered signal set so that their Euclidean distance from the transmitted signal ($\mathbf{s}_0$) is δ_k. We note that for BPSK modulated signals where $\mathbf{s} = (2\mathbf{c} - \underline{1})\sqrt{E_\text{s}}$, the Euclidean distance between the two signals $\mathbf{s}_i$ and $\mathbf{s}_0$ is directly linked to the Hamming weight of the codeword $\mathbf{c}_i$. Let the Hamming distance between the two codewords be k, i.e., $w_\text{H}(\mathbf{c}_i) = k$, then the Euclidean distance between the two BPSK modulated signals is equal to $\delta_k = 2\sqrt{kE_\text{s}}$. In the latter case, S_k is the number of codewords of the code $\mathcal{C}$ with Hamming weight k (i.e., $\{S_k\}$ is the distance spectrum of the linear code $\mathcal{C}$).

The combination of Eqs. (3.4) and (3.5) gives

$$\begin{aligned}\text{Prob}\left(E(z_1)|\ z_1\right) \leq \sum_k \Big\{ & S_k \text{ Prob}\left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right)\Big\} \\ & + \text{Prob}\left(\mathbf{r} \notin C_n(\theta) \mid z_1\right).\end{aligned} \tag{3.6}$$

The second term in the right hand side of (3.6) is easily handled:

$$\begin{aligned}\text{Prob}\left(\mathbf{r} \notin C_n(\theta)|\ z_1\right) &= \text{Prob}\left(Y > r_{z_1}^2|\ z_1\right) \\ &= \int_{r_{z_1}^2}^{+\infty} f_Y(y)dy \\ &= \int_{r_{z_1}^2}^{+\infty} \frac{y^{\frac{n-3}{2}} e^{-\frac{y}{2\sigma^2}}}{2^{\frac{n-1}{2}}\ \sigma^{n-1}\ \Gamma\left(\frac{n-1}{2}\right)}\ dy.\end{aligned}$$

This integral is expressible in terms of the incomplete Gamma function

$$\gamma(a,x) \triangleq \frac{1}{\Gamma(a)} \int_0^x t^{a-1} e^{-t}\ dt, \quad a > 0,\ x \geq 0 \tag{3.7}$$

so we obtain that

$$\text{Prob}\left(\mathbf{r} \notin C_n(\theta)|\ z_1\right) = 1 - \gamma\left(\frac{n-1}{2},\ \frac{r_{z_1}^2}{2\sigma^2}\right). \tag{3.8}$$

Let z_2 be the tangential component of the noise vector $\mathbf{z}$ which is on the plane connecting the two signals $\mathbf{s}_0$ and $\mathbf{s}_i$, and orthogonal to the radial component z_1 (see Fig. 3.1). Referring to the first term in the right hand side of (3.6), it follows from the geometry in Fig. 3.1 that

$$\begin{aligned}\text{Prob}\ \left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) &= \text{Prob}\ \left(E_k(z_1), Y \leq r_{z_1}^2 \mid z_1\right) \\ &= \text{Prob}\ \left(\beta_k(z_1) \leq z_2 \leq r_{z_1}, Y \leq r_{z_1}^2\right).\end{aligned}$$

Let $V \triangleq \sum_{i=3}^n z_i^2$, then $V = Y - z_2^2$, and

$$\text{Prob}\ \left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) = \text{Prob}\ \left(\beta_k(z_1) \leq z_2 \leq r_{z_1}, V \leq r_{z_1}^2 - z_2^2\right).$$

The random variable V is χ^2 distributed with $n-2$ degrees of freedom, so its *pdf* is equal to

$$f_V(v) = \frac{v^{\frac{n-4}{2}}\ e^{-\frac{v}{2\sigma^2}}}{2^{\frac{n-2}{2}} \sigma^{n-2}\ \Gamma\left(\frac{n-2}{2}\right)}\ , \quad v \geq 0$$

and since the random variables z_2 and V are statistically independent, then

$$\mathrm{Prob}\left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) = \int_{\beta_k(z_1)}^{r_{z_1}} \frac{e^{-\frac{z_2^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} \int_0^{r_{z_1}^2 - z_2^2} f_V(v)\, dv\, dz_2. \tag{3.9}$$

The calculation of the statistical expectation of both sides of (3.6) w.r.t. the radial noise component (z_1) gives

$$\begin{aligned} P_{\mathrm{e}} &= E_{z_1}\left[\mathrm{Prob}\left(E(z_1)|\ z_1\right)\right] \\ &\leq \sum_{k:\, \beta_k(z_1) < r_{z_1}} \left\{ S_k\, E_{z_1}\Big[\mathrm{Prob}\left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right)\Big]\right\} \\ &\qquad + E_{z_1}\Big[\mathrm{Prob}\left(\mathbf{r} \notin C_n(\theta) \mid z_1\right)\Big] \end{aligned}$$

where the condition $\beta_k(z_1) < r_{z_1}$ in the above sum follows directly from the condition $\zeta < \theta$ (based on the geometry in Fig. 3.1 in p. 18), and therefore is independent of z_1. From (3.8) and (3.9), we obtain the following upper bound on the block error probability under ML decoding which only depends on the distance spectrum of the code:

$$\begin{aligned} P_{\mathrm{e}} \leq \int_{-\infty}^{+\infty} \frac{e^{-\frac{z_1^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} \Bigg\{ \sum_{k:\, \frac{\delta_k}{2} < \alpha_k} \left\{ S_k \int_{\beta_k(z_1)}^{r_{z_1}} \frac{e^{-\frac{z_2^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} \int_0^{r_{z_1}^2 - z_2^2} f_V(v)\, dv\, dz_2 \right\} \\ + 1 - \gamma\left(\frac{n-1}{2},\ \frac{r_{z_1}^2}{2\sigma^2}\right) \Bigg\}\, dz_1. \end{aligned} \tag{3.10}$$

The determination of the indices k which are taken into account in the summation in the RHS of (3.10) relies on the equivalence between the inequalities $\beta_k(z_1) < r_{z_1}$ and $\frac{\delta_k}{2} < \alpha_k$ where

$$\alpha_k \triangleq r\sqrt{1 - \frac{\delta_k^2}{4nE_{\mathrm{s}}}}. \tag{3.11}$$

The upper bound (3.10) on the ML decoding error probability is valid for all positive values of r. Hence, the optimal radius r (in the sense of achieving the tightest upper bound) is determined by setting to zero the partial derivative of the right side in (3.10) w.r.t. r_{z_1}. After tedious but straightforward algebra, we obtain the following optimization equation[1] for the optimal value of the radius r of the TSB:

$$\begin{cases} \displaystyle\sum_{k:\, \frac{\delta_k}{2} < \alpha_k} S_k \int_0^{\varphi_k} \sin^{n-3}\phi\, d\phi = \frac{\sqrt{\pi}\,\Gamma\left(\frac{n-2}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)} \\ \varphi_k \triangleq \cos^{-1}\left(\frac{\delta_k}{2\alpha_k}\right) \end{cases} \tag{3.12}$$

where α_k is given in (3.11). The nice property of (3.12) is that the optimized value of r *does not depend on the signal to noise ratio, but only on the distance spectrum of the code.* This enables to calculate the optimal value of r for a specific binary linear block code, and then to calculate the TSB in (3.10) for different values of $\frac{E_{\mathrm{b}}}{N_0}$. We note that the integrals in the left hand side of

[1] We note that in [259], the $\sqrt{\pi}$ in the first equation of (3.12) was written by mistake in the denominator instead of the numerator, which yields a sub-optimal choice of r in (3.17), and hence a looser upper bound in the final form (3.10) of the TSB.

the optimization equation (3.12) can be handled by invoking the identities in [129, Eqs. (2.511.2) & (2.511.3) (see p. 159)]. It is evident that the optimized TSB does not exceed 1 (since if we let $\theta \to 0$, then the n-dimensional cone $C_n(\theta)$ tends to an empty set; the second term of the RHS in (3.4) tends therefore to 1, which implies the trivial bound $P_{\mathrm{e}} \leq 1$). This is in contrast to the union bound which diverges at low values of E_{b}/N_0. A proof for the existence and uniqueness of a solution r to the optimization equation (3.12) was provided in [283, Appendix B], together with an efficient algorithm to solve numerically this equation. Sason and Shamai have adapted the TSB to obtain upper bounds on the *bit error probability* of a binary linear code $\mathcal{C}$ (see [283, Appendix C]). To this end, one replaces the Euclidean distance spectrum $\{S_k\}$ in the RHS of (3.10) by

$$S'_k = \sum_{w=1}^{K} \left(\frac{w}{K}\right) A_{w,k} \ , \quad k = d_{\min}, \ldots, n \tag{3.13}$$

where K designates the dimension of the linear code $\mathcal{C}$, and $A_{w,k}$ designates the number of the codewords which are encoded by information bits whose Hamming weight is w and mapped to signals of Euclidean distance δ_k from the transmitted signal.[2] By this replacement, the right hand side of (3.10) becomes an upper bound on the bit error probability of the code $\mathcal{C}$. Since

$$S_k = \sum_{w=1}^{K} A_{w,k} \ , \quad k = d_{\min}, \ldots, n \tag{3.14}$$

then it immediately follows from (3.13) and (3.14) that $S'_k \leq S_k$ for all integer values of k, so the resulting upper bound on the bit error probability is clearly smaller than the upper bound on the block error probability, as could be expected. For further observations on the derivation of the TSB-based upper bound on the bit error probability, we refer the reader to [283, Appenix C].

In the following, we suggest a slightly looser upper bound which is considerably easier for calculation. The inner integral in the RHS of (3.9) can be upper bounded with the aid of the incomplete Gamma function, as follows:

$$\begin{aligned}
&\mathrm{Prob}\left(E_k(z_1),\ \mathbf{r} \in C_n(\theta) \mid z_1\right) \\
&= \int_{\beta_k(z_1)}^{r_{z_1}} \frac{1}{\sqrt{2\pi}\sigma}\, e^{-\frac{z_2^2}{2\sigma^2}}\, \gamma\left(\frac{n-2}{2}, \frac{r_{z_1}^2 - z_2^2}{2\sigma^2}\right)\, dz_2 \\
&\leq \gamma\left(\frac{n-2}{2}, \frac{r_{z_1}^2 - \beta_k^2(z_1)}{2\sigma^2}\right)\, \int_{\beta_k(z_1)}^{r_{z_1}} \frac{1}{\sqrt{2\pi}\sigma}\, e^{-\frac{z_2^2}{2\sigma^2}}\, dz_2 \\
&= \gamma\left(\frac{n-2}{2}, \frac{r_{z_1}^2 - \beta_k^2(z_1)}{2\sigma^2}\right) \left[Q\left(\frac{\beta_k(z_1)}{\sigma}\right) - Q\left(\frac{r_{z_1}}{\sigma}\right)\right].
\end{aligned} \tag{3.15}$$

The suggested upper bound circumvents the need to numerically compute the inner integral which appears inside the sum of the RHS of (3.10) (such a numerical computation is required for every integer k for which $\delta_k < \frac{\alpha_k}{2}$ and $S_k \neq 0$). Finally, we obtain from (3.10) and (3.15) that the slightly looser version of the TSB upper bound on the block error probability (P_{e}) is only based on the

[2] We note that for BPSK modulated signals, $A_{w,k}$ designates the number of codewords of Hamming weight k which are encoded by a block of information bits of Hamming weight is w.

Euclidean distance spectrum $\{S_k\}$, and it reads:

$$\begin{aligned} P_{\mathrm{e}} \;\leq\; & \int_{-\infty}^{+\infty} \frac{dz_1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{z_1^2}{2\sigma^2}} \left\{ \begin{array}{l} 1-\gamma\left(\frac{n-1}{2}\,,\,\frac{r_{z_1}^2}{2\sigma^2}\right) \\ +\sum\limits_{k:\,\frac{\delta_k}{2}<\alpha_k} S_k\left[Q\left(\frac{\beta_k(z_1)}{\sigma}\right)-Q\left(\frac{r_{z_1}}{\sigma}\right)\right] \\ \qquad\qquad \cdot\gamma\left(\frac{n-2}{2}\,,\,\frac{r_{z_1}^2-\beta_k^2(z_1)}{2\sigma^2}\right) \end{array} \right\} \\ & + \; Q\left(\sqrt{\frac{2nRE_{\mathrm{b}}}{N_0}}\right) \end{aligned} \tag{3.16}$$

where

$$\left\{ \begin{array}{lcl} \sigma^2 & = & \frac{N_0}{2} \\ r_{z_1} & = & \left(1-\frac{z_1}{\sqrt{nE_{\mathrm{s}}}}\right) r \\ \beta_k(z_1) & = & \frac{r_{z_1}}{\sqrt{1-\frac{\delta_k^2}{4nE_{\mathrm{s}}}}} \cdot \frac{\delta_k}{2r} \end{array} \right. \tag{3.17}$$

and α_k is given in (3.11). We note that the need for the last summand in the RHS of (3.16) was explained in [283, Appendix A]; it is due to the possibility that the radial component of the noise vector (z_1) is large enough so that we refer to the second side of the cone (i.e., if $z_1 > \sqrt{nE_{\mathrm{s}}}$). This term didn't appear in the original derivation of the TSB in [141, 259], but it has typically a negligible effect on the upper bound (see [283, Appendix A]).

In order to significantly reduce the complexity which is involved with the calculation of the TSB and paying a very minor loss in the tightness of this bound, we refer in the continuation to the form of the TSB in (3.16) and the optimization equation in (3.12).

We emphasize that the TSB is not limited to binary linear codes (see e.g. [102] where the TSB is applied to the performance analysis of Reed-Solomon codes). The single property of the coding scheme which is required for the validity of the bound is the equal-energy property, so that the signals are represented by points on an n-dimensional sphere (see Fig. 3.1 in p. 18). We note that the constellation is not restricted to be geometrically uniform; in this case, if the signal set is still of equal energy, then one can use the proposed bounding technique to evaluate the conditional error probability given a particular transmitted signal, provided that the Euclidean distance spectrum w.r.t. that signal point is available.

In [371], Yousefi and Khandani generalize the derivation of the TSB. To this end, they choose to consider geometrical regions $\mathcal{R}$ in (3.1) which have an azimuthal symmetry w.r.t. the radial component of the noise. For equal-energy (sphere) codes, they prove that the optimal geometrical region is the conical region of the TSB, and therefore the generalized TSB is equal to the TSB of Poltyrev. This proves the optimality of the cones as geometrical regions for signals of equal energy, and it therefore justifies the tightness of the TSB, based on its geometrical interpretation.

In many communication systems, data is divided into different importance levels, and unequal error protection is desirable for taking into account their different error sensitivities. In [11], the TSB is applied to obtain performance bounds on the ML decoding error probability when binary linear block codes are transmitted over an AWGN channel and the bits are unequally error protected. These bounds are applied to uniformly interleaved turbo codes under ML decoding, and compared

with computer simulation results under iterative decoding; this comparison shows a good match, also at a portion of the rate region between the cutoff rate and the channel capacity.

The TSB of Poltyrev [259] happens often to be one of the tightest reported upper bounds for block codes which are transmitted over a binary-input AWGN channel and ML decoded (see e.g., [86, 140, 141, 236, 259, 282, 283, 284, 349, 376]). However, in the random coding setting, it fails to reproduce the random coding exponent (especially for high code rates) while the 1965 Gallager bound [123] (which is introduced in Section 4.2.1 in the continuation of this chapter) achieves the channel capacity for fully random block codes. For further details on the random coding error exponent of the TSB over the binary-input AWGN channel, we refer the reader to [259, Section 5].

3.2.2 Improvements on the TSB

Zangl and Herzog [376] improved the TSB on the bit error probability which was derived earlier by Sason and Shamai [283] (as explained in Section 3.2.1). The improvement in [376] refers to the term of the TSB which corresponds to the case where the noise is large enough so that the received signal vector falls outside the conical region around the transmitted signal vector (see Fig. 3.1). The basic idea in the improved upper bound of Zangl and Herzog is to replace the worst case assumption which assumes that if the received signal vector falls outside the conical region, then all the decoded information bits are wrong by a more refined calculation which takes into account the actual fraction of information bits which are decoded incorrectly in the latter case. We note that for the parallel and serial concatenated codes which are exemplified in [376], the improvement achieved by the new bound is rather small.

Yousefi and Khandani [372] derived a new upper bound on the block error probability of binary linear block codes whose transmission takes place over a binary-input AWGN channel, and where they are coherently detected and ML decoded. To this end, they used a Bonferroni-type inequality of the second degree [120, 154] (instead of the union bound) to get an upper bound on the joint probability of decoding error and the event that the received signal vector falls within the corresponding conical region around the transmitted signal vector. The basic idea in [372] relies on the inequality which states that if $\{A_i\}_{i=1}^M$ designates a set of M events, and A_i^C designates the complementary of the event A_i, then

$$\begin{aligned}
&\text{Prob}\left(\bigcup_{i=1}^{M} A_i\right) \\
&= \text{Prob}\,(A_1) + \text{Prob}\,(A_2 \cap A_1^C) + \ldots + \text{Prob}\,(A_M \cap A_1^C \ldots \cap A_{M-1}^C) \\
&\leq \text{Prob}\,(A_1) + \sum_{i=2}^{M} \text{Prob}\,(A_i \cap A_{\hat{i}}^C). \qquad (3.18)
\end{aligned}$$

where the indices $\hat{i} \in \{1, 2, \ldots, i-1\}$ are chosen arbitrarily for $2 \leq i \leq M$. Clearly, inequality (3.18) gives a tighter bound on the probability of a union of events than the union bound in (2.1). The concept of the inequality in (3.18) is applied to the first term in the right hand side of (3.4) (instead of the union bound in (3.5) which gives a looser upper bound on the joint probability of decoding error and the event where the received signal vector is inside the corresponding conical region in Fig. 3.1). Since the resulting upper bound in [372, Eq. (25)] cannot be calculated in terms of the distance spectrum of the code (or ensemble of codes), they needed to loosen the upper bound so that it will only depend on the distance spectrum. From the final form of the bound in [372, Eqs. (28), (30)–(33)] which involves the enlargement of the original codebook by all n-tuples of Hamming weight w (the parameter w is optimized numerically in the final form of the bound),

it is not clear that the latter bound is uniformly better than the TSB. Yousefi and Khandani claim that their new bound is tighter than the TSB [372], but it seems that proving such a claim is not trivial. We note that for two short BCH codes, it was shown numerically that the TSB is slightly worse than the new bound in [372], but a general proof for this claim is missing. Moreover, the upper bound in [372] is more complex for calculation than the TSB because of the introduction of the additional parameter (w) in the former bound which is optimized numerically (due to the enlargement of the original codebook by all vectors of Hamming weight w).

Recently, Mehrabian and Yousefi proposed in [220] an improved version of the TSB. The derivation of their proposed bound follows along the lines of the bound derived in [372], and the numerical results provided by the authors shows that the resulting improvement of the new bound as compared to the TSB is very marginal, but in parallel its computational complexity as compared to the TSB is significantly higher.

3.2.3 Tangential Bound (TB)

The TB was derived by Berlekamp [38, pp. 572–574]. It can be seen as a particular case of the TSB (see Section 3.2.1) where we let the radius r of the cone in Fig. 3.1 (see p. 18) tend to infinity. The derivation of the bound relies on the inequality

$$P_{\mathrm{e}} \leq \mathrm{Prob}\left(\mathrm{error}, z_1 \leq \gamma_0\right) + \mathrm{Prob}\left(z_1 > \gamma_0\right) \tag{3.19}$$

where z_1 is the radial component of the noise vector (see Fig. 3.1), and γ_0 is an arbitrary constant which is later optimized, as to obtain the tightest bound within this form. To proceed, a union bound like the one in (3.5) (see p. 19) is invoked in order to obtain an upper bound on the first term in the RHS of (3.19). The second term in the RHS of (3.19) is easy to calculate: since $z_1 \sim N(0, \sigma^2)$, then $\mathrm{Prob}\left(z_1 > \gamma_0\right) = Q\left(\frac{\gamma_0}{\sigma}\right)$. Referring to Fig. 3.1, we obtain the following inequality from (3.19):

$$P_{\mathrm{e}} \leq \sum_k \Big\{ S_k \, \mathrm{Prob}\left(z_1 \leq \gamma_0, \; z_2 \geq \beta_k(z_1)\right) \Big\} + Q\left(\frac{\gamma_0}{\sigma}\right) \tag{3.20}$$

and then from (3.2) (see p. 17) and (3.20), we obtain that

$$P_{\mathrm{e}} \leq \sum_k \left\{ S_k \int_{-\infty}^{\gamma_0} Q\left(\frac{\sqrt{nE_{\mathrm{s}}} - z_1}{\sqrt{nE_{\mathrm{s}} - \frac{\delta_k^2}{4}}} \, \frac{\delta_k}{2\sigma} \right) \frac{1}{\sqrt{2\pi}\sigma} \, e^{-\frac{z_1^2}{2\sigma^2}} \, dz_1 \right\} + Q\left(\frac{\gamma_0}{\sigma}\right). \tag{3.21}$$

The particular choice of $r \to \infty$ in Fig. 3.1 is sub-optimal for the minimization of the TSB (i.e., this choice does not necessarily lead to the tightest upper bound in (3.16)). It therefore follows that the TB is inherently looser than the TSB (as was proved in [141, Lemma 2]). The optimal value of γ_0 in the RHS of (3.21) is calculated by setting its partial derivative w.r.t. γ_0 to zero, which gives the following equation:[3]

$$\sum_k \left\{ S_k \, Q\left(\frac{\sqrt{nE_{\mathrm{s}}} - \gamma_0}{\sqrt{nE_{\mathrm{s}} - \frac{\delta_k^2}{4}}} \, \frac{\delta_k}{2\sigma} \right) \right\} = 1. \tag{3.22}$$

In the following, we derive a slightly looser bound which is easier for calculation than the TB in (3.21). Let

$$u = z_1 \, \sin\zeta + z_2 \, \cos\zeta, \quad v = -z_1 \, \cos\zeta + z_2 \, \sin\zeta \tag{3.23}$$

[3] Due to the monotonicity of the Q function, the existence and uniqueness of a solution γ_0 in the implicit equation (3.22) is always insured; the monotonicity property makes the numerical solution of γ_0 in (3.22) an easy task (e.g., by using the bisection method).

be the noise components which are received by rotating z_1 and z_2, respectively, in the clockwise direction by an angle $\xi \triangleq \frac{\pi}{2} - \zeta$; it is evident from Fig. 3.1 that $\sin\zeta = \frac{\delta_k}{2\sqrt{nE_{\rm s}}}$. The condition for an erroneous pairwise ML decoding in Fig. 3.1 is equivalent to the satisfiability of the condition $u \geq \frac{\delta_k}{2}$. It is therefore easy to verify from (3.23) that if the ML decoder decides in favor of $\mathbf{s}_i$ (instead of the transmitted signal point $\mathbf{s}_0$) and $z_1 \leq \gamma_0$, then

$$u \geq \frac{\delta_k}{2}\ , \quad v \geq \frac{\frac{\delta_k^2}{4} - \gamma_0\sqrt{nE_{\rm s}}}{\sqrt{nE_{\rm s} - \frac{\delta_k^2}{4}}}. \tag{3.24}$$

Since the random variables u and v in (3.23) are un-correlated and jointly Gaussian distributed with zero mean and variance σ^2, then they are statistically independent. Based on (3.20) and (3.24), we obtain the following upper bound on the ML decoding error probability:

$$P_{\rm e} \leq \sum_k \left\{ S_k\, Q\left(\frac{\delta_k}{2\sigma}\right)\ Q\left(\frac{\frac{\delta_k^2}{4} - \gamma_0\sqrt{nE_{\rm s}}}{\sqrt{nE_{\rm s} - \frac{\delta_k^2}{4}}\ \sigma}\right)\right\} + Q\left(\frac{\gamma_0}{\sigma}\right) \tag{3.25}$$

The bound stated in (3.25), together with the corresponding optimized γ_0 (computed by setting to zero the partial derivative of the function in the RHS of (3.25) w.r.t. γ_0), form a simplified version of the TB. Although the TB (and hence, its simplified version) are always looser than the TSB, they are both uniformly tighter than the union bound (as was proved in [141, Lemma 1]). The latter thing is easily proved by letting γ_0 in the simplified version of the TB (3.25) tend to infinity; if $\gamma_0 \to \infty$, then we obtain from (3.25) that $P_{\rm e} \leq \sum_k S_k\, Q\left(\frac{\delta_k}{2\sigma}\right)$ which coincides with the union bound, and therefore shows that the simplified version of the TB (3.25) (and hence, the TB in (3.21) and (3.22)) are always tighter than the union bound. We note that in the case where $\gamma_0 \to -\infty$, then (3.25) gives the trivial bound $P_{\rm e} \leq 1$; this shows that like the TSB (but in contrast to the union bound), the TB and its simplified version do not diverge at low SNR values.

3.2.4 Divsalar Bound

In [86], Divsalar derived a simple upper bound on the ML decoding error probability of linear block codes whose transmission takes place over an AWGN channel. The simplicity of the bound stems from the fact that it is given in closed form, and its calculation does not involve numerical integration and parameter optimization. It is therefore widely used for calculating bounds on the thresholds of turbo-like code ensembles under ML decoding, and for assessing the decoding error probability of linear block codes which are communicated over a binary-input AWGN channel (see, e.g., [1, 86, 87, 156, 157, 189, 190, 204, 289, 296]).

In the following, we present Divsalar bound [86]. In the derivation of the bound here, we will prove the final expressions of the Chernoff bounds in [86, Eqs. (14)–(17)] (their proofs are missing in [86]); on the other hand, we will skip the technical details related to the derivation of the final closed form expressions for the optimized parameters in this bound, and refer the reader to [86].

Consider an (n, k) binary linear code $\mathcal{C}$ with rate $R = \frac{k}{n}$, and let $\{S_d\}$ designate its distance spectrum. Suppose the code is BPSK modulated and transmitted over an AWGN channel. Let $\mathbf{c}_i \in \mathcal{C}$ be an arbitrary codeword, and let $\mathbf{x}_i$ designate the vector where zeros and ones in the codeword $\mathbf{c}_i$ are mapped to $+1$ and -1, respectively (due to the BPSK modulation). Let $\mathbf{c}_i$ and $\mathbf{r}$ be an arbitrary transmitted codeword and the corresponding n dimensional vector at the output of the channel, respectively. By scaling the vector $\mathbf{r} = (r_1, r_2, \ldots, r_n)$, it is modeled as

$$r_j = \gamma\, x_{i,j} + n_j\ , \quad j = 1, 2, \ldots, n$$

where $\gamma \triangleq \sqrt{\frac{2RE_{\rm b}}{N_0}}$, $x_{i,j}$ is the j-th component of $\mathbf{x}_i$, and n_j is a zero mean and unit-variance Gaussian noise sample. Due to the symmetry of the channel and the linearity of the code, the conditional error probability does not depend on the transmitted codeword, so we assume that the all-zero codeword ($\mathbf{c}_0$) is the transmitted codeword. An error occurs under ML decoding if there exists $i \in \{1, 2, \cdots, 2^{nR}-1\}$ such that $\sum_{j=1}^{n} r_j \; x_{i,j} > \sum_{j=1}^{n} r_j \; x_{0,j}$. Now, let $\mathcal{C}_d$ (for $d = 1, 2, \cdots, n$) designate the subset of all the codewords of $\mathbf{c} \in \mathcal{C}$ that their Hamming weight is d; the size of the subset $\mathcal{C}_d$ is clearly equal to S_d. We designate by E_d the error event where there exists a codeword $\mathbf{c} \in \mathcal{C}_d$ which is chosen by the ML decoder in preference to the $\mathbf{c}_0$ codeword. Based on the union bound, the decoding error probability is bounded by $P_{\rm e} \leq \sum_{d>0} \text{Prob}\left\{E_d|\mathbf{x}_0\right\}$. Divsalar relied on the inequality

$$\text{Prob}\left\{E_d|\mathbf{x}_0\right\} \leq \text{Prob}\left\{E_d, \mathbf{r} \in \mathcal{R}|\; \mathbf{x}_0\right\} + \text{Prob}\left\{\mathbf{r} \notin \mathcal{R}|\; \mathbf{x}_0\right\} \tag{3.26}$$

which was used earlier for the derivation of the TSB and the TB, and where (3.26) holds for any region $\mathcal{R}$ in the n-dimensional observation space. Referring to Fig. 3.2, Divsalar defined the region $\mathcal{R}$ to be an n-dimensional sphere with radius $\sqrt{nR^2}$ (R is later optimized). The center of that sphere is at $\mu\gamma\mathbf{x_0}$, a point along the line connecting the origin to the transmitted codeword $\mathbf{x_0}$ (see Fig. 3.2). Based on the union bound and the choice of $\mathcal{R}$ in Fig. 3.2, we obtain the inequality

$$\begin{aligned}
&\text{Prob}\left\{E_d|\mathbf{x}_0\right\} \\
&\leq \sum_{\mathbf{c}_i \in \mathcal{C}_d} \text{Prob}\left\{\sum_{j=1}^{n} r_j \; x_{i,j} > \sum_{j=1}^{n} r_j \; x_{0,j}, \; \sum_{j=1}^{n} (r_j - \eta x_{0,j}\gamma)^2 \leq nR^2 \mid \mathbf{x}_0\right\} \\
&\quad + \text{Prob}\left\{\sum_{j=1}^{n} (r_j - \eta x_{0,j}\gamma)^2 > nR^2 \mid \mathbf{x}_0\right\}.
\end{aligned} \tag{3.27}$$

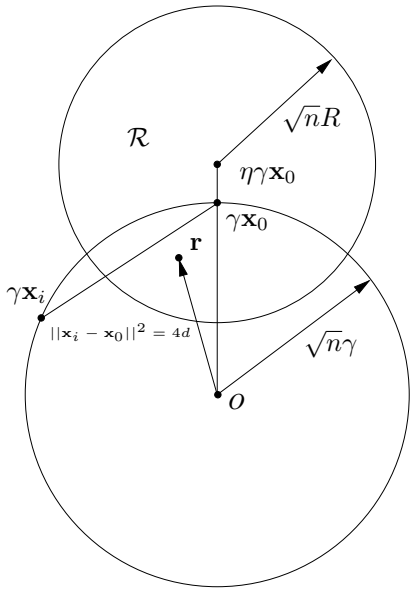


Figure 3.2: The geometric interpretation of Divsalar bound [86].

Let

$$Z = \sum_{j=1}^{n} r_j \; x_{i,j} - \sum_{j=1}^{n} r_j \; x_{0,j}, \quad W = \sum_{j=1}^{n} (r_j - \eta x_{0,j}\gamma)^2 - nR^2$$

then, based on the Chernoff bounds

$$\text{Prob}\left\{Z \geq 0, W \leq 0\right\} \leq E\left\{e^{tZ+rW}\right\}, \quad \forall\, t \geq 0,\ r \leq 0 \tag{3.28}$$

and

$$\text{Prob}\left\{W \geq 0\right\} \leq E\left\{e^{sW}\right\}, \quad \forall\, s \geq 0 \tag{3.29}$$

Divsalar obtained in [86] Chernoff upper bounds for the two terms in the right-hand side of (3.27). We derive here the two Chernoff bounds. To this end, one relies on the equality which states that if $X \sim N(m, \sigma^2)$, then

$$E\left[e^{sX^2}\right] = \frac{e^{\frac{m^2 s}{1-2\sigma^2 s}}}{\sqrt{1-2\sigma^2 s}}, \quad \forall\, s < \frac{1}{2\sigma^2}. \tag{3.30}$$

Since it is assumed that the all-zero codeword is transmitted, then $\mathbf{x}_0 = (1, 1, \ldots, 1)$, and from (3.29) and (3.30) we obtain that

$$\begin{aligned}
&\text{Prob}\left\{\sum_{j=1}^{n}(r_j - \eta x_{0,j}\gamma)^2 > nR^2 \mid \mathbf{x}_0\right\} \\
&\leq E\left[e^{s\left(\sum_{k=1}^{n}(r_k - \eta x_{0,k}\gamma)^2 - nR^2\right)} \mid \mathbf{x}_0\right] \qquad s \geq 0 \\
&= e^{-nsR^2}\left(f_1(\gamma, s, \eta)\right)^n \triangleq e^{-nsR^2}\, A, \quad 0 < s < \frac{1}{2}
\end{aligned} \tag{3.31}$$

where

$$f_1(\gamma, s, \eta) = \frac{e^{\frac{(1-\eta)^2\gamma^2 s}{1-2s}}}{\sqrt{1-2s}}. \tag{3.32}$$

For $\mathbf{c}_i \in \mathcal{C}_d$, there are d indices $j \in \{1, 2, \ldots, n\}$ where $x_{i,j} = -1$, and for the other indices of j: $x_{i,j} = 1$. We therefore obtain from the Chernoff bound in (3.28) and the moment generating function in (3.30) that for an arbitrary codeword $\mathbf{c}_i \in \mathcal{C}_d$, and for arbitrary $t \geq 0$ and $r \leq 0$

$$\begin{aligned}
&\text{Prob}\left\{\sum_{j=1}^{n} r_j\, x_{i,j} > \sum_{j=1}^{n} r_j\, x_{0,j},\ \sum_{j=1}^{n}(r_j - \eta x_{0,j}\gamma)^2 \leq nR^2 \mid \mathbf{x}_0\right\} \\
&\leq E\left[e^{t\left(\sum_{k=1}^{n} r_k x_{i,k} - \sum_{k=1}^{n} r_k x_{0,k}\right) + r\left(\sum_{k=1}^{n}(r_k - \eta x_{0,k}\gamma)^2 - nR^2\right)} \mid \mathbf{x}_0\right] \\
&= e^{-nrR^2}\left(f_1(\gamma, r, \eta)\right)^{n-d} \left(\frac{e^{\frac{r}{1-2r}\left[(1-\eta)\gamma - \frac{t}{r}\right]^2 - 2\eta\gamma t - \frac{t^2}{r}}}{\sqrt{1-2r}}\right)^d.
\end{aligned} \tag{3.33}$$

The minimization of the exponent in (3.33) w.r.t. t gives that

$$t = \frac{\gamma}{2}\,(1 - 2r\eta)$$

(t is indeed non-negative, since $r \leq 0$), so the substitution of t in (3.33) yields that for all codewords $\mathbf{c}_i \in \mathcal{C}_d$

$$\begin{aligned}
&\text{Prob}\left\{\sum_{j=1}^{n} r_j\, x_{i,j} > \sum_{j=1}^{n} r_j\, x_{0,j},\ \sum_{j=1}^{n}(r_j - \eta x_{0,j}\gamma)^2 \leq nR^2 \mid \mathbf{x}_0\right\} \\
&\leq \left(f_1(\gamma, r, \eta)\right)^{n-d}\left(f_2(\gamma, r, \eta)\right)^d e^{-nrR^2}
\end{aligned} \tag{3.34}$$

where the function f_1 is introduced in (3.32) and

$$f_2(\gamma, r, \eta) = \frac{e^{-\frac{\gamma^2(1-2r\eta^2)}{2}}}{\sqrt{1-2r}} \ . \tag{3.35}$$

Eqs. (3.34) and (3.35) yield therefore the upper bound

$$\begin{aligned} &\sum_{\mathbf{c}_i \in \mathcal{C}_d} \text{Prob} \left\{ \sum_{j=1}^{n} r_j \ x_{i,j} > \sum_{j=1}^{n} r_j \ x_{0,j}, \ \sum_{j=1}^{n} (r_j - \eta x_{0,j}\gamma)^2 \le nR^2 \mid \mathbf{x}_0 \right\} \\ &\le S_d \Big(f_1(\gamma, r, \eta) \Big)^{n-d} \ \Big(f_2(\gamma, r, \eta) \Big)^d \ e^{-nrR^2} \triangleq e^{-nrR^2} B, \ \ r \le 0. \end{aligned} \tag{3.36}$$

From (3.27), (3.31) and (3.36), the minimization of the resulting upper bound on Prob $\{E_d|\mathbf{x}_0\}$ with respect to e^{nR^2} gives

$$\text{Prob} \ \{E_d|\mathbf{x}_0\} \le 2^{h(\frac{s}{s-r})} \ A^{-\frac{r}{s-r}} \ B^{\frac{s}{s-r}} \ , \quad 0 < s < \frac{1}{2}, \ r \le 0$$

where $h(x) = -x \log_2(x) - (1-x) \log_2(1-x)$ is the binary entropy function to the base 2. Instead of the parameters s, r and η, Divsalar introduced in [86] the three new parameters

$$\rho = \frac{s}{s-r}, \ \beta = \rho(1-2r), \ \xi = \rho(1-2r\eta) \quad (0 \le \rho \le 1, \ \beta \ge 0, \ \xi \ge 0) \tag{3.37}$$

Let us define $c \triangleq \frac{E_\text{s}}{N_0}$, the normalized Hamming weight as $\delta \triangleq \frac{d}{n}$, and the growth rate of the distance spectrum as $r_n(\delta) \triangleq \frac{\ln S_d}{n}$. The resulting upper bound gets the form

$$\text{Prob} \ \{E_d|\mathbf{x}_0\} \le 2^{h(\rho)} \ e^{-nE(c,\delta,\beta,\rho,\xi)}$$

where

$$\begin{aligned} E(c, \delta, \beta, \rho, \xi) \ &= \ -\rho \ r_n(\delta) - \frac{\rho}{2} \ \ln\left(\frac{\rho}{\beta}\right) - \frac{1-\rho}{2} \ \ln\left(\frac{1-\rho}{1-\beta}\right) \\ &\quad + c \left[1 - (1-\delta) \ \frac{\xi^2}{\beta} - \frac{(1-\xi)^2}{1-\beta} \right]. \end{aligned} \tag{3.38}$$

The parameters ρ, ξ and β were optimized in [86, pp. 6–7] as to get the maximal error exponent in (3.38) (and hence, the tightest upper bound within this form). Their optimal values are

$$\rho = \frac{\beta}{\beta + (1-\beta)e^{2r_n(\delta)}} \ , \qquad \xi = \frac{\beta}{\beta + (1-\beta)(1-\delta)}$$

where the optimal value of β is given by

$$\beta = \sqrt{\frac{c(1-\delta)}{\delta} \frac{2}{1 - e^{-2r_n(\delta)}} + \left(\frac{1-\delta}{\delta}\right)^2 \ \Big[(1+c)^2 - 1\Big]} - \frac{1-\delta}{\delta} \cdot (1+c) \tag{3.39}$$

and $c \triangleq \frac{E_\text{s}}{N_0} = \frac{RE_\text{b}}{N_0}$.

Let d_min and d_max designate, respectively, the minimal and the maximal Hamming weight (d) so that $S_d \neq 0$. The final form of Divsalar bound on the block error probability is

$$P_\text{e} \le \sum_{d=d_\text{min}}^{d_\text{max}} \ \min \left\{ e^{-nE(\delta, \beta, \frac{E_\text{s}}{N_0})}, S_d \ Q\left(\sqrt{\frac{2dE_\text{s}}{N_0}}\right) \right\} \tag{3.40}$$

where

$$E(\delta,\beta,\frac{E_{\mathrm{s}}}{N_0}) = -r_n(\delta) + \frac{1}{2}\,\ln\Big(\beta + (1-\beta)e^{2r_n(\delta)}\Big) + \frac{\beta\delta}{1-(1-\beta)\delta}\,\frac{E_{\mathrm{s}}}{N_0} \tag{3.41}$$

and β is given in (3.39). We note that if $\beta = 1$, the bound in (3.40) reduces to the union bound, so since β in (3.39) is the optimized value, then it follows immediately that Divsalar bound is inherently tighter the union bound. To compute an upper bound on the bit error probability, the distance spectrum S_d is replaced by $\sum_{w=1}^{nR}\big\{\big(\frac{w}{nR}\big)\ A_{w,d}\big\}$ where $A_{w,d}$ designates the number of codewords whose Hamming weight is equal to d, and which are encoded by information bits of Hamming weight w (i.e., $A_{w,d}$ designates the input-output weight enumerator of the code $\mathcal{C}$).

Divsalar bound provides a closed-form upper bound on the threshold of the signal to noise ratio which yields vanishing block error probability under ML decoding. If we let the block length tend to infinity, then the upper bound on the $\frac{E_{\mathrm{b}}}{N_0}$ threshold [86, Eq. (33)] is given by

$$\left(\frac{E_{\mathrm{b}}}{N_0}\right)_{\mathrm{threshold}} \leq \frac{1}{R}\ \max_{0<\delta<1}\ \frac{(1-\delta)(1-e^{-2r(\delta)})}{2\delta} \tag{3.42}$$

where $r(\delta) = \lim_{n\to\infty} r_n(\delta)$ designates the asymptotic growth rate of the distance spectrum.

3.2.5 Sphere Upper Bound

Herzberg and Poltyrev derived the sphere upper bound [140, Section 2A]. This bound relies on the basic inequality in (3.1) where the region $\mathcal{R}$ is chosen to be a sphere whose center coincides with the signal point ($\mathbf{s}_0$) which represents the transmitted signal. Let $\mathbf{z}$ designate the AWGN vector, then we obtain from the (3.1) and the union bound w.r.t. the sphere region that

$$\begin{aligned} P_{\mathrm{e}} &\leq \mathrm{Prob}\,(\mathrm{error},\ ||\mathbf{z}|| \leq r) + \mathrm{Prob}\,(||\mathbf{z}|| > r) \\ &\leq \sum_k \Big\{ S_k \cdot \mathrm{Prob}\,(E_k,\ ||\mathbf{z}|| \leq r)\Big\} + \mathrm{Prob}\,(||\mathbf{z}|| > r) \end{aligned} \tag{3.43}$$

where E_k stands for the error event at the output of the ML decoder, while the decoded signal is at an Euclidean distance δ_k from $\mathbf{s}_0$, and S_k designates the number of signal points at distance δ_k from $\mathbf{s}_0$. Since the joint probability $\mathrm{Prob}\,(E_k,\ ||\mathbf{z}|| \leq r)$ is equal to zero for $r \leq \frac{\delta_k}{2}$, then the summation in (3.43) can be restricted to values of k such that $r > \frac{\delta_k}{2}$. Let $\delta_1, \delta_2, \ldots, \delta_k$ be the sequence of Euclidean distances from $\mathbf{s}_0$ which are ordered in an increasing order, and let $N(r)$ designate the maximal positive integer k so that $r > \frac{\delta_k}{2}$. Then, we obtain from (3.43) that

$$P_{\mathrm{e}} \leq \min_{r>0}\left\{ \sum_{k=1}^{N(r)} \Big\{ S_k \cdot \mathrm{Prob}\,(E_k,\ ||\mathbf{z}|| \leq r)\Big\} + \mathrm{Prob}\,(||\mathbf{z}|| > r)\ \right\} \tag{3.44}$$

Referring to Fig. 3.2, the sphere bound is obtained for the particular case where $\eta = 1$, but the calculations of the probabilities in the right-hand side of (3.44) are done exactly, without the need for Chernoff bounds. It is also easily observed that in the limit where $r \to \infty$, the sphere bound in (3.43) coincides with the union bound, so the sphere bound with the optimized radius r in (3.44) cannot be worse than the union bound. On the other hand, in the limit where $r \to 0$, the sphere bound becomes the trivial bound of unity (where on the other hand, the union bound may exceed unity at low signal to noise ratios).

Let z_1 be the component of the noise vector $\mathbf{z}$ in the direction of the line connecting $\mathbf{s}_0$ with $\mathbf{s}_k$, and let $Y \triangleq \sum_{i=1}^{n} z_i^2$ be the magnitude of the noise vector $\mathbf{z}$. Due to the spherical symmetry of

the AWGN, we obtain that

$$\begin{aligned}\text{Prob}\left(E_k,\ ||\mathbf{z}|| \le r)\right\} &= \text{Prob}\left(\frac{\delta_k}{2} \le z_1 \le r,\ Y \le r^2\right) \\ &= \int_0^{r^2} \int_{\frac{\delta_k}{2}}^{r} p_{z_1,Y}(z_1, y) dz_1 dy \end{aligned} \tag{3.45}$$

where

$$p_{z_1,Y}(z_1, y) = \frac{(y - z_1^2)^{\frac{n-3}{2}} e^{-\frac{y}{2\sigma^2}}\ U(y - z_1^2)}{\sqrt{\pi} 2^{\frac{n}{2}} \sigma^n \Gamma\left(\frac{n-1}{2}\right)} \tag{3.46}$$

is the joint probability density function of z_1 and Y; since z_1 and $Y - z_1^2 = \sum_{i=2}^{n} z_i^2$ are i.i.d., the pdf in (3.46) is calculated by multiplying the normal pdf of z_1 and the pdf of $Y - z_1^2$ (which is a χ^2 distribution with $n-1$ degrees of freedom). The functions $U(\cdot)$ and $\Gamma(\cdot)$ in (3.46) are the unit step function and the complete Gamma function (see (3.3)), respectively. Similarly, since the random variable Y is χ^2 distributed with n degrees of freedom, then we obtain that the second probability in the right-hand side of (3.44) is equal to

$$\begin{aligned}\text{Prob}\left(||\mathbf{z}|| > r\right) &= \text{Prob}\left(Y > r^2\right) \\ &= \int_{r^2}^{\infty} \frac{y^{\frac{n-2}{2}}\ e^{-\frac{y}{2\sigma^2}}}{2^{\frac{n}{2}}\ \sigma^n\ \Gamma\left(\frac{n}{2}\right)}\ dy \\ &= 1 - \gamma\left(\frac{n}{2},\ \frac{r^2}{2\sigma^2}\right) \end{aligned} \tag{3.47}$$

where $\gamma(\cdot, \cdot)$ designates the incomplete Gamma function introduced in (3.7). Finally, the sphere upper bound follows by the substitution of (3.45)–(3.47) into (3.44). The optimization of the radius r in the sphere upper bound (3.44) is obtained by a numerical solution of the equation

$$\sum_{k=1}^{N(r)} S_k \int_0^{\theta_k} \sin^{n-2}\phi\ d\phi = \frac{\sqrt{\pi}\ \Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \tag{3.48}$$

where

$$\theta_k \triangleq \cos^{-1}\left(\frac{\delta_k}{2r}\right)\ , \quad k = 1, 2, \cdots, N(r).$$

3.2.6 Engdahl and Zigangirov (EZ) Bound

In [103], Engdahl and Zigangirov derived new upper bounds on the bit and burst error probabilities, when a convolutional code is used over an AWGN channel. To this end, they break the convolutional code into a collection of block codes with increasing block lengths l (i.e., the lengths of unmerged paths in the trellis diagram of the code). Without any loss of generality, it is assumed that the all zero sequence is transmitted, and that the coded binary symbols 0 and 1 corresponds to the antipodal signaling $+\sqrt{E_s}$ and $-\sqrt{E_s}$, respectively. Let us assume that the convolutional code is of rate $R_c = \frac{b}{c}$, and let $\mathbf{r}_t = (r_t^{(1)}, r_t^{(2)}, \ldots, r_t^{(c)})$ designate the received symbols at the output of the AWGN channel at time t. By introducing the random variable

$$X = \sum_{t=0}^{l-1} \sum_{i=1}^{c} r_t^{(i)}\ , \quad l = m+1, m+2, \ldots$$

which corresponds to a path in the trellis diagram which merges to the all-zero state after l unit times, then we obtain that $X \sim N(cl\sqrt{E_{\rm s}}, \frac{clN_0}{2})$. Let $\mathcal{D}$ be the set of $r_t^{(i)}$ (where $t = 0, 1, \ldots, l-1$ and $i = 1, 2, \ldots, c$), such that the corresponding code symbols which corresponds to the trellis path are non-zero, and let

$$Y = \sum_{r_t^{(i)} \in \mathcal{D}} r_t^{(i)}.$$

The ML decoder (a Viterbi decoder) will favor another code sequence and hence, will make a erroneous decision if $Y < 0$. The EZ bounds on the bit and burst error probabilities of convolutional codes are based on the basic inequality in (3.1) where the region $\mathcal{R}$ (i.e., the "good region") corresponds to the case where the random variable X (whose mean is equal to $cl\sqrt{E_{\rm s}}$) is above a certain threshold value u_l (which is later optimized, and its value depends on the length l of the unmerged paths).

Since the considered channel is Gaussian, the conditional and un-conditional pairwise error probabilities for every such block code (conditioned on the value of the random variable X) are calculated exactly without the need for Chernoff bounds. The EZ bounds on the burst and bit error probabilities are stated in [103, Theorem 1] and [103, Theorem 2], respectively, and these bounds are uniformly tighter than union bounds.

We note that Craig's identity (2.9) for the Q-function (see [72]) can be very useful in expressing the EZ bound in terms of the weight enumerator of the convolutional code, if such a function is available. This approach provides an exact upper bound, instead of the truncation which is used for the numerical calculation of the bound in [103, Section 4]. In this respect, we note that McEliece derived a general technique for calculating the weight enumerators of convolutional codes (see [214]). Finally, we note that the adaptation of the bounding technique of Engdahl and Zigangirov in order to obtain upper bounds on the block and bit error probabilities of linear block codes is straightforward, and even leads to simpler bounds; for linear block codes, there is no need to sum over all the possible lengths of unmerged paths in the trellis diagram (as all the codewords have the same block length). For linear block codes, it yields upper bounds which are subject to one parameter optimization, and in fact, this optimized parameter can be computed numerically as the single root of a related optimization equation.

Consider the case where the codewords of a binary linear block code are BPSK modulated and transmitted over a binary-input AWGN channel. The concept of the derivation of the EZ bound in [103] yields a conceptually similar bound where the region $\mathcal{R}$ in (3.1) forms a *plane* in the n-dimansional Euclidean space, where the "good region" is associated with the case where the correlation between the received vector and the transmitted codeword exceeds a certain threshold (to be optimized in order to obtain the tightest bound within this form). The resulting final form of this bound is the following:

$$\begin{aligned} P_{\rm e} \leq & \sum_{d=1}^{n} \left\{ S_d \sqrt{\frac{E_{\rm s}}{\pi n N_0}} \int_{\eta}^{\infty} Q\left(\sqrt{\frac{2E_{\rm s}}{N_0} \frac{d}{n(n-d)}}\ t \right) \exp\left(-\frac{E_{\rm s}}{N_0} \frac{(t-n)^2}{t} \right) dt \right\} \\ & + Q\left(\sqrt{\frac{2E_{\rm s}}{N_0}} \frac{n-\eta}{\sqrt{n}} \right) \end{aligned} \tag{3.49}$$

where the optimal value of the parameter η, associated with the threshold for the correlation between the received vector and the transmitted codeword which determines in this case the region $\mathcal{R}$ in (3.1), is calculated by solving numerically the equation

$$\sum_{d=1}^{n} S_d\, Q\left(\sqrt{\frac{2E_{\rm s}}{N_0} \frac{d}{n(n-d)}}\ \eta \right) = 1. \tag{3.50}$$

The last equation is equivalent to [103, Eq. (27)]. We note that in [103], Engdahl and Zigangirov obtained (3.50) in a different approach, but their choice for η happends to yield the optimal value.

Since the geometrical regions $\mathcal{R}$ in (3.1) which correpond to the EZ bound and the tangential bound (TB) are both planes in the n-dimensional Euclidean space whose free parameter is optimized in order to minimize the resulting upper bound, they actually can be shown to yield an equivalent bound [195]. The novelty in the derivation of the EZ bound in [103] is in the adaptation of their bounding tecnique for obtaining upper bounds on the bit error probability and burst error probability of ML decoded convolutional codes (where on the other hand, the TB was derived by Berlekamp [38] as an upper bound on the block error probability of binary linear block codes).

3.2.7 The 1966 Viterbi Bound

Viterbi bound was derived in [341, Chapter 8]. The general idea of this bound is that the decoding error probability of a code becomes worse by increasing the correlation between the codewords. Hence, if we define $\rho_{\max}$ to be the maximal correlation between any two codewords of the code, then the decoding error probability of the code is upper bounded by the error probability which results in if the correlation between any two codewords of the code was equal to $\rho_{\max}$. If the communication takes place over an AWGN channel, then we get the following bound on the decoding error probability

$$P_{\mathrm{e}} \leq 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} \left[1 - Q\left(x + \sqrt{\frac{2nRE_{\mathrm{b}}(1-\rho_{\max})}{N_0}} \right) \right]^{M-1} dx \tag{3.51}$$

where n and R are the block length and the rate of the code, and $M = 2^{nR}$ is the number of codewords. If we assume that the codewords of a binary linear block code $\mathcal{C}$ are BPSK modulated, then the maximal correlation between any two codewords of $\mathcal{C}$ is directly linked to the minimum distance ($d_{\min}$) of the code, and it satisfies the simple relation

$$\rho_{\max} = 1 - \frac{2d_{\min}}{n}. \tag{3.52}$$

The bound is actually the exact decoding error probability for binary linear block codes with fixed composition (i.e., codes whose all codewords (except the all-zero codeword) have constant Hamming weight) and in particular, it gives the exact decoding error probability for bi-orthogonal codes and the dual of Hamming codes.

We note that if X is a zero mean, unit-variance, normal random variable, then the following equality holds (see [339, Eq. (3.66)])

$$E\Big[Q(\mu + \lambda X)\Big] = Q\left(\frac{\mu}{\sqrt{1+\lambda^2}} \right) \ , \quad \forall\, \mu, \lambda \in \mathbb{R}$$

and by setting $\lambda = 1$ in the latter equality, we obtain that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{x^2}{2}}\, Q(x+\mu)\, dx = Q\left(\frac{\mu}{\sqrt{2}} \right) \ , \quad \forall\, \mu \in \mathbb{R}.$$

Since from Jensen's inequality

$$E\Big[1 - (1-y)^{M-1}\Big] \leq 1 - \Big(1 - E[y]\Big)^{M-1}$$

then, similarly to the approach in [153], we obtain from the Viterbi bound (3.51) a looser but a simplified upper bound on the ML decoding error probability:

$$P_{\mathrm{e}} \leq 1 - \left[1 - Q\left(\sqrt{\frac{nRE_{\mathrm{b}}(1-\rho_{\max})}{N_0}}\right)\right]^{M-1}. \tag{3.53}$$

The bound in (3.53) is useful for the case of a small error probability, and it circumvents the need for the numerical integration in (3.51).

3.2.8 Hughes' Bound

In [152], Hughes derived an upper bound on the ML decoding error probability for *any signal set* which is communicated over the AWGN channel. We note that as opposed to upper bounds which were earlier presented in this section, the transmitted signals here are not necessarily expected to have equal energy, but are only assumed to be equally probable. Let the signal set be represented by the signal points $\mathbf{s}_0, \mathbf{s}_1, \ldots, \mathbf{s}_{M-1}$ where M designates the cardinality of the signal set. The basic idea in the derivation of the bound stems from the geometry of the decision regions under ML decoding; each decision region is a polyhedron which is formed by at most $M-1$ supporting hyper-planes. Now, the complementary of the decision region which corresponds to the transmitted signal is decomposed into a disjoint union of $M-1$ truncated polyhedral cones with vertices at the transmitted signal point ($\mathbf{s}_0$) (see [152, Section 2]). Then, it is shown that the conditional error probability is increased by replacing these $M-1$ polyhedral cones by $M-1$ circular cones with the same solid angles, and with positive axes which are coincident with the rays from $\mathbf{s}_0$ through $\mathbf{s}_i$ (where $i = 1, 2, \ldots, M-1$). Since the exact geometry of the decision regions is not available in general, then the upper bound on the conditional ML decoding error probability is finally obtained by maximizing over all feasible solid angles of the circular cones.

In order to state the bound explicitly, let n be the number of coordinates of each signal point $\mathbf{s}_i$ ($i = 0, 1, \ldots, M-1$), and let us define the function

$$B_n(\theta, x) \triangleq \frac{\Gamma\left(\frac{n}{2}\right)}{\sqrt{\pi}\,\Gamma\left(\frac{n-1}{2}\right)} \int_0^{\theta} \sin^{n-2}\phi \cdot \left(1 - \gamma\left(\frac{n}{2}, \frac{x^2 \sec^2\phi}{2}\right)\right) d\phi \tag{3.54}$$

where $\Gamma(x)$ designates the Gamma function in (3.3), and $\gamma(a, x)$ designates the incomplete Gamma function in (3.7). Intuitively, $B_n(\theta, x)$ represents the probability that an n-tuple of i.i.d. Gaussian random variables with zero mean and unit variance, lies in a circular cone of half-angle θ truncated at distance x from the origin (we refer here to the resulting infinite circular cone whose axis is the ray connecting the two signal points $\mathbf{s}_0$ and $\mathbf{s}_i$, and which starts at distance x from $\mathbf{s}_0$). Useful recursive formulas for the calculation of the function $B_n(\theta, x)$ are provided in [152, Section 3].

Let $\delta_i \triangleq ||\mathbf{s}_i - \mathbf{s}_0||$ be the Euclidean distance between $\mathbf{s}_i$ and $\mathbf{s}_0$ (where $i = 1, 2, \ldots, M-1$). The final form of the upper bound on the conditional error probability is stated in [152, Proposition 2], and it reads:

$$P_{\mathrm{e}|0} \leq \sum_{i=1}^{M-1} A_n\left(\frac{\delta_i}{\alpha_0}, \frac{\delta_i}{2\sigma}\right) \tag{3.55}$$

where $\alpha = \alpha_0$ is the unique positive solution of the equation

$$\sum_{i=1}^{M-1} A_n\left(\frac{\delta_i}{\alpha_0}, 0\right) = 1 \tag{3.56}$$

and

$$A_n(x,y) \triangleq B_n\Big(\Theta(x), y\Big), \qquad \Theta(x) \triangleq \begin{cases} \cos^{-1}(x) & , \ |x| \leq 1 \ , \\ \\ 0 & , \ |x| > 1 \ . \end{cases}$$

A nice property which follows easily from (3.56) is that the value of α_0 does not depend on the noise variance σ^2, so the value of α_0 needs to be calculated only one time for a particular code. It is also noted in [152] that any value of α for which $\sum_{i=1}^{M-1} A_n\left(\frac{\delta_i}{\alpha}, 0\right) \geq 1$ yields a valid upper bound in (3.55).

The bound of Hughes is always tighter than the union bound. Hughes noted in [152] that his bound provides a significant improvement over the union bound if $M \gg n$ (i.e., when the cardinality of the signal set is much larger than the blocklength of the code). In light of this, the improvement of the his bound over the union bound seems to be primarily useful for lattice codes which are densely packed (the reader is referred to the examples in [152, Section 4]).

3.2.9 Voronoi Regions for Binary Linear Block Codes and Expurgated Bounds

We begin our discussion by summarizing briefly the preliminaries and results on Voronoi regions for binary linear block codes which are provided in [3, 4, 10], and then we apply these results to obtain expurgated upper bounds on the ML decoding error probability for the AWGN channel.

Consider a set of M points in $\mathbb{R}^n$, $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_M\}$, and the distance measure $d(\mathbf{x}, \mathbf{y}) = ||\mathbf{x} - \mathbf{y}||$. If all vectors in $\mathbb{R}^n$ are grouped together according to which of the points in $\mathcal{C}$ they are closest to, then the Euclidean space $\mathbb{R}^n$ is partitioned into Voronoi regions. The Voronoi region of a point $\mathbf{c}_i \in \mathcal{C}$ (say Ω_i) is defined as the set of vectors in $\mathbb{R}^n$ which are closest to this point, i.e.,

$$\Omega_i \triangleq \{\mathbf{x} \in \mathbb{R}^n \mid d(\mathbf{x}, \mathbf{c}_i) \leq d(\mathbf{x}, \mathbf{c}), \ \forall \ \mathbf{c} \in \mathcal{C}\} \ . \tag{3.57}$$

In case of a tie between two or more points in $\mathcal{C}$, the vectors belong to more than one Voronoi region, but no vector belongs to the interior of more than one Voronoi region. The Voronoi regions of a block code govern many aspects of the code performance on an AWGN channel, and they play a central role in the performance analysis and soft-decision decoding algorithms. A Voronoi region, as defined by the set of inequalities in (3.57), is the intersection of $M-1$ half-spaces, and is therefore a convex region in n dimensions. It has usually fewer facets than M, which means that part of the inequalities in (3.57) are redundant. In this case, the region Ω_i can be represented by a subset of these inequalities, i.e., there exists a subset $\mathcal{N}_i \subseteq \mathcal{C}$ such that

$$\Omega_i \triangleq \{\mathbf{x} \in \mathbb{R}^n \mid d(\mathbf{x}, \mathbf{c}_i) \leq d(\mathbf{x}, \mathbf{c}), \ \forall \ \mathbf{c} \in \mathcal{N}_i\} \tag{3.58}$$

and the *minimal set* $\mathcal{N}_i$ for which the right-hand side of (3.58) is equal to the Voronoi region in (3.57) is called *the set of the Voronoi neighbors* of $\mathbf{c}_i$. An equivalent definition of Voronoi neighbors which is given in [3] states that the two points $\mathbf{c}_i, \mathbf{c}_j \in \mathcal{C}$ are neighbors if

$$\exists \ \mathbf{x} \in \mathbb{R}^n : d(\mathbf{x}, \mathbf{c}_i) = d(\mathbf{x}, \mathbf{c}_j) < d(\mathbf{x}, \mathbf{c}) \ , \ \forall \ \mathbf{c} \in \mathcal{C} \backslash \{\mathbf{c}_i, \mathbf{c}_j\} \tag{3.59}$$

which means that the Voronoi regions of $\mathbf{c}_i$ and $\mathbf{c}_j$ have an $(n-1)$-dimensional facet in common. A related concept which is considered in [3] is *Gabriel neighborhood* which depends in general on a looser condition. These two points are said to be Gabriel neighbors if the condition in (3.59) is satisfied for the vector $\mathbf{x}$ which is the halfway between $\mathbf{c}_i$ and $\mathbf{c}_j$, i.e., if

$$d(\mathbf{m}_{i,j}, \mathbf{c}_i) = d(\mathbf{m}_{i,j}, \mathbf{c}_j) < d(\mathbf{m}_{i,j}, \mathbf{c}) \ , \ \forall \ \mathbf{c} \in \mathcal{C} \backslash \{\mathbf{c}_i, \mathbf{c}_j\} \tag{3.60}$$

where $\mathbf{m}_{i,j} \triangleq \frac{1}{2}\,(\mathbf{c}_i + \mathbf{c}_j)$. The geometrical interpretation of a pair of Gabriel neighbors is that the straight line connecting these two points goes directly from one Voronoi region to the other, without touching a third Voronoi region. By comparing (3.59) and (3.60), it follows immediately that Gabriel neighbors are also Voronoi neighbors. The central theorem in [3] states that all Voronoi neighbors of a codeword in a binary linear block code are also its Gabriel neighbors. The proof of this theorem relies on both the linearity of the code and its binary alphabet. Next, it is shown in [3] that for a binary code, the condition for two codewords to be Gabriel neighbors depends only on the positions where both codewords have the same values; two codewords are Gabriel neighbors if and only if there is no other codeword which has the same values in these positions. Therefore, a codeword $\mathbf{c}$ is a Gabriel neighbor of the all-zero codeword if and only if there is no other codeword having zeros where $\mathbf{c}$ has zeros. Since for a binary linear code whose transmission takes place over an AWGN channel, one can assume without any loss of generality that the all-zero codeword is the transmitted codeword, then we can use the latter property to determine all the Gabriel neighbors of the all-zero codeword. The obvious way to perform this test for the Gabriel neighborhood of an arbitrary codeword $\mathbf{c}$ to the all-zero codeword is to generate all the codewords in the code, comparing each one of them to $\mathbf{c}$ and checking if there exists a codeword with zeros in all the positions where $\mathbf{c}$ has zeros. The computational complexity of this approach is $O(2^k n)$ which is not so feasible for practical codes. A more efficient algorithm for binary linear block codes is presented in the appendix of [3]. The general concept of this algorithm is that instead of generating all codewords, the algorithm finds the number of codewords having zeros in certain positions; it is done by linear row operations of the generator matrix G of the code. This matrix is brought into another valid generator matrix of the same code, and it includes two parts: the last r rows contain codewords with the specified zero pattern, and the first $k-r$ rows of the new generator matrix are such that no linear combination of these rows gives this zero pattern. Then, a total of 2^r codewords including the all-zero codeword and the codeword $\mathbf{c}$ will contain the zero pattern. According to the test above for Gabriel neighborhood, the codeword $\mathbf{c}$ is a neighbor of the all-zero codeword if and only if $r = 1$. The computational complexity of this algorithm for deciding if a single codeword is a neighbor of the all-zero codeword is $O(n^2 k)$, which forms a considerable improvement over $O(2^k n)$, unless the dimension (k) of the code is very small. Therefore, because of the equivalence between Gabriel neighbors and Voronoi neighbors for a binary linear block code, the computational complexity which is involved with this algorithm for the determination of the Voronoi region of the all-zero codeword is $O(2^k \cdot n^2 k)$. A further simplification in the characterization of the Voronoi neighbors of the all-zero codeword stems from the observation in [3] that all the codewords of Hamming weight up to $2d_{\min} - 1$ (where $d_{\min}$ designates the minimum distance of the binary linear block code) are necessarily neighbors of the all-zero codeword. It is also proved in [4] that for a general binary linear block code, all codewords of Hamming weight above $n - k + 1$ are necessarily not neighbors of the all-zero codeword; this yields that for the AWGN channel, the union bound (or any improved upper bound on the ML decoding error probability which relies on the distance spectrum of the code) can be truncated so that codewords of Hamming weight above $n - k + 1$ are not taken into consideration.

The distance spectrum (or weight distribution) of a binary linear code served earlier in this chapter for the derivation of upper bounds on the ML decoding error probability. Since in many cases the set of neighbors of a codeword is considerably smaller than the total number of codewords, then it is useful to look at the set of neighbors $\mathcal{N}_0$ of the all-zero codeword (see Eq. (3.58)), and to define the *local distance spectrum* to be the cardinality of the codewords in the set $\mathcal{N}_0$ as a function of their Hamming weights. For a binary linear block code $\mathcal{C}$, let

$$S_d \triangleq |\,\{\mathbf{c} \in \mathcal{C} :\ w_{\mathrm{H}}(\mathbf{c}) = d\}\,|\,, \quad d = 0, \ldots, n \tag{3.61}$$

and

$$L_d \triangleq |\,\{\mathbf{c} \in \mathcal{N}_0 :\ w_{\mathrm{H}}(\mathbf{c}) = d\}\,|\,, \quad d = 0, \ldots, n \tag{3.62}$$

designate the *(full) distance spectrum* and the *local distance spectrum* of the code $\mathcal{C}$, respectively, where $w_H(\mathbf{x})$ stands for the Hamming weight of a vector $\mathbf{x}$. Clearly, it follows immediately that $L_d \leq S_d$ for $d = 0, 1, \ldots, n$. Since all codewords of Hamming weight up to $2d_{\min} - 1$ are neighbors of the all-zero codeword, then the equality $L_d = S_d$ holds for $d = 0, 1, 2, \ldots, 2d_{\min} - 1$. Finally, since all the codewords of Hamming weight above $n - k + 1$ are not neighbors of the all-zero codeword, then $L_d \equiv 0$ for $d = n - k + 2, n - k + 3, \ldots, n$.

The *expurgated union bound* of a binary linear block code which is BPSK modulated and transmitted over the AWGN channel gets the form (see [4, Eq. (10)])

$$P_{\mathrm{e}} \leq \sum_{d=d_{\min}}^{n} L_d \, Q\left(\sqrt{\frac{2RE_{\mathrm{b}}}{N_0}}\right) \ . \tag{3.63}$$

Even without knowing the exact set of neighbors of the all-zero codeword, since the expurgated union bound does not include codewords of Hamming weight above $n - k + 1$, then the error probability is upper-bounded by

$$P_{\mathrm{e}} \leq \sum_{d=d_{\min}}^{n-k+1} S_d \, Q\left(\sqrt{\frac{2RE_{\mathrm{b}}}{N_0}}\right) \tag{3.64}$$

which is still a better upper bound than the original (i.e., the non-expurgated) union bound.

Considering the Voronoi regions for binary linear block codes, it is natural to ask what is the fraction of codewords of a binary linear block code which are neighbors of a certain codeword (e.g., the all-zero codeword). This question is especially interesting for sufficiently long codes where it is of interest to characterize the asymptotic fraction of the codewords which are neighbors of the all-zero codeword. For an arbitrary binary linear block code $\mathcal{C}$, the fraction of codewords which are neighbors of the all-zero codeword is called the *neighbor ratio*, and like the notation in [4], it will be designated here by $\Gamma(\mathcal{C})$. According to [4, Theorem 5], for any binary linear block code $\mathcal{C}$ whose rate satisfies the inequality $R > \frac{1}{2} + \frac{1}{n}$ (i.e., its rate is only slightly above one-half), the neighbor ratio is upper bounded by

$$\Gamma(\mathcal{C}) \leq \left(R - \frac{1}{2} - \frac{1}{n}\right)^{-2} \frac{1}{4n}$$

so if $R > \frac{1}{2}$, and if we let the block length tend to infinity, then the neighbor ratio clearly tends to zero. On the other hand, it was proved in [4, Theorem 8] that if $R < \frac{1}{2}$ and $\varepsilon > 0$ (ε can be chosen arbitrarily small), then the fraction of binary linear block codes of length n and rate R whose neighbor ratio is above $1 - \varepsilon$ tends to 1 as $n \to \infty$. These observations show the nice property that $R = \frac{1}{2}$ forms a threshold w.r.t. the asymptotic neighbor ratio of binary linear block codes; for code rates above this ratio, the neighbor ratio tends to zero when we let the block length tend to infinity, and on the other hand, for code rates below this threshold, the neighbor ratio tends asymptotically to unity with probability 1. Therefore, it is expected that the expurgated upper bounds on the ML decoding error probability will be especially effective for *high* code rates.

The characterization of the local distance spectrum of random linear block codes, Hamming codes and the second-order Reed Muller codes is presented in [10, Theorems 2.2, 2.7 and 2.9]. For the ensemble of random q-ary linear block codes whose parity-check matrices have independent and equiprobable entries, the average local distance spectrum is given in [10, Theorem 2.2], and it reads

$$L_d = \begin{cases} \binom{n}{d} \frac{(q-1)^d}{q^{n-k}} \prod_{i=0}^{d-2} (1 - q^{-(n-k-i)}) & \text{for } \ d \leq n - k + 1 \ , \\ 0 & \text{otherwise.} \end{cases} \tag{3.65}$$

The average distance spectrum of this ensemble can be easily derived. Since the probability that a given vector satisfies a random parity-check equation is $\frac{1}{q}$, then the probability that this vector is contained in a random linear block code with $n-k$ parity-check equations is $q^{-(n-k)}$, and the average distance spectrum of such an ensemble is therefore given by

$$S_d = \binom{n}{d} \frac{(q-1)^d}{q^{n-k}}. \tag{3.66}$$

In the continuation of this section we will check the effect of such an expurgation in the distance spectrum on the tightness of the bound, and we will also compare the tightness of the expurgated union bound to the expurgated TSB. We can study from such comparisons between the union bound and the TSB with and without expurgation how much we gain because of the expurgation of non-neighbor codewords and how much we gain because of the improved tangential-sphere bounding technique (as compared to the ubiquitous union bound). The effect of both improvements is expected to be more pronounced in the low SNR region, as is shown in Figs. 3.3 and 3.4.

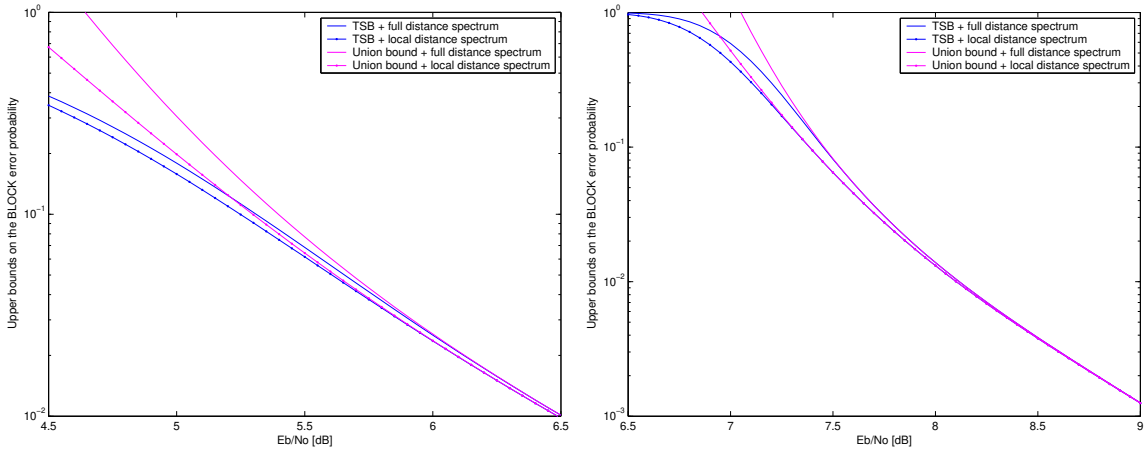


Figure 3.3: Comparison between upper bounds on the block error probability under soft-decision ML decoding. The performance is evaluated for the ensemble of random binary linear block codes whose parity-check matrices have independent and equiprobable entries. The codes are BPSK modulated, transmitted over a binary-input AWGN channel, and coherently detected. The union bounds are compared to the tangential-sphere bounds (TSBs) with and without expurgation (i.e., w.r.t. the local and full distance spectrum, respectively). We refer to the ensembles of codes whose block length is $n = 100$ and the rate is $R = 0.95$ bits per channel use (left plot), and the ensemble whose block length is $n = 1000$ and the rate is $R = 0.995$ bits per channel use (right plot).

In Fig. 3.3, we refer to ensembles of random binary linear block codes with fixed block length (n) and rate (R), and whose parity-check matrices have independent and equiprobable entries. The average *local* distance spectrum of random binary linear block codes is given in (3.65) for $q = 2$, so we rely on this result for the evaluation of the expurgated union bound and the expurgated TSB. From the two plots in Fig. 3.3, it follows that for both bounds (i.e., the union bound and the TSB), the expurgation of non-neighbor codewords of the all-zero codeword yields a certain an improvement on their tightness. For the ensemble of binary linear block codes with block length of $n = 100$ and code rate $R = 0.95$ bits per channel use, the expurgated TSB and the expurgated union bound provide at a block error probability of 0.1 a gain of 0.07 dB and 0.10 dB over the non-expurgated TSB and the non-expurgated union bound, respectively (see the left plot in Fig. 3.3). At a block error probability of 10^{-2}, the expurgated bounds give a gain of 0.02 dB as compared to

the non-expurgated bounds (it is shown in this figure that the TSB and the union bound coincide at a block error probability of 10^{-2}, and the only improvement on the tightness of these bounds is made by the expurgation of the distance spectrum). For the ensemble of random binary linear block codes of length $n = 1000$ and code rate $R = 0.995$ bits per channel use, then even at a high block error probability (e.g., a block error probability of 0.1), the union bound and the TSB coincide (see the right plot in Fig. 3.3); however, the expurgated bounds provide a gain of 0.08 dB over the non-expurgated bounds at a block error probability of 0.1. We note that for linear block codes, the expurgation of the distance spectrum yields an improvement on the tightness of the upper bounds only for high rate codes; to this end, for longer binary linear block codes, their rate should be made closer to 1 (in Fig. 3.3, we consider the case where the number of the redundant bits in both ensembles is equal to 5).

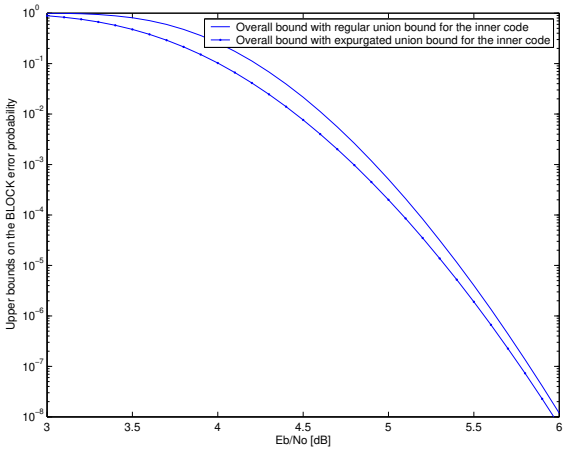


Figure 3.4: A comparison between two upper bounds which refer to the ensemble performance of serially concatenated codes where the outer code is the primitive $(127, 99, 29)$ RS code, and the inner code is a random binary linear (8, 7) block code. The encoded bits are BPSK modulated, transmitted over the AWGN channel and coherently detected. The decoding is done in two stages: the (8, 7) binary linear block code is soft-decision ML decoded, and the output of its decoder is provided to the RS decoder for a hard-decision ML decoding . The looser bound is based on a union bound w.r.t. the inner code, and the improvement is achieved by expurgating the union bound.

Fig. 3.4 presents a possible application of the concept of expurgation. Let us consider an ensemble of serially concatenated codes where the inner code is a primitive Reed-Solomon (RS) code with symbols from the Galois field $GF(2^m)$ (so, its block length is $N = 2^m - 1$ symbols), and the inner code is chosen at random from the ensemble of binary linear block codes which extend every symbol of m bits to a codeword of $m + 1$ bits (we set the dimension of the inner code to m bits, and the block length of this inner code is equal to $m + 1$). Let us choose $m = 7$, so the block length of the primitive RS code is $N = 127$ symbols, and we set the dimension of this code to $K = 99$. In this way, we obtain a 14-error-correcting RS code, and the inner code of the ensemble of serially concatenated codes is chosen at random from the ensemble of $(8, 7)$ binary linear block codes. We assume that the encoded bits are BPSK modulated, transmitted over the AWGN channel and coherently detected. The decoding is done in two stages: the $(8, 7)$ binary linear block code which is soft-decision ML decoded, and then the output of the inner decoder is forwarded to the RS decoder for a hard-decision ML decoding (hence, the RS decoder can correct up to $t = \lfloor \frac{d_{\min}-1}{2} \rfloor = \lfloor \frac{N-K}{2} \rfloor = 14$ symbols). We use here the union bound on the decoding error probability of the inner code which then serves as an upper bound on the a-priori error probability of the symbols of the RS code. The decoding error probability of the considered ensemble of serially

concatenated codes therefore satisfies

$$P_{\mathrm{e}} \leq 1 - \sum_{i=0}^{t} \binom{N}{i} p^i (1-p)^{N-i} \tag{3.67}$$

where p designates the value of the union bound which refers to the inner code under soft-decision ML decoding. We compare here the bound in (3.67) with the regular union bound w.r.t. the inner code, and an improved bound which is obtained by combining (3.67) with the expurgation of the union bound. The local distance spectrum of the ensemble of random binary linear block codes is provided by in (3.65) for $q = 2$, and we use this result (as before) for the calculation of the expurgated union bound. The improved bound in Fig. 3.4, which relies on the expurgated union bound, provides a gain of 0.17 dB, 0.12 dB and 0.10 dB for a block error probability of $10^{-2}, 10^{-3}$ and 10^{-4}, respectively.

3.2.10 Error Exponents of the Chernoff Version of Various Upper Bounds

In his paper [86], Divsalar derived simplified Chernoff versions of some upper bounds on the ML decoding error probability. In Chapter 4, these bounds will be demonstrated to be some particular cases of the 1961 Fano-Gallager bounding technique. These simplified versions include the Chernoff version of the tangential-sphere bound (TSB) of Poltyrev (see Section 3.2.10), the Chernoff version of the tangential bound (see Section 3.2.10) and the Chernoff version of the sphere bound (see Section 3.2.10). It will be demonstrated in this section (with calculations which are self-contained, as opposed to the presentation in [86]) that the Chernoff versions of the tangential bound and the sphere bound incur error exponents which are looser than the error exponent of the simple bound of Divsalar, and that the error exponent of the TSB actually coincides with the error exponent of the simple bound of Divsalar [86]. The former result actually indicates that for ensembles of codes with large block lengths, the bound of Divsalar is significantly tighter that the tangential bound and the sphere bound, and the latter result indicates that the TSB and Divsalar's bound possess the same exponential behavior when we let the block length tend to infinity.

Error Exponent of a Simplified TSB and its Connection to Divsalar Bound

In [86, Section 4.D], Divsalar derived a simplified TSB which is based on the Chernoff bounding technique, and he showed that the error exponent of the simplified TSB coincides with the error exponent of his bound (see Section 3.2.4 here). The derivation of the simplified TSB and the comparison between its error exponent and the error exponent of Divsalar bound are outlined in [86, Section 4.D]; due to the importance of this result, we will prove it here in details.

For the derivation of a simplified TSB, we will calculate Chernoff bounds on the two types of probabilities in the right-hand side of (3.6). We note that these probabilities were calculated exactly in Section 3.2.1, but the final form of the TSB in (3.16) involves numerical integrations and also a numerical solution of an associated optimization equation (3.12), and it is therefore not expressed in closed form.

We start now the derivation of a Chernoff version of the TSB. Let us substitute $c \triangleq \frac{E_{\mathrm{s}}}{N_0}$ and $\eta \triangleq \tan^2(\theta)$. Based on the geometrical interpretation of the TSB in Fig. 3.1 (see Section 3.2.1), we

obtain that

$$\begin{aligned}
\text{Prob}\left(\mathbf{r} \notin C_n(\theta)\right) &= \text{Prob}\left(\textstyle\sum_{i=2}^n z_i^2 > r_{z_1}^2\right) \\
&\leq E\left[e^{s'\left(\sum_{i=2}^n z_i^2 - r_{z_1}^2\right)}\right] \qquad s' \geq 0 \\
&= \textstyle\prod_{i=2}^n E\left[e^{s' z_i^2}\right] \cdot E\left[e^{-s' r_{z_1}^2}\right] \\
&\overset{\text{(a)}}{=} \left(\frac{1}{\sqrt{1-2\sigma^2 s'}}\right)^{n-1} \cdot E\left[e^{-s' r_{z_1}^2}\right] \\
&\overset{\text{(b)}}{=} \left(\frac{1}{\sqrt{1-2\sigma^2 s'}}\right)^{n-1} \cdot E\left[e^{-s'\eta\left(z_1 - \sqrt{nE_{\text{s}}}\right)^2}\right] \\
&\overset{\text{(c)}}{=} \left(\frac{1}{\sqrt{1-2\sigma^2 s'}}\right)^{n-1} \cdot \frac{e^{-\frac{n\eta E_{\text{s}} s'}{1+2\eta\sigma^2 s'}}}{\sqrt{1+2\eta\sigma^2 s'}} \\
&= \left(\frac{1}{\sqrt{1-2s}}\right)^{n-1} \cdot \frac{e^{-\frac{n\eta E_{\text{s}} s/\sigma^2}{1+2\eta s}}}{\sqrt{1+2\eta s}} \qquad s \triangleq \sigma^2 s' \geq 0 \\
&= \left(\frac{1}{\sqrt{1-2s}}\right)^{n-1} \cdot \frac{e^{-\frac{2n\eta cs}{1+2\eta s}}}{\sqrt{1+2\eta s}} \\
&= \sqrt{\tfrac{1-2s}{1+2\eta s}} \left(\frac{1}{\sqrt{1-2s}}\right)^{n} e^{-\frac{2n\eta cs}{1+2\eta s}}
\end{aligned}$$

where equality (a) follows from (3.30) and since $z_i \sim N(0, \sigma^2)$ for $i = 1, 2, \ldots, n$ (where $\sigma^2 = \frac{N_0}{2}$), equality (b) follows from (3.2) and the substitution $\eta = \tan^2(\theta)$, and equality (c) follows from (3.30) and since $z_1 - \sqrt{nE_{\text{s}}} \sim N(-\sqrt{nE_{\text{s}}}, \frac{\sigma^2}{2})$. We therefore obtain the following Chernoff upper bound on the probability that the received vector $\mathbf{r}$ falls outside the cone $C_n(\theta)$ in Fig. 3.1

$$\text{Prob}\left(\mathbf{r} \notin C_n(\theta)\right) \leq \sqrt{\frac{1-2s}{1+2\eta s}} \cdot e^{-nE_1(c,s,\eta)} \tag{3.68}$$

where

$$E_1(c, s, \eta) \triangleq \frac{2\eta cs}{1+2\eta s} + \frac{1}{2}\ln(1-2s)\,, \qquad 0 \leq s < \frac{1}{2}\,. \tag{3.69}$$

Based on the notation in Section 3.2.1 and Fig. 3.1, and with the Chernoff bounding technique, we will obtain an upper bound on the joint probability

$$\begin{aligned}
&\text{Prob}\left(E_k(z_1), \mathbf{r} \in C_n(\theta)\right) \\
&= \text{Prob}\left(\beta_k(z_1) \leq z_2,\ \sum_{i=2}^n z_i^2 \leq r_{z_1}^2\right) \\
&= E_{z_1}\left[\text{Prob}\left(\beta_k(z_1) \leq z_2,\ \sum_{i=2}^n z_i^2 \leq r_{z_1}^2\ |z_1\right)\right] \\
&\leq E_{z_1}\left[e^{s'(z_2 - \beta_k(z_1)) + t'\left(\sum_{i=2}^n z_i^2 - r_{z_1}^2\right)} | z_1\right] \qquad s' \geq 0,\ t' \leq 0 \\
&= E\left[e^{-s'\beta_k(z_1) - t' r_{z_1}^2}\right] E\left[e^{s' z_2 + t' z_2^2}\right] \prod_{i=3}^n E\left[e^{t' z_i^2}\right].
\end{aligned} \tag{3.70}$$

Since $z_i \sim N(0,\sigma^2)$, then we obtain from (3.30) that

$$\begin{aligned} E\Big[e^{s'z_2+t'z_2^2}\Big] &= e^{-\frac{(s')^2}{4t'}}\; E\Big[e^{t'(z_2+\frac{s'}{2t'})^2}\Big] \\ &= e^{-\frac{(s')^2}{4t'}} \cdot \frac{e^{\frac{\left(\frac{s'}{2t'}\right)^2 t'}{1-2\sigma^2 t'}}}{\sqrt{1-2\sigma^2 t'}} \\ &= \frac{1}{\sqrt{1-2\sigma^2 t'}} \cdot e^{\frac{\sigma^2 (s')^2}{2(1-2\sigma^2 t')}} \end{aligned} \tag{3.71}$$

and

$$E[e^{t'z_i^2}] = \frac{1}{\sqrt{1-2\sigma^2 t'}} \qquad i = 3,4,\ldots,n, \;\; t' \leq 0. \tag{3.72}$$

By substituting (3.71) and (3.72) into (3.70), we obtain that for $t' \leq 0$ and $s' \geq 0$

$$\begin{aligned} \text{Prob}\Big(E_k(z_1), \mathbf{r} \in C_n(\theta)\Big) &\leq E\Big[e^{-s'\beta_k(z_1)-t'r_{z_1}^2}\Big] \\ &\quad \cdot \left(\frac{1}{\sqrt{1-2\sigma^2 t'}}\right)^{n-1} e^{\frac{\sigma^2 (s')^2}{2(1-2\sigma^2 t')}} \end{aligned} \tag{3.73}$$

so it only remains to calculate the expectation w.r.t. $z_1 \sim N(0,\sigma^2)$ which appears in the right-hand side of (3.73). Let us define $\delta = \frac{k}{n}$ as the normalized Hamming weight of a competitive codeword (where we assume without any loss of generality that the all-zero codeword is transmitted). Then, for BPSK modulated signals, the Euclidean distance between the considered pair of codewords is $\delta_k = 2\sqrt{kE_{\rm s}}$, and with the substitution $\eta \triangleq \tan^2(\theta)$, we obtain from (3.2) that

$$\beta_k(z_1) = \sqrt{\frac{\delta}{1-\delta}}\,(\sqrt{nE_{\rm s}} - z_1)\,, \qquad r_{z_1} = \sqrt{\eta}\,(\sqrt{nE_{\rm s}} - z_1). \tag{3.74}$$

By substituting (3.74) in the exponent inside the expectation w.r.t. z_1, we obtain that the following equality holds

$$-s'\beta_k(z_1) - t'r_{z_1}^2 = -t'\eta(z_1 - A)^2 + B$$

where

$$A \triangleq \frac{s'}{2t'\eta}\sqrt{\frac{\delta}{1-\delta}} + \sqrt{nE_{\rm s}}\,, \quad B \triangleq \frac{(s')^2}{4t'\eta}\frac{\delta}{1-\delta} \tag{3.75}$$

and then it follows from (3.30) that the expectation w.r.t. z_1 gives the equality

$$E\Big[e^{-s'\beta_k(z_1)-t'r_{z_1}^2}\Big] = \frac{e^{B-\frac{A^2\eta t'}{1+2\eta\sigma^2 t'}}}{\sqrt{1+2\eta\sigma^2 t'}} \qquad 1+2\eta\sigma^2 t' > 0. \tag{3.76}$$

From (3.73) and (3.76), we obtain the Chernoff upper bound

$$\begin{aligned} \text{Prob}\Big(E_k(z_1), \mathbf{r} \in C_n(\theta)\Big) \leq \quad & \frac{1}{\sqrt{1+2\eta\sigma^2 t'}}\left(\frac{1}{\sqrt{1-2\sigma^2 t'}}\right)^{n-1} \\ & \cdot e^{B-\frac{A^2 t'\eta}{1+2\eta\sigma^2 t'}+\frac{\sigma^2(s')^2}{2(1-2\sigma^2 t')}} \end{aligned} \tag{3.77}$$

where $t' \leq 0$, $s' \geq 0$ and $1+2\eta\sigma^2 t' > 0$, and the parameters A and B are given in (3.75). We will now calculate the optimal value of $s' \geq 0$ which achieves a minimal value for the upper bound in (3.77). By substituting A and B in (3.75) into the exponent in the right-hand side of (3.77), and by introducing the new parameter $v \triangleq \sigma^2 t'$, we obtain that the following equality holds

$$B - \frac{A^2 t'\eta}{1+2\eta\sigma^2 t'} + \frac{\sigma^2(s')^2}{2(1-2\sigma^2 t')} = \alpha(s')^2 - \beta s' - \gamma \tag{3.78}$$

where

$$\left\{\begin{array}{lcl} \alpha & \triangleq & \frac{\sigma^2}{2}\left(\frac{1}{1+2v\eta}\,\frac{\delta}{1-\delta}+\frac{1}{1-2v}\right) \\ \beta & \triangleq & \frac{1}{1+2v\eta}\sqrt{\frac{\delta}{1-\delta}}\,\sqrt{nE_{\mathrm{s}}} \\ \gamma & \triangleq & \frac{nE_{\mathrm{s}}}{\sigma^2}\,\frac{v\eta}{1+2v\eta}. \end{array}\right. \tag{3.79}$$

We note that since $v \leq 0$ and $1+2v\eta > 0$, then $\alpha, \beta > 0$. Hence, the minimal value of the parabola in the right-hand side of (3.78) is achieved at $s' = \frac{\beta}{2\alpha} \geq 0$, and the value of the parabola at this point is equal to

$$-\frac{\beta^2}{4\alpha} - \gamma = -nc\left(\frac{2v\eta + \frac{\delta}{1-\delta}\,(1-2v)}{1+2v\eta+\frac{\delta}{1-\delta}\,(1-2v)}\right), \quad c \triangleq \frac{E_{\mathrm{s}}}{N_0}\,. \tag{3.80}$$

From Eqs. (3.77), (3.78) and (3.80), we finally obtain the Chernoff upper bound

$$\mathrm{Prob}\left(E_k(z_1), \mathbf{r} \in C_n(\theta)\right) \leq \sqrt{\frac{1-2v}{1+2v\eta}}\,\left(\frac{1}{\sqrt{1-2v}}\right)^n \cdot e^{-nc\cdot\left(\frac{2v\eta+\frac{\delta}{1-\delta}\,(1-2v)}{1+2v\eta+\frac{\delta}{1-\delta}\,(1-2v)}\right)}$$

so

$$S_k\ \mathrm{Prob}\left(E_k(z_1), \mathbf{r} \in C_n(\theta)\right) \leq \sqrt{\frac{1-2v}{1+2v\eta}}\ e^{-nE_2(c,v,\delta,\eta)} \tag{3.81}$$

where

$$\left\{\begin{array}{l} E_2(c,v,\delta,\eta) \triangleq -r_n(\delta) + \frac{1}{2}\ln(1-2v) + c\cdot\left(\frac{2v\eta+\frac{\delta}{1-\delta}\,(1-2v)}{1+2v\eta+\frac{\delta}{1-\delta}\,(1-2v)}\right) \\ \delta \triangleq \frac{k}{n}, \quad r_n(\delta) \triangleq \frac{\ln(S_k)}{n} \end{array}\right. \tag{3.82}$$

and the constraints we got for the error exponents E_1 and E_2 in (3.69) and (3.82), respectively, are

$$s \geq 0, \quad v \leq 0, \quad 1-2s > 0, \quad 1+2v\eta > 0. \tag{3.83}$$

Rather than taking the derivative w.r.t. η in order to minimize the upper bound, for large n (the block length of the code), Divsalar simply solved the equation

$$E_1(c,s,\eta) = E_2(c,v,\delta,\eta). \tag{3.84}$$

In order to simplify the calculations in the continuation of the analysis, Divsalar introduced a new set of four parameters (ρ, β, ξ and d) instead of the old set of four parameters (s, v, η and δ):

$$\rho \triangleq \frac{s}{s-v}, \quad \beta \triangleq \rho(1-2v), \quad \xi \triangleq \frac{1}{1+2s\eta}, \quad d \triangleq \frac{\delta}{1-\delta} \tag{3.85}$$

and from the constraints in (3.83), the following constraints on the new set of parameters in (3.85) follow easily

$$0 \leq \rho \leq 1, \quad 0 \leq \beta \leq 1, \quad \frac{1}{1+\eta} \leq \xi \leq 1, \quad 0 \leq d \leq \eta \tag{3.86}$$

where the constraint on d follows from the fact that the parameter δ (i.e., the normalized Hamming weight) appears only in the exponent E_2 in (3.82); the latter exponent corresponds to the case where $r_{z_1} \geq \beta_k(z_1)$, and then the inequality $d \leq \eta$ follows directly from (3.74). Armed with the new set of parameters in (3.85), it is straightforward to show that the exponents E_1 and E_2 in (3.69) and (3.82), respectively, are transformed to

$$E_1(c,\beta,\rho,\xi) = c - c\xi + \frac{1}{2}\ln\left(\frac{1-\beta}{1-\rho}\right) \tag{3.87}$$

and

$$E_2(c,\beta,\rho,\xi,d) = c - \frac{c\rho\xi}{(1+\beta d)\xi - (1-\rho)} + \frac{1}{2}\ln\left(\frac{\beta e^{-2r_n(\delta)}}{\rho}\right) . \tag{3.88}$$

From (3.84), (3.87) and (3.88), it is straightforward to show that

$$c\xi - \frac{c\rho\xi}{(1+\beta d)\xi - (1-\rho)} + a = 0, \qquad a \triangleq \frac{1}{2}\ \ln\left(\frac{\beta(1-\rho)e^{-2r_n(\delta)}}{\rho(1-\beta)}\right). \tag{3.89}$$

By solving the quadratic equation in ξ, we choose ξ (which should be non-negative) to be

$$\xi = \frac{c-(1+\beta d)a + \sqrt{(c-(1+\beta d)a)^2 + 4c(1-\rho)(1+\beta d)a}}{2c(1+\beta d)} \tag{3.90}$$

This implies the coincidence of the two exponents in (3.87) and (3.88) (i.e., the requirement in (3.84) is fulfilled for the value of ξ in (3.90)).

The optimization of the parameter ρ in obtained by maximizing the exponent E_1 in (3.87) w.r.t. ρ, and showing that $\frac{\partial E_1}{\partial \rho} = 0$ if $a = 0$ (where a is introduced in (3.89)). This implies that the optimal value of ρ is given by

$$\rho = \frac{\beta e^{-2r_n(\delta)}}{1-\beta+\beta e^{-2r_n(\delta)}} . \tag{3.91}$$

Since the optimization w.r.t. ρ yields that $a = 0$, then we obtain from (3.90) that $\xi = \frac{1}{1+\beta d}$, and it follows from (3.85) and (3.87) that

$$E_1 = \frac{c\beta d}{1+\beta d} + \frac{1}{2}\ln\left(1-\beta+\beta e^{-2r_n(\delta)}\right), \quad d \triangleq \frac{\delta}{1-\delta} . \tag{3.92}$$

Finally, the maximization of the exponent E_1 w.r.t. β (i.e., solving the equation $\frac{\partial E_1}{\partial \beta} = 0$) gives

$$\begin{aligned} \beta &= \frac{1-\delta}{\delta}\left[\sqrt{\frac{c}{c_0(\delta)} + (1+c)^2 - 1} - (c+1)\right], \\ c_0(\delta) &\triangleq \left(1-e^{-2r_n(\delta)}\right)\ \left(\frac{1-\delta}{2\delta}\right). \end{aligned} \tag{3.93}$$

Surprisingly, we obtain that *the error exponent of the Chernoff version of the TSB in* (3.92) *is equal to the error exponent of Divsalar bound in* (3.41), and this is true since also the optimized values of β in (3.39) and (3.93) coincide. This nice result was made by Divsalar in [86, Section 4.D].

Error Exponent of a Simplified TB

In [86, Section 4.B], Divsalar derived the error exponent of the Chernoff version of the tangential bound (TB). The error exponent of this simplified bound is of the form

$$E(c,\delta,\rho) = -\rho\, r_n(\delta) + \frac{\delta\rho c}{1-\delta+\delta\rho} , \quad c \triangleq \frac{E_s}{N_0}, \ \ 0 \le \rho \le 1 \tag{3.94}$$

where δ is the normalized Hamming weight, and $r_n(\delta)$ is the normalized logarithm of the distance spectrum (see (3.82)). The optimal value of ρ which maximizes the error exponent is given by

$$\rho = \begin{cases} 1 & , \ \ 0 \le \frac{r_n(\delta)}{c} \le \delta(1-\delta) , \\ \\ \sqrt{\frac{(1-\delta)c}{\delta\, r_n(\delta)}} - \frac{1-\delta}{\delta} & , \ \ \delta(1-\delta) \le \frac{r_n(\delta)}{c} \le \frac{\delta}{1-\delta} . \end{cases}$$

Finally, the substitution of the optimal value of ρ in (3.94) gives the error exponent of the simplified TB

$$E(c,\delta) = \begin{cases} c\delta - r_n(\delta) & , \quad 0 \leq \frac{r_n(\delta)}{c} \leq \delta(1-\delta) \, , \\ \\ \left(\sqrt{c} - \sqrt{\frac{r_n(\delta)\,(1-\delta)}{\delta}}\right)^2 & , \quad \delta(1-\delta) \leq \frac{r_n(\delta)}{c} \leq \frac{\delta}{1-\delta} \, . \end{cases} \tag{3.95}$$

We note that the error exponent in (3.95) coincides with the error exponent of the Viterbi & Viterbi bound [347] which will be considered later (see Section 4.4.2).

In the asymptotic case where the block length goes to infinity, we obtain from (3.95) the following upper bound on the threshold under ML decoding

$$\left(\frac{E_{\mathrm{b}}}{N_0}\right)_{\mathrm{threshold}} \leq \frac{1}{R} \max_{0<\delta<1} \frac{(1-\delta)\, r(\delta)}{\delta} \tag{3.96}$$

where $r(\delta) \triangleq \lim_{n\to\infty} r_n(\delta)$. Since $1 - e^{-x} < x$ for $x > 0$, then we obtain from a comparison between (3.42) and (3.96) that the upper bound on the threshold which is given by the simplified TSB and the Divsalar bound is better than the bound which results in from the simplified TB.

Error Exponent of a Simplified Sphere Bound and Comparison with Divsalar Bound

Referring to Fig. 3.2, the sphere bound refers to the geometrical region $\mathcal{R}$ which is a sphere whose center is at $\gamma \mathbf{x}_0$. Based on the notation in Fig. 3.2, then $\eta = 1$ for the sphere bound, and the optimization is only w.r.t. the radius of the sphere $\mathcal{R}$. Since $\eta = 1$, then we obtain from (3.37) that $\xi = \beta$, and therefore the error exponent in (3.38) becomes

$$E(c,\delta,\beta,\rho) = -\rho r_n(\delta) - \frac{\rho}{2} \ln\left(\frac{\rho}{\beta}\right) - \frac{1-\rho}{2} \ln\left(\frac{1-\rho}{1-\beta}\right) + \delta\beta c, \quad c \triangleq \frac{E_{\mathrm{s}}}{N_0}. \tag{3.97}$$

The maximization of the error exponent in (3.97) w.r.t. ρ gives the optimized value

$$\rho = \frac{1}{1 + \left(\frac{1-\beta}{\beta}\right)\, e^{2r_n(\delta)}}$$

and by substituting the optimized value of ρ in (3.97), we obtain the error exponent

$$E = -r_n(\delta) + \frac{1}{2}\, \ln\Big(\beta + (1-\beta)e^{2r_n(\delta)}\Big) + \delta\beta c. \tag{3.98}$$

Finally, the maximization of the error exponent in (3.98) gives the optimized value of β

$$\beta^* = \frac{2\delta c - \left[1 - e^{-2r_n(\delta)}\right]}{2\delta c \left[1 - e^{-2r_n(\delta)}\right]} \, .$$

We note that $\beta \geq 0$ in (3.37), and the error exponent in (3.98) vanishes at $\beta = 0$ and $\frac{\partial E}{\partial \beta} > 0$ for $0 < \beta < \beta^*$. Therefore, in order to have a positive error exponent, we need that $\beta^* > 0$, or equivalently

$$c > \max_{0<\delta<1} \frac{1 - e^{-2r_n(\delta)}}{2\delta} \, .$$

In the asymptotic case where the block length goes to infinity, we obtain an upper bound on the threshold under ML decoding which follows directly from the lower bound on c, and is given by

$$\left(\frac{E_{\mathrm{b}}}{N_0}\right)_{\mathrm{threshold}} \leq \frac{1}{R} \max_{0<\delta<1} \frac{1 - e^{-2r(\delta)}}{2\delta} \tag{3.99}$$

where $r(\delta) \triangleq \lim_{n\to\infty} r_n(\delta)$. From the comparison of the upper bounds on the threshold values in (3.42) and (3.99), we see that Divsalar bound yields a tighter upper bound on the threshold then the Chernoff version of the sphere bound (due to the factor of $1-\delta$ which appears in the right-hand side of (3.42)). This improvement is attributed to the extra free parameter η in Fig. 3.2 which is optimized in the derivation of Divsalar bound [86] (see Section 3.2.4), in contrast to the case with the sphere bound where the value of η is chosen *a-priori* to be 1.

3.2.11 Numerical Results on the Error Probability of ML Decoded Binary Linear Block Codes

We present in this section upper bounds on the decoding error probability of specific binary linear block codes and structured ensembles of turbo-like codes (e.g., regular LDPC codes [121, 122] and regular RA codes [85]).

For short block codes, we compare the upper bounds with the exact performance of these codes under ML decoding, as to provide an indication about the tightness of the improved upper bounds, and the coding gain they provide as compared to union bounds (see Fig. 3.5). The performance of the $(24, 12, 8)$ Golay code under soft-decision ML decoding is estimated by computer simulations (i.e., the decoder decides on the codeword which corresponds to the largest correlation among the $2^{12} = 4096$ matched filters). The exact performance of the (63, 6, 32) bi-orthogonal code is calculated from (3.51), since it is known that the 1966 Viterbi bound holds in equality for binary linear block codes with fixed composition, such as orthogonal or bi-orthogonal codes (see [341, Section 8] and Section 3.2.7 here).

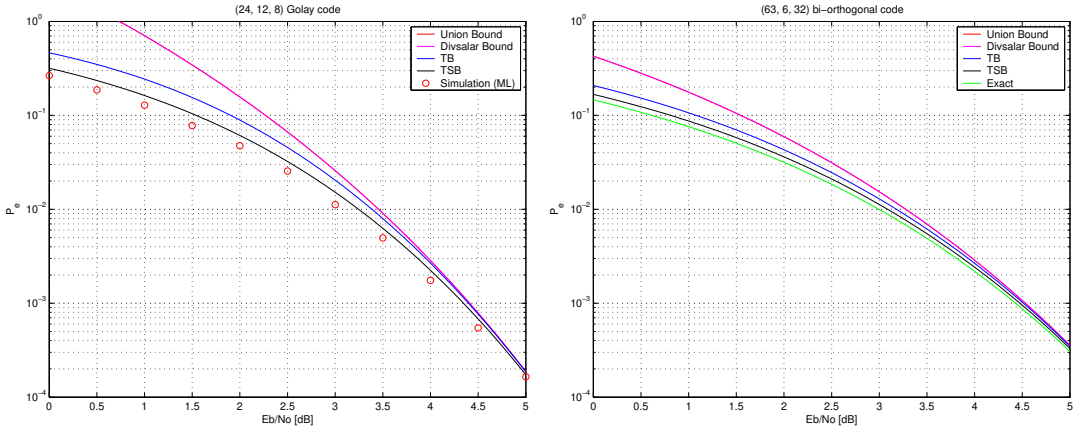


Figure 3.5: Comparison between upper bounds on the decoding error probability of binary linear block codes with short length. The (24, 12, 8) Golay code and the (63, 6, 32) bi-orthogonal code are examined. Their transmission takes place over a binary-input AWGN channel, and the codes are ML decoded with soft-decision. The union bound, Divsalar bound, the tangential bound (TB) and the tangential-sphere bound (TSB) are compared. The upper bounds on the block error probability for the two codes are compared with the exact decoding error probability. For the (24, 12, 8) Golay code, we obtain simulation results of a soft-decision ML decoder. For the bi-orthogonal code, the exact error probability is calculated from Eqs. (3.51) and (3.52) (see Section 3.2.7). The union bound and Divsalar bound coincide for these two plots.

The plots in Fig. 3.5 rely on the distance spectrum of the (24, 12, 8) Golay code and the (63, 6, 32) bi-orthogonal code, respectively. We note that the distance spectra of the binary (23, 12, 7)

Golay code (which is a perfect code) and the extended (24, 12, 8) Golay code are well-known and given for example in [345, Table 2.2]. The (63, 6, 32) bi-orthogonal code includes $2^6 = 64$ codewords; except of the all-zero codeword, all the other 63 codewords of this code have a Hamming weight of 32. From the two plots in Fig. 3.5, one can see that the TSB bound is very tight, and it provides a significant improvement over the union bound. Although the error exponent of the Chernoff version of the TSB and the error exponent of Divsalar bound coincide (see Section 3.2.10), it is obvious from Fig. 3.5 that especially for short codes, the TSB bound is significantly better than the Divsalar bound.

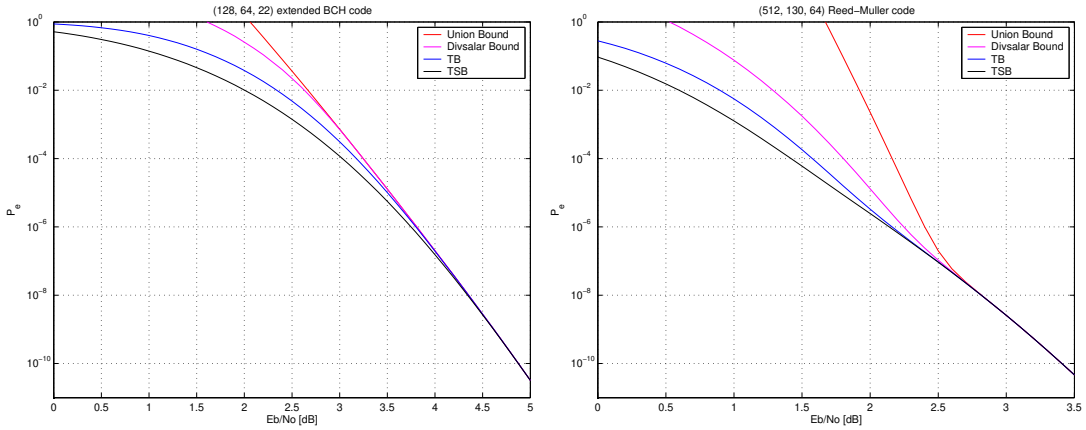


Figure 3.6: Comparison between upper bounds on the decoding error probability of binary linear block codes with moderate length. The (128, 64, 22) extended BCH code (left plot) and the third order (512, 130, 64) Reed-Muller code (right plot) are examined. Their transmission takes place over a binary-input AWGN channel, and the codes are ML decoded with soft-decision. The union bound, Divsalar bound, tangential bound (TB) and the tangential-sphere bound (TSB) are compared.

Upper bounds on the decoding error probability of the extended (128, 64, 22) BCH code and the third-order (512, 130, 64) Reed-Muller code are depicted in Fig. 3.6; it is assumed that the codes are BPSK modulated, transmitted over the AWGN channel, coherently detected and ML decoded. The plots in Fig. 3.6 rely on the distance spectra of the extended (128, 64, 22) BCH code and the third-order (512, 130, 64) Reed-Muller code which are provided in [76] and [315], respectively (see also [173]).

Upper bounds on the performance of ML decoding for some ensembles of (n, j, k) LDPC codes with rate one-half are presented in Figs. 3.7. The effect of the block length n on the closeness to the Shannon capacity limit is also demonstrated. The error bounds are based on Gallager's upper bound on the ensemble distance spectrum of regular LDPC codes (see [122, Theorem 2.3]) where based on the upper sub-matrices of the parity-check matrices of Gallager's ensembles of regular LDPC codes (see [122, Fig. 2.1]), it follows directly that the Hamming distance of all codewords of these ensembles should be *even*; the average distance spectrum of these ensembles is therefore equal to zero for all the Hamming weights which are odd. In order to obtain upper bounds on the performance of these ensembles, Gallager's upper bound on the distance spectrum is combined here with two possible upper bounds on the ML decoding error probability: the TSB, and the union bound in its Q-form. It is also shown in Figs. 3.7 that for large block lengths, the union bounds are essentially useless at rates above the cutoff rate (for a binary-input AWGN channel and a rate one half, the cutoff rate corresponds to $\frac{E_b}{N_0} = 2.45$ dB). In both plots of Figs. 3.7, the same values of j and k (i.e., the degree of the variable nodes and the degree of the parity-check nodes,

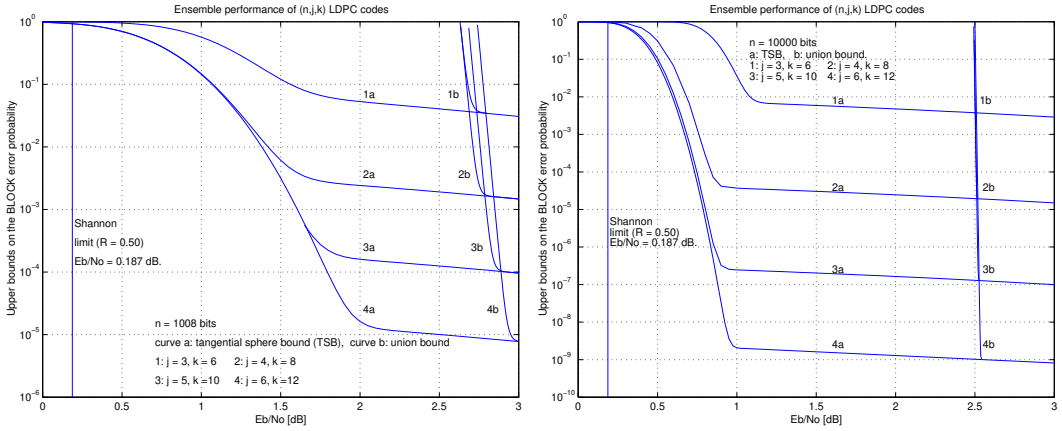


Figure 3.7: Upper bounds on the block error probability of ML decoding for ensembles of (n, j, k) regular LDPC codes in a binary-input AWGN channel where $n = 1008$ (left plot) and $n = 10000$ (right plot), $j = 3, 4, 5, 6$ and $k = 2j$ (rate of one half). The upper bounds are based on Gallager's upper bound on the ensemble distance spectrum of (n, j, k) LDPC codes and the TSB. Union bounds in Q-form appear for comparison.

respectively) are compared for ensembles of (n, j, k) LDPC codes with block lengths of $n = 1$K and 10K coded bits; it reflects the dramatic improvement that results by increasing n by a factor of 10. For example, for the case of $j = 6$, $k = 12$, the values of $\frac{E_b}{N_0}$ required by the TSB for achieving a *block error probability* of 10^{-4} are 1.80 dB and 0.73 dB for $n = 1$K and 10K respectively (while the Shannon capacity in this case corresponds to 0.19 dB). The considerable advantage of the tangential sphere bounding technique over the union bounds (especially for large block lengths) is demonstrated in Figs. 3.7 (see also [282, 283]). For example, the gain achieved for a block error probability of 10^{-5} is 1.72 dB in the case that $n = 10$K, $j = 6$ and $k = 12$. As exhibited in Fig. 3.7, there is only a slight improvement in the ensemble performance of (n, j, k) LDPC codes, by increasing the value of j above 6 (while keeping $k = 2j$ for maintaining the code rate fixed). Therefore, the ML block error probabilities of ensembles of (n, j, k) LDPC codes, are investigated here for $j = 6$ for a variety of rates and block lengths of $n = 5$K, 10K, 20K and 40K. The results of this comparison are summarized in Table 3.1 (see p. 48), demonstrating the impressive potential performance of ensembles of LDPC codes of length in the range 5K–40K. This table is created by comparing an upper bound on the value of required $\frac{E_b}{N_0}$ for achieving a block error probability of 10^{-5} by ML decoding with a lower bound on the Shannon capacity (as R is lower bounded by $1 - j/k$ [122], the value of $\frac{E_b}{N_0}$ that corresponds to the channel capacity with this lower bound on the code rate, is actually a lower bound on the Shannon capacity that corresponds to the exact code rate. However, for large block length n, this lower bound on the rate is very tight). Fig. 3.8 presents results for ensembles of (n, j, k) LDPC codes of rate 0.250. For example in the case of $j = 6$, $k = 8$ and $n = 40$K, an upper bound on the value of $\frac{E_b}{N_0}$ required to achieve a block error probability of 10^{-5} with ML decoding is -0.46 dB, that is only 0.33 dB away from the channel capacity (see also Table 3.1).

The ensemble performance of uniformly interleaved repeat-accumulate (RA) codes was considered by Divsalar, Jin and McEliece [85]. The structure of these codes and the expressions for the input-output weight enumerator function (IOWEF) of such ensembles of codes is introduced in [85] (see Fig. 3.9). The average number of codewords in the ensemble of uniformly interleaved (qN, N)

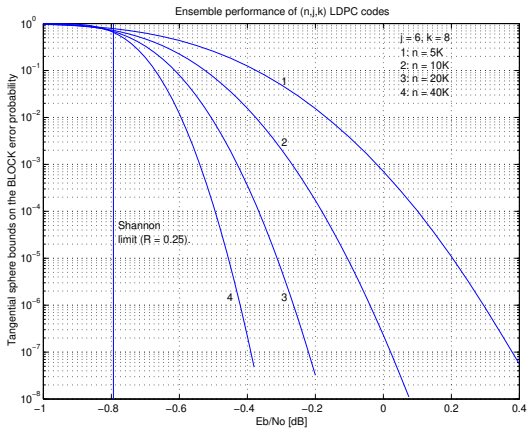


Figure 3.8: Upper bounds on the block error probability of ML decoding for some ensembles of (n, j, k) regular LDPC codes of rate 1/4 in a binary-input AWGN channel (where $j = 6$ and $k = 8$). The upper bounds are based on Gallager's upper bound on their ensemble distance spectrum [122, Section 2.2] and the TSB of Poltyrev.

The number of ones (k) in each row of the parity matrix H of the ensemble of codes and a tight lower bound on the rate (R).	$k = 8$ $R = 0.250$	$k = 12$ $R = 0.500$	$k = 24$ $R = 0.750$
The block length n	The value of $\frac{E_b}{N_0}$ (and the gap to capacity)		
$n = 5$K bits	0.20 dB (0.99 dB)	1.00 dB (0.81 dB)	2.57 dB (0.94 dB)
$n = 10$K bits	−0.11 dB (0.68 dB)	0.79 dB (0.60 dB)	2.39 dB (0.76 dB)
$n = 20$K bits	−0.31 dB (0.48 dB)	0.64 dB (0.45 dB)	2.26 dB (0.63 dB)
$n = 40$K bits	−0.46 dB (0.33 dB)	0.54 dB (0.35 dB)	2.18 dB (0.55 dB)

Table 3.1: The value of energy per bit to noise spectral density $\left(\frac{E_b}{N_0}\right)$ required for an upper bound on the block error probability of 10^{-5} with soft-decision ML decoding for the Gallager ensemble of (n, j, k) regular LDPC codes with $j = 6$. The bounds are based on the TSB. The gaps (in dB) between the values of $\frac{E_b}{N_0}$ achieving the upper bound on the block error probability of 10^{-5} to the Shannon capacity of a binary-input AWGN channel are written in parenthesis.

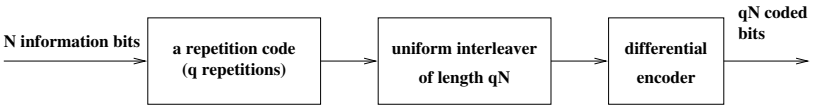


Figure 3.9: Block diagram of the ensemble of uniformly interleaved RA codes. The N information bits are repeated q times, then the qN symbols are uniformly interleaved and differentially encoded.

RA codes having an information weight w and an overall Hamming weight ℓ is

$$A_{w,\ell}^{(N)} = \frac{\binom{N}{w} \binom{qN-\ell}{\lfloor qw/2 \rfloor} \binom{\ell-1}{\lceil qw/2 \rceil - 1}}{\binom{qN}{qw}} \tag{3.100}$$

where $0 \leq w \leq N$, $0 \leq \ell \leq qN$, and $\lfloor x \rfloor$, $\lceil x \rceil$ denote the maximal and minimal integers, respectively, satisfying the inequality $\lfloor x \rfloor \leq x \leq \lceil x \rceil$.

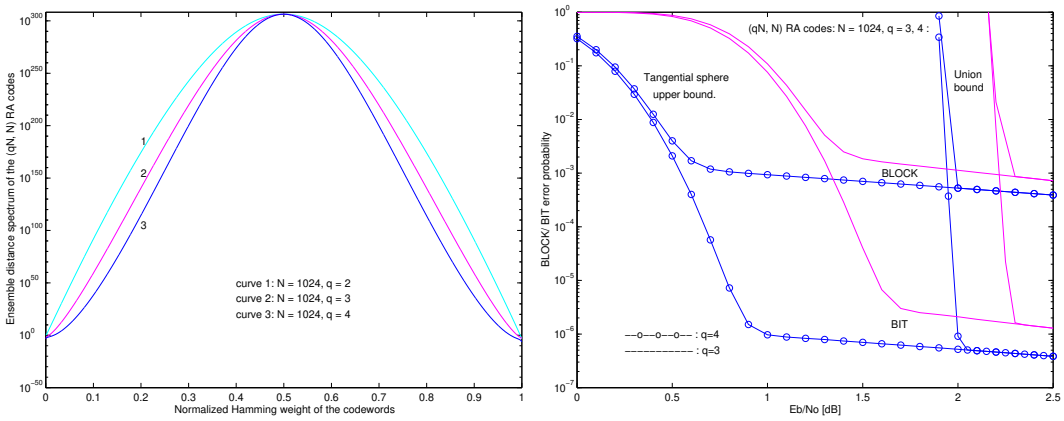


Figure 3.10: The left plot depicts the distance spectrum of some ensembles of uniformly interleaved (qN, N) RA codes versus the normalized Hamming weights of the codewords (normalized with respect to the block length qN) where $N = 1024$ and $q = 2, 3, 4$. The right plot shows Upper bounds on the block and bit error probabilities of two ensembles of uniformly interleaved and serially concatenated repeat-accumulate (RA) codes [85]. The two ensembles in the right plot refer to $q = 3, 4$ repetitions of the information bits in Fig. 3.9. The considered upper bounds are the TSB and the union bound in Q-form.

Based on the IOWEF of uniformly interleaved RA codes in (3.100), the distance spectrum of some ensembles of (qN, N) RA codes, are illustrated in the upper plot of Fig. 3.10, versus the normalized Hamming weight of the codewords (the normalization relates to block length qN), where $N = 1024$ and $q = 2, 3, 4$. It is observed there that for a fixed value of N, the number of codewords with relatively low Hamming weights is decreased by increasing the value of q (note that the overall number of codewords (2^N) remains constant). Therefore, it is reasonable (as indicated in the lower plot of Fig. 3.10) that the ensemble performance of the uniformly interleaved and serially concatenated RA codes is improved by increasing the value of q; by increasing the value of q, the interleaver length (qN) is increased which enhances the interleaver gain of such an ensemble of interleaved and serially concatenated codes. Upper bounds on the block and bit error probability of uniformly interleaved RA codes under soft-decision ML decoding are illustrated in the right plot of Fig. 3.10. The usefulness of the TSB is demonstrated at rates considerably above the channel cutoff rate: for $q = 3, 4$ (the code rate of the RA code is $\frac{1}{3}$ or $\frac{1}{4}$, respectively), the cutoff rate of the binary-input AWGN channel corresponds to $\frac{E_b}{N_0} = 2.03$ dB and 1.85 dB, respectively. From the

right plot of Fig. 3.10, we obtain that for $q = 3$, the TSB on the bit error probability [283] is equal to 10^{-5} at $\frac{E_b}{N_0} = 1.58$ dB (i.e., 0.45 dB below the value of $\frac{E_b}{N_0}$ that corresponds to the channel cutoff rate). For $q = 4$, the TSB is equal to 10^{-5} at $\frac{E_b}{N_0} = 0.79$ dB (i.e., 1.06 dB below the value of $\frac{E_b}{N_0}$ which corresponds to the cutoff rate, but 1.59 dB above the value of $\frac{E_b}{N_0}$ which corresponds to the channel capacity). It is also shown in the right plot of Fig. 3.10 that the union bounds in Q-form are useless at rates beyond the cutoff rate of the channel, as expected for long enough block codes (since the interleaver length for these RA codes is 3072 and 4096 for $q = 3, 4$ respectively).

3.3 Improved Bounds for Fully Interleaved Fading Channels

In this section, we present rigorous analytical upper bounds on the ML decoding error probability of binary linear block codes, operating over fully interleaved (memoryless) Rician fading channels. These bounds are applied to several ensembles of turbo-like codes, demonstrating their significant advantage over the ubiquitous union bounds in a portion of the rate region above the channel cut-off rate (R_0). Throughout this section, we assume a perfect side information on the channel i.i.d. fading samples which are also available to the receiver. The theoretical implications of this assumption are discussed in [181].

3.3.1 System Model, Channel Capacity and Cut-off Rate, and Union Bounds

The System Model

The model of the communication system is the following: the information bits are encoded, fully interleaved and BPSK modulated. Then, the modulated signal is transmitted through a frequency non-selective fading channel. We discuss in this section fully interleaved Rician fading channels and also Rayleigh fading channels where the latter may be combined with space diversity of order L, based on the maximum ratio combining (MRC) principle. As a consequence of a perfect channel interleaver (which clearly differs from the interleaver of the code), the fading samples which correspond to the interleaved coded bits are assumed to be i.i.d. The noise added to the signal is an additive white Gaussian noise (AWGN) with a zero mean and a double-side spectral density of N_0. At the receiver, assumed to be equipped with ideal channel state information (realizations of the fading values), we assume that the fading samples received by these L antennas are statistically independent. The received signal is coherently detected. Finally, the demodulated bits are deinterleaved (according to the inverse permutation of the channel interleaver) and ML decoded. Considering a turbo code, we assume a termination to the all-zero state at the end of each frame (block). Clearly, no termination is required for block codes in general and for LDPC codes and RA codes in particular.

We denote the pdf of the non-negative[4] fading a by $p(a)$. Clearly, with a (ideally given to the receiver) interpreted as part of the measurements and independent of the transmitted signals, then

$$\begin{array}{rcl} p_0(y,a) & = & \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(y - a\sqrt{\frac{2E_\mathrm{s}}{N_0}}\right)^2\right] \cdot p(a) \\ p_1(y,a) & = & \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(y + a\sqrt{\frac{2E_\mathrm{s}}{N_0}}\right)^2\right] \cdot p(a) \end{array} \quad , \quad \begin{array}{c} -\infty < y < \infty \\ a \geq 0 \end{array} \tag{3.101}$$

where $\frac{E_\mathrm{s}}{N_0}$ stands for the energy per symbol to the spectral noise density.

For MBIOS channels in (3.101), the capacity-achieving distribution is clearly symmetric: Prob $(x = 0) =$ Prob $(x = 1) = \frac{1}{2}$. A straightforward calculation of the channel capacity (C) of the fully interleaved fading channels in (3.101) gives that

$$\begin{aligned} C &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty}\int_{0}^{\infty} p(a) \exp\left[-\frac{1}{2}\left(y + a\sqrt{\frac{2E_\mathrm{s}}{N_0}}\right)^2\right] \\ &\qquad \cdot \log_2\left(\frac{2}{1+\exp\left(2ay\sqrt{\frac{2E_\mathrm{s}}{N_0}}\right)}\right) da\, dy \end{aligned} \tag{3.102}$$

in units of bits per channel use.

For a fully interleaved Rician fading channel, the probability density function of the amplitudes of the i.i.d. fading samples is

$$p(a) = \begin{cases} 2a(K+1)\exp(-(K+1)a^2 - K)\, I_0\left(2\sqrt{K(K+1)}\, a\right), & a \geq 0 \\ 0 & a < 0 \end{cases} \tag{3.103}$$

where the Rician parameter K stands for the power ratio of the direct to the diffused received path of the fading channel, and $I_0(x)$ designates the Modified Bessel function of order zero:

$$I_0(x) = \frac{1}{\pi}\int_0^\pi \exp(x\cos\theta)d\theta\ , \quad x \in \mathbb{R}.$$

The probability density function of the i.i.d. fading samples of a fully interleaved Rayleigh fading channel with MRC diversity of order L admits the form

$$p(a) = \begin{cases} \dfrac{2L^L a^{2L-1}\exp(-La^2)}{(L-1)!} & a \geq 0 \\ 0 & a < 0 \end{cases} .$$

Clearly, for coded communications: $E_\mathrm{s} = RE_\mathrm{b}$ where E_s and E_b designate the energies per coded symbol and per information bit, respectively, and R is the rate of the code (in bits per channel use). The value of $\frac{E_\mathrm{b}}{N_0}$ which corresponds to the channel capacity for a certain code rate R is calculated numerically by solving the implicit equation $C = R$, where C is expressed in (3.102), associated with the appropriate expression for the probability density function $p(a)$ of the i.i.d. fading samples.

[4]The effect of the phases of the fading measurements is eliminated at the receiver and the fades during each symbol are treated as non-negative random variables.

When the bit error probability is restricted not to exceed a certain value P_{b}, then by applying the rate-distortion theory with the Hamming distance between two binary codewords as a distortion measure, the minimal theoretical value of $\frac{E_{\mathrm{b}}}{N_0}$ is found by solving numerically the implicit equation $C = R\,\Big(1 - h(P_{\mathrm{b}})\Big)$ where $h(x)$ designates the binary entropy function to the base 2, and the capacity C of the considered fully interleaved fading channels is expressed in (3.102).

The cutoff rate (in bits per channel use) of a fully interleaved Rayleigh fading channel with MRC space diversity of order L admits the form

$$R_0 = 1 - \log_2\left(1 + \left(1 + \frac{E_{\mathrm{s}}}{LN_0}\right)^{-L}\right)$$

or $\frac{E_{\mathrm{b}}}{N_0}$ is given in terms of R as

$$\frac{E_{\mathrm{b}}}{N_0} = \frac{L}{R}\left[(2^{1-R} - 1)^{-\frac{1}{L}} - 1\right]\,. \tag{3.104}$$

A straightforward calculation of the cutoff rate of a fully interleaved Rician fading channel yields that

$$R_0 = 1 - \log_2(1 + Z) \tag{3.105}$$

bits per channel use, where Z is given in closed form in [286, Eq. (8)].

The Pairwise Error Probability and Union Bounds

For a fully interleaved Rician fading channel with a Rician factor K, the expression of the pairwise ML decoding error probability for two codewords which differ in d symbols was derived in [307]. This derivation relied on Craig's identity for the Q-function in (2.9).

Under the assumptions in Section 3.3.1, the following expressions for the pairwise error probability refer to the case where the Hamming distance between the transmitted codeword (x) and another competitive codeword ($\hat{x}$) is d. For a fully interleaved Rician fading channel whose Rician factor is equal to K, the expression for the pairwise error probability results in by combining [307, Eqs. (25), (30)]:

$$\begin{aligned}&\mathrm{Prob}\left(x \to \hat{x} \mid W_H(x,\hat{x}) = d\right)\\ &= \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \left(\frac{1+K}{1+K+\frac{E_{\mathrm{s}}/N_0}{\sin^2\theta}}\right)^d e^{-\frac{Kd}{1+\frac{(1+K)\sin^2\theta}{Es/N_0}}}\,d\theta\end{aligned} \tag{3.106}$$

where $W_H(x,\hat{x})$ designates the Hamming distance between the binary codewords x and $\hat{x}$. We note that an alternative expression for the pairwise error probability of fully interleaved Rayleigh fading channels with perfect CSI is given in [155] (hence, it is an alternative expression to (3.106) for the particular case where $K = 0$).

For fully interleaved Rayleigh fading channels with MRC space diversity of order L (where the fading samples received by the L antennas are assumed to be i.i.d.), the expression for the pairwise ML decoding error probability results in by substituting $K = 0$ in (3.106), and by also replacing E_{s} and d in the left hand side of (3.106) with $\frac{E_{\mathrm{s}}}{L}$ and dL, respectively:

$$\mathrm{Prob}\left(x \to \hat{x} \mid W_{\mathrm{H}}(x,\hat{x}) = d\right) = \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \left(\frac{1}{1+\frac{E_{\mathrm{s}}/N_0}{L\sin^2\theta}}\right)^{dL} d\theta\,. \tag{3.107}$$

The union bound on the block error probability for an ML decoded linear and binary block code admits the form

$$P_{\mathrm{e}} \leq \sum_{d=d_{\min}}^{n} S_d \, \mathrm{Prob}\Big(x \to \hat{x} \mid W_{\mathrm{H}}(x,\hat{x}) = d\Big) \tag{3.108}$$

where $d_{\min}$ denotes the minimal Hamming weight of the nonzero codewords of the code $\mathcal{C}$, and the expressions of the pairwise error probability for the fully interleaved fading channels are provided in (3.106) and (3.107). Here, S_d is the appropriate coefficient of the distance spectrum of the code, which designates the number of codewords whose Hamming weight equals d.
Similarly, the union bound on the bit error probability admits the form

$$P_{\mathrm{b}} \leq \sum_{d=d_{\min}}^{n} S'_d \, \mathrm{Prob}\Big(x \to \hat{x} \mid W_{\mathrm{H}}(x,\hat{x}) = d\Big) \tag{3.109}$$

where S'_d is introduced in (3.13).

3.3.2 Generalization of the EZ Bound for Fully Interleaved Fading Channels

The Engdahl and Zigangirov (EZ) bound in [103] provides an upper bound on the ML decoding error probability when the communications takes place over the AWGN channel (see Section 3.2.6). Sason and Shamai generalized the EZ bound for fully interleaved fading channels [286, Section 3.B]. In the following, we present the guidelines for the derivation of the generalized bound. For further details, we refer the interested reader to [286, Section III.B and Appendices B and C].

Let's assume a binary linear block code $\mathcal{C}$ of length n and dimension k which is BPSK modulated, and transmitted over a fully interleaved fading channel. As noted in Section 3.3.1, we assume here that a perfect channel side information (CSI) is available at the receiver. Let the vector $\mathbf{a} = (a_1, a_2, , \ldots, a_n)$ denote the perfect measurements of the i.i.d. fading samples, where $E\Big[a_i^2\Big] = 1$. We assume that the symbols 0 and 1 are mapped by the BPSK modulator to $+\sqrt{E_{\mathrm{s}}}$ and $-\sqrt{E_{\mathrm{s}}}$, respectively. As before, we also assume without any loss of generality that the all-zero codeword is transmitted (due to the linearity of the code and the symmetry of the channel). The components of the demodulated signal ($\mathbf{y}$) are

$$y_i = a_i \sqrt{E_{\mathrm{s}}} + \nu_i \ , \quad i = 1, 2, \ldots, n$$

where $\{a_i\}_{i=1}^n$ are the i.i.d fading samples, and $\{\nu_i\}_{i=1}^n$ are the i.i.d. zero-mean Gaussian noise samples with variance $E[\nu_i^2] = \frac{N_0}{2}$. Let us define the random variable (RV)

$$z \triangleq \sum_{i=1}^{n} a_i \, y_i = \sum_{i=1}^{n} \Big(a_i^2 \sqrt{E_{\mathrm{s}}} + a_i \, \nu_i\Big). \tag{3.110}$$

We now partition the binary and linear block code $\mathcal{C}$ into constant Hamming weight subcodes ($\mathcal{C}_d$), where the subcode $\mathcal{C}_d$ ($d = 1, 2, \ldots, n$) includes all the codewords of $\mathcal{C}$ with a Hamming weight d and also includes the all-zero codeword.

The fading samples of the channel are i.i.d. RVs, so the pairwise ML decoding error probability doesn't change for all the competitive codewords of the same Hamming weight (d). Since we consider an MBIOS channel, then we assume without any loss of generality that a competitive codeword of Hamming weight d differs from the transmitted all-zero codeword in its first d coordinates, i.e, the first d coordinated of the competitive codeword are ones and the succeeding $n - d$ components are zeros.

Let W_d (where $d = 1, 2 \ldots n$) be the RV

$$W_d \triangleq \sum_{i=1}^{d} a_i \, y_i = \sum_{i=1}^{d} \Big(a_i^2 \, \sqrt{E_{\mathrm{s}}} + a_i \, \nu_i \Big).$$

If $W_d < 0$, then such a codeword of Hamming weight d is preferred by the ML decoder (rather than the all-zero codeword which was transmitted).

The case where z is small enough may yield an unreliable decision of the ML decoder (it may happen, for example, when the channel undergoes a severe fade). We therefore introduce a threshold η_z, and we wish to optimize it in order to derive the tightest upper bound among the following family of upper bounds on the ML decoding error probability:

$$P_{\mathrm{e}} \leq \sum_{d=1}^{n} \Big\{ S_d \cdot \mathrm{Prob} \Big(W_d < 0 \, , \; z \geq \eta_z \Big) \Big\} + \mathrm{Prob} \Big(z < \eta_z \Big) \tag{3.111}$$

where $\{S_d\}_{d=0}^{n}$ is the distance spectrum of the block code $\mathcal{C}$. We note that (3.111) follows as a particular case of the general inequality in (3.1) where the region $\mathcal{R}$ in the latter inequality is chosen to be a plane (see (3.110)), and where we apply the union bound to the first term in the right-hand side of (3.1). We also note that in the limit where $\eta_z \to -\infty$, the upper bound in (3.111) turns to be the well known union bound in (3.108). Therefore, an optimization of the right-hand side of (3.111) over the parameter η_z yields an upper bound which is uniformly tighter than the union bound, and also cannot exceed unity. The final version of this bound is expressed explicitly in [286, Eq. (38)] (with some algebraic simplifications in [286, Appendix C], and the optmized free parameter of this bound is given implictly as a solution to the optimization equation [286, Eq. (39)]. Note that the existence and uniqueness of a solution to the latter equation for the optimized free parameter in (3.111) is proved in [286, Appendix B]. In particular, we show in [286, Section III.B] that the generalized EZ bound for fully interlwaved fading channels particularizes to the EZ bound in [103] (see Section 3.2.6) for the case of a binary-input AWGN channel.

An upper bound on the *bit error probability* for an ensemble of linear, binary block codes results in by simply replacing the average distance spectrum $\{S_d\}_{d=0}^{n}$ of the ensemble of codes by the sequence $\{S'_d\}_{d=0}^{n}$, as defined in (3.13).

The generalized EZ bound is later applied to ensembles of turbo-like codes whose transmission takes place over fully interleaved fading channels, and the tightness of this bound is compared to other reported bounds (see Section 4.7.2).

In the block-fading channel model, an n-length codeword is affected by a number (M) of independent fading gains. In this case, an n-length codeword is split into a number of blocks M which is a fraction of n, and over each block the channel fading is so highly correlated so that it is considered to be constant. The EZ bound was recently generalized for the performance analysis under ML decoding of block codes whose transmission takes place over block-fading channels. For further details, the interested reader is referred to [361]. For small values of M, the generalized EZ bound for block-fading channels shows an improvement over previously reported bounds (see, e.g., the Malkamaki-Leib bound in [211]); this improvement is especially pronounced in [361] for small values of M.

3.3.3 Generalization of the Viterbi and Viterbi Bound

The Viterbi and Viterbi bound is an upper bound on the ML decoding error probability of binary linear block codes operating over a binary-input AWGN channel [347]. A generalization of this

bound for fully interleaved Rician fading channels with perfect CSI at the receiver was derived in [288] (see Section 2 and the Appendix in [288] where the generalization of the Viterbi and Viterbi bound for fully interleaved Rician fading channels was derived in two alternative ways). The generalized bound admits the form

$$P_{\mathrm{e}} \leq \sum_{d} P_{\mathrm{e}}(d)$$

where

$$\begin{aligned} P_{\mathrm{e}}(d) \leq (S_d)^{\rho} & \left(\frac{1+K}{1+K+\beta_1}\right)^{d\rho} \cdot \exp\left(-\frac{K\beta_1 d\rho}{1+K+\beta_1}\right) \\ & \cdot\left(\frac{1+K}{1+K+\beta_2}\right)^{(n-d)\rho} \cdot \exp\left(-\frac{K\beta_2 (n-d)\rho}{1+K+\beta_2}\right) \\ & \cdot\left(\frac{1+K}{1+K+\beta_3}\right)^{n(1-\rho)} \cdot \exp\left(-\frac{K\beta_3 n(1-\rho)}{1+K+\beta_3}\right) \end{aligned}$$

and

$$\beta_1 = \frac{E_{\mathrm{s}}}{N_0}, \quad \beta_2 = \frac{E_{\mathrm{s}}}{N_0} \cdot \left[1 - \Big(1+\zeta(1-\rho)\Big)^2\right], \quad \beta_3 = \frac{E_{\mathrm{s}}}{N_0} \cdot \left[1 - (1-\zeta\rho)^2\right].$$

The optimization of the bound is performed with respect to the two parameters ρ and ζ where $0 \leq \rho \leq 1$, and $1 + K + \beta_i > 0$ for $i = 1, 2, 3$.

For a binary-input AWGN channel, since $K \to \infty$, it can be verified (see [288]) that the generalized bound above reduces to the Viterbi and Viterbi bound in [347] for the optimal value $\zeta = \frac{\delta}{1-\delta+\delta\rho}$ where $\delta \triangleq \frac{d}{n}$ is the normalized weight of codewords with Hamming weight d and block length n.

We note that the generalization of the Viterbi and Viterbi bound for fully interleaved fading channels with perfect CSI, as reported in [13, 14], seems to be problematic. Our reservation stems from the fact that the Viterbi and Viterbi bounding technique in [347] is invalidated once the parameter $H = \exp(-\frac{E_{\mathrm{s}}}{N_0})$ (which is the Bhattacharyya constant for a binary-input AWGN channel) is replaced by the one corresponding to fast Rician fading channels (see [13, Eq. (12)]). The specific correlations in the derivation of the bound in [347] demand special care when generalized to fading channels. The problem with the generalization in [13, 14] is that in contrast to the particular case of a binary-input AWGN channel, for fully interleaved fading channels, we obtain that the covariance between the outputs of the two correlators which correspond to the all-zero codeword and to an arbitrary codeword of Hamming weight d does not depend only on d (i.e., as a result of the i.i.d. fading samples, this covariance does not stay constant w.r.t. all the codewords of Hamming weight d). A generalization of the Viterbi and Viterbi bound for fully interleaved fading channels with imperfect CSI at the receiver is considered in [301]. Again, the derivation in [301] is problematic since it relies on the arguments of [13, 14], when now H is replaced by the one corresponding to imperfect CSI.

We note that the generalized Viterbi and Viterbi bound for fully interleaved Rician fading channels is derived in [288, Section 2] as a particular case of the Gallager-type bounds; these bounds are discussed extensively in the next chapter.

Chapter 4

Gallager-Type Upper Bounds: Variations, Connections and Applications

Chapter overview: In addressing the Gallager bounds and their variations, we focus on the Duman and Salehi variation which originates from the standard Gallager bound. A large class of efficient recent bounds (or their Chernoff versions) is demonstrated to be a special case of the generalized second version of the Duman and Salehi bounds. Implications and applications of these observations are pointed out, including the fully interleaved fading channel, resorting to either matched or mismatched decoding. The proposed approach can be generalized to geometrically uniform non-binary codes, finite state channels, bit interleaved coded modulation systems, and to upper bounds on the conditional decoding error probability.

4.1 Introduction

Since the error performance of coded communication systems rarely admits exact expressions, tight analytical bounds serve as a useful theoretical and engineering tool for assessing performance and for gaining insight into the effect of the main system parameters. As specific good codes are hard to identify, the performance of ensembles of codes is usually considered. Fano [107] and Gallager [123] upper bounds were introduced as efficient tools to determine the error exponents of the ensemble of random codes, providing informative results up to the ultimate capacity limit. Since the advent of information theory, the search for efficient coding systems has motivated the introduction of efficient bounding techniques tailored to specific codes or some carefully chosen ensembles of codes. A classical example is the adaptation of the Fano upper bounding technique [107] to specific codes, as reported in the seminal dissertation by Gallager [122] (to be referred to as the 1961 Gallager-Fano bound). The incentive for introducing and applying such bounds has increased with the recent introduction of turbo codes [40] and the rediscovery of low-density parity-check (LDPC) codes [208]. Clearly, the desired bounds must not be subject to the union bound limitation, since for long blocks these ensembles of turbo-like codes perform reliably at rates which are considerably above the cutoff rate (R_0) of the channel (recalling that union bounds for long codes are not informative at the portion of the rate region above R_0, where the performance of these capacity-approaching codes is most appealing). Although maximum-likelihood (ML) decoding is in general prohibitively complex for long codes, the derivation of upper bounds on the ML decoding error probability is of interest, providing an ultimate indication of the system performance. Further, the structure of efficient codes is usually not available, necessitating efficient bounds on performance to rely only on basic features, such as the distance spectrum and input-output weight enumeration function (IOWEF) of the examined codes. These latter features can be found by analytical methods (e.g, [214]).

In classical treatments, due to the difficulty in the analytic characterization of optimal codes, random codes were introduced ([107], [123], [124]). This is also the case with modern approaches and practical coding techniques, where ensembles of codes (e.g., turbo codes [40], repeat-accumulate codes [1, 85], and LDPC codes [122]) lend themselves to analytical treatment, while this is not necessarily the case for specifically chosen codes within these families. A desirable feature is to identify efficient bounding techniques encompassing both specific codes and ensembles.

In continuation to the previous chapter, we consider here various reported upper bounds on the ML decoding error probability and demonstrate the underlying connections that exist between them; the bounds are based on the distance spectrum or the input-output weight enumerators of the considered codes. In the following, we consider variations on the Gallager bounds, and we focus on the second version of the recently introduced bounds by Duman and Salehi ([94], [95]) whose derivation is based on the 1965 Gallager bounding technique ([123], [124]). Though originally derived for binary signaling over an additive white Gaussian noise (AWGN) channel, we demonstrate here its considerable generality and show that it provides the natural bridge between the 1961 and 1965 Gallager bounds ([122], [123]). It is suitable for both random and specific codes [86], as well as for either bit or block error probability analysis. It is also demonstrated here that a large class of efficient recent bounds or their Chernoff versions are special cases of the generalized second version of the Duman and Salehi (DS2) bound. We exemplify the use of this generalized bound in various settings, such as the fully interleaved fading channel ([286],[288]). In an important recent contribution, Divsalar [86] has introduced some efficient and easily applicable bounds, and has also provided insightful observations on the Duman and Salehi bounding technique ([94], [95]) in view of other bounds. In our setting, we shall rely on some of the interesting observations in [86].

The chapter is organized as follows: The 1965 Gallager bound ([123], [124]) and the DS2 bound [95] are presented as preliminary material in Section 4.2. These two upper bounds form the underlying bounding technique in this chapter, as we rely on them throughout. In Section 4.3, the 1961 Gallager-Fano bound [122] is presented, and some interconnections among the Gallager bounds and the DS2 bound are demonstrated. It is shown in Section 4.3 that the DS2 bound provides the natural bridge between the 1961 and 1965 Gallager bounds. In Section 4.4, it is demonstrated that many reported bounds on the ML decoding error probability (which were originally derived independently) can be considered as special cases of the DS2 bound. The 1965 Gallager random coding bound is extended in Section 4.5 to the mismatched decoding regime, where the decoder operates in a ML fashion, but may use a mismatched metric. These Gallager-type bounds which are derived for the mismatched decoding regime can be applied to deterministic codes and ensembles. Some reported results ([127], [164], [221]) are derived in Section 4.5, based on an alternative approach which appropriately limits the code ensemble. Some applications and examples of these bounds are presented in Section 4.7, which include fully interleaved fading channels and mismatched metrics. Finally, Section 4.8 concludes the chapter.

4.2 Gallager Bounds for Symmetric Memoryless Channels

4.2.1 The 1965 Gallager Bound

Suppose an arbitrary codeword $\underline{x}^m$ (of length-N) is transmitted over a channel. Let $\underline{y}$ designate the observation vector (of N components), and $p_N(\underline{y}|\underline{x}^m)$ be the channel transition probability

measure. Then, the conditional ML decoding error probability is given by

$$P_{\mathrm{e}|m} = \sum_{\underline{y}\colon \left\{\exists\, m' \neq m\colon\, p_N(\underline{y}|\underline{x}^{m'}) \geq p_N(\underline{y}|\underline{x}^m)\right\}} p_N(\underline{y}|\underline{x}^m)$$

If the observation vector $\underline{y}$ is such that there exists $m' \neq m$ so that $p_N(\underline{y}|\underline{x}^{m'}) \geq p_N(\underline{y}|\underline{x}^m)$, then for arbitrary $\lambda, \rho \geq 0$, the value of the expression

$$\left(\sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}$$

is clearly lower bounded by 1, and in general, it is always non-negative. The 1965 Gallager bound [123, 124] therefore states that

$$P_{\mathrm{e}|m} \leq \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \left(\sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}, \quad \lambda,\, \rho \geq 0\,. \tag{4.1}$$

The upper bound (4.1) is usually not easily evaluated in terms of basic features of particular codes, except for example, orthogonal codes and the special case of $\rho = 1$ and $\lambda = \frac{1}{2}$ (which yields the Bhattacharyya-union bound).

For discrete, memoryless and output-symmetric channels, the upper bound (4.1) is not directly applicable for actual code performance calculation, since it cannot be factored into single-letter expressions (because of the ρth power which operates on the *inner* summation of (4.1)). Therefore, the bound does not lend itself to code performance calculation in terms of the distance spectrum of the ensemble of codes. This difficulty could be circumvented had the ρth power in (4.1) been taken over the expression $\sum_{\underline{y}} \left\{ p_N(\underline{y}|\underline{x}^m) \sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right\}$ (instead of its actual location in the inner summation of the bound (4.1)).

For ensembles of random block codes, where the codewords are independently selected with an arbitrary probability distribution $q_N(\underline{x})$, Gallager derived an upper bound on the average ML decoding error probability (where the average is over the randomly and independently selected codewords). By setting $\lambda = \frac{1}{1+\rho}$ in (4.1), we obtain that for $\rho \geq 0$, the average ML decoding error probability satisfies

$$P_{\mathrm{e}} \leq \sum_{\underline{x}^1} \ldots \sum_{\underline{x}^M} q_N(\underline{x}^1) \ldots q_N(\underline{x}^M) \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)^{\frac{1}{1+\rho}} \left(\sum_{m' \neq m} p_N(\underline{y}|\underline{x}^{m'})^{\frac{1}{1+\rho}} \right)^{\rho}.$$

Without any loss of generality, it is assumed that $m = 1$ (since all the codewords are chosen randomly and independently according to the same input distribution $q_N(\underline{x})$), so by changing the order of summation in the last inequality, we obtain that

$$\begin{aligned} P_{\mathrm{e}} \;\leq\; & \sum_{\underline{y}} \sum_{\underline{x}^1} q_N(\underline{x}^1)\, p_N(\underline{y}|\underline{x}^1)^{\frac{1}{1+\rho}} \sum_{\underline{x}^2} \ldots \sum_{\underline{x}^M} q_N(\underline{x}^2) \ldots q_N(\underline{x}^M) \\ & \cdot \left(\sum_{m'=2}^{M} p_N(\underline{y}|\underline{x}^{m'})^{\frac{1}{1+\rho}} \right)^{\rho}, \quad \rho \geq 0. \end{aligned}$$

By invoking the Jensen inequality in the last inequality, $E[z^\rho] \leq (E[z])^\rho$ for $0 \leq \rho \leq 1$, we obtain that for $0 \leq \rho \leq 1$

$$\begin{aligned} P_{\mathrm{e}} &\leq \sum_{\underline{y}} \sum_{\underline{x}^1} q_N(\underline{x}^1)\, p_N(\underline{y}|\underline{x}^1)^{\frac{1}{1+\rho}} \left(\sum_{\underline{x}^2} \ldots \sum_{\underline{x}^M} q_N(\underline{x}^2) \ldots q_N(\underline{x}^M) \right. \\ &\qquad \left. \sum_{m'=2}^{M} p_N(\underline{y}|\underline{x}^{m'})^{\frac{1}{1+\rho}} \right)^\rho \\ &= \sum_{\underline{y}} \sum_{\underline{x}^1} q_N(\underline{x}^1)\, p_N(\underline{y}|\underline{x}^1)^{\frac{1}{1+\rho}} \left(\sum_{m'=2}^{M} \sum_{\underline{x}^{m'}} q_N(\underline{x}^{m'})\, p_N(\underline{y}|\underline{x}^{m'})^{\frac{1}{1+\rho}} \right)^\rho \\ &= \sum_{\underline{y}} \sum_{\underline{x}^1} q_N(\underline{x}^1)\, p_N(\underline{y}|\underline{x}^1)^{\frac{1}{1+\rho}} \left((M-1) \cdot \sum_{\underline{x}} q_N(\underline{x})\, p_N(\underline{y}|\underline{x})^{\frac{1}{1+\rho}} \right)^\rho . \end{aligned}$$

The 1965 Gallager random coding bound therefore reads

$$P_{\mathrm{e}} \leq (M-1)^\rho \sum_{\underline{y}} \left(\sum_{\underline{x}} q_N(\underline{x})\, p_N(\underline{y}|\underline{x})^{\frac{1}{1+\rho}} \right)^{1+\rho}, \quad 0 \leq \rho \leq 1 \tag{4.2}$$

where P_{e} designates the average decoding error probability, and M is the number of the codewords. For the particular case of a memoryless channel $\left(p_N(\underline{y}|\underline{x}) = \prod_{i=1}^{N} p(y_i|x_i)\right)$ and a memoryless input distribution $\left(q_N(\underline{x}) = \prod_{i=1}^{N} q(x_i)\right)$, the Gallager random coding bound (4.2) admits the form

$$P_{\mathrm{e}} \leq e^{-N \cdot E_{\mathrm{r}}(R, \mathbf{q})} \tag{4.3}$$

where $R = \frac{\ln M}{N}$ is the code rate (in nats per channel use) and the associated error exponent is

$$E_{\mathrm{r}}(R, \mathbf{q}) = \max_{0 \leq \rho \leq 1} \{E_0(\rho, \mathbf{q}) - \rho R\} \tag{4.4}$$

where

$$E_0(\rho, \mathbf{q}) \triangleq -\ln \left(\sum_{y} \left(\sum_{x} q(x)\, p(y|x)^{\frac{1}{1+\rho}} \right)^{1+\rho} \right) . \tag{4.5}$$

From the Gallager random coding bound, ensembles of fully random block codes achieve the channel capacity with an optimal soft-decision ML decoder. However, the randomness of the codewords is not crucial for proving the channel coding theorem, and a requirement of pairwise independence of the codewords is sufficient. The latter condition holds, e.g., for multilevel codes when the encoding at each level is performed independently. Using multiuser information theory, it is proved in [351, Section 3] that ensembles of multilevel codes achieve the channel capacity with an optimal ML decoder by properly choosing the rates of the component codes at each level (see [351, Section 3]). We also note that the random coding error exponent (4.4) serves as one of the efficient design rules for multilevel codes, in order to properly select the rates of the component codes at each level (see [351, Section 4]).

4.2.2 The DS2 Bound

The bounding technique of Duman and Salehi [94, 95] originates from the 1965 Gallager bound. Let $\psi_N^m(\underline{y})$ designate an arbitrary probability measure (which may also depend on the transmitted codeword $\underline{x}^m$). The 1965 Gallager bound (4.1) then yields that

$$\begin{aligned} P_{\mathrm{e}|m} &\leq \sum_{\underline{y}} \psi_N^m(\underline{y})\, \psi_N^m(\underline{y})^{-1}\, p_N(\underline{y}|\underline{x}^m) \left(\sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho} \\ &= \sum_{\underline{y}} \psi_N^m(\underline{y}) \left(\psi_N^m(\underline{y})^{-\frac{1}{\rho}}\, p_N(\underline{y}|\underline{x}^m)^{\frac{1}{\rho}} \sum_{m' \neq m} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho} \\ &\qquad \forall\, \lambda, \rho \geq 0. \end{aligned} \tag{4.6}$$

By invoking the Jensen inequality in (4.6), the DS2 bound results

$$P_{\mathrm{e}|m} \leq \left(\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)^{\frac{1}{\rho}}\, \psi_N^m(\underline{y})^{1-\frac{1}{\rho}} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}, \qquad 0 \leq \rho \leq 1,\ \lambda \geq 0. \tag{4.7}$$

Let $G_N^m(\underline{y})$ be an arbitrary non-negative function of $\underline{y}$, and let the probability density function $\psi_N^m(\underline{y})$ be

$$\psi_N^m(\underline{y}) = \frac{G_N^m(\underline{y})\, p_N(\underline{y}|\underline{x}^m)}{\sum\limits_{\underline{y}} G_N^m(\underline{y})\, p_N(\underline{y}|\underline{x}^m)}\,. \tag{4.8}$$

The functions $G_N^m(\underline{y})$ and $\psi_N^m(\underline{y})$ are referred to as the un-normalized and normalized tilting measures, respectively. The substitution of (4.8) into (4.7) yields the following upper bound on the conditional ML decoding error probability

$$\begin{aligned} P_{\mathrm{e}|m} &\leq \left(\sum_{\underline{y}} G_N^m(\underline{y})\, p_N(\underline{y}|\underline{x}^m) \right)^{1-\rho} \\ &\quad \cdot \left\{ \sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\, G_N^m(\underline{y})^{1-\frac{1}{\rho}} \left(\frac{p_N(\underline{y}|\underline{x}^{m'})}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right\}^{\rho}, \\ &\qquad 0 \leq \rho \leq 1\,,\ \lambda \geq 0. \end{aligned} \tag{4.9}$$

The upper bound (4.9) was also derived in [86, Eq. (62)].

For the case of memoryless channels, and for the choice of $\psi_N^m(\underline{y})$ as $\psi_N^m(\underline{y}) = \prod\limits_{i=1}^{N} \psi^m(y_i)$ (recalling that the function ψ^m may depend on the transmitted codeword $\underline{x}^m$), the upper bound (4.7) is relatively easily evaluated (similarly to the standard union bounds) for linear block codes. In that case, (4.7) is calculable in terms of the distance spectrum of the code, not requiring the fine details of the code structure. Moreover, (4.7) is also amenable to some generalizations, such as for the class of discrete memoryless channels with arbitrary input and output alphabets.

4.2.3 The 1961 Gallager-Fano Bound

In his monograph on LDPC codes, Gallager introduced an upper bound on the ML decoding error probability for block codes operating over an arbitrary MBIOS channel (see [122, Chapter 3]). This bound depends on the average distance spectrum of the block code (or ensemble of codes), and therefore, it can be applied to ensembles as well as specific codes (as opposed to the 1965 Gallager bound which applies essentially to the ensemble of random block codes). The derivation of this bound is based on Fano's 1961 bounding technique [107] for random codes, which is adapted in [122] to specific codes. The concept of this bound is detailed in the following. Let $\mathcal{C}$ be a block code of length N which is communicated over an MBIOS channel, and let $\underline{x}^m$ be the transmitted codeword (where $m \in \{0, 1, \ldots, M-1\}$). We define the tilted metric

$$D_m(\underline{x}^{m'}, \underline{y}) \triangleq \ln \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^{m'})} \right) \tag{4.10}$$

where $\underline{x}^{m'}$ is an arbitrary codeword of the code $\mathcal{C}$ and $\underline{y}$ is the received vector (of length N) at the output of the channel, $p_N(\underline{y}|\underline{x}) = \prod_{i=1}^N p(y_i|x_i)$ is the conditional transition probability of the MBIOS channel, and $f_N^m(\underline{y})$ is an arbitrary function which is positive if $p_N(\underline{y}|\underline{x}^{m'})$ is positive for any m' (and may also depend on the transmitted message). Note that $D_m(\cdot, \cdot)$ is in general not computable at the receiver; it is used here as a conceptual tool to evaluate the upper bound on the decoding error probability. If ML decoding is applied, then an error occurs if there exists $m' \neq m$ so that

$$D_m(\underline{x}^{m'}, \underline{y}) \leq D_m(\underline{x}^m, \underline{y}).$$

By increasing the value of $D_m(\underline{x}^m, \underline{y})$, then if the above condition holds, it is likely to hold for more than a single value of m' (where $m' \neq m$). The union bound which upper bounds the probability of the union of events by the sum of the probabilities of the individual events does not therefore yield a tight upper bound, since it counts one decoding error many times in the bound. Gallager therefore chose to separate the set of the observation vectors (of length N), $\mathcal{Y}^N$, into two disjoint subsets: the good subset ($\mathcal{Y}_g^N$) corresponds to the case where the value of $D_m(\underline{x}^m, \underline{y})$ is not larger than a certain threshold which was chosen to be linearly proportional to the block length N (say Nd, where $d \in \mathbb{R}$ is arbitrary), and the bad subset ($\mathcal{Y}_b^N$) corresponds to the case where the value of $D_m(\underline{x}^m, \underline{y})$ is larger than Nd. In the derivation of his bound, Gallager applied the union bound only to the case where the received vector is inside the good region (i.e., if $\underline{y} \in \mathcal{Y}_g^N$). Therefore, we obtain that

$$\begin{aligned} \mathcal{Y}^N &= \mathcal{Y}_g^N \bigcup \mathcal{Y}_b^N \\ \mathcal{Y}_g^N &\triangleq \left\{ \underline{y} \in \mathcal{Y}^N : D_m(\underline{x}^m, \underline{y}) \leq Nd \right\} \\ \mathcal{Y}_b^N &\triangleq \left\{ \underline{y} \in \mathcal{Y}^N : D_m(\underline{x}^m, \underline{y}) > Nd \right\} \end{aligned} \tag{4.11}$$

where d is an arbitrary real number. The parameter d is later optimized, as to get the tightest upper bound within this family.

The conditional ML decoding error probability can be expressed as a sum of two terms

$$P_{\mathrm{e}|m} = \mathrm{Prob}\left(\text{error}\,, \underline{y} \in \mathcal{Y}_b^N\right) + \mathrm{Prob}\left(\text{error}\,, \underline{y} \in \mathcal{Y}_g^N\right) \tag{4.12}$$

which leads to the following upper bound:

$$P_{\mathrm{e}|m} \leq \mathrm{Prob}\left(\underline{y} \in \mathcal{Y}_b^N\right) + \mathrm{Prob}\left(\text{error}\,, \underline{y} \in \mathcal{Y}_g^N\right). \tag{4.13}$$

The arbitrary function $f_N^m(\cdot)$ in (4.10) has clearly no effect on the RHS of (4.12) because it does not affect which codeword is decoded when $\underline{x}^m$ is transmitted. However, the function $f_N^m(\cdot)$ affects the upper bound on the conditional ML decoding error probability in (4.13). As was explained in Section 3.1, an inequality like (4.13) forms the starting point of many efficient bounds, e.g., the TB of Berlekamp [38] (see Section 3.2.3 here), Hughes' bound [152] (see Section 3.2.8 in p. 33), the TSB of Poltyrev [259] (see Section 3.2.1 in p. 17), and the EZ bound of Engdahl and Zigangirov [103] (see Section 3.2.6 in 30). In the Gallager-Fano approach, the regions $\mathcal{Y}_g^N$, $\mathcal{Y}_b^N$ are related to the choice of the arbitrary function $f_N^m(\cdot)$ in (4.10). At this stage, Fano [107] proceeded with the Chernoff bounding technique and the random coding approach, while Gallager bound [122] is better suited to treat particular codes.

Based on the Chernoff bound

$$\text{Prob}\,(\underline{y} \in \mathcal{Y}_b^N) \leq E(e^{sW})\,, \quad s \geq 0 \tag{4.14}$$

where

$$W \triangleq D_m(\underline{x}^m, \underline{y}) - Nd = \ln\left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right) - Nd. \tag{4.15}$$

The second term in (4.13) is also upper bounded by a combination of the union and the Chernoff bounds, which then yields

$$\begin{aligned}
&\text{Prob}\,(\text{decoding error},\ \underline{y} \in \mathcal{Y}_g^N) \\
&= \text{Prob}\Big(D_m(\underline{x}^{m'}, \underline{y}) \leq D_m(\underline{x}^m, \underline{y}) \text{ for some } m' \neq m\,, \quad \underline{y} \in \mathcal{Y}_g^N\Big) \\
&\leq \sum_{m' \neq m} \text{Prob}\Big(D_m(\underline{x}^{m'}, \underline{y}) \leq D_m(\underline{x}^m, \underline{y})\,,\ D_m(\underline{x}^m, \underline{y}) \leq Nd\Big) \\
&\leq \sum_{m' \neq m} E\Big(\exp(t\ Z_{m'} + r\ W)\Big), \quad \forall\, t, r \leq 0
\end{aligned} \tag{4.16}$$

where, based on (4.10)

$$Z_{m'} \triangleq D_m(\underline{x}^{m'}, \underline{y}) - D_m(\underline{x}^m, \underline{y}) = \ln\left(\frac{p_N(\underline{y}|\underline{x}^m)}{p_N(\underline{y}|\underline{x}^{m'})}\right)$$

and W is defined in (4.15). In [122], Gallager considered the case where the functions $f_N^m(\underline{y})$ can be expressed in the product form

$$f_N^m(\underline{y}) = \prod_{i=1}^{N} f(y_i) \tag{4.17}$$

(where the function f does not depend here on the index i). For simplifying the derivation of the 1961 Gallager-Fano bound [122], it was also assumed that the non-negative function f is *even*, i.e., $f(y) = f(-y)$ for all $y \in \mathcal{Y}$. Let $p_x(y) \triangleq p(y|x)$ be the probability transition measure of the MBIOS channel (so $p_1(y) = p_0(-y)$ for all $y \in \mathcal{Y}$).

Let $N_m(l)$ be the number of codewords in the code $\mathcal{C}$ with Hamming distance l from the transmitted codeword $\underline{x}^m$ (where $l \in \{0, 1, \ldots, N\}$). For ensembles of block codes, we refer to the expected number of codewords with Hamming distance l from the transmitted codeword, where the expectation is taken over all the codes from this ensemble. We note that for geometrically

uniform block codes (e.g., linear block codes or fully random block codes), the value of $N_m(l)$ does not depend on the index of the codeword m. For the general case, let $S_l \triangleq \frac{1}{M} \sum_{m=0}^{M-1} N_m(l)$ be the average distance spectrum of the code (or ensemble) $\mathcal{C}$. Based on (4.13)–(4.16) and by optimally setting $t = \frac{r-1}{2}$, the 1961 Gallager-Fano bound on the average ML decoding error probability (see [122, Eqs. (3.28)–(3.30)]) reads

$$P_{\mathrm{e}} \leq g(s)^N \ \exp(-Nsd) + \sum_{l=0}^{N} \Big\{ S_l \ [h(r)]^l \ [g(r)]^{N-l} \Big\} \exp(-Nrd) \tag{4.18}$$

where $s \geq 0$, $r \leq 0$ and $d \in \mathbb{R}$. The term in (4.18) for $l = 0$ accounts for the pathological possibility that another codeword in $\mathcal{C}$ is identical to the transmitted codeword; S_0 is the (average) number of codewords in $\mathcal{C}$, other than the transmitted codeword, which are identical to it.

From the symmetry of the channel, and since the function $f(\cdot)$ is assumed to be even, Gallager obtained with the Chernoff bounding technique (see (4.14)–(4.16)) that

$$\begin{aligned} g(s) &= \sum_{y \in \mathcal{Y}} p_0(y)^{1-s} \ f(y)^s \\ &= \frac{1}{2} \sum_{y \in \mathcal{Y}} \Big\{ \Big([p_0(y)]^{1-s} + [p_1(y)]^{1-s} \Big) \ f(y)^s \Big\} \\ h(r) &= \sum_{y \in \mathcal{Y}} \Big\{ [p_0(y) \ p_1(y)]^{\frac{1-r}{2}} \ f(y)^r \Big\} \end{aligned} \tag{4.19}$$

and where the sums above are replaced by integrals when the output alphabet is continuous. The upper bound (4.18) on the ML decoding error probability depends on the distance spectrum of the considered block code, and it therefore applies to fixed codes and also structured ensembles of codes. It can be verified (though not mentioned in [122]) that the optimization of the parameter d in (4.18) yields that

$$e^{-nd} = \left(-\frac{r}{s} \ \frac{1}{[g(s)]^N} \ \sum_{l=0}^{N} S_l [h(r)]^l \ [g(r)]^{N-l} \right)^{\frac{1}{s-r}} \tag{4.20}$$

and by the substitution of (4.20) in the RHS of (4.18), one obtains that the tightest bound within this family gets the expression

$$P_{\mathrm{e}} \leq 2^{H(\rho)} \ [g(s)]^{N(1-\rho)} \ \left(\sum_{l=0}^{N} S_l \ [h(r)]^l \ [g(r)]^{N-l} \right)^{\rho}, \quad 0 \leq \rho \leq 1. \tag{4.21}$$

Here $\rho \triangleq \frac{s}{s-r}$ (so it follows that $0 \leq \rho \leq 1$, since $s \geq 0$ and $r \leq 0$), and $H(\rho) \triangleq -\rho \log_2(\rho) - (1-\rho) \log_2(1-\rho)$ is the binary entropy function to the base 2. The optimization of the function $f(\cdot)$ in (4.18) yields an implicit solution (as indicated in [122, Eq. (3.40)]). The following approximation to the optimal non-negative and symmetric function $f(\cdot)$ was proposed by Gallager (see [122, Eq. (3.41)]):

$$f(y) = k \ \left\{ \frac{\Big[p_0(y)^{\frac{1-r}{2}} + p_1(y)^{\frac{1-r}{2}} \Big]^2}{p_0(y)^{1-s} + p_1(y)^{1-s}} \right\}^{\frac{1}{s-r}} \tag{4.22}$$

where the constant k in (4.22) is arbitrary and cancels out in the bound. For the ensemble of fully random binary block codes where all the binary block codes of a fixed rate and a fixed blocklength are equiprobable, Gallager noted in his monograph (see [122, p. 30]) that the function $f(\cdot)$ in (4.22) is indeed the optimal one (so in this case, it is not just an approximation); the optimality of the

function $f(\cdot)$ is in the sense of minimizing the 1961 Gallager-Fano upper bound in (4.18) among all the non-negative and even functions (since these properties on the arbitrary function $f(\cdot)$ were required for the derivation of this bound). The optimality of the function $f(\cdot)$ in (4.22) is proved rather easily with calculus of variations, starting from Eqs. (4.18) and (4.19); to this end, one relies on the distance spectrum of the ensemble of fully random binary block codes with rate R bits per channel use and blocklength N, which is given by

$$S_l = 2^{-N(1-R)} \binom{N}{l}, \quad l = 0, 1, \ldots, N.$$

4.3 Interconnections Between Bounds

4.3.1 The Random Coding Version of the DS2 Bound

We show here that the random coding version of the DS2 bound coincides with the well known 1965 Gallager bound for random codes. For the ensemble of random codes, where the N-length codewords are randomly and independently selected with respect to the input distribution $q_N(\underline{x})$, the DS2 bound yields the following upper bound on the conditional decoding error probability:

$$\begin{aligned} P_{\mathrm{e}|m} \quad \leq \quad & (M-1)^{\rho} \left(\sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\, G_N^m(\underline{y}) \right)^{1-\rho} \\ & \cdot \left\{ \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\, G_N^m(\underline{y})^{1-\frac{1}{\rho}} \sum_{\underline{x}'} q_N(\underline{x}') \left(\frac{p_N(\underline{y}|\underline{x}')}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right\}^{\rho}, \\ & 0 \leq \rho \leq 1, \ \lambda \geq 0. \end{aligned} \tag{4.23}$$

The optimal non-negative function $G_N^m(\underline{y})$ minimizing (4.23) is

$$G_N^m(\underline{y}) = \left(\sum_{\underline{x}'} q_N(\underline{x}') \left(\frac{p_N(\underline{y}|\underline{x}')}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}. \tag{4.24}$$

The substitution of (4.24) into (4.23) gives

$$P_{\mathrm{e}|m} \leq (M-1)^{\rho} \sum_{\underline{y}} \left\{ p_N(\underline{y}|\underline{x}^m) \left(\sum_{\underline{x}'} q_N(\underline{x}') \left(\frac{p_N(\underline{y}|\underline{x}')}{p_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho} \right\}. \tag{4.25}$$

After averaging over the transmitted codeword $\underline{x}^m$, which is randomly selected with respect to the input distribution $q_N(\underline{x}^m)$, (4.25) yields the following upper bound on the ML decoding error probability:

$$P_{\mathrm{e}} \leq (M-1)^{\rho} \sum_{\underline{y}} \left\{ \left(\sum_{\underline{x}} q_N(\underline{x})\, p_N(\underline{y}|\underline{x})^{1-\lambda\rho} \right) \left(\sum_{\underline{x}'} q_N(\underline{x}')\, p_N(\underline{y}|\underline{x}')^{\lambda} \right)^{\rho} \right\}. \tag{4.26}$$

Letting $\lambda = \frac{1}{1+\rho}$, the standard random coding Gallager bound [123] results. It is hence demonstrated that in the standard random coding setting, no penalty is incurred for invoking the Jensen inequality in the optimized DS2 bound.

4.3.2 Relations to the 1961 Gallager-Fano Bound

In this subsection we study the relation of the 1961 Gallager-Fano bound [122] to the DS2 bound.

Let us assume that a codeword $\underline{x}^m$ was transmitted. From the Chernoff inequality in Eqs. (4.14) and (4.15) we obtain that for arbitrary $s \geq 0$ and $d \in \mathbb{R}$:

$$\begin{aligned} \text{Prob}\,(\underline{y} \in \mathcal{Y}_b^N) &\leq \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ \exp\Big(s\ (D_m(\underline{x}^m, \underline{y}) - Nd)\Big) \\ &= e^{-nsd} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right)^s . \end{aligned} \tag{4.27}$$

Similarly, we obtain from the Chernoff upper bound in (4.16) that given the codeword $\underline{x}^m$ was transmitted, then the conditional joint probability of having the observation vector in the good region which is associated with the 1961 Gallager-Fano bound (i.e., the event that the received vector $\underline{y}$ is inside $\mathcal{Y}_g^N$), and also that the ML decoder makes an error, satisfies the following inequality for arbitrary $t, r \leq 0$ and $d \in \mathbb{R}$:

$$\begin{aligned} &\text{Prob}\,(\text{decoding error}, \underline{y} \in \mathcal{Y}_g^N) \\ &\leq \sum_{m' \neq m} p_N(\underline{y}|\underline{x}^m)\ \exp\left(tZ_{m'} + rW\right) \\ &= e^{-nrd} \sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \left(\frac{p_N(\underline{y}|\underline{x}^m)}{p_N(\underline{y}|\underline{x}^{m'})}\right)^t \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right)^r . \end{aligned} \tag{4.28}$$

Based on the upper bound on the conditional ML decoding error probability (4.13), and from Eqs. (4.27), (4.28), we obtain that

$$\begin{aligned} P_{\text{e}|m} &\leq e^{-nsd} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right)^s \\ &\quad + e^{-nrd} \sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \left(\frac{p_N(\underline{y}|\underline{x}^m)}{p_N(\underline{y}|\underline{x}^{m'})}\right)^t \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right)^r \end{aligned}$$

where $s \geq 0$, $r \leq 0$, $t \leq 0$, and $d \in \mathbb{R}$. In a similar way to the optimization of the parameter d in Section 4.2.3, by doing the same kind of optimization in the bound above, and by substituting $\rho = \frac{s}{s-r}$ $(0 \leq \rho \leq 1)$ in the resulting optimized bound (w.r.t. the parameter d), one obtains the following upper bound on the conditional ML decoding error probability:

$$\begin{aligned} P_{\text{e}|m} &\leq 2^{H(\rho)} \left(\sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)}\right)^s\right)^{1-\rho} \\ &\quad \left(\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)\ \left(\frac{p_N(\underline{y}|\underline{x}^m)}{p_N(\underline{y}|\underline{x}^{m'})}\right)^t\ \left(\frac{f_N^m(\underline{y})}{p_N^m(\underline{y}|\underline{x}^m)}\right)^{s\left(1-\frac{1}{\rho}\right)}\right)^{\rho}, \\ &\quad 0 \leq \rho \leq 1 \end{aligned} \tag{4.29}$$

as was first indicated by Divsalar [86, Eqs. (71), (72)].

Divsalar [86] has renamed $-t$ by λ (where $t \leq 0$ is introduced in (4.16), so $\lambda \geq 0$ as required in (4.9)), and has also set

$$G_N^m(\underline{y}) = \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)} \right)^s \tag{4.30}$$

deriving then the DS2 bound (4.9) with an additional factor of $2^{H(\rho)}$ (where $1 \leq 2^{H(\rho)} \leq 2$). This demonstrates the superiority of the DS2 bound over the 1961 Gallager-Fano bounding technique when applied to a particular code or ensemble of codes. It has been demonstrated in [281] that the 1961 Gallager-Fano bound equals the 1965 Gallager random coding bound (4.26) up to the $2^{H(\rho)}$ coefficient, where as shown before, the latter bound agrees with the optimized DS2 bound.

For the derivation of the 1961 Gallager-Fano bound for arbitrary MBIOS channels, the function $f_N^m(\underline{y})$ was assumed to be expressible in the product form of (4.17) (see [122, Chapter 3]). From the interesting connection between the DS2 bound and the 1961 Gallager-Fano bound, we arrive to the following conclusion w.r.t. the application of these two bounds to MBIOS channels. Since the un-normalized tilting measure $G_N^m(\underline{y})$ in the DS2 bound (4.9) is an arbitrary non-negative function, and also based on the relation in (4.30), then we conclude that the one-dimensional function $f(\cdot)$ in the RHS of (4.17) should not be in fact an even function (but should be only a non-negative function); this therefore makes the general form of the DS2 bound when applied to MBIOS channels valid under wider conditions, as compared to the 1961 Gallager-Fano bound.

4.3.3 Geometric Interpretation of the Gallager-Type Bounds

The connections between the Gallager-Fano tilting measure and the Duman and Salehi normalized and un-normalized tilting measures (which are designated here by $f_N^m(\underline{y})$, $\psi_N^m(\underline{y})$ and $G_N^m(\underline{y})$, respectively) are indicated in (4.8) and (4.30). We will see that these connections also provide some geometric interpretations of various reported bounds. The measure $f_N^m(\underline{y})$ in the 1961 Gallager-Fano bound, which in general does not imply a product form, entails a geometric interpretation associated with the conditions in the inequalities (4.11), specifying the disjoint regions $\mathcal{Y}_g^N$, $\mathcal{Y}_b^N \subseteq \mathcal{Y}^N$. The geometric interpretation of the 1961 Gallager-Fano bound is not necessarily unique, as measures $f_N^m(\underline{y})$ of different functional structure may imply equivalent conditions in the inequality

$$\ln \left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)} \right) \leq Nd. \tag{4.31}$$

The non-uniqueness of the measures $f_N^m(\underline{y})$ in this respect is due to the shifting and factoring invariance of inequality (4.31), and since the parameter d in (4.18) is subjected to optimization. We demonstrate here the connection between the non-unique measure $f_N^m(\underline{y})$ and its associated decision region, and exemplify the non-uniqueness property by focusing on the Divsalar bound [86] (see also Section 3.2.4 in p. 25).

The conditional pdf for the binary-input AWGN channel is

$$p_N(\underline{y}|\underline{x}^m) = \prod_{l=1}^{N} \left\{ \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} \, (y_l - \gamma \, x_l^m)^2 \right] \right\} \tag{4.32}$$

where $\gamma \triangleq \sqrt{\frac{2RE_b}{N_0}}$. Here x_l^m designates the l-th symbol of the m-th codeword, where 0 and 1 are mapped to 1 and -1, respectively.

Divsalar bound [86] is specified by the associated decision region

$$\mathcal{Y}_g^N = \left\{ \underline{y} \, | \, \sum_{l=1}^{N} (y_l - \eta\gamma \, x_l^m)^2 \leq N \, r^2 \right\} \tag{4.33}$$

which is an N-dimensional sphere whose center that is located along the line connecting the origin to the codeword $\underline{x}^m$ (see Fig. 3.2 in p. 26). The parameters r, η are analytically optimized in Divsalar bound [86]. It can be verified that the following Gallager-Fano tilting measures in (4.31) imply the same decision region in (4.33):

$$f_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \exp\left(\frac{(1-\eta)\, y_l^2}{2\eta} \right) \right\}, \tag{4.34a}$$

$$f_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \exp\Big(\gamma\, (1-\eta)\, x_l^m\, y_l \Big) \right\}, \tag{4.34b}$$

$$f_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \exp\Big(\theta\, (y_l - \phi\, x_l^m)^2 \Big) \right\} \tag{4.34c}$$

$$\text{where } \phi \triangleq \gamma \left(\eta + \frac{\eta - 1}{2\theta} \right),\ \theta > -\frac{1}{2}.$$

In the following, we choose to prove that the measure $f_N^m(\underline{y})$ in (4.34c) yields the geometrical region in (4.33).

Proof. From Eqs. (4.32) and (4.34c), we get

$$\begin{aligned}
&\ln\left(\frac{f_N^m(\underline{y})}{p_N(\underline{y}|\underline{x}^m)} \right) \\
&= \ln\left(\frac{\displaystyle\prod_{l=1}^{N} \Big\{ \exp\Big(\theta\, (y_l - \phi\, x_l^m)^2 \Big) \Big\}}{\displaystyle\prod_{l=1}^{N} \left\{ \frac{1}{\sqrt{2\pi}}\, \exp\Big(-\frac{1}{2}\, (y_l - \gamma\, x_l^m)^2 \Big) \right\}} \right) \\
&= \frac{N}{2}\, \ln(2\pi) + \sum_{l=1}^{N} \Big\{ \theta\, (y_l - \phi\, x_l^m)^2 + \frac{1}{2}\, (y_l - \gamma x_l^m)^2 \Big\} \\
&= \left(\theta + \frac{1}{2} \right) \sum_{l=1}^{N} y_l^2 - (2\theta\phi + \gamma) \sum_{l=1}^{N} x_l^m y_l + N \left(\theta\, \phi^2 + \frac{\gamma^2}{2} + \frac{\ln(2\pi)}{2} \right) \\
&= \left(\theta + \frac{1}{2} \right) \sum_{l=1}^{N} \left(y_l - \frac{2\theta\phi + \gamma}{2\theta + 1} \cdot x_l^m \right)^2 + N \left[\theta\, \phi^2 + \frac{\gamma^2}{2} + \frac{\ln(2\pi)}{2} \right. \\
&\qquad\qquad \left. - \frac{(2\theta\phi + \gamma)^2}{2(2\theta + 1)} \right].
\end{aligned} \tag{4.35}$$

The equivalence between the geometrical regions defined in (4.31) and (4.33) in invariant to a shift in the value of the parameter d in (4.31), and to a factoring by a positive constant in (4.33). The reason for this invariance is because the two parameters d and r in (4.31) and (4.33), respectively, are subjected to optimizations. This implies that we need $\theta + \frac{1}{2}$ to be positive, and by comparing (4.33) with (4.35), then we also require the following equation to be satisfied:

$$\frac{2\theta\phi + \gamma}{2\theta + 1} = \gamma\eta$$

which yields the conditions in (4.34c). □

We note that although the three tilting measures $f_N^m(\underline{y})$ in (4.34a)–(4.34c) imply the same decision region $\mathcal{Y}_g^N$, the second and third measures (in (4.34b) and (4.34c), respectively) are amenable to generalizations for the class of fully interleaved fading channels, as opposed to the first measure (since the exponential term of the first tilting measure in (4.34a) is quadratic in $\{y_l\}_{l=1}^N$, but does not depend on the coordinates of the transmitted codeword $\{x_l^m\}_{l=1}^N$); it therefore demonstrates a different functional behavior of these tilting measures.

The geometrical region which is associated with the sphere bound (see Section 3.2.5 in p. 29) is an N-dimensional sphere whose center coincides with the transmitted codeword (so this region is a particular case of (4.33) where $\eta = 1$). For the sphere bound, the tilting measures in (4.34a) and (4.34b) are identical (i.e., $f_N^m(\underline{y}) \equiv 1$), and the third tilting measure in (4.34c) gets the form

$$f_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \exp\Big(\theta \, (y_l - \gamma \, x_l^m)^2 \Big) \right\} , \quad \theta > -\frac{1}{2} \, .$$

4.4 Special Cases of the DS2 Bound

In this section we demonstrate that *many reported upper bounds can be considered as special cases of the DS2 bound* [1] (see Section 4.2.2 in p. 60). These observations rely on the material presented in [86, 281, 296].

4.4.1 Shulman and Feder Bound

In [303], Shulman and Feder derived an upper bound on the ML decoding error probability of linear block codes. Their bound forms a generalization of the random coding bound (see (4.3) and (4.4) in p. 59), and is valid for an arbitrary MBIOS channel. Unlike the random coding bound, the Shulman and Feder bound can be also applied to structured ensembles of linear block codes or specific linear block codes. For structured ensembles of linear block codes, the Shulman and Feder bound resembles to the random coding bound of Gallager, up to a multiplicative term which depends on the maximum ratio between the average distance spectrum of the considered ensemble and the average distance spectrum of fully random block codes with the same rate and block length (note that the distance spectrum of fully random block codes is binomially distributed).

Unlike the original derivation of the Shulman and Feder bound in [303], we derive this bound here as a particular case of the DS2 bound. In the continuation, we discuss the implications of the bound. Finally, we consider a modified bound which properly combines the Shulman and Feder bound with the union bound, and discuss the tightness of the resulting bound.

Let $\mathcal{C}$ be a binary linear block code of length N and rate $R = \frac{\log_2 M}{N}$ bits per channel use, with a distance spectrum $\{S_k\}_{k=0}^N$ (if we consider an ensemble of codes, then we refer to the average distance spectrum of this ensemble). Suppose that the communication takes place over an MBIOS channel, so due to the linearity of the binary code, then the average ML decoding error probability is equal to the conditional error probability, given that an arbitrary codeword of the code $\mathcal{C}$ is transmitted. Without any loss of generality, we can assume for the simplicity of the analysis that the all-zero codeword is transmitted, and we let $p(y|x)$ designate the transition probability of the MBIOS channel. In order to express the upper bound (4.9) in terms of the distance spectrum of

[1] The first version of the Duman and Salehi bounds which is introduced in this section is shown to be a particular case of the DS2 bound. Therefore, we introduced the latter bound in Section 4.2, before the presentation of their first version bound in Section 4.4.

the block code $\mathcal{C}$, we restrict our attention here to the special case where the un-normalized Duman and Salehi tilting measure $G_N^0(\underline{y})$ is expressed in a product form $G_N^0(\underline{y}) = \prod_{i=1}^{N} g(y_i)$. From (4.9), we obtain the following upper bound on the conditional ML decoding error probability which is valid for $0 \leq \rho \leq 1$:

$$\begin{aligned}
P_{\mathrm{e}|0} &\leq \left(\sum_y g(y)\, p(y|0)\right)^{N(1-\rho)} \\
&\quad \cdot \left\{\sum_{l=0}^{N} S_l \left(\sum_y p(y|0) g(y)^{1-\frac{1}{\rho}}\right)^{N-l} \left(\sum_y p(y|0)^{1-\lambda} p(y|1)^{\lambda} g(y)^{1-\frac{1}{\rho}}\right)^{l}\right\}^{\rho} \\
&= \left(\sum_y g(y)\, p(y|0)\right)^{N(1-\rho)} 2^{-N(1-R)\rho} \\
&\quad \left\{\sum_{l=0}^{N} \left(\frac{S_l}{2^{-N(1-R)}\binom{N}{l}}\right) \binom{N}{l} \left(\sum_y p(y|0) g(y)^{1-\frac{1}{\rho}}\right)^{N-l} \right. \\
&\qquad \left. \cdot \left(\sum_y p(y|0)^{1-\lambda} p(y|1)^{\lambda} g(y)^{1-\frac{1}{\rho}}\right)^{l}\right\}^{\rho} \\
&\leq \left(\sum_y g(y)\, p(y|0)\right)^{N(1-\rho)} \left(\max_{0\leq l\leq N} \frac{S_l}{2^{-N(1-R)}\binom{N}{l}}\right)^{\rho} 2^{-N(1-R)\rho} \\
&\quad \left\{\sum_{l=0}^{N} \binom{N}{l} \left(\sum_y p(y|0) g(y)^{1-\frac{1}{\rho}}\right)^{N-l} \left(\sum_y p(y|0)^{1-\lambda} p(y|1)^{\lambda} g(y)^{1-\frac{1}{\rho}}\right)^{l}\right\}^{\rho} \\
&= \left(\sum_y g(y)\, p(y|0)\right)^{N(1-\rho)} \left(\max_{0\leq l\leq N} \frac{S_l}{2^{-N(1-R)}\binom{N}{l}}\right)^{\rho} 2^{-N(1-R)\rho} \\
&\quad \left(\sum_y p(y|0)\, g(y)^{1-\frac{1}{\rho}} + \sum_y p(y|0)^{1-\lambda}\, p(y|1)^{\lambda}\, g(y)^{1-\frac{1}{\rho}}\right)^{N\rho}. \qquad (4.36)
\end{aligned}$$

By setting

$$g(y) = \left[\frac{1}{2}\, p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\, p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho} p(y|0)^{-\frac{\rho}{1+\rho}}\ ,\ \lambda = \frac{1}{1+\rho} \qquad (4.37)$$

we obtain from (4.36), (4.37) and the symmetry of the channel that

$$\begin{aligned}
\sum_y g(y)\, p(y|0) &= \sum_y \left[\frac{1}{2}\, p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\, p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho} p(y|0)^{\frac{1}{1+\rho}} \\
&= \sum_y \left[\frac{1}{2}\, p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\, p(y|1)^{\frac{1}{1+\rho}}\right]^{1+\rho} \qquad (4.38)
\end{aligned}$$

and

$$\begin{aligned}
&\sum_y p(y|0)\, g(y)^{1-\frac{1}{\rho}} + \sum_y p(y|0)^{1-\lambda}\, p(y|1)^{\lambda}\, g(y)^{1-\frac{1}{\rho}} \\
&= \sum_y p(y|0)^{\frac{2}{1+\rho}} \cdot \left[\frac{1}{2}\; p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\; p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho-1} \\
&\quad + \sum_y \left[p(y|0)\, p(y|1)\right]^{\frac{1}{1+\rho}} \left[\frac{1}{2}\; p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\; p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho-1} \\
&= \frac{1}{2}\sum_y \left[p(y|0)^{\frac{2}{1+\rho}} + p(y|1)^{\frac{2}{1+\rho}}\right] \cdot \left[\frac{1}{2}\; p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\; p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho-1} \\
&\quad + \sum_y \left[p(y|0)\, p(y|1)\right]^{\frac{1}{1+\rho}} \left[\frac{1}{2}\; p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\; p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho-1} \\
&= \frac{1}{2}\sum_y \left[p(y|0)^{\frac{1}{1+\rho}} + p(y|1)^{\frac{1}{1+\rho}}\right]^2 \cdot \left[\frac{1}{2}\; p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\; p(y|1)^{\frac{1}{1+\rho}}\right]^{\rho-1} \\
&= 2\sum_y \left[\frac{1}{2}\, p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\, p(y|1)^{\frac{1}{1+\rho}}\right]^{1+\rho} . \qquad (4.39)
\end{aligned}$$

Finally, the substitution of (4.38) and (4.39) into (4.36) gives the following upper bound for $0 \leq \rho \leq 1$:

$$\begin{aligned}
P_{\mathrm{e}|0} \;\leq\; & 2^{NR\rho} \left(\sum_y \left[\frac{1}{2}\, p(y|0)^{\frac{1}{1+\rho}} + \frac{1}{2}\, p(y|1)^{\frac{1}{1+\rho}}\right]^{1+\rho}\right)^N \\
& \cdot \left(\max_{0\leq l\leq N} \frac{S_l}{2^{-N(1-R)}\binom{N}{l}}\right)^{\rho} \qquad (4.40)
\end{aligned}$$

which agrees with Shulman and Feder bound [303]. Based on the DS2 bound, a possible generalization of the Shulman and Feder bound is detailed in [296, Appendix A].

We will consider now the implications of the Shulman and Feder bound. First, it can be easily verified that for the ensemble of fully random and binary block codes, the Shulman and Feder bound (4.40) coincides with the random coding bound of Gallager (see (4.3) and (4.4) in p. 59). We consider now the Shulman and Feder bound for a general ensemble of binary linear block codes $\mathcal{C}$ where we let the block length tend to infinity. Let $r_{\mathcal{C}}(\delta)$ designate the asymptotic growth rate of the average distance spectrum of the considered ensemble. Based on our notation

$$S_l = e^{N\cdot r_{\mathcal{C}}(\delta)}, \quad \delta \triangleq \frac{l}{N}.$$

For the ensemble of fully random binary block codes of length N and rate R, the asymptotic growth rate of its average distance spectrum is given by

$$\begin{aligned}
&\lim_{N\to\infty} \frac{\ln\left(2^{-N(1-R)}\binom{N}{l}\right)}{N} \\
&= -(1-R)\ln 2 + \lim_{N\to\infty} \frac{1}{N}\; \ln\binom{N}{N\delta} \\
&= h(\delta) - (1-R)\ln 2
\end{aligned}$$

where $h(x) = -x \ln x - (1-x) \ln(1-x)$ designates the binary entropy function to base e. In the asymptotic case where the block length of $\mathcal{C}$ tends to infinity, the last term in the RHS of (4.40) is equal to

$$\left(\max_{0 \le l \le N} \frac{S_l}{2^{-N(1-R)} \binom{N}{l}} \right)^{\rho} = e^{N\rho \max\limits_{0 \le \delta \le 1} \Big[r_{\mathcal{C}}(\delta) - \big(h(\delta) - (1-R) \ln 2 \big) \Big]}. \tag{4.41}$$

From (4.2)–(4.4) and (4.41), we obtain that the Shulman and Feder bound in (4.40) gets the form

$$\overline{P_{\mathrm{e}}} = P_{\mathrm{e}|0} \le e^{-N \cdot E(R + \Delta R, \, \underline{q})} \tag{4.42}$$

where $\underline{q} = (\frac{1}{2}, \frac{1}{2})$ is the input distribution which achieves the capacity of an arbitrary MBIOS channel, and

$$\Delta R \triangleq \max_{0 \le \delta \le 1} \Big[r_{\mathcal{C}}(\delta) - \big(h(\delta) - (1-R) \ln 2 \big) \Big]. \tag{4.43}$$

We therefore conclude that for a general ensemble of binary linear block codes $\mathcal{C}$, the effective rate in the error exponent (4.42) of the Shulman and Feder bound is increased by the amount ΔR given in (4.43). The all-zero vector is always a valid codeword of a linear code, so $r(\delta = 0) \ge 0$ for an arbitrary ensemble of binary linear block codes; it readily follows from (4.43) that $\Delta R \ge (1 - R) \ln 2$ where we express here the rates in units of bits per channel use. In order to compare our result with the random coding bound in (4.2), we express R in units of nats per channel use, and obtain that $\Delta R \ge \ln 2 - R$. This implies a severe penalty in the increase of the effective rate in (4.42) due to the fact that the average distance spectrum of the considered ensemble of linear codes deviates from the binomial distribution of the average distance spectrum of the ensemble of fully random block codes; the two distance spectra typically deviate considerably at the portions of low and high Hamming weights (i.e., the exponential growth rate of the two distance spectra differ considerably when the normalized Hamming weight, δ, is close to 0 or 1, and on the other hand, they are close enough near $\delta = \frac{1}{2}$). As a consequence of the shift of the effective rate by ΔR, the error exponent of the random coding bound $E(R)$ in (4.3) is changed to $E(\ln 2)$ where unfortunately the latter error exponent is equal to zero (since a rate of $\ln 2$ nats per channel use is equivalent to a rate of 1 bit per channel use). The fact that the error exponent of the Shulman and Feder bound in (4.40) is equal to zero for an arbitrary ensemble of binary linear block codes is an artifact of the tightness of this bound. In order to overcome this obstacle, Miller and Burshtein suggested in [222, Theorem 1] to combine the ubiquitous union bound and the Shulman and Feder bound in a sophisticated manner. The idea of combining the union bound and the Shulman and Feder bound is based on using the union bound at the portion of the Hamming weights where the asymptotic growth rate of the distance spectrum of the considered ensemble deviates considerably from the one which corresponds to the ensemble of fully random block codes, and on the other hand, using the Shulman and Feder bound at the complementary portion of the Hamming weights where the asymptotic exponents of the two distance spectra of are close enough. An upper bound on the ML decoding error probability which combines the Shulman and Feder bound with the union bound was used for the analysis of LDPC codes under ML decoding (see [222] and [290, Theorem 2]). This generalized bound was used in order to prove that for an arbitrary MBIOS channel, properly chosen ensembles of regular LDPC codes achieve under ML decoding rates which can be made arbitrarily close to the channel capacity.[2] Based on the same bounding technique which combines the union bound with the Shulman and Feder bound, the performance of punctured LDPC codes under ML decoding was studied in [146]. It is shown in this paper that under ML decoding, capacity-achieving codes of

[2] We note that a parallel result was not proved yet for any ensemble of LDPC codes under a sub-optimal iterative message-passing decoding algorithm. For ensembles of regular LDPC codes, it is well known that by increasing the degrees of the variable nodes and the parity-check nodes so that the design rate stays constant, the achievable rates of ensembles of regular LDPC codes improve under ML decoding, though their achievable rates under iterative message-passing decoding degrade.

any rate and for any MBIOS channel can be constructed by puncturing some original LDPC codes with small enough rate; this indicates the high potential of rate-compatible puncturing to be used in designing capacity-achieving codes for an arbitrary MBIOS channel (see also [133] for the design and analysis of punctured LDPC codes under iterative message-passing decoding). These results demonstrate that the generalized bound which properly combines the Shulman and Feder bound with the union bound provides a rather powerful bounding technique for the ML analysis of various ensembles of LDPC codes (or other capacity-approaching ensembles of linear codes). We refer the reader to [36] where Burshtein and Bennatan stated this bounding technique for ensembles of non-binary codes, and applied their bounds to various ensembles of non-binary LDPC codes which are communicated over an arbitrary discrete memoryless channel.

4.4.2 Gallager-Type Bounds for the Binary-Input AWGN Channel

In this section, we present various upper bounds on the ML decoding error probability for the binary-input AWGN channels; the derivation of these bounds rely on the DS2 bound which is a variation of the Gallager bounding technique. We show that these reported bounds are special cases of the DS2 bound. In general, we prove it by choosing appropriate probability tilting measures which enable to derive these bounds as special cases of the DS2 bound.

Let $\mathcal{C}$ be a binary linear block code of length N. Throughout this section, $\delta \triangleq \frac{d}{N}$ (where $0 \leq \delta \leq 1$) designates the normalized Hamming weight of an arbitrary codeword of $\mathcal{C}$ whose Hamming weight is equal to d, and $r_N(\delta) \triangleq \frac{\ln S_d}{N}$ designates the exponential growth rate of the distance spectrum of $\mathcal{C}$ as a function of δ.

The derivation of the upper bounds in this section relies on the partitioning of the code $\mathcal{C}$ into constant Hamming weight subcodes $\{\mathcal{C}_d\}_{d=0}^{N}$ where the subcode $\mathcal{C}_d$ includes all the codewords of Hamming weight d, and also the all-zero codeword. We note that although the subcodes $\{\mathcal{C}_d\}$ are not necessarily linear (since the sum of two codewords of Hamming weight d is not necessarily a codeword of such a Hamming weight), then for every subcode $\mathcal{C}_d$, the upper bound on the *conditional* error probability (4.7) (see p. 60) is still valid; the reason is that the property of the linearity of the code is not required for the validity of (4.7).

Due to the channel symmetry and the linearity of the code, an overall union bound over the subcodes yields that

$$P_{\mathrm{e}} = P_{\mathrm{e}|0} \leq \sum_{d=d_{\min}}^{N} P_{\mathrm{e}|0}(d) \tag{4.44}$$

where $d_{\min}$ is the minimum Hamming distance of the code $\mathcal{C}$, and $P_{\mathrm{e}|0}(d)$ is the conditional ML decoding error probability of the subcode $\mathcal{C}_d$, given the all-zero codeword is transmitted. In the continuation to this section, we will apply the DS2 bound with different tilting measures in order to re-derive previously reported upper bounds as special cases of the former bound. From (4.44), it suffices to obtain bounds on $P_{\mathrm{e}|0}(d)$.

1. **Duman and Salehi Bound (First Version)**

   The Duman and Salehi bound in [94] (which is named here as the first version of the Duman and Salehi bounds) is a special case of the DS2 bound [95]. More specifically, the bound in [94] is a special case of the generalized DS2 bound in (4.7) where the normalized tilting measure is

given by

$$\psi_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \sqrt{\frac{\alpha}{2\pi}} \exp\left[-\frac{\alpha}{2} \left(y_l - \frac{\beta}{\alpha} \sqrt{\frac{2E_{\mathrm{s}}}{N_0}}\, x_l^m \right)^2 \right] \right\}$$
$$\alpha > 0, \quad \beta \in \mathbb{R}. \tag{4.45}$$

We assume here that the components of the codeword $\underline{x}^m$ are either $+1$ or -1 for a '0' or '1' input, respectively. By (4.7), we obtain the following upper bound on $P_{\mathrm{e}|0}(d)$:

$$P_{\mathrm{e}|0}(d) \leq (S_d)^{\rho} \left(\int_{-\infty}^{+\infty} p_0(y)^{\frac{1}{\rho}}\, \psi(y)^{1-\frac{1}{\rho}}\, dy \right)^{(N-d)\rho}$$
$$\cdot \left(\int_{-\infty}^{+\infty} p_0(y)^{\frac{1-\lambda\rho}{\rho}}\, p_1(y)^{\lambda}\, \psi(y)^{1-\frac{1}{\rho}}\, dy \right)^{d\rho}$$
$$0 \leq \rho \leq 1, \ \lambda > 0 \tag{4.46}$$

where from (4.45), and given that the all-zero codeword is transmitted, then we obtain that

$$\phi(y) = \sqrt{\frac{\alpha}{2\pi}} \exp\left[-\frac{\alpha}{2} \left(y - \frac{\beta}{\alpha} \sqrt{\frac{2E_{\mathrm{s}}}{N_0}} \right)^2 \right], \ y \in \mathbb{R} \tag{4.47}$$

and the probability density functions

$$p_0(y) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(y - \sqrt{\frac{2RE_{\mathrm{b}}}{N_0}} \right)^2 \right],$$
$$p_1(y) = p_0(-y), \quad y \in \mathbb{R} \tag{4.48}$$

refer to the binary-input AWGN channel. The substitution of Eqs. (4.45) and (4.48) into (4.46), and the optimization over the parameters $\lambda \geq 0$ and $\beta \in \mathbb{R}$ in this bound gives (see [94])

$$\beta^* = \frac{1 - \frac{d}{N}}{\frac{1}{\alpha} - \frac{d}{N}(1-\rho)}, \quad \lambda^* = \frac{1}{2} \left(\beta^* + \frac{1-\beta^*}{\rho} \right). \tag{4.49}$$

The closed-form expressions of the integrals in (4.46) for the specific probability measures $\psi(\cdot)$ and $p_0(\cdot)$ in (4.47) and (4.48), respectively, finally gives the following upper bound:

$$P_{\mathrm{e}|0}(d) \leq (S_d)^{\rho}\, \alpha^{\frac{N(1-\rho)}{2}} \left(\alpha - \frac{\alpha - 1}{\rho} \right)^{-\frac{N\rho}{2}}$$
$$\exp\left\{ \frac{NRE_{\mathrm{b}}}{N_0} \left[-1 + \frac{(\beta^*)^2(1-\rho)}{\alpha} \right.\right.$$
$$\left.\left. + \frac{\rho\left(1 - \frac{d}{N}\right)\left(\beta^* + \frac{1-\beta^*}{\rho}\right)^2}{\alpha - \frac{\alpha-1}{\rho}} \right] \right\}. \tag{4.50}$$

The bound in (4.50) is the first version of the Duman and Salehi bounds [94], and it is numerically optimized over the range $0 < \alpha < \frac{1}{1-\rho}$, $0 < \rho \leq 1$ (we note that the reason for the range of α as above is because we get from (4.50) that $\alpha - \frac{\alpha-1}{\rho}$ should be positive, and also $\alpha > 0$ is a requirement in (4.45)).

Based on our notation for the normalized Hamming weight (δ) and the exponential growth rate of the distance spectrum ($r_N(\delta)$) of a linear block code, then the bound in (4.46) is also expressible in the following equivalent form:

$$P_{\mathrm{e}|0}(d) \leq \left\{ e^{\rho r_N(\delta)} \left(\int_{-\infty}^{+\infty} p_0(y)^{\frac{1}{\rho}} \, \psi(y)^{1-\frac{1}{\rho}} \, dy \right)^{(1-\delta)\rho} \right. \\ \left. \cdot \left(\int_{-\infty}^{+\infty} p_0(y)^{\frac{1-\lambda\rho}{\rho}} \, p_1(y)^{\lambda} \, \psi(y)^{1-\frac{1}{\rho}} \, dy \right)^{\delta\rho} \right\}^N \\ 0 \leq \rho \leq 1, \ \lambda > 0. \tag{4.51}$$

This form will be later useful for the discussion on the $\frac{E_\mathrm{b}}{N_0}$–thresholds of linear codes (see Section 4.7.3 in p. 87).

2. **Viterbi and Viterbi Bound (First Version)**

The Viterbi and Viterbi bound [347] is an upper bound on the ML decoding error probability for BPSK modulated block codes operating over a binary-input AWGN channel. It can be verified that it is also a particular case of the first version of Duman and Salehi bounds by substituting $\alpha = 1$ in (4.49) and (4.50), which yields the following conditional upper bound on the ML decoding error probability:

$$P_{\mathrm{e}|0}(d) \leq (S_d)^\rho \, \exp\left(-\frac{NRE_\mathrm{b}}{N_0} \, \frac{\left(\frac{d}{N}\right) \rho}{1 - \frac{d}{N}\,(1-\rho)} \right), \quad 0 \leq \rho \leq 1 \, .$$

The optimization over the parameter ρ then gives the Viterbi & Viterbi upper bound [347], which reads

$$P_{\mathrm{e}|0}(d) \leq \exp\Big(-N \, E_{v_1}(\delta) \Big)$$

where

$$E_{v_1}(\delta) = \begin{cases} \delta c - r(\delta) & , \quad 0 \leq \dfrac{r(\delta)}{c} \leq \delta\,(1-\delta) \\ \left(\sqrt{c} - \sqrt{\dfrac{(1-\delta)\, r(\delta)}{\delta}} \right)^2 & , \quad \delta\,(1-\delta) \leq \dfrac{r(\delta)}{c} \leq \dfrac{\delta}{1-\delta} \end{cases}$$

and

$$\delta \triangleq \frac{d}{N} \, , \ \ r(\delta) \triangleq \frac{\ln(S_d)}{N} \, , \ \ c \triangleq \frac{E_\mathrm{s}}{N_0} = \frac{RE_\mathrm{b}}{N_0}.$$

3. **Viterbi and Viterbi Bound (Second Version)**

The second version of the Viterbi and Viterbi bounds [348] is based on the 1961 Gallager-Fano bound, and is valid for an arbitrary MBIOS channel. This bound reads

$$P_{\mathrm{e}|0}(d) \leq \exp\Big(-N \, E_{v_2}(\delta) \Big)$$

where

$$E_{v_2}(\delta) = \max_{0 \leq \rho \leq 1} \Big\{ -\rho \, r(\delta) + \delta \, \ln\Big(\bar{h}(\rho)\Big) + (1-\delta) \, \ln\Big(\bar{g}(\rho)\Big) \\ -(1-\rho) \, \ln\Big(\bar{h}(\rho) + \bar{g}(\rho)\Big) \Big\} \, .$$

From (4.19) and (4.22), and the substitution $\rho = \frac{s}{s-r}$ (so since, $s \geq 0$, $r \leq 0$, then $0 \leq \rho \leq 1$), we obtain that

$$\bar{h}(\rho) \triangleq h(r) = \sum_{y} \left\{ \left[p_0(y)^{\frac{1}{1+\rho}} + p_0(-y)^{\frac{1}{1+\rho}} \right]^{-(1-\rho)} \left[p_0(y)\, p_0(-y) \right]^{\frac{1}{1+\rho}} \right\}$$

$$\bar{g}(\rho) \triangleq g(r) = \sum_{y} \left\{ \left[p_0(y)^{\frac{1}{1+\rho}} + p_0(-y)^{\frac{1}{1+\rho}} \right]^{-(1-\rho)} p_0(y)^{\frac{2}{1+\rho}} \right\} .$$

For the binary-input AWGN channel, $p_0(\cdot)$ is introduced in (4.48). Clearly, for channels with continuous output, the sums above should be replaced by integrals.

The second version of the Viterbi & Viterbi bound is again a special case of the DS2 bound, as noticed by the substitution in (4.9) of the un-normalized tilting measure

$$G_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \left(p(y_l|0)^{\frac{1}{1+\rho}} + p(y_l|1)^{\frac{1}{1+\rho}} \right)^{\rho} p(y_l|0)^{-\frac{\rho}{1+\rho}} \right\} .$$

4. **Divsalar Bound**

The geometric interpretation of the bound of Divsalar [86] is shown in Fig. 3.2 (see p. 26). We show in Section 4.3.3 (see p. 66) that the spherical region which is shown in Fig. 3.2 is equivalently represented by the condition in (4.31) with an appropriate (and non-unique) tilting measure of the 1961 Gallager-Fano bound (for some possible choices of this tilting measure, see e.g., (4.34a)–(4.34c)).

The following connection between the Gallager-Fano tilting measure ($f_N^m(\underline{y})$) and the normalized tilting measure ($\psi_N^m(\underline{y})$) in the DS2 bound follows from (4.8) and (4.30):

$$\psi_N^m(\underline{y}) = \frac{[f_N^m(\underline{y})]^s \, [p_N^m(\underline{y}|\underline{x}^m)]^{1-s}}{\sum_{\underline{y}} \left\{ [f_N^m(\underline{y})]^s \, [p_N^m(\underline{y}|\underline{x}^m)]^{1-s} \right\}} . \tag{4.52}$$

Substituting the tilting measure (4.34c) and the *pdf* (4.32) into (4.52), yields that the corresponding normalized tilting measure (ψ_N^m) for the Divsalar bound coincides with the one in (4.45). This result is verified by setting the parameters as follows:

$$\alpha = 1 - (2\theta + 1)s \,, \quad \beta = 1 - (2\theta + 1)\eta s. \tag{4.53}$$

This is the grounds for the observation in [86] that Divsalar bound is a closed form expression of the Duman and Salehi (first version) bound [94].

5. **The EZ Bound**

For the EZ bound [103] (see also Section 3.2.6 in p. 30) which was derived for a binary-input AWGN channel, the decision region $\mathcal{Y}_g^N$ associated with the transmitted codeword $\underline{x}^m$ is an N-dimensional plane:

$$\mathcal{Y}_g^N = \left\{ \underline{y} \,|\, \sum_{l=1}^{N} y_l \, x_l^m \geq Nd \right\} \tag{4.54}$$

where $d \in \mathbb{R}$ is a parameter of the 1961 Gallager-Fano bound (without abuse of notation, we make it clear in our discussion when the parameter d stands for the real parameter in the 1961

Gallager-Fano bound (see, e.g., (4.18) and (4.31)), and when d designates the Hamming weights of codewords of a linear code).

The motivation for (4.54) is that inside the good region (i.e, $\underline{y} \in \mathcal{Y}_g^N$), the correlation between the received vector $\underline{y}$ and the transmitted codeword $\underline{x}^m$ is not expected to be too small. Therefore, it is above a certain threshold (Nd), where the parameter d is to be optimized, as to get the tightest upper bound within the family of bounds (4.31) and the associated decision region $\mathcal{Y}_g^N$ in (4.54). The following Gallager-Fano tilting measure can be associated with the same decision region $\mathcal{Y}_g^N$:

$$f_N^m(\underline{y}) = \prod_{l=1}^{N} \left\{ \exp\left(-\frac{1}{2}\, y_l^2 + \beta\, y_l\, x_l^m \right) \right\} \tag{4.55}$$

where $\beta \neq \gamma$ (γ is introduced in (4.32)). The free parameter β in (4.55) demonstrates again that there might be some functionally different Gallager-Fano tilting measures which imply the same decision region (based on the definition of the regions in (4.11)). It is interesting to note, that for this specific choice of a decision region (4.54), it was demonstrated in [103] that there is no need to invoke the Chernoff bounds for the binary-input AWGN channel, and the two terms in the right hand side of (4.13) can be exactly calculated.

6. The Chernoff Version of Various Bounds are Gallager-Type Bounds

In his paper [86], Divsalar derived simplified Chernoff versions of some upper bounds, which result as particular cases of the 1961 Gallager-Fano bounding technique. These simplified Chernoff versions include the following bounds:

The Chernoff version of the TSB (see Section 3.2.10 in p. 39): The TSB of Poltyrev [259] happens often to be the tightest reported upper bound for block codes which are transmitted over a binary-input AWGN channel and ML decoded (see e.g., [283], [284]). However, in the random coding setting, it fails to reproduce the random coding exponent [259] while the DS2 bound does. This bound involves a numerical solution of an associated optimization equation ([283, Eq. (4.7)]), and it is therefore not expressed in closed form. In his paper [86], Divsalar derived a simplified Chernoff version of the TSB. That upper bound is shown in [86] to have the same error exponent as the TSB of Poltyrev, and therefore the loosening of the TSB by invoking the Chernoff bounding technique, does not carry any implications on the tightness of the bound for asymptotically infinite block length.

The Chernoff version of the TB (see Section 3.2.10 in p. 43): This version coincides with the first version of the Viterbi and Viterbi bound [347]. It can be shown that the relevant Gallager-Fano decision region is a plane, which is also the case for the EZ bound [103]. However, as noted above, in the latter bound the Chernoff bounding technique is not invoked, improving thus the bound for a finite block length.

The Chernoff version of the sphere bound (see Section 3.2.10 in p. 44): This bound results as a particular case of the decision region in Divsalar bound (4.34), where $\eta = 1$.

Due to the connection of the Gallager-Fano tilting measure to the Duman and Salehi variation, it is evident that the Chernoff versions of these bounds can be also viewed as special cases of the DS2 bound. It should be emphasized that the mentioned bounds above were originally developed without resorting to Chernoff bounding technique, yielding thus tighter versions of these bounds.

4.4.3 Gallager-Type Bounds for Fully Interleaved Fading Channels with Perfect CSI

We demonstrate here various variations of the DS2 bound when applied to the class of fully interleaved Rayleigh fading channels with perfect channel state information at the receiver. This problem is treated in detail in [87], [288], [286] and here we present in a comparative fashion some insightful results. The model is:

$$y = ax + n \tag{4.56}$$

where y stands for the received signal, x stands for the BPSK modulated input signal (that is $\pm\sqrt{2E_{\mathrm{s}}}$) and n designates the additive zero mean and $\frac{N_0}{2}$ variance Gaussian noise component. The fading a is assumed to be perfectly known at the receiver and hence is considered to be real valued, as the receiver compensates for any phase rotation. Due to the ideal interleaving, the channel is assumed to be memoryless. The bounds are based on first decomposing the code to constant-weight subcodes (where every subcode also includes the all-zero codeword), over which a union bound is invoked as in (4.44).

The Optimized DS2 Bound

In a similar way to Section 4.4.2 (see p. 72), the derivation of the upper bounds in this section relies on the partitioning of the code $\mathcal{C}$ into constant Hamming weight subcodes $\{\mathcal{C}_d\}_{d=0}^{N}$ where the subcode $\mathcal{C}_d$ includes all the codewords of Hamming weight d, and also the all-zero codeword. Due to the channel symmetry and the linearity of the code, an overall union bound over the subcodes yields (4.44).

In [286], the measure

$$\Psi(\underline{y},\, \underline{a}) = \prod_{l=1}^{N} \psi(y_l,\, a_l) \tag{4.57}$$

is optimized to yield the tightest conditional DS2 bound (4.7) w.r.t. the subcode $\mathcal{C}_d$ (given the all-zero codeword is transmitted), where (y, a) are interpreted as the available measurements at the receiver.

$$\begin{aligned} P_{\mathrm{e}|0}(d) \leq (S_d)^{\rho} \; \Bigg\{ & \left(\int_{-\infty}^{\infty} \int_{0}^{\infty} \psi(y,a)^{1-\frac{1}{\rho}}\, p_0(y,a)^{\frac{1}{\rho}}\, da\, dy \right)^{(1-\delta)\rho} \\ & \cdot \left(\int_{-\infty}^{\infty} \int_{0}^{\infty} \psi(y,a)^{1-\frac{1}{\rho}}\, p_0(y,a)^{\frac{1-\lambda\rho}{\rho}}\, p_1(y,a)^{\lambda}\, da\, dy \right)^{\delta\rho} \Bigg\}^{N} \\ & 0 < \rho \leq 1, \quad \lambda \geq 0 \end{aligned} \tag{4.58}$$

where $\delta \triangleq \frac{d}{n}$ designates the normalized Hamming weight ($0 \leq \delta \leq 1$), the probability density functions $p_0(\cdot,\cdot)$ and $p_1(\cdot,\cdot)$ are introduced in (3.101) for fully interleaved fading channels, and $\psi(\cdot,\cdot)$ is an arbitrary non-negative function which satisfies the condition

$$\int_{-\infty}^{\infty} \int_{0}^{\infty} \psi(y,a)\, da\, dy = 1\ . \tag{4.59}$$

The function $\psi(\cdot,\cdot)$ can be regarded as a tilting measure depending on the measurements y, a, which are perfectly available to the receiver. Resorting to calculus of variations, we obtain the following

form for the optimal function ψ, as to provide the tightest upper bound of the family above:

$$\psi(y,a) = \beta\, p_0(y,a) \cdot \left(1 + \alpha \left[\frac{p_1(y,a)}{p_0(y,a)}\right]^{\lambda}\right)^{\rho}, \qquad -\infty < y < \infty, \;\; a \geq 0 \tag{4.60}$$

where the parameters α, β are non-negative (see [286, Appendix A]). For specific values of λ, ρ and δ, the parameter α is optimally determined as to satisfy the implicit equation

$$\frac{\displaystyle\int_{-\infty}^{\infty}\int_{0}^{\infty} p_0(y,a) \cdot \left(1 + \alpha \left[\frac{p_1(y,a)}{p_0(y,a)}\right]^{\lambda}\right)^{\rho-1} da\, dy}{\displaystyle\int_{-\infty}^{\infty}\int_{0}^{\infty} p_0(y,a) \cdot \left(1 + \alpha \left[\frac{p_1(y,a)}{p_0(y,a)}\right]^{\lambda}\right)^{\rho} da\, dy} = 1 - \delta. \tag{4.61}$$

The parameter β is then optimally determined by the following relation (which stems directly from (4.59) and (4.60)):

$$\beta = \left\{\int_{-\infty}^{\infty}\int_{0}^{\infty} p_0(y,a) \cdot \left(1 + \alpha \left[\frac{p_1(y,a)}{p_0(y,a)}\right]^{\lambda}\right)^{\rho} da\, dy\right\}^{-1}. \tag{4.62}$$

We note here that the left hand side of (4.61) is a decreasing function of the non-negative parameter α, and it also admits every value between zero and unity (corresponding to $\alpha \to \infty$ and $\alpha = 0$, respectively). Therefore, the existence and uniqueness of a solution α for (4.61) is assured for any δ (as $0 < \delta < 1$), and this solution can be determined numerically (e.g, by the bisection method).

We observe from our discussion so far that the minimization of the above upper bound on the block error probability (which is based on the minimization of the upper bound on $P_{\mathrm{e}|0}(d)$ in (4.58) and the calculation of the union bound in (4.44)) involves a numerical minimization of (4.58) over the parameters λ and ρ (where $\lambda \geq 0$ and $0 < \rho \leq 1$), for every particular subcode $\mathcal{C}_d$ ($d = 0, 1, \cdots, N$). The optimal values of α and β which are related to the optimal tilting measure ψ in (4.60) are numerically evaluated from (4.61) and (4.62) as a function of the two optimized parameters λ and ρ. That minimization is performed separately for every subcode (where the number of the subcodes $\mathcal{C}_d$ doesn't exceed the length N of the linear block code $\mathcal{C}$, and clearly we are interested only on the subcodes $\mathcal{C}_d$ for which $S_d > 0$, as otherwise $P_{\mathrm{e}|0}(d) = 0$).

Suppose we wish to calculate an upper bound on the block error probability for an ensemble of linear block codes whose *average* distance spectrum is calculable. By invoking the Jensen inequality to the right hand side of (4.58), then $E[(S_d)^{\rho}] \leq (E[S_d])^{\rho}$, where $0 \leq \rho \leq 1$. Hence, for ensembles of codes, the upper bound (4.58) therefore stays valid by replacing the distance spectrum in the right hand side of (4.58) with the statistical expectation of the distance spectrum (as was first noted in [94]).

Exponential Tilting Measure

In [288], a sub-optimal selection for ψ in (4.57) is suggested which in fact is motivated by the Duman and Salehi (first version) bound [94]. An exponential tilting is also applied in [288] to the fading sample a (treated as a measurement), which gives rise to the exponential tilting measure

$$\psi(y,a) = \frac{\sqrt{\frac{\alpha}{2\pi}}\, \exp\left[-\frac{\alpha}{2}\left(y - au\sqrt{\frac{2E_{\mathrm{s}}}{N_0}}\right)^2 - \frac{\alpha v^2 a^2 E_{\mathrm{s}}}{N_0}\right] p(a)}{\displaystyle\int_{0}^{+\infty} p(a)\, \exp\left(-\frac{\alpha v^2 a^2 E_{\mathrm{s}}}{N_0}\right) da} \tag{4.63}$$

where $\alpha \geq 0$, $-\infty < u < +\infty$, $-\infty < v < +\infty$, and $p(a)$ designates the probability density function of the independent fading samples. That yields a closed form upper bound in (4.58), which reads (see [288])

$$\begin{aligned} P_{\mathrm{e}|0}(d) &\leq (S_d)^{\rho}\, \alpha^{\frac{-N(1-\rho)}{2}} \left(\alpha - \frac{\alpha-1}{\rho}\right)^{-\frac{N\rho}{2}} \\ &\quad \cdot \left(\frac{1}{1+t}\right)^{N(1-\rho)} \left(\frac{1}{1+\varepsilon}\right)^{d\rho} \left(\frac{1}{1+v}\right)^{(N-d)\rho} \end{aligned} \tag{4.64}$$

where

$$t = \frac{\alpha\nu^2 E_{\mathrm{s}}}{N_0} \tag{4.65a}$$

$$v = \frac{E_{\mathrm{s}}}{N_0}\left[\alpha(u^2+\nu^2)\left(1-\frac{1}{\rho}\right) + \frac{1}{\rho} - \frac{\left(\alpha u - \frac{\alpha u - 1}{\rho}\right)^2}{\alpha - \frac{\alpha-1}{\rho}}\right] \tag{4.65b}$$

$$\varepsilon = v + \frac{E_{\mathrm{s}}}{N_0}\,\frac{\left(\alpha u - \frac{\alpha u-1}{\rho}\right)^2 - \left(\alpha u - \frac{\alpha u-1}{\rho} - 2\lambda\right)^2}{\alpha - \frac{\alpha-1}{\rho}}\,. \tag{4.65c}$$

This bound is in fact equivalent to the Divsalar and Biglieri bound [87] which has been derived via a geometric extension of the associated decision region in Divsalar bound [86] (by rotating the displaced sphere region in [86]). The bound (4.63)–(4.65) also yields to an extension of the Viterbi and Viterbi (first version) bound for fully interleaved fading channels and perfect CSI at the receiver (see [288]), by setting

$$\alpha = 1\,, \quad u = 1 - \xi\rho\,, \quad \nu = \sqrt{1-(1-\xi\rho)^2}\,, \quad \lambda = \frac{1+\xi\,(1-\rho)}{2}\,. \tag{4.66}$$

In [288], the DS2 bound associated with the exponential tilting measure (4.63) is also applied to the fully interleaved Rician fading channel. As a special case, it yields the Duman and Salehi (first version) bound for a binary-input AWGN channel, where the Rician parameter K standing for the power ratio of the direct and the diffused received paths, goes to infinity. The bounds are depicted in Fig. 4.4 for the 'repeat and accumulate' (RA) codes (an ensemble of turbo-like codes which was introduced by Divsalar, Jin and McEliece [85]) operating over a fully interleaved Rayleigh fading channel. The penalty for the gradual specialization of the DS2 bound by constraining the selection ψ in (4.57) is explicitly indicated.

4.5 Gallager-Type Bounds for the Mismatched Decoding Regime

In this section we generalize the DS2 bound for a mismatched decoding metric. The basic setting is as before, that is a codeword $\underline{x}$ is conveyed via a channel of transition probability $p_N(\underline{y}|\underline{x})$. The decoder operates in a ML fashion, but may use a mismatched metric $Q_N(\underline{y}|\underline{x})$. Namely, code $\underline{x}^j$ out of the M possible equiprobable codewords is declared if

$$Q_N(\underline{y}|\underline{x}^j) > Q_N(\underline{y}|\underline{x}^i)\,, \quad \forall\, i \neq j\,, \quad i = 1, 2, \ldots, M\,, \tag{4.67}$$

and ties are randomly resolved. The matched case results when $Q_N(\underline{y}|\underline{x}) = p_N(\underline{y}|\underline{x})$. In general, $Q_N(\underline{y}|\underline{x})$ is not necessarily normalized to yield a probability measure.

We first employ the Duman and Salehi bounding technique in this setting and then examine the performance of random ensembles of codes.

4.5.1 The Mismatched Duman and Salehi Bound

The standard Gallager upper bound on the conditional decoding error probability for the mismatched case $P_{\mathrm{e}|m}$ (conditioned on the transmitted codeword $\underline{x}^m$) is given by

$$P_{\mathrm{e}|m} \leq \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \left(\sum_{m' \neq m} \left(\frac{Q_N(\underline{y}|\underline{x}^{m'})}{Q_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}, \quad \lambda, \rho \geq 0. \tag{4.68}$$

By invoking the Duman and Salehi bounding technique as described in section 4.2.2, then starting from (4.68) yields in parallel to (4.7),

$$P_{\mathrm{e}|m} \leq \left(\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m)^{\frac{1}{\rho}} \, \psi_N^m(\underline{y})^{1-\frac{1}{\rho}} \left(\frac{Q_N(\underline{y}|\underline{x}^{m'})}{Q_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}, \quad \lambda \geq 0, \;\; 0 \leq \rho \leq 1 \tag{4.69}$$

where as in Section 4.2.2, $\psi_N^m(\underline{y})$ is the normalized Duman and Salehi tilting measure. In parallel to (4.9), the DS2 bound with the un-normalized tilting measure reads

$$P_{\mathrm{e}|m} \leq \left(\sum_{\underline{y}} G_N^m(\underline{y}) \, p_N(\underline{y}|\underline{x}^m) \right)^{1-\rho} \left(\sum_{m' \neq m} \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \, G_N^m(\underline{y})^{1-\frac{1}{\rho}} \left(\frac{Q_N(\underline{y}|\underline{x}^{m'})}{Q_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \right)^{\rho}, \quad 0 \leq \rho \leq 1, \;\; \lambda \geq 0 \,. \tag{4.70}$$

The advantage of this bound, as compared to the original Gallager mismatched bound (4.68), is as in the matched case: The possible utilization for specific codes and ensembles. This is demonstrated in Section 4.7.2 for a fully interleaved fading channel with a faulty channel state information.

4.5.2 Ensembles of Random Codes

In this subsection, we examine the bound for a random coding strategy. As in the matched regime, for a general distribution of codewords $q_N(\underline{x})$, the DS2 bound with the optimized tilting measure reconstructs the 1965 Gallager random coding bound. We therefore continue with the mismatched Gallager bound (4.68) and restrict our attention to ensembles of codes where each codeword satisfies the inequality

$$N\varepsilon - \delta \leq \Gamma_N(\underline{x}^j) < N\varepsilon \,, \quad \delta > 0 \,, \quad j = 1, 2, \ldots, M \tag{4.71}$$

where $\Gamma_N(\underline{x}^j)$ stands for an arbitrary cost function involved in the transmission of the codeword $\underline{x}^j$ and ε is a positive constant, both to be specified. For a codebook satisfying the inequality (4.71) (for every codeword), we further loosen the bound (4.68) letting

$$P_{\mathrm{e}|m} \leq \sum_{\underline{y}} p_N(\underline{y}|\underline{x}^m) \left[\sum_{m' \neq m} \left(\frac{Q_N(\underline{y}|\underline{x}^{m'})}{Q_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \left(\frac{e^{\Gamma_N(\underline{x}^{m'})}}{e^{\Gamma_N(\underline{x}^m)}} \right) e^{\delta} \right]^{\rho}, \quad \lambda, \rho \geq 0 \,. \tag{4.72}$$

This is since for every value of j ($j = 1, 2, \ldots, M$), the inequality $\frac{e^{\Gamma_N(\underline{x}^{m'})}}{e^{\Gamma_N(\underline{x}^m)}} e^{\delta} > 1$ holds for any codebook satisfying (4.71). Now, we select randomly and independently each codeword in such a

codebook by the probability law $\alpha_N(\underline{x})$. The upper bound on the decoding error probability over this ensemble equals

$$P_{\mathrm{e}} \leq \sum_{\underline{y}} \sum_{\underline{x}^m} \alpha_N(\underline{x}^m) p_N(\underline{y}|\underline{x}^m) \sum_{\underline{x}^1} \cdots \sum_{\underline{x}^{m-1}} \sum_{\underline{x}^{m+1}} \cdots \sum_{\underline{x}^M} \left\{ \prod_{\substack{i=1 \\ i \neq m}}^{M} \alpha_N(\underline{x}^i) \right.$$
$$\left. \cdot \left(\sum_{j \neq m} \left(\frac{Q_N(\underline{y}|\underline{x}^j)}{Q_N(\underline{y}|\underline{x}^m)} \right)^{\lambda} \frac{e^{\Gamma_N(\underline{x}^j)}}{e^{\Gamma_N(\underline{x}^m)}} e^{\delta} \right)^{\rho} \right\} \tag{4.73}$$

where $\lambda, \rho \geq 0$.

As indicated in (4.71), the measure $\alpha_N(\underline{x})$ satisfies the condition

$$\alpha_N(\underline{x}) \equiv 0 \quad \forall\, \underline{x} \text{ where } \Gamma_N(\underline{x}) \notin [N\varepsilon - \delta,\, N\varepsilon) \,. \tag{4.74}$$

In principle, the parameter δ introduced in (4.71) may depend on N. Invoking the Jensen inequality in (4.73) yields the following upper bound on the decoding error probability:

$$P_{\mathrm{e}} \leq (M-1)^{\rho}\, e^{\delta\rho} \sum_{\underline{y}} \sum_{\underline{x}} \alpha_N(\underline{x})\, p_N(\underline{y}|\underline{x}) \left\{ \sum_{\underline{x}'} \alpha_N(\underline{x}') \left(\frac{Q_N(\underline{y}|\underline{x}')}{Q_N(\underline{y}|\underline{x})} \right)^{\lambda} \frac{e^{\Gamma_N(\underline{x}')}}{e^{\Gamma_N(\underline{x})}} \right\}^{\rho} \tag{4.75}$$

where $0 \leq \rho \leq 1$, $\lambda \geq 0$, and the superscript of $\underline{x}^m$ was removed (as the bound is clearly invariant to the transmitted codeword). Let $q_N(\underline{x})$ be an arbitrary probability measure and we set

$$\alpha_N(\underline{x}) = \frac{q_N(\underline{x})}{\mu_\alpha} \tag{4.76}$$

where based on (4.71)

$$\mu_\alpha = \mu_\alpha(\Gamma, \delta, \varepsilon) = \sum_{\underline{x}\,:\,\Gamma_N(\underline{x}) \in [N\varepsilon - \delta, N\varepsilon)} q_N(\underline{x}) \,. \tag{4.77}$$

The substitution of (4.76) into (4.75) gives

$$P_{\mathrm{e}} \leq \frac{(M-1)^{\rho}\, e^{\delta\rho}}{(\mu_\alpha)^{1+\rho}} \sum_{\underline{y}} \sum_{\underline{x}} q_N(\underline{x})\, p_N(\underline{y}|\underline{x}) \left\{ \sum_{\underline{x}'} q_N(\underline{x}') \left(\frac{Q_N(\underline{y}|\underline{x}')}{Q_N(\underline{y}|\underline{x})} \right)^{\lambda} \frac{e^{\Gamma_N(\underline{x}')}}{e^{\Gamma_N(\underline{x})}} \right\}^{\rho} \tag{4.78}$$

where $\lambda \geq 0$, $0 \leq \rho \leq 1$ and $\delta > 0$. We may further loosen the bound (4.78) by replacing $\frac{e^{\delta\rho}}{(\mu_\alpha)^{1+\rho}}$ by $\frac{e^{\delta}}{\mu_\alpha^2}$, as $\delta > 0$, $0 \leq \rho \leq 1$ and $\mu_\alpha \leq 1$. This yields

$$P_{\mathrm{e}} \leq \frac{e^{\delta} M^{\rho}}{\mu_\alpha^2} \sum_{\underline{y}} \sum_{\underline{x}} q_N(\underline{x})\, p_N(\underline{y}|\underline{x}) \left\{ \sum_{\underline{x}'} q_N(\underline{x}') \left(\frac{Q_N(\underline{y}|\underline{x}')}{Q_N(\underline{y}|\underline{x})} \right)^{\lambda} \frac{e^{\Gamma_N(\underline{x}')}}{e^{\Gamma_N(\underline{x})}} \right\}^{\rho} \tag{4.79}$$

where we also upper bounded $M-1$ by M. Let the channel probability law $p_N(\underline{y}|\underline{x})$ and the mismatched metric $Q_N(\underline{y}|\underline{x})$ be memoryless, i.e.,

$$p_N(\underline{y}|\underline{x}) = \prod_{k=1}^{N} p(y_k|x_k),\; Q_N(\underline{y}|\underline{x}) = \prod_{k=1}^{N} Q(y_k|x_k) \tag{4.80}$$

and we also set an i.i.d. probability measure $q_N(\underline{x})$

$$q_N(\underline{x}) = \prod_{k=1}^{N} q(x_k). \tag{4.81}$$

Let the cost function $\Gamma_N(\underline{x})$ be an additive function

$$\Gamma_N(\underline{x}) = \sum_{k=1}^{N} \gamma(x_k). \tag{4.82}$$

Substituting (4.80)-(4.82) into (4.79) then yields the single letter expression

$$P_{\mathrm{e}} \leq \frac{e^{\delta}}{\mu_\alpha^2}\, e^{\rho\, RN} \left[\sum_y \sum_x q(x)\, p(y|x) \left(\sum_{x'} q(x') \left(\frac{Q(y|x')}{Q(y|x)} \right)^{\lambda} \frac{e^{\gamma(x')}}{e^{\gamma(x)}} \right)^{\rho} \right]^{N} \tag{4.83}$$

where the rate R equals $R = \frac{\ln(M)}{N}$. Alternatively, we obtain that

$$P_{\mathrm{e}} \leq \frac{e^{\delta}}{\mu_\alpha^2} \cdot e^{-N[E_0(q,\gamma,\rho,\lambda)-\rho R]} \tag{4.84}$$

where

$$E_0(q,\gamma,\rho,\lambda) = -\ln \left[\sum_y \sum_x q(x)\, p(y|x) \left(\sum_{x'} q(x') \left(\frac{Q(y|x')}{Q(y|x)} \right)^{\lambda} \frac{e^{\gamma(x')}}{e^{\gamma(x)}} \right)^{\rho} \right]. \tag{4.85}$$

Before turning to treat the exponent in (4.84), let us estimate μ_α for fixed (N-independent) δ. Based on (4.77) and (4.82)

$$\mu_\alpha = \mathrm{Prob} \left(-\delta \leq \sum_{k=1}^{N} \gamma(x_k) - N\varepsilon < 0 \right) \tag{4.86}$$

where $x_k,\ k = 1, 2, \ldots, N$, are i.i.d. random variables governed by the single-letter probability measure $q(x)$. Now, we choose ε to satisfy the equality

$$\varepsilon = \sum_x q(x)\, \gamma(x) \tag{4.87}$$

for a fixed $\gamma(\cdot)$. Similarly to Gallager [124], one can estimate μ_α for $N \to \infty$, yielding

$$\lim_{N\to\infty} \sqrt{N}\, \mu_\alpha = \frac{\delta}{\sqrt{2\pi\sigma^2}} \tag{4.88}$$

where

$$\sigma^2 = \sum_x q(x)\, \gamma(x)^2 - \left(\sum_x q(x)\, \gamma(x) \right)^2. \tag{4.89}$$

This is directly found based on (4.86) by the central limit property as follows:

$$\begin{aligned} \mu_\alpha \quad &= \quad \mathrm{Prob} \left(\frac{-\delta}{\sqrt{N}} \leq \frac{1}{\sqrt{N}} \sum_{k=1}^{N} \gamma(x_k) - \sqrt{N}\varepsilon < 0 \right) \\ &\xrightarrow[N\to\infty]{} \quad \mathrm{Prob} \left(\frac{-\delta}{\sqrt{N}} \leq N(0,\sigma^2) < 0 \right) \\ \frac{\delta}{\sqrt{N}} \ll 1 \quad &\approx \quad \frac{\delta}{\sqrt{N}}\, \frac{1}{\sqrt{2\pi\sigma^2}}\,. \end{aligned} \tag{4.90}$$

The pre-exponent factor in (4.84) can be optimized over $\delta \geq 0$ to yield

$$\min_{\delta>0} \lim_{N\to\infty} \frac{1}{N} \frac{e^{\delta}}{\mu_{\alpha}^2} = \min_{\delta>0} (2\pi\sigma^2) \cdot \frac{e^{\delta}}{\delta^2} = (2\pi\sigma^2) \cdot \frac{e^2}{4} \tag{4.91}$$

which demonstrates that the behavior of the pre-exponent in (4.84) is asymptotically proportional to N (with $N \to \infty$). Thus for finite ε (4.87) and σ^2 (4.89), the pre-exponent in (4.84) has no exponential implications as it behaves asymptotically like a $\frac{\ln(N)}{N}$ term in the exponent. For a fixed input distribution $q(x)$, we then attempt to maximize the error exponent

$$E(R,q) = \sup_{\substack{0\leq\rho\leq 1 \\ \lambda\geq 0 \\ \gamma(x)}} \Big(E_0(q,\gamma,\rho,\lambda) - \rho R \Big) \tag{4.92}$$

where $\gamma(x)$ is a real function yielding finite ε and σ^2 in (4.87) and (4.89) respectively. Consider the rate equation

$$\begin{aligned} R &= \frac{\partial}{\partial\rho} \Big(E_0(q,\gamma,\rho,\lambda) \Big) \\ &= e^{E_0(q,\gamma,\rho,\lambda)} \\ &\quad \cdot \Bigg\{ - \sum_y \sum_x q(x)\, p(y|x) \left(\sum_{x'} q(x') \left(\frac{Q(y|x')}{Q(y|x)} \right)^{\lambda} \frac{e^{\gamma(x')}}{e^{\gamma(x)}} \right)^{\rho} \\ &\qquad\qquad \cdot \ln \left[\frac{\sum\limits_{x'} q(x')\, Q^{\lambda}(y|x')\, e^{\gamma(x')}}{Q^{\lambda}(y|x)\, e^{\gamma(x)}} \right] \Bigg\} \end{aligned} \tag{4.93}$$

where the last equality results from (4.85). The maximal rate R_H is calculated by substituting $\rho = 0$ in (4.93), yielding

$$R_H = \sup_{\gamma(x)} \max_{\lambda\geq 0} \sum_y \sum_x q(x)\, p(y|x)\, \ln \left(\frac{Q^{\lambda}(y|x)\, e^{\gamma(x)}}{\sum\limits_{x'} q(x')\, Q^{\lambda}(y|x')\, e^{\gamma(x')}} \right) \tag{4.94}$$

This, in fact, equals what is known as the Csiszár-Körner-Hui lower bound on the mismatched capacity, since it yields the dual representation in the terminology of [127]. It is rather straightforwardly verified that the error exponent $E(R)$ in (4.92) is positive for $R < R_H$. The input distribution $q(x)$ can now be chosen to maximize R_H, in the usual sense [127], [221]. In the case of continuous input and output alphabets, the relevant sums should be replaced by integrals and it should be verified that the optimized $\gamma(x)$ and $q(x)$ yield finite (though arbitrary) ε and σ in (4.87) and (4.89) respectively.

It can be also verified that at $\rho = 1$

$$R = \sup_{\lambda\geq 0} E_0(q, \gamma = 0\,,\, \rho = 1, \lambda) \tag{4.95}$$

yields the generalized cutoff rate [164], which cannot be further improved by optimizing over the real function $\gamma(x)$. In terms of ensembles, it is worth emphasizing that the upper bound (4.84) resulted by restricting our attention to ensembles of codewords satisfying (4.71). This restriction is necessary, as if a purely random ensemble is attempted, then it was concluded in [127] that the

results associated with $\gamma(x) = 0$ cannot be surpassed. The maximal rate then corresponds to the generalized mutual information (GMI) [221].

In [256], the 1965 Gallager bounding technique was applied to derive upper bounds for i.i.d. random and fixed composition codes, operating over memoryless channels with finite input alphabets and arbitrary output alphabets, and whose decoding metric is matched to the channel. The metric which is at the beginning general, was then optimized to yield an equivalent metric to ML decoding, and yet the results in [256] reproduced the error exponent for fixed composition codes [73]. Here we deal with a mismatched metric and the code ensemble is restricted by introducing a generalized energy constraint (4.71), which is subject to optimization.

4.5.3 Ensembles of Structured Codes

We apply here the generalization of the DS2 bound for the mismatched decoding regime (4.68) to ensembles of structured codes. We assume that the transition probabilities of the channel $p_N(\underline{y}|\underline{x})$ and the mismatched metric $Q_N(\underline{y}|\underline{x})$ are MBIOS (i.e., memoryless, binary-input and output-symmetric).

The optimization of the normalized Duman and Salehi tilting measure $\psi_N^m(\underline{y})$ in (4.69) is restricted here to the case where $\psi_N^0(\underline{y})$ can be expressed in the product form $\psi_N^0(\underline{y}) = \prod_{i=1}^{N} \psi(y_i)$. In that case, the partitioning of the code (or ensemble of codes) to constant Hamming weight subcodes yields (4.44), where

$$P_{\mathrm{e}|0}(d) \leq (S_d)^{\rho} \left(\sum_y g_1(y) \, \psi(y)^{1-\frac{1}{\rho}} \right)^{(N-d)\rho} \left(\sum_y g_2(y) \, \psi(y)^{1-\frac{1}{\rho}} \right)^{d\rho} \tag{4.96}$$

and

$$g_1(y) = p(y|0)^{\frac{1}{\rho}} \, , \quad g_2(y) = p(y|0)^{\frac{1}{\rho}} \left(\frac{Q(y|1)}{Q(y|0)} \right)^{\lambda} . \tag{4.97}$$

For continuous output channels, the sums in (4.96) should be replaced by integrals. With the aid of the calculus of variations, the optimal normalized Duman and Salehi tilting measure ψ, in terms of minimizing the upper bound (4.96)-(4.97), admits the form

$$\psi(y) = \frac{p(y|0) \left(1 + \alpha \left(\frac{Q(y|1)}{Q(y|0)} \right)^{\lambda} \right)^{\rho}}{\sum_y p(y|0) \left(1 + \alpha \left(\frac{Q(y|1)}{Q(y|0)} \right)^{\lambda} \right)^{\rho}} \, , \quad \lambda \geq 0 \, , \; 0 \leq \rho \leq 1 \, . \tag{4.98}$$

As demonstrated in [296, Appendix B], the parameter α is optimally determined by a numerical solution of the equation

$$\frac{\sum_y p(y|0) \left(1 + \alpha \left(\frac{Q(y|1)}{Q(y|0)} \right)^{\lambda} \right)^{\rho-1}}{\sum_y p(y|0) \left(1 + \alpha \left(\frac{Q(y|1)}{Q(y|0)} \right)^{\lambda} \right)^{\rho}} = 1 - \delta \tag{4.99}$$

where $\delta \equiv \frac{d}{N}$ is the normalized Hamming weight of the N-length codewords possessing a Hamming weight d. The existence and uniqueness of a solution α in (4.99) is proved in [296, Appendix B].

From the discussion above, the upper bound for the mismatched decoding regime in (4.44), (4.96)-(4.99) involves numerical optimizations over the two parameters $\lambda \geq 0$, $0 \leq \rho \leq 1$, for at most the $N+1$ constant Hamming weight subcodes of the considered ensemble of codes. The necessary information on the ensemble of codes comprises the average distance spectrum (or even a tight upper bound on the ensemble distance spectrum), and a refined information on the algebraic structure of the codes is not required.

4.6 Gallager-Type Bounds for Parallel Channels

The error performance analysis in the case where each codeword is partitioned and each part is transmitted over one of independent channels in parallel is of interest. Code partitioning is employed in transmission over block-fading channels (for performance bounds of coded communication systems over block-fading channels, see [131, 360, 379, 380]), incremental redundancy retransmission schemes, cooperative coding, multi-carrier signaling (for performance bounds of coded orthogonal-frequency division multiplexing (OFDM) systems, see [188, 377]), etc.

In his thesis [101], Ebert considered the problem of communicating over parallel discrete time channels, distributed by arbitrary additive Gaussian noise with a total power constraint on the set of channels. He found explicit upper and lower bounds on the ML decoding error probability, which decreased exponentially with block length. The exponents of the upper and lower bounds coincide for rates between the critical rate (R_{crit}) and capacity. The results were also shown to be applicable to colored Gaussian noise channels with an average power constraint on the channel.

In [199], Liu et al. derive upper bounds on the ML decoding error probability of structured ensembles of codes whose transmission takes place over (independent) parallel channels. The analysis in [199] modifies the 1961 Gallager-Fano bound [122, Chapter 3] and adapts this bounding technique for the communication over parallel channels. As special cases of this modified bound, a generalization of the union-Bhattacharyya bound, the Shulman-Feder bound [303], sphere bound, and a combination of the two former bounds are derived for parallel channels. In order to make the calculation of the weight enumerators feasible when taking into account the partitioning of codewords and their transmission over parallel channels, the authors study the case where the bits of the codewords are randomly assigned to J parallel channels, where the fraction of bits transmitted over each channel is determined in advance. The upper bounds on the ML decoding error probability are applied to ensembles of codes defined on graphs (e.g., uniformly interleaved repeat-accumulate codes and turbo codes). The comparison between upper bounds under ML decoding and computer simulations of the performance of such ensembles under iterative decoding shows a good match in several cases. For a given ensemble of codes and a given codeword-symbol to channel assignment rule, a reliable channel region is defined as the closure of the set of parallel-channel transition probabilities for which the decoding error probability vanishes as the codeword length goes to infinity. The upper bounds on the block error probability derived in [199] enable to derive achievable regions for ensuring reliable communications under ML decoding.

4.7 Some Applications of the Gallager-Type Bounds

In Section 4.4, a large class of efficient recent bounds (or their Chernoff versions) was demonstrated to be a special case of the generalized DS2 bound. Implications and applications of these observations are pointed out here, including the fully interleaved fading channel, resorting to either matched or mismatched decoding. The proposed approach can be also generalized to geometrically

uniform non-binary codes, finite state channels, bit interleaved coded modulation systems, and it can be also used for the derivation of upper bounds on the conditional decoding error probability.

4.7.1 AWGN Channels

We apply here some variants of the Gallager bounds and other reported bounds to the ensemble of un-punctured turbo codes of rate $\frac{1}{3}$ with a uniform interleaver of length 1000 and two recursive systematic convolutional (RSC) component codes whose generators are $(1, \frac{21}{37})$ in octal form (see Fig. 4.1). The considered ensemble of codes is also terminated to the all-zero state at the end

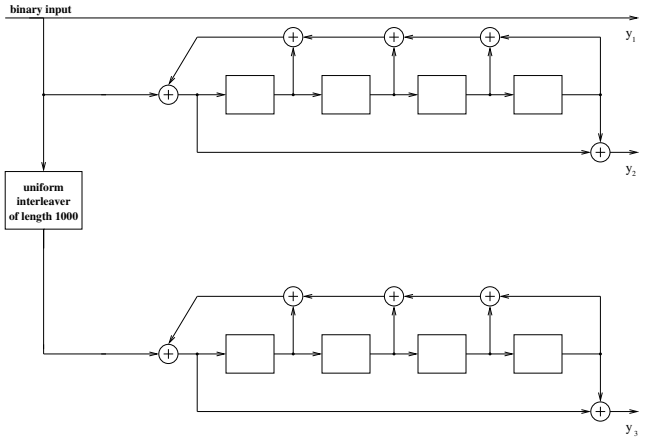


Figure 4.1: Block diagram of an ensemble of uniformly interleaved turbo codes of rate-$\frac{1}{3}$ and interleaver length 1000. The generator of the two identical component codes (16–states, recursive systematic convolutional (RSC) codes) is $\left[1, \frac{1+D^4}{1+D+D^2+D^3+D^4}\right]$. A termination to the all-zero state is assumed at the end of each block.

of each frame by additional four bits (having thus an overall of 3012 coded bits). The following upper bounds on the bit error probability are depicted in Fig. 4.2: The original TSB of Poltyrev ([259, 283]) (differing from the loosened Chernoff version bound derived in [86]), the Engdahl and Zigangirov (EZ) bound [103] (the non-Chernoff version in Section 5 here), Duman and Salehi bounds [94, 95] (Sections 4.2.2 and 1 here), Divsalar bound [86] (Section 4 here), Viterbi and Viterbi bounds [347, 348] (Sections 2 and 3 here), and finally the ubiquitous union bound. It is demonstrated in Fig. 4.2, that the difference between the two versions of the Duman and Salehi bounds for the binary-input AWGN channel is very small (about 0.01 dB) and that also the Duman and Salehi (first version) bound coincides with Divsalar bound. The first observation is consistent with [95] and the second observation verifies numerically the fact that Divsalar bound is indeed a closed form of Duman and Salehi (first version) bound [94], as was first indicated in [86] (the negligible difference between the two bounds depicted in Fig. 4.2 is attributed to the numerical optimizations associated with the calculation of the first version of the Duman and Salehi bound (4.50)). The TSB [259] and the EZ bound [103] were derived without invoking the Chernoff bounding technique, explaining therefore their advantage over Duman and Salehi bounds and some other related bounds for block codes of moderate block-length, as the latter bounds rely on the Chernoff bounding technique (see Fig. 4.2). However, as indicated in [86] (and Section 4.4.2), the loosening of the TSB which turns it into a particular case of the DS2 bound, does not carry any implications on its associated error exponent for the asymptotically infinite block length.

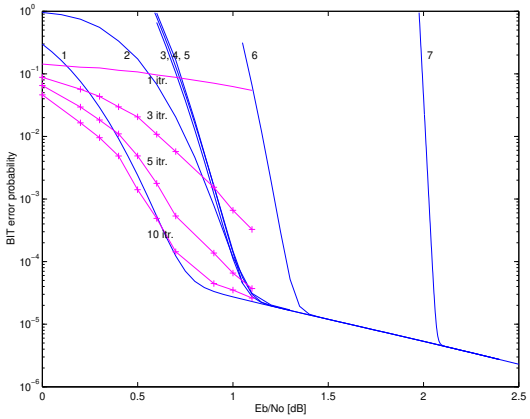


Figure 4.2: A comparison between upper bounds on the bit error probability with ML decoding. The bounds refer to the ensemble of turbo codes in Fig. 4.1, operating over a binary-input AWGN channel.
1. Tangential-sphere bound (TSB) ([259], [283]).
2. Engdahl and Zigangirov (EZ) bound [103].
3. DS2 bound [95].
4. Divsalar bound [86].
5. Duman and Salehi (first version) bound [94].
6. Viterbi & Viterbi (second version) bound [348].
7. Union bound in Q-form.

The bounds 2–6 are combined with the union bound in its Q-form for every constant Hamming-weight subcode of the considered ensemble. We also plot here simulation results of Log-MAP iterative decoding with 1, 3, 5 and 10 iterations.

4.7.2 Fully Interleaved Fading Channels

We apply here various variants of the generalized DS2 bound to the fully interleaved Rayleigh fading channel with perfect channel state information (see Section 4.4.3). These bounds are evaluated for the ensemble of rate $\frac{1}{3}$ uniformly interleaved turbo codes depicted in Fig. 4.1. The optimization of the generalized DS2 bound (combined with the tight version of the union bound) is demonstrated as the tightest reported bound (see also [288],[286]). It also approximately replicates here the performance of these turbo codes when iteratively decoded with 10 iterations of the Log-MAP iterative decoding algorithm (see Fig. 4.3). These bounds are also evaluated for the ensemble of rate-$\frac{1}{4}$ uniformly interleaved RA codes [85], with a block length of $N = 4096$ (see Figs. 3.9 and 4.4). In [286, 288], these bounds were applied to some ensembles of efficient codes (turbo [40], Gallager-LDPC [122] and RA codes [85]). For moderate values of energy per bit to spectral noise density $\left(\frac{E_b}{N_0}\right)$, the optimized DS2 upper bound (under ML decoding) falls below the computer simulation results for the sum-product iterative decoding algorithm (see Fig. 4.4), demonstrating the mild sub-optimality of the latter decoding algorithm.

4.7.3 Thresholds of Codes

Let $\mathcal{C}$ be an ensemble of binary, linear (or geometrically uniform) block codes of block-length N and rate R. The threshold is defined as the minimal value of $\frac{E_b}{N_0}$ ensuring a block decoding error probability that decays to zero for asymptotically infinite block length. Obviously, the threshold depends

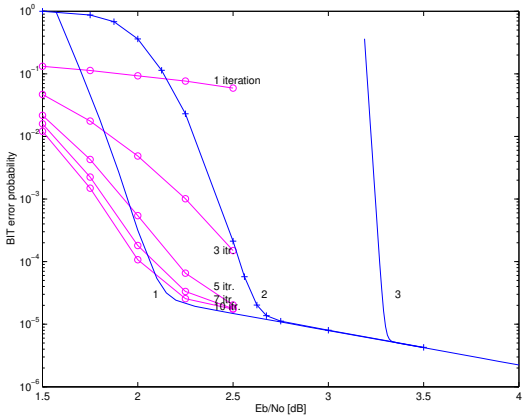


Figure 4.3: A comparison between upper bounds on the bit error probability with ML decoding. The bounds refer to the ensemble of turbo codes in Fig. 4.1, operating over a fully interleaved Rayleigh fading channel with perfect channel state information.
1. The generalized DS2 bound [286] combined with the union bound in its tight form.
2. The generalization of the Engdahl and Zigangirov (EZ) bound [286].
3. The union bound in its tight form.
These upper bounds are also compared to computer simulation results of the Log-MAP iterative decoding algorithm with up to 10 iterations.

on three parameters: The channel model, the applied decoding algorithm and the considered code (or ensemble of codes). The thresholds associated with some decoding algorithms were recently studied: an analytical upper bound on the thresholds of ML decoded block codes operating over the binary-input AWGN channel was derived by Divsalar [86]. Upper bounds on the thresholds of RA [85] and Gallager-LDPC codes [122] under "typical pairs" decoding algorithm, were derived by McEliece et. al. [5] for the binary symmetric channel (BSC) and for the binary-input AWGN channel. Exact thresholds for the BSC under a specific iterative decoding algorithm were derived by Bazzi et. al. [28]. Based on the optimized DS2 bound, an upper bound on the thresholds which are associated with ML decoding were numerically calculated for the fully interleaved fading channels with perfect channel state information (see [286, Section 3.3]). As predicted, these thresholds meet the ultimate Shannon capacity limit for the ensemble of fully random block codes of rate R. Thresholds for the ensembles of RA and (j, k) Gallager-LDPC codes are depicted in Table 4.1. These thresholds refer to the fully interleaved Rayleigh fading channel with perfect channel state information and a maximum ratio combining space diversity of order four. As expected, due to the simple structure of RA codes, the calculated thresholds for the ensemble of RA codes are worse than the corresponding thresholds for the ensembles of Gallager-LDPC codes.

In general, the calculation of the upper bounds on these thresholds depends on the asymptotic exponent of the distance spectrum of the considered codes (or ensembles). Recently, some techniques for the calculation of the asymptotic exponents of the distance spectra of turbo codes and irregular LDPC codes where derived in [289] and [57], respectively. These techniques can be applied to calculate upper bounds on the $\frac{E_b}{N_o}$-thresholds for turbo codes and LDPC codes under optimal ML decoding.

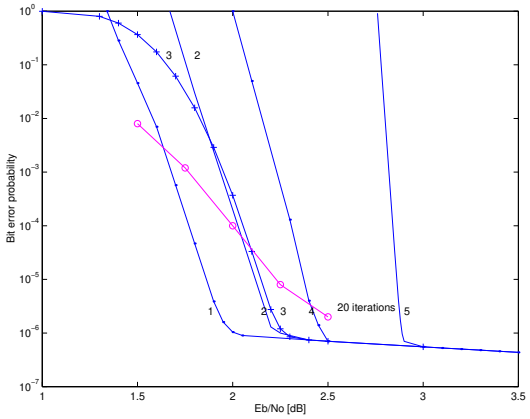


Figure 4.4: Comparison between upper bounds on the bit error probability with ML decoding. The bounds refer to the RA codes in Fig. 3.9, operating over a fully interleaved Rayleigh fading channel with perfect channel state information.
1. The generalized DS2 bound [286].
2. Divsalar and Biglieri bound [87].
3. The generalization of the Engdahl and Zigangirov (EZ) bound [286].
4. The generalization of Viterbi and Viterbi bound [288].
5. The tight form of the union bound.

The upper bounds 1–4 are combined with the tight version of the union bound for every constant Hamming-weight subcode of the considered ensemble of codes. The bounds are also compared with computer simulation results of the sum-product iterative decoding algorithm with 20 iterations.

4.7.4 Mismatched Decoding

We apply here the generalized DS2 bound introduced in Section 4.5.3, to study the robustness of a mismatched decoder that is based on ML decoding with respect to the faulty channel measurements. We examine here a BPSK modulated signal, transmitted through a fully interleaved Rayleigh fading channel. For simplicity, we apply our bounds to the case of a perfect phase estimation of the i.i.d. fading samples (in essence reducing the problem to a real channel). We also assume that the estimated and real magnitudes of the Rayleigh fading samples satisfy a joint distribution of two correlated bivariate Rayleigh variables with an average power of unity. We note that if the phase estimation is not perfect (as opposed to what we have assumed here, that the error occurs only in the amplitude estimation), assuming that the actual fading coefficient (a) and its estimate ($\hat{a}$) are correlated Gaussian circularly symmetric random variables, the implication of the correlation decoder manifest itself in just some enhanced additive independent Gaussian noise. This is in fact the best (matched) decoder given the perfect side information at hand. This is since:

$$y = ax + n = \hat{a}x + (a - \hat{a})x + n.$$

In the Gaussian regime the estimated error $(a - \hat{a})$ is Gaussian distributed and independent of the estimator $\hat{a}$, even when conditioned on x (as neither the mean nor the variance of $(a - \hat{a})x$ does not depend on x for the binary case, as $||x||^2 = E_s$ [301]). The perfectly known phase treated here is a different case, since the results are not captured in terms of scaling the additive noise.

Rate	$\frac{E_b}{N_0}$ thresholds for the ensembles of RA codes (j,k) Gallager-LDPC codes		The Shannon capacity limit
$\frac{1}{2}$	-	(4,8) : 0.76 dB, (5,10) : 0.69 dB	0.58 dB
$\frac{1}{3}$	1.02 dB	(4,6) : -0.11 dB, (8,12) : -0.21 dB	-0.25 dB
$\frac{1}{4}$	0.12 dB	(3,4) : -0.29 dB, (6,8) : -0.58 dB	-0.62 dB
$\frac{1}{5}$	-0.30 dB	(4,5) : -0.73 dB	-0.82 dB

Table 4.1: Upper bounds on the $\frac{E_b}{N_o}$–thresholds for ensembles of codes operating over a fully interleaved Rayleigh fading channel with space diversity (based on the maximum ratio combining principle) of order four, and perfect channel state information at the receiver.

Based on the notation in Section 4.5, we therefore obtain for the MBIOS channel

$$\begin{aligned} Q(y,\hat{a}|0) &= Q(-y,\hat{a}|1) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(y-\hat{a}\sqrt{\frac{2RE_b}{N_0}}\right)^2\right] 2\hat{a}\,\exp(-\hat{a}^2)\,, \\ p(y,\hat{a}|0) &= p(-y,\hat{a}|1) \\ &= \int_0^{+\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(y-a\sqrt{\frac{2RE_b}{N_0}}\right)^2\right] p_{a,\hat{a}}(a,\hat{a})\,da\,, \\ & \qquad\qquad -\infty < y < +\infty\,,\ \hat{a} \geq 0 \end{aligned}$$

where $a, \hat{a}$ are jointly Rayleigh distributed according to the following *pdf*:

$$p_{a,\hat{a}}(r_1,\, r_2) = \frac{4r_1r_2}{1-\rho^*}\, I_0\left(\frac{2\sqrt{\rho^*}\, r_1r_2}{1-\rho^*}\right) \exp\left(-\frac{r_1^2+r_2^2}{1-\rho^*}\right)\,, \quad r_1,\, r_2 \geq 0 \tag{4.100}$$

$E(a^2) = E(\hat{a}^2) = 1$. Here, the parameter ρ^* designates the correlation coefficient between the pairs of squared Rayleigh random variables $a^2, \hat{a}^2$, i.e.,

$$\rho^* = \frac{\mathrm{Cov}\,(a^2,\, \hat{a}^2)}{\sqrt{\mathrm{Var}(a^2)\,\mathrm{Var}(\hat{a}^2)}}\,. \tag{4.101}$$

The integral expressed in (4.100),(4.100) can be transformed to another integral which is easily calculated with the aid of the Gaussian numerical integration formula (see [296, Appendix C]).

By partitioning the considered linear and binary block code $\mathcal{C}$ of length N and rate R to constant Hamming weight subcodes, the tight version of the union bound on the ML decoding error probability, corresponding to the subcode of Hamming weight d $(0 \leq d \leq N)$, is given in this case by the expression

$$S_d\, E\left[Q\left(\sqrt{\frac{2RE_b}{N_0}}\, \frac{\sum_{i=1}^{d} a_i\,\hat{a}_i}{\sqrt{\sum_{i=1}^{d} \hat{a}_i^2}}\right)\right] \tag{4.102}$$

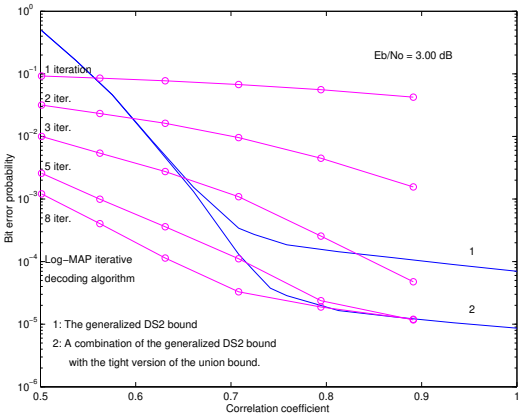


Figure 4.5: A comparison between two upper bounds on the bit error probability. The bounds refer to the ensemble of turbo codes depicted in Fig. 4.1, operating over a fully interleaved Rayleigh fading channel with mismatched decoding. The generalized DS2 bound is compared with the improved bounding technique which combines the DS2 bound with the tight form of the union bound (for every constant Hamming weight subcode). The bounds are demonstrated for $\frac{E_{\rm b}}{N_0} = 3$ dB, and are depicted as a function of the correlation coefficient (in the range $\frac{1}{2}$ to 1) between the squares of the i.i.d. jointly Rayleigh fading samples and their estimates.

where S_d designates the number of codewords of Hamming weight d, the Q-function is introduced in (2.8) and the notation E stands for the statistical expectation with respect to the i.i.d. Rayleigh fading samples $\{\underline{a}_i\}$ and their Rayleigh distributed estimations $\{\underline{\hat{a}}_i\}$. The expressions in (4.102) (where $0 \leq d \leq N$) are calculated via the Monte-Carlo method, by generating the correlated bivariate Rayleigh random variables with a certain correlation coefficient ρ^* (expressed in (4.101)), based on the algorithm proposed in [325]. The upper bound on the bit error probability based on the generalized DS2 bound in (4.96)-(4.99) is compared in Fig. 4.5 to the improved upper bound that combines the DS2 bound with the tight version of the union bound for every constant Hamming weight subcode. The comparison between these two bounds refers to the ensemble of uniformly interleaved turbo codes depicted in Fig. 4.1 where $\frac{E_{\rm b}}{N_0} = 3$ dB. It is reflected from Fig. 4.5 that the latter bound yields a considerably tighter error floor (the error floor in Figs. 4.5 and 4.6 is observed for a sufficiently high correlation coefficient ρ^*, and it reflects the robustness of the system to faulty measurements of the fading samples). The error floor exhibited by the improved upper bound in Fig. 4.5 also predicts reliably the error floor of the turbo codes associated with the suboptimal and efficient Log-MAP iterative decoding algorithm (based on computer simulations with 10 iterations). The upper bounds on the bit error probability that are based on the combination of the generalized DS2 bound in (4.96)-(4.99) and the tight version of the union bound (4.102) are depicted in Fig. 4.6 for several fixed values of $\frac{E_{\rm b}}{N_0}$. These curves are plotted as a function of the correlation coefficient ρ^* between the squares of the i.i.d. jointly Rayleigh fading samples and their estimates. The bounds in Fig. 4.6 refer to the ensemble without puncturing of rate $\frac{1}{3}$ turbo codes depicted in Fig. 4.1. Since for a fully interleaved Rayleigh fading channel with *perfect* side information on the fading samples, the channel cutoff rate corresponds to $\frac{E_{\rm b}}{N_0} = 3.23$ dB, then according to the upper bounds depicted in Fig. 4.6, the ensemble performance of these turbo codes (associated with the hypothetical ML decoding) is sufficiently robust in case of mismatched decoding, even in a portion of the rate region exceeding the channel cutoff rate.

The proposed upper bounds depicted in Figs. 4.5 and 4.6 were efficiently implemented in Matlab software, indicating their applicability in terms of complexity and the practical running time involved in their calculations.

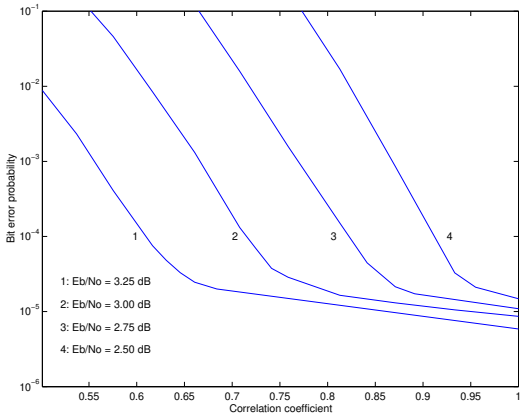


Figure 4.6: A comparison between upper bounds on the bit error probability for the ensemble of turbo codes depicted in Fig. 4.1. The codes operate over a fully interleaved Rayleigh fading channel with mismatched decoding. The bounds are based on combining the generalized DS2 bound with the tight form of the union bound, and it is applied to every constant Hamming weight subcode. The bounds are plotted for $\frac{E_b}{N_0} = 2.50, 2.75, 3.00$ and 3.25 dB, as a function of the correlation coefficient (in the range $\frac{1}{2}$ to 1) between the squares of the i.i.d. jointly Rayleigh fading samples and their estimates.

4.8 Summary and Conclusions

In this chapter, we discuss numerous efficient bounds on the decoding error probability of specific codes and ensembles under ML decoding, and we demonstrate the underlying connections that exist between them. In addressing the Gallager bounding techniques and their variations, we focus here on the Duman and Salehi variation, which originates from the classical 1965 Gallager bound. The considered upper bounds on the block and bit error probabilities under ML decoding rely on the distance spectrum and the input-output weight distributions of the codes, respectively, which are in general calculable. By generalizing the second version of the Duman and Salehi (DS2) bound which was originally derived for the binary-input AWGN channel [95], we demonstrate its remarkable generality and show that it provides the natural bridge between the 1961 and 1965 Gallager bounds (see Section 4.3). It is applicable for both random and specific codes, as well as for either bit or block error analysis. Some observations and interconnections between the Gallager and Duman and Salehi bounds are presented in Section 4.3, which partially rely on insightful observations made by Divsalar [86]. In particular, it is demonstrated in Section 4.3 that the 1965 Gallager random coding bound can be re-derived from the DS2 bound. The geometric interpretations of these Gallager-type bounds are introduced in Section 4.3, reflecting the non-uniqueness of their associated probability tilting measures. In Section 4.4, many reported upper bounds on the ML decoding error probability (or their Chernoff versions) are shown to be special cases of the DS2 bound. This is done by choosing an appropriate probability tilting measure and calculating the resulting bound with the specific chosen measure. This framework also facilitates to generalize the Shulman and Feder bound [303] (see [296, Appendix A]).

The proposed approach can be generalized to geometrically uniform non-binary codes, finite state channels, bit interleaved coded modulation systems, and it can be also used for the derivation of upper bounds on the conditional decoding error probability (as to account for a possible partitioning of the original code to subcodes). The DS2 bound is generalized in Section 4.5 to the mismatched decoding regime, where the decoder performs ML decoding with respect to a mis-

matched metric. The generalized bound is applicable to the analysis of the error performance of deterministic codes and ensembles. We address in particular the random coding version, which matches the 1965 Gallager random coding setting, and reproduces in Section 4.5 some known results (see [127], [164], [221]), hinging on an alternative approach which appropriately limits the considered ensemble of codes. Implications and applications of these observations are pointed out in Section 4.7, which include the fully interleaved fading channel with either matched or mismatched decoding.

The generalization of the DS2 upper bound for the analysis of the bit error probability yields the replacement of the distance spectrum $\{S_d\}_{d=1}^{N}$ (which appears in the upper bound on the block error probability) by the sequence $\{S'_d\}_{d=1}^{N}$ where: $S'_d = \sum_{w=1}^{NR} \left\{ \left(\frac{w}{NR} \right) A_{w,d} \right\}$, and $A_{w,d}$ designates the number of codewords in a systematic block code with an information Hamming weight w and a total Hamming weight d (where $0 \leq w \leq NR$ and $0 \leq d \leq N$). The derivation of the generalized DS2 bound on the bit error probability is detailed in [286], and it yields a considerable improvement in the tightness of the bound, as compared to [94, 95]. The computational complexity which is involved in calculating the DS2 bound is moderate.

The interconnections between many reported upper bounds are depicted in Fig. 4.7, where it is reflected that the DS2 bound yields many reported upper bounds as special cases.

The TSB [259] happens often to be the tightest reported upper bound for block codes which are transmitted over the binary-input AWGN channel and ML decoded (see Fig. 4.2 and [282, 283, 284, 287]). However, in the random coding setting, it fails to reproduce the random coding error exponent (see [259]), while the DS2 bound does. In fact, also Shulman and Feder bound which is a special case of the latter bound (see Section 4.4.1) achieves capacity for the ensemble of fully random block codes. This substantiates the claim that there is no uniformly best bound, providing the incentive for the generalization of Shulman and Feder bound (see [296, Appendix A]). However, we note that the loosened version of the TSB [86] (which involves the Chernoff inequality) maintains the asymptotic (i.e, for infinite block length) exponential tightness of the TSB of Poltyrev [259], and this loosened version is a special case of the DS2 bound (see Section 4.4.2).

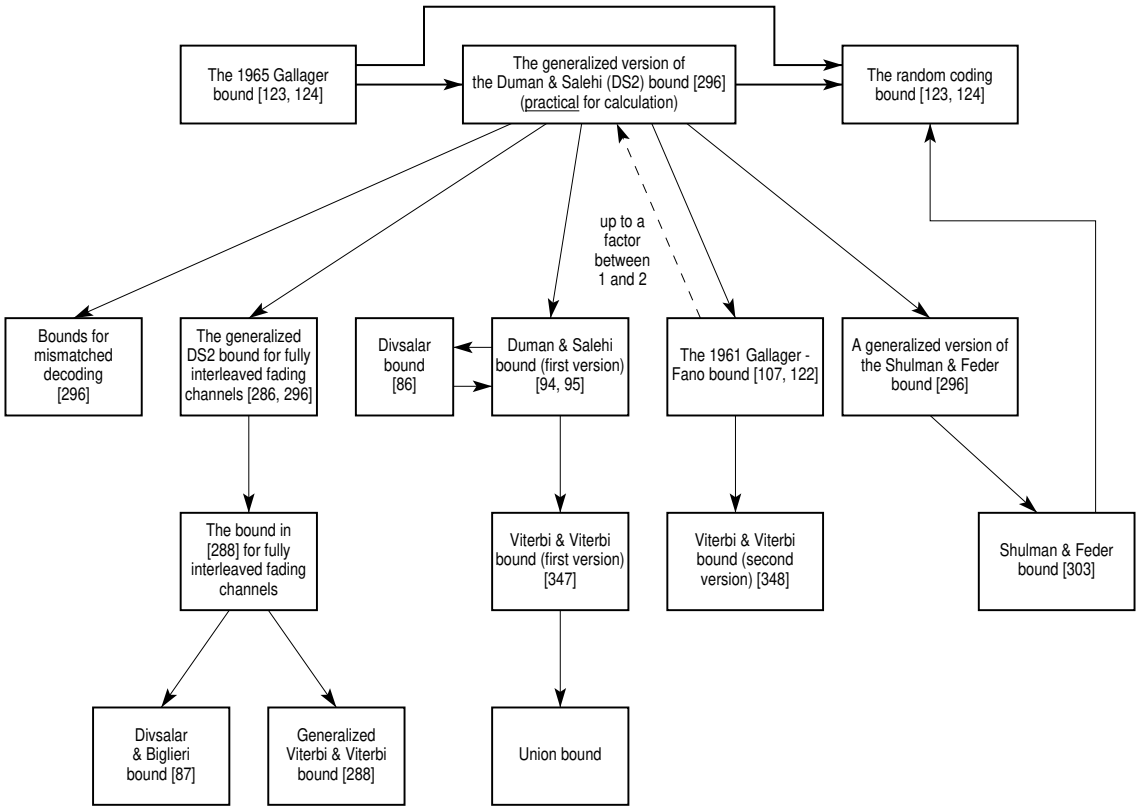


Figure 4.7: A diagram which shows the interconnections among the various upper bounds on the error probability with the ML decoding algorithm.

Chapter 5

Sphere-Packing Bounds on the Decoding Error Probability: Classical and Recent Results

Chapter overview: This chapter reviews the concepts used for the derivation of the classical 1967 sphere-packing bound which was derived by Shannon, Gallager and Berlekamp for discrete memoryless channels, and the 1959 sphere-packing bound of Shannon for the Gaussian channel. These ideas serve for presenting the recent improvements on the tightness of the sphere-packing bound, as suggested by Valembois and Fossorier. The new bound provides an improved version of the classical sphere-packing bound which is suitable for block codes of moderate block lengths, and it also applies to memoryless continuous-output channels.

5.1 Introduction

In the asymptotic case where the block length of a code tends to infinity, the best known lower bound on the decoding error probability for high levels of noise is the 1967 sphere-packing (SP67) bound [300]. Like the random coding bound of Gallager [123], the sphere-packing bound decreases exponentially with the block length. Further, the error exponent of the SP67 bound is a convex function of the rate which is known to be tight at the portion of the rate region between the critical rate (R_{c}) and the channel capacity; for this important rate region, the error exponent of the SP67 bound coincides with the error exponent of the random coding bound [300, Part 1]. For the AWGN channel, the 1959 sphere-packing (SP59) bound was derived by Shannon [299] by first showing that the error probability of any code whose codewords lie on a sphere must be greater than the error probability of a code of the same length and rate whose codewords are uniformly distributed over that sphere.

The reason that the SP67 bound fails to provide useful results for codes of small to moderate block length is due to the original focus in [300] on asymptotic analysis. In their paper [334], Valembois and Fossorier have recently revisited the SP67 bound in order to make it applicable for codes of moderate block lengths, and also to extend its field of application to continuous output channels (e.g, the AWGN channel which is the communication channel model of the SP59 bound of Shannon [299]). The motivation for the study in [334] was strengthened due to the outstanding performance of codes defined on graphs even of moderate block length. The remarkable improvement in the tightness of the SP67 was exemplified in [334] for the case of the AWGN channel with BPSK signaling, and it was shown that the improved version of the SP67 is an interesting alternative to the SP59 bound [299].

This chapter is structured as follows: Section 5.2 presents the SP59 bound of Shannon [299]

which provides a lower bound on the ML decoding error probability for the Gaussian channel in terms of the code rate and the block length of the code; this section also addresses some tips from [334] which are provided for simplifying the computation of the SP59 bound. Section 5.3 presents the classical SP67 bound which was originally derived by Shannon, Gallager and Berlekamp for a general discrete memoryless channel (DMC) [300], and it explains the concepts and the main ideas which were used for the derivation of the SP67 bound. Section 5.4 provides an improved version of the SP67 bound, as suggested by Valembois and Fossorier in [334], which makes the bound tighter for codes of moderate block lengths; the new bound extends the validity of this bounding technique also to memoryless continuous-output channels (e.g., the Gaussian channel, for which the SP59 bound applies); for the Gaussian channel, some numerical results from [334], comparing between the SP59 bound and the improved SP67 bound, are presented. Concluding remarks on sphere-packing bounds are finally addressed in Section 5.5.

5.2 The 1959 Shannon Lower Bound for the AWGN Channel

The 1959 sphere-packing (SP59) bound of Shannon [299] provides a lower bound on the decoding error probability of an arbitrary block code whose transmission takes place over an AWGN channel. Consider a block code $\mathcal{C}$ where each of its codewords forms a sequence of N input symbols, and assume that the rate of the code is R bits per channel use per dimension. It is assumed that all the codewords are mapped to signals which lie on an N-dimensional sphere whose center is located at the origin; hence, all the modulated signals have the same energy (e.g., PSK modulation), but the bound does not take into account the particular modulation which is used. Since all the signals lie on the surface of a sphere whose center coincides with the origin of $\mathbb{R}^N$, every Voronoi cell is a polyhedric cone which is limited by at most $2^{NR} - 1$ hyper planes intersecting at the origin.

Definition 5.1 (Solid Angle of a Cone)**.** The solid angle of a cone is defined as the area of a sphere of unit radius (in the same Euclidean space) cut out by the cone.

Since the 2^{NR} Voronoi cells partition the Euclidean space $\mathbb{R}^N$, the sum of their solid angles is equal to the solid angle of the whole space, i.e., the it is equal to the total area of a sphere of unit radius in the N-dimensional space.

The main idea used for the derivation of the SP59 bound is that the decoding error probability of an arbitrary code under optimal ML decoding is lower bounded by the decoding error probability which corresponds to the case where the Voronoi regions are circular cones of solid angles which are all equal to a fraction 2^{-NR} of the solid angle of $\mathbb{R}^N$. This comes as a corollary of the two following propositions [299]:

- Among the cones of a given solid angle, the circular cone provides the lowest decoding error probability under ML decoding.
- In order to minimize the average decoding error probability, it is best to share the total solid angle equally between the 2^{NR} Voronoi cells.

Lemma 5.1 (Solid Angle of a Circular Cone)**.** The solid angle of a circular cone of half angle θ in $\mathbb{R}^N$ is given by

$$\Omega_N(\theta) = \frac{2\pi^{\frac{N-1}{2}}}{\Gamma(\frac{N-1}{2})} \int_0^\theta (\sin\phi)^{N-2}\, d\phi \,. \tag{5.1}$$

In particular, the solid angle of the whole space is given by

$$\Omega_N(\pi) = \frac{2\pi^{\frac{N-1}{2}}}{\Gamma(\frac{N-1}{2})} \int_0^{\pi} (\sin\phi)^{N-2}\, d\phi = \frac{2\pi^{\frac{N}{2}}}{\Gamma(\frac{N}{2})}\ . \tag{5.2}$$

Theorem 5.1 (The 1959 Sphere-Packing (SP59) Bound)**.** Let

$$f_n(x) = \frac{1}{2^{(n-1)/2}\Gamma(\frac{n+1}{2})} \int_0^{\infty} z^{n-1} \exp(-z^2 + zx)\, dz\ , \quad \forall\, x \in \mathbb{R},\ n \in \mathbb{N} \tag{5.3}$$

and

$$\begin{aligned} P_{\mathrm{SPB}}(N,\theta,A) &= \frac{(N-1)e^{-\frac{NA^2}{2}}}{\sqrt{2\pi}} \int_{\theta}^{\frac{\pi}{2}} (\sin\phi)^{N-2}\ f_N(\sqrt{N}A\cos\phi)\ d\phi \\ &\quad + Q(\sqrt{N}A). \end{aligned} \tag{5.4}$$

Then, the ML decoding error probability of any code of block length N and rate R satisfies the following lower bound:

$$P_{\mathrm{e}}(\mathrm{ML}) > P_{\mathrm{SPB}}(N,\theta,A)\ , \quad A \triangleq \sqrt{\frac{2E_{\mathrm{s}}}{N_0}} \tag{5.5}$$

for all $\theta \in [0,\pi]$ so that $2^{-NR} \le \frac{\Omega_N(\theta)}{\Omega_N(\pi)}$.

Lemma 5.2 (Bounds on the Solid Angle)**.** The solid angle of a circular cone of half angle θ in the Euclidean space $\mathbb{R}^N$ satisfies the inequality

$$\frac{\Gamma(\frac{N}{2})(\sin\theta)^{N-1}}{2\Gamma(\frac{N+1}{2})\sqrt{\pi}\cos\theta}\left(1 - \frac{(\tan\theta)^2}{N}\right) \le \frac{\Omega_N(\theta)}{\Omega_N(\pi)} \le \frac{\Gamma(\frac{N}{2})(\sin\theta)^{N-1}}{2\Gamma(\frac{N+1}{2})\sqrt{\pi}\cos\theta}\ . \tag{5.6}$$

Corollary 5.1 (SP59 Bound (Cont.))**.** If θ^* satisfies the equation

$$\frac{\Gamma(\frac{N}{2})(\sin\theta^*)^{N-1}}{2\Gamma(\frac{N+1}{2})\sqrt{\pi}\cos\theta^*}\left(1 - \frac{(\tan\theta^*)^2}{N}\right) = 2^{-NR} \tag{5.7}$$

then $\frac{\Omega_N(\theta^*)}{\Omega_N(\pi)} \ge 2^{-NR}$, and thus

$$P_{\mathrm{e}}(\mathrm{ML}) > P_{\mathrm{SPB}}(N,\theta^*,A) \tag{5.8}$$

In [334, Section 2], the SP59 bound was reviewed, and an algorithm which simplifies the calculation of the bound is given in the following theorem:

Theorem 5.2 (Recursive Equations for Simplifying the Calculation of the SP59 Bound)**.** [334, Theorem 3]: The set of functions $\{f_n\}$ introduced in (5.3) can be rewritten in the alternative form

$$f_n(x) = P_n(x) + Q_n(x)\exp(\frac{x^2}{2}) \int_{-\infty}^{x} \exp(-\frac{t^2}{2})\, dt\ , \quad x \in \mathbb{R},\ n \in \mathbb{N} \tag{5.9}$$

where P_n and Q_n are two polynomials which can be determined by the same recursive equation for all $n \ge 5$:

$$\begin{aligned} P_n(x) &= \frac{2n-5+x^2}{n-1}\ P_{n-2}(x) - \frac{n-4}{n-1}\ P_{n-4}(x)\ , \\ Q_n(x) &= \frac{2n-5+x^2}{n-1}\ Q_{n-2}(x) - \frac{n-4}{n-1}\ Q_{n-4}(x) \end{aligned} \tag{5.10}$$

with the initial conditions

$$\begin{aligned}
&P_1(x) = 0, \quad Q_1(x) = 1 \\
&P_2(x) = \sqrt{\frac{2}{\pi}}, \quad Q_2(x) = \sqrt{\frac{2}{\pi}}\, x \\
&P_3(x) = \frac{x}{2}, \quad Q_3(x) = \frac{1+x^2}{2} \\
&P_4(x) = \sqrt{\frac{2}{\pi}}\frac{2+x^2}{3}, \quad Q_4(x) = \sqrt{\frac{2}{\pi}}\frac{3x+x^3}{3}.
\end{aligned} \tag{5.11}$$

Proof. See [334, Appendix A]. □

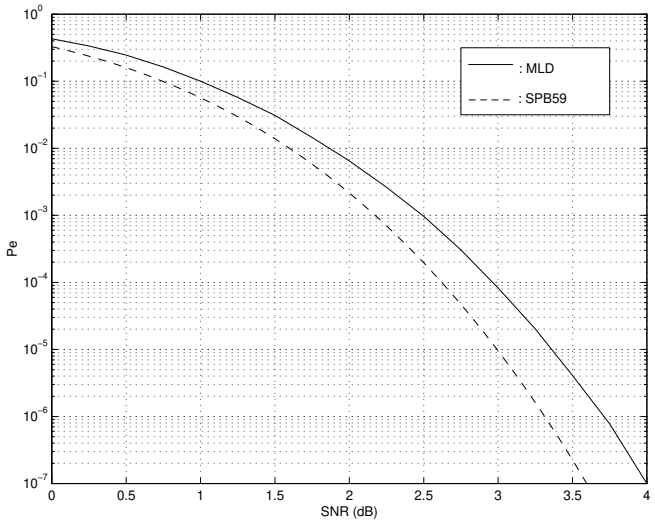


Figure 5.1: A comparison between the 1959 sphere-packing (SP59) bound of Shannon [299] for $(128, 64)$ block codes and the ML performance of the extended $(128, 64, 22)$ BCH code over the AWGN channel with BPSK modulation. The performance of the extended BCH code relies on the BMA algorithm from [333] which is a near-ML decoding algorithm with reduced complexity. This figure is reproduced (with permission) from [334]. For further details, the interested reader is referred to [333, Section 6B].

Note that the algorithm can be applied to the calculation of the SP59 bound for values of N not exceeding 1,000 (due to numerical problems of overflow which are related to the calculation of the recursive equations (5.10) for larger values of N. For numerical accuracy purposes, it was suggested in [334] to use the two recursive equations (5.10) with their initial values in (5.11) in order to pre-compute the coefficients of the two polynomials P_N and Q_N (i.e., by simply performing convolution operations); note that the function

$$y(x) = \exp(\frac{x^2}{2}) \int_{-\infty}^{x} \frac{\exp\left(-\frac{t^2}{2}\right)}{\sqrt{2\pi}}\, dt$$

appearing as a multiplicative function at the second term of the RHS in (5.9) can be easily calculated using standard programming languages; e.g., with the MATLAB software, the function y can be expressed in the form

$$y(x) = \mathrm{erfcx}(-\frac{x}{\sqrt{2}})\,, \quad x \in \mathbb{R}$$

and for $x \ll -1$, it is well approximated by $y(x) \approx -\frac{1}{\sqrt{2\pi}\, x}$.

In [299], Shannon provided good asymptotic formula for the SP59 bound with large values of N. This asymptotic expression was not proved yet as a rigorous lower bound, but it gives very accurate results for the SP59 bound as we let the block length N become large (see also [91]).

Fig. 5.1 which was reproduced from [334] refers to the $(128, 64, 22)$ extended BCH codes; it compares the SP59 bound with the actual performance achieved by a near-ML decoding algorithm when transmission takes place over an AWGN channel and the codewords are BPSK modulated (for further details about this sub-optimal decoding algorithm which is referred in Fig. 5.1, the interested reader is referred to [333, Section 6B]). This figure shows a small gap of only 0.34 dB for a target block error probability of $P_{\mathrm{t}} = 10^{-4}$ between the SP59 bound and the performance of the considered rate-$\frac{1}{2}$ code of block length 128 bits.

A comparison of the performance of turbo codes under iterative sum-product decoding algorithm with the SP59 lower bound shows (see [91, 182, 210, 293]) the existence of good turbo codes of rate between $\frac{1}{6}$ and $\frac{1}{2}$ which are about 0.7 dB away of the SP59 bound for a target error probability of 10^{-4} and moderate block lengths.

5.3 The 1967 Sphere-Packing Bounds

In the following, we outline the concepts and the main ideas which serve for the derivation of the 1967 sphere-packing (SP67) bound, as was originally introduced by Shannon, Gallager and Berlekamp [300]. The SP67 bound applies to all discrete memoryless channels, and it possesses the very pleasing feature that its error exponent is exact for all rates between the critical rate and channel capacity; it also has this nice feature in the limit where the code rate tends to zero. The analysis in [300] was primarily focused on the exponential behavior of the SP67 bound, and the aim of the authors was to simply the derivation of the bound as much as possible as long as no penalty in the exponential behavior of the resulting lower bound is incurred.

Apart of outlining the classical derivation of the SP67 bound, the discussion in this section also serves as a preparatory step towards the consideration of possible refinements of the SP67 bound for making the new bound attractive for moderate block lengths. Hence, in the next section, we rely on the proof concepts introduced in this section, and use them to explain the improvements on the SP67 bound, as suggested by Valembois and Fossorier [334]. By doing so, Valembois and Fossorier have managed to adapt their new sphere-packing bound so that it also applies to memoryless channels with continuous output (and not only to discrete memoryless channels); their new bound competes well with the 1959 sphere-packing bound of Shannon [299]. Their recent study in [334], extending the applicability of the sphere-packing bounding technique for relatively short block lengths, was very highly motivated by the outstanding performance of turbo-like codes which obtain reliable communication at rates which are close to capacity with tolerable latency and complexity (even under sub-optimal iterative decoding algorithms). To conclude, in this section we start our discussion by outlining the concepts of the proofs used for the classical derivation of the SP67 bound [300], and then move in the next section to improved versions of this bound for moderate block lengths and for channels with continuous outputs; we consider this study of primary significance in the evaluation of how close to optimal are turbo-like codes together with their (sub-optimal) iterative decoding algorithms (in this respect, the interested reader is also referred to [91], [182] and [210]).

For the derivation of the classical SP67 bound [300], let $\mathbf{x}_1, \ldots, \mathbf{x}_M$ be a set of M codewords each

of length N, and assume that their transmission takes place over a discrete memoryless channel. Also assume a list decoding scheme where for each received sequence $\mathbf{y}$, it outputs a list of at most L integers from 1 to M; these integers correspond to the indices of the codewords (or the corresponding messages before encoding). If message m is transmitted over the channel, an error is declared if the index m does not appear in the list. In [300], the authors derive a lower bound on the decoding error probability of a code with M codewords of block length N, and for an arbitrary list decoder scheme whose size is limited to L. The particular case where $L = 1$ clearly provides a lower bound on the decoding performance under optimal ML decoding where the list size is limited to 1 for each received sequence at the channel output. Referring to a list decoder of size at most L, the code rate (in nats per symbol use) is given by $R = \frac{\ln\left(\frac{M}{L}\right)}{N}$; the motivation for this definition is that for the case where $L = 1$, this definition particularizes to the standard definition of code rate, where in the other extreme where $L = M$, all the codewords are in the list for any output sequence $\mathbf{y}$, and hence there is no need for a communication channel.

Let $\mathcal{Y}_m$ be the set of output sequences $\mathbf{y}$ for which message m is on the decoding list, then the probability of list decoding error when message m is sent over a DMC is given by

$$P_{\mathrm{e},m} = \sum_{y \in \mathcal{Y}_m^c} \mathrm{Prob}\left(\mathbf{y}|\mathbf{x}_m\right) \tag{5.12}$$

where the superscript c designates the complement of a set. Let $P_{\mathrm{e,max}}$ designate the maximum over m of $P_{\mathrm{e},m}$ for the code and list decoding under consideration. Assuming that the codewords are equally likely to be transmitted, the average decoding error probability is given by

$$P_{\mathrm{e}} = \frac{1}{M} \sum_{m=1}^{M} P_{\mathrm{e},m}. \tag{5.13}$$

In [300], the derivation of the SP67 bound was done in three steps, where the authors first derive upper and lower bounds on the decoding error probability for a block code with simply two codewords, later derive a lower bound on the error probability for fixed composition codes, and finally derive the SP67 bound for an arbitrary block code under the assumption of list decoding. In the following, we outline the derivation of the SP67 bound according to these steps.

Upper and Lower Bounds for a Code with Two Codewords

Let $P_m(\mathbf{y})$ (where $m = 1, 2$) be the probability of receiving the sequence $\mathbf{y}$ when message m is transmitted. For initial motivation, it is assumed that the decoder is ML, and let $\mathcal{Y}_m$ be the set of sequences decoded into message m. Summing (5.12) over m gives the equality

$$P_{\mathrm{e},1} + P_{\mathrm{e},2} = \sum_{\mathbf{y}} \min_{m=1,2} P_m(\mathbf{y}).$$

In order to obtain upper and lower bounds on the two conditional error probabilities $P_{\mathrm{e},1}$ and $P_{\mathrm{e},2}$ and to study the tradeoff between these two error probabilities, the authors rely on a simple and useful inequality which is later shown to be asymptotically tight in the limit where N tends to infinity, and which states that

$$\min_{m=1,2} P_m(\mathbf{y}) \leq P_1(\mathbf{y})^{1-s} P_2(\mathbf{y})^s \leq \max_{m=1,2} P_m(\mathbf{y}), \quad 0 < s < 1$$

hence, the combination of the last two equations gives

$$P_{\mathrm{e},1} + P_{\mathrm{e},2} \leq e^{\mu(s)}, \quad 0 < s < 1. \tag{5.14}$$

where

$$\mu(s) \triangleq \ln\Big(\sum_{\mathbf{y}} P_1(\mathbf{y})^{1-s} P_2(\mathbf{y})^s\Big). \tag{5.15}$$

By extending the definition of the function μ to the interval [0,1]:

$$\mu(0) \triangleq \lim_{s\to 0^+} \mu(s), \quad \mu(1) \triangleq \lim_{s\to 1^-} \mu(s)$$

so as to cover the endpoints $s = 0$ and $s = 1$, one obtains from (5.14) the inequality

$$P_{\mathrm{e},1} + P_{\mathrm{e},2} \leq \min_{0\leq s\leq 1} e^{\mu(s)}. \tag{5.16}$$

Let the codewords be denoted by $\mathbf{x}_m = (k_{m,1}, \ldots, k_{m,N})$ where $m = 1, 2$, and let the received sequence be $\mathbf{y} = (j_1, \ldots, j_N)$. Then, since the communication channel is memoryless, then $P_m(\mathbf{y}) = \prod_{n=1}^{N} \mathrm{Prob}\,(j_n|k_{m,n})$, and the function μ is expressible in the form

$$\mu(s) = \sum_{n=1}^{N} \mu_n(s), \quad \mu_n(s) \triangleq \ln\Big(\sum_{j=1}^{J} \mathrm{Prob}\,(j|k_{1,n})^{1-s} \mathrm{Prob}\,(j|k_{2,n})^s\Big). \tag{5.17}$$

The next step in [300] is the derivation of upper and lower bounds on $P_{\mathrm{e},1}$ and $P_{\mathrm{e},2}$, where the authors allow themselves the flexibility of making $P_{\mathrm{e},1}$ very much larger than $P_{\mathrm{e},2}$ or vice versa. The authors prove the following theorem:

Theorem 5.3 (Upper and Lower Bounds for the Pairwise Error Probability)**.** [300, Theorem 5]: Let $P_1(\mathbf{y})$ and $P_2(\mathbf{y})$ be two probability assignments on a discrete set of sequences, let $\mathcal{Y}_1$ and $\mathcal{Y}_2$ be disjoint decision regions for these sequences, let $P_{\mathrm{e},1}$ and $P_{\mathrm{e},2}$ be given by (5.12), and assume that $P_1(\mathbf{y})P_2(\mathbf{y}) \neq 0$ for at least one sequence $\mathbf{y}$. Then, for all $s \in (0, 1)$, either

$$P_{\mathrm{e},1} > \frac{1}{4} \exp\Big(\mu(s) - s\mu'(s) - s\sqrt{2\mu''(s)}\Big) \tag{5.18}$$

or

$$P_{\mathrm{e},2} > \frac{1}{4} \exp\Big(\mu(s) + (1-s)\mu'(s) - (1-s)\sqrt{2\mu''(s)}\Big) \tag{5.19}$$

where the function μ is introduced in (5.15). Furthermore, for an appropriate choice of $\mathcal{Y}_1$ and $\mathcal{Y}_2$

$$P_{\mathrm{e},1} \leq \exp\Big(\mu(s) - s\mu'(s)\Big) \tag{5.20}$$

and

$$P_{\mathrm{e},2} \leq \exp\Big(\mu(s) + (1-s)\mu'(s)\Big). \tag{5.21}$$

Finally, the function μ is non-positive and convex over the interval $(0, 1)$. The convexity of μ is strict unless $\frac{P_1(\mathbf{y})}{P_2(\mathbf{y})}$ is constant over all the sequences $\mathbf{y}$ for which $P_1(\mathbf{y})P_2(\mathbf{y}) \neq 0$. Moreover, the function μ is strictly negative over the interval $(0, 1)$ unless $P_1(\mathbf{y}) = P_2(\mathbf{y})$ for all $\mathbf{y}$.

Concept of the proof: Taking the first and second derivatives of the function μ gives

$$\mu'(s) = \sum_{\mathbf{y}} Q_s(\mathbf{y}) D(y), \quad \mu''(s) = \left(\sum_{\mathbf{y}} Q_s(\mathbf{y}) D(y)^2\right) - \mu'(s)^2 \tag{5.22}$$

where Q_s is defined to be the probability measure

$$Q_s(\mathbf{y}) = \frac{P_1(\mathbf{y})^{1-s} P_2(\mathbf{y})^s}{\sum_{\mathbf{y}'} P_1(\mathbf{y}')^{1-s} P_2(\mathbf{y}')^s}$$

and $D(y) \triangleq \ln \frac{P_2(\mathbf{y})}{P_1(\mathbf{y})}$ designates the log-likelihood ratio. If we consider $D(\mathbf{y})$ to be a RV with probability assignment $Q_s(\mathbf{y})$, then it follows from (5.22) that the first and second derivatives of μ are equal to the expectation and the variance of the RV $D(y)$, respectively, i.e.,

$$\mu'(s) = \mathbb{E}_{Q_s}\big\{D(\mathbf{y})\big\}, \quad \mu''(s) = \mathrm{Var}_{Q_s}\big\{D(\mathbf{y})\big\}.$$

Hence, $\mu''(s) \geq 0$, so the function μ is convex (and it is strictly convex if and only if $D(y)$ is not a constant for all the sequences $\mathbf{y}$ for which $P_1(\mathbf{y})P_2(\mathbf{y}) \neq 0$). It is direct to show that the probability assignments P_1 and P_2 are expressible in the form

$$P_1(\mathbf{y}) = \exp\big(\mu(s) - sD(\mathbf{y})\big)\ Q_s(\mathbf{y}) \tag{5.23}$$

$$P_2(\mathbf{y}) = \exp\big(\mu(s) + (1-s)D(\mathbf{y})\big)\ Q_s(\mathbf{y}). \tag{5.24}$$

For the derivation of the lower bounds, the authors define in [300] the parameterized set

$$\begin{aligned} \widetilde{\mathcal{Y}}_s &\triangleq \Big\{\mathbf{y} : |D(\mathbf{y}) - \mu'(s)| \leq \sqrt{2\mu''(s)}\Big\} \\ &= \big\{\mathbf{y} : |D(\mathbf{y}) - \mathbb{E}_{Q_s}\big\{D(\mathbf{y})\big\}|^2 \leq 2\mathrm{Var}_{Q_s}\big\{D(\mathbf{y})\big\}\big\} \end{aligned} \tag{5.25}$$

hence from the Chebychev inequality, the probability that the RV $D(y)$ does not deviate from its statistical expectation by more than twice its standard deviation is greater than one-half, i.e., $\sum_{\mathbf{y}\in\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}) > \frac{1}{2}$. The idea of the derivation of the lower bounds on $P_{\mathrm{e},1}$ and $P_{\mathrm{e},2}$ is to take into account only the error events where the received sequence $\mathbf{y}$ falls within the set $\widetilde{\mathcal{Y}}_s$ defined in (5.25). From (5.23) and (5.25)

$$\begin{aligned} P_{\mathrm{e},1} &= \sum_{\mathbf{y}\in\mathcal{Y}_1^{\mathrm{c}}} P_1(\mathbf{y}) \\ &\geq \sum_{\mathbf{y}\in\mathcal{Y}_1^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} P_1(\mathbf{y}) \\ &\geq \exp\Big(\mu(s) - s\mu'(s) - s\sqrt{2\mu''(s)}\Big) \sum_{\mathcal{Y}_1^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}) \end{aligned} \tag{5.26}$$

and similarly

$$P_{\mathrm{e},2} \geq \exp\Big(\mu(s) + (1-s)\mu'(s) - (1-s)\sqrt{2\mu''(s)}\Big) \sum_{\mathcal{Y}_2^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}). \tag{5.27}$$

Finally, since the sets $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are assumed to be disjoint, then

$$\sum_{\mathcal{Y}_1^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}) + \sum_{\mathcal{Y}_2^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}) = \sum_{\mathbf{y}\in\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y}) > \frac{1}{2}$$

so at least one of the terms $\sum_{\mathcal{Y}_1^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y})$ and $\sum_{\mathcal{Y}_2^{\mathrm{c}}\cap\widetilde{\mathcal{Y}}_s} Q_s(\mathbf{y})$ should be at least $\frac{1}{4}$ which finally leads to the validity of at least one of the lower bounds in (5.18) and (5.19).

For the derivation of the upper bounds in (5.20) and (5.21), let

$$\mathcal{Y}_1 \triangleq \big\{\mathbf{y} | D(\mathbf{y}) < \mu'(s)\big\}, \quad \mathcal{Y}_2 \triangleq \mathcal{Y}_1^{\mathrm{c}} \tag{5.28}$$

so that the sets $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are disjoint. The derivation of the upper bounds in (5.20) and (5.21) follows from (5.23) and (5.24), and it relies on the definition of the disjoint sets $\mathcal{Y}_1$ and $\mathcal{Y}_2$ in (5.28). To obtain the upper bounds, one simply replaces the sums $\sum_{\mathbf{y}\in\mathcal{Y}_m} Q_s(\mathbf{y})$ by 1.

Concepts of Proof for the Derivation of the SP67 Bound for Fixed Composition Codes

Following the derivation of upper and lower bounds on the pairwise error probability (see Theorem 5.3), Shannon, Gallager and Berlekamp derived in [300] a lower bound on the decoding error probability for *fixed composition codes*. Let $\mathcal{C}$ be a block code of M codewords, each of length N, and assume that the transmission of this code takes place over a DMC whose probability law is given by $P(j|k)$ where $k \in \{1, \ldots, K\}$ and $j \in \{1, \ldots, J\}$ designate the input and output symbols, respectively. For the decoding of the code $\mathcal{C}$, an arbitrary list decoding scheme is assumed where the size of the list is limited to L for any output sequence $\mathbf{y}$. By assumption, since the code $\mathcal{C}$ has a fixed composition, then each symbol $k \in \{1, \ldots, K\}$ appears in every one of the codewords $\mathbf{x}_1, \ldots, \mathbf{x}_M$ in $\mathcal{C}$ an equal number of times. Let us define the vector $\mathbf{q} \triangleq (q_1, \ldots, q_K)$ to be a K-length vector where q_k $(k = 1, \ldots, K)$ designates the fraction of appearances of the symbol k in each one of the codewords $\mathbf{x}_1, \ldots, \mathbf{x}_M$. Let $P_{\mathrm{e},m}$, as given in (5.12), be the error probability for the list decoder under consideration given message m is transmitted, and let $P_{\mathrm{e,max}}$ designate the maximum over m of $P_{\mathrm{e},m}$ (where $m = 1, \ldots, M$). For a given m, one can reduce the value of $P_{\mathrm{e},m}$ by enlarging the set $\mathcal{Y}_m$ of output sequences for which message m is on the decoding list. However, since the size of the list is limited to L for any output sequence, this will decrease the size of the set $\mathcal{Y}_{m'}$ for some $m' \neq m$, and hence, this will increase the value of $P_{\mathrm{e},m'}$. In order to keep some control on the size of the set $\mathcal{Y}_m$ without specifically considering the other codewords, the authors in [300] define an arbitrary tilting measure which can be expressed in the product form

$$f_N(\mathbf{y}) = \prod_{n=1}^{N} f(j_n) \tag{5.29}$$

for any output sequence $\mathbf{y} = (j_1, \ldots, j_N)$, and define accordingly the *size* of the set $\mathcal{Y}_m$ as

$$F(\mathcal{Y}_m) \triangleq \sum_{\mathbf{y} \in \mathcal{Y}_m} f_N(\mathbf{y}). \tag{5.30}$$

Next they rely on Theorem 5.3 in order to relate the conditional error probability $P_{\mathrm{e},m}$ and $F(\mathcal{Y}_m)$; to this end, they let $\mathrm{Prob}\,(\mathbf{y}|\mathbf{x}_m)$ correspond to $P_1(\mathbf{y})$ and let $f_N(\mathbf{y})$ correspond to $P_2(\mathbf{y})$ in the proof of Theorem 5.3. In light of these substitutions, the function μ in (5.15) gets the form

$$\mu(s) \triangleq \ln\Big(\sum_{\mathbf{y}} \mathrm{Prob}\,(\mathbf{y}|\mathbf{x}_m)^{1-s} f_N(\mathbf{y})^s\Big) \tag{5.31}$$

and then, similarly to (5.17), since the tilting measure f_N is expressed in the product form (5.29) and the code is assumed to have a fixed composition, then the function μ in (5.31) is expressible by the sum

$$\mu(s) = N \sum_{k=1}^{K} q_k \mu_k(s, f), \quad \mu_k(s, f) \triangleq \ln\left(\sum_{j=1}^{J} P(j|k)^{1-s} f(y)^s\right). \tag{5.32}$$

By doing so, Theorem 5.3 provides a lower bound on either $P_{\mathrm{e},m}$ and $F(\mathcal{Y}_m)$ which is valid for every $m \in \{1, \ldots, M\}$. Next, since the size of the list decoding is limited to L, then the sum of the terms $F(\mathcal{Y}_m)$ in (5.30) satisfies

$$\sum_{m=1}^{M} F(\mathcal{Y}_m) = \sum_{m=1}^{M} \sum_{\mathbf{y} \in \mathcal{Y}_\mathbf{m}} f_N(\mathbf{y}) \leq \sum_{l=1}^{L} \sum_{\mathbf{y}} f_N(\mathbf{y}) = L \tag{5.33}$$

so there should exist an integer $m \in \{1, \ldots, M\}$ for which $F(\mathcal{Y}_m) \leq \frac{L}{M}$. Moreover, for this value of m, one can simply replace a lower bound on $P_{\mathrm{e},m}$ with a lower bound on $P_{\mathrm{e,max}}$. Based on the

additivity property of the function μ in (5.17), the term $\sqrt{\mu''(s)}$ in the exponents of the lower bounds in (5.18) and (5.19) is proportional to the square root of N while the other terms $\mu(s)$ and $\mu'(s)$ in these exponents are linearly proportional to N. Hence, since Shannon, Gallager and Berlekamp mainly focused in their paper [300] on the exponential behavior of the upper and lower bounds for large values of N, they chose for the simplicity of their analysis to obtain a loose but general upper bound on $\sqrt{\mu''(s)}$. By doing so, this does not affect the exponential behavior of the bounds in the asymptotic case where N tends to infinity; however, as we will see in the next section, this bound is loose when considering codes of moderate block lengths, and motivated Valembois and Fossorier [334] to improve the tightness of the bound on $\sqrt{\mu''(s)}$ in order to improve the sphere-packing bound for finite length codes. A loose but simple bound on the second derivative of the function μ was derived in [300, Appendix B], and it get the form

$$s\sqrt{\mu_k''(s,f)} \le \ln\Big(\frac{e}{\sqrt{P_{\min}}}\Big) \tag{5.34}$$

where $P_{\min}$ designates the smallest non-zero transition probability of the DMC.

Following this approach and using the additivity property of the function μ in (5.32) and inequality (5.34), one obtains from Theorem 5.3 that if there exists a sequence $\mathbf{y}$ for which $\text{Prob}\,(\mathbf{y}|\mathbf{x}_m) f_N(\mathbf{y}) \neq 0$, then either

$$P_{\text{e,max}} > \exp\left\{ N \left[\sum_{k=1}^{K} q_k \big(\mu_k(s,f) - s\mu_k'(s,f)\big) + O_1\left(\frac{1}{\sqrt{N}}\right)\right]\right\} \tag{5.35}$$

or

$$\frac{L}{M} \ge F(\mathcal{Y}_m) > \exp\left\{ N \left[\sum_{k=1}^{K} q_k \big(\mu_k(s,f) + (1-s)\mu_k'(s,f)\big) + O_2\left(\frac{1}{\sqrt{N}}\right)\right]\right\} \tag{5.36}$$

where, for large N, the terms $O_1\left(\frac{1}{\sqrt{N}}\right)$ and $O_2\left(\frac{1}{\sqrt{N}}\right)$ scale like the inverse of the square root of N; hence, they asymptotically do not affect the exponents of these two bounds for large N. The inequalities in (5.35) and (5.36) provide a parametric lower bound on $P_{\text{e,max}}$ for a given $\frac{L}{M}$ in terms of the parameter s (where $0 < s < 1$) in the same way Theorem 5.3 provides a parametric lower bound on $P_{\text{e},1}$ for a given $P_{\text{e},2}$. This lower bound is valid for an arbitrary fixed composition block code of composition $\mathbf{q}$ with M codewords of length N, and for an arbitrary list decoder whose lists are limited to size L. Since the lower bound is a function of an arbitrary tilting measure f which can be expanded in the form (5.29), one would in general wishes to find the function f which *maximizes* the lower bound on $P_{\text{e,max}}$ for a given composition. In addition, one may look for the best composition in the sense that the composition $\mathbf{q}$ *minimizes* this lower bound. This is a kind of a min-max optimization problem. The authors of [300] find the following solution for this min-max problem: For $0 < s < 1$, let $\mathbf{q}_s = (q_{1,s}, \ldots, q_{K,s})$ satisfy the condition

$$\sum_{j=1}^{J} P(j|k)^{1-s} (\alpha_{j,s})^{\frac{s}{1-s}} \ge \sum_{j=1}^{J} (\alpha_{j,s})^{\frac{1}{1-s}} \ , \quad \forall\, k = 1, \ldots, K \tag{5.37}$$

where

$$\alpha_{j,s} \triangleq \sum_{k=1}^{K} q_{k,s} P(j|k)^{1-s} \tag{5.38}$$

and let $\mathbf{f}_s \triangleq (f_s(1), \ldots, f_s(J))$ in the product form (5.29) be given by

$$f_s(j) = \frac{(\alpha_{j,s})^{\frac{1}{1-s}}}{\sum_{j'=1}^{J} (\alpha_{j',s})^{\frac{1}{1-s}}} \ . \tag{5.39}$$

Note that by multiplying both sides of (5.37) by $q_{k,s}$ and summing over $k \in \{1, \ldots, K\}$, the condition in (5.37) should be satisfied in equality for all k for which $q_{k,s} > 0$. This form of solution is reminiscent of the form of solution given in [124, Theorem 5.6.5] for the random coding error exponent. To this end, one simply needs to substitute $\rho = \frac{s}{1-s}$ in order to make Eqs. (5.37) and (5.38) look similar to those referring to the random coding error exponent in [124, Eqs. (5.6.37) and (5.6.38)]. In light of the analysis of the random coding error exponent [123], and the similarity of the two expressions, it was ensured in [300] that a solution to the equations given in (5.37) and (5.38) necessarily exist. Since the parameter s lies within the interval $(0, 1)$, then from the following substitution, the range of ρ is $[0, \infty)$; this is in contrast to the random coding error exponent where ρ is restricted to lie within the interval $[0, 1]$ (see Section 4.2.1). The methodology of the bounding technique in [300], used for the derivation of the sphere-packing bound for fixed composition codes, was to verify that unless the lower bound on $P_{\text{e,max}}$ becomes trivial (i.e., the case where it is equal to zero), there exists a value of $s \in (0, 1)$ for which inequality in (5.36) is violated, and hence for this value of s, the lower bound on $P_{\text{e,max}}$ should be necessarily satisfied. This approach gives the following lower bound on the maximal decoding error probability:

$$P_{\text{e,max}} > \exp\left[-N\left(E_0\Big(\frac{s}{1-s}, \mathbf{q}_s\Big) - \frac{s}{1-s}\, R\right) + O\Big(\frac{1}{\sqrt{N}}\Big)\right] \tag{5.40}$$

where the term $O\Big(\frac{1}{\sqrt{N}}\Big)$ scales like the inverse of the square root of N and vanishes as N tends to infinity. With the substitution $\rho = \frac{s}{1-s}$, as mentioned above (so that $\rho \geq 0$), and by taking the maximization of the error exponent with respect to ρ within the interval $[0, \infty)$, the authors end up with a sphere-packing lower bound for fixed composition codes. The final form of this bound is stated in the following theorem:

Theorem 5.4 (Sphere-Packing Lower Bound on the Decoding Error Probability for Fixed Composition Codes)**.** [300, Theorem 6]: Let $P(j|k)$ be the set of transition probabilities which characterizes a DMC, and let $\mathcal{C}$ be a *fixed composition code* which is communicated over the DMC and which forms a set of M codewords of length N. Assume an arbitrary list decoding scheme where the size of the list is limited to L. Then, the *maximal error probability* is lower bounded by

$$P_{\text{e,max}} \geq \exp\left[-N\left(E_{\text{sp}}\Big(R - \frac{\ln 4}{N} - \varepsilon\Big) + \sqrt{\frac{8}{N}} \ln\Big(\frac{e}{\sqrt{P_{\min}}}\Big) + \frac{\ln 4}{N}\right)\right] \tag{5.41}$$

where $R \triangleq \frac{\ln\left(\frac{M}{L}\right)}{N}$, the parameter ε is an arbitrary small positive number, and the function E_{sp} is given by

$$E_{\text{sp}}(R) \triangleq \sup_{\rho \geq 0}\big(E_0(\rho) - \rho R\big) \tag{5.42}$$

$$E_0(\rho) \triangleq \max_{\mathbf{q}} E_0(\rho, \mathbf{q}) \tag{5.43}$$

$$E_0(\rho, \mathbf{q}) \triangleq -\ln\left(\sum_{j=1}^{J}\Big[\sum_{k=1}^{K} q_k P(j|k)^{\frac{1}{1+\rho}}\Big]^{1+\rho}\right). \tag{5.44}$$

The maximum in (5.43) is over all probability vectors $\mathbf{q} = (q_1, \ldots, q_K)$, i.e., over all $\mathbf{q}$ with K non-negative components summing to 1.

In the following, we outline the ideas which finally lead to the generalization of the sphere-packing bound for arbitrary block codes (i.e., without imposing the restriction that the codes possess a fixed composition).

Concepts Used for the Generalization of the SP67 Bound for General Block Codes

The sphere-packing bound was finally generalized in [300] for general block codes which do not necessarily have a fixed composition. One fact which serves for this generalization is that the number of different ways to choose the composition of a code word is equal to the number of ways of picking K non-negative integers which sum to N. Hence, there are $\binom{N+K-1}{K-1}$ different compositions, and since the total number of codewords is equal to M, there must be some composition which contains a number of codewords M' bounded by

$$M' \geq \frac{M}{\binom{N+K-1}{K-1}} \,. \tag{5.45}$$

Since the authors in [300] are mainly focused on the analysis of the error exponent, they use the upper bound

$$\binom{N+K-1}{K-1} \leq N^K \tag{5.46}$$

for simplifying the final form of their bound; this loosening does not incur any penalty on the error exponent, since the number of different compositions only grows *polynomially* with N. Clearly, this loosening of the bound is not in place while considering a moderate value of N, as noted in [334] (see the next section which explains the improvements of the SP67 bound for moderate block lengths). By Considering the set of messages which corresponds to this set of M' codewords as a fixed composition code, and assuming that the same list decoder is used as for the original code, then for each m in this fixed composition set, $P_{\mathrm{e},m}$ stays the same as in the original code. Theorem 5.4 now applies here by replacing the rate R with $\frac{\ln\left(\frac{M'}{L}\right)}{N}$; since the error exponent E_{sp} is a decreasing function of its argument, one can loosen the lower bound by replacing M' with $\frac{M}{N^K}$, and obtain the following lower bound on the maximal error probability from Theorem 5.4:

$$P_{\mathrm{e,max}} \geq \exp\left[-N\left(E_{\mathrm{sp}}\Big(\frac{\ln(\frac{M}{L})}{N} - \frac{K\ln N}{N} - \frac{\ln 4}{N}\Big) + \sqrt{\frac{8}{N}}\ln\Big(\frac{e}{\sqrt{P_{\min}}}\Big) + \frac{\ln 4}{N}\right)\right] \tag{5.47}$$

where the $\varepsilon > 0$ in (5.41) was chosen to absorb the inequality in (5.46).

In order to obtain a lower bound on the average error probability (rather than the maximal error probability), the authors in [300] use the concept of expurgation of half of the bad codewords order to get the following inequality for an arbitrary block code with M codewords of length N and list decoding of size L:

$$P_{\mathrm{e}}(N,M,L) \geq \frac{1}{2}P_{\mathrm{e,max}}(N,M/2,L).$$

This finally gives the following sphere-packing lower bound:

Theorem 5.5 (1967 Sphere-Packing Bound for Discrete Memoryless Channels)**.** [300, Theorem 2]: Let $\mathcal{C}$ be an arbitrary block code whose transmission takes place over a DMC. Assume that the DMC is specified by the set of transition probabilities $P(j|k)$ where $k \in \{1,\ldots,K\}$ and $j \in \{1,\ldots,J\}$ designate the channel input and the channel output alphabets, respectively. Assume that the code $\mathcal{C}$ forms a set of M codewords of length N (i.e., each codeword is a sequence of N letters from the input alphabet), and assume that the code is decoded with an arbitrary list decoding scheme where the size of the list is limited to L. Then, the *average decoding error probability* of the code $\mathcal{C}$ is lower bounded by

$$P_{\mathrm{e}}(N,M,L) \geq \exp\left\{-N\Big[E_{\mathrm{sp}}\Big(R - O_1\Big(\frac{1}{N}\Big)\Big) + O_2\Big(\frac{1}{\sqrt{N}}\Big)\Big]\right\} \tag{5.48}$$

where $R \triangleq \frac{\ln\left(\frac{M}{L}\right)}{N}$, the error exponent $E_{\rm sp}(R)$ is introduced in (5.42), and the terms

$$\begin{aligned} O_1\Big(\frac{1}{N}\Big) &= \frac{\ln 8}{N} + \frac{K \ln N}{N} \\ O_2\Big(\frac{1}{\sqrt{N}}\Big) &= \sqrt{\frac{8}{N}}\ \ln\Big(\frac{e}{\sqrt{P_{\min}}}\Big) + \frac{\ln 8}{N} \end{aligned} \tag{5.49}$$

with

$$P_{\min} \triangleq \min_{1 \le k \le K, 1 \le J \le J} P(j|k)$$

scale like the inverse of N and the inverse of the square root of N, respectively, and hence, both terms vanish as we let N tend to infinity.

Comparing the expression for the sphere-packing error exponent ($E_{\rm sp}$) with (4.4) and (4.5), one verifies that the only difference between $E_{\rm sp}(R)$ and $E_{\rm r}(R)$ which form upper and lower bounds on the error exponents, respectively, is only the range of the parameter ρ. For the upper bound, ρ can take values in the semi-infinite interval $\rho \ge 0$, where for the lower bound, ρ is restricted to the finite interval $0 \le \rho \le 1$. Hence, the two bounds on the error exponents coincide as long as the value of ρ which maximizes the RHS of (5.42) lies within the interval $[0, 1]$. Since, the optimal parameter ρ forms a non-increasing function of the rate R, once the upper and lower bound on the error exponent coincide at a certain rate R, they should also coincide at higher rates. The authors of [300] should have been excited from the fact that the two bounds coincide at the portion of the rate region between the critical rate, defined by $R_{\rm c} \triangleq E_0'(1)$ and the channel capacity. This is the rate region where the performance of capacity-approaching codes (e.g, turbo-like codes) is most appealing.

5.4 Sphere-Packing Bounds Revisited for Moderate Block Lengths

The analysis related to the derivation of the SP67 bound was mainly asymptotic; the purpose of the authors in [300] (see also Section 5.3) was to try simplifying their analysis as long as there was no penalty with respect to the error exponent of the resulting bound. It was thus very encouraging that the error exponent of the SP67 bound [300] coincides with the error exponent of the random coding bound [123] at the whole rate region between the critical rate of a DMC channel and its capacity.

The introduction of turbo-like codes (e.g., turbo codes, LDPC codes and RA codes) which closely approach the channel capacity with feasible complexity (see e.g., [270] and [271]) revolutionized the field of coding theory. During the last decade, there is a large interest in the construction of structured codes with moderate block lengths which closely approach the capacity limit. Based on these exciting developments, Valembios and Fossorier [334] were motivated to review the derivation of the SP67 bound, and suggested several improvements which made the classical SP67 bound tighter for moderate block lengths. They have also made this bound applicable for memoryless continuous-output channels (and not only for DMC channels); as was exemplified in [334] (and will be shown later in this section), the modified SP67 bound when being particularized for the Gaussian channel competes well with the SP59 bound [299] (see Section 5.2) for practical rates and block lengths.

The derivation of the improved version of the SP67 bound in [334] deviates from the classical derivation of the SP67 bound [300] (see Section 5.3) in the following respects:

- Instead of the region $\mathcal{Y}_s$ which was introduced in (5.25), the derivation of the modified SP67 bound introduces an additional free parameter x, and relies on the Chebychev inequality to obtain

$$\begin{aligned}\widetilde{\mathcal{Y}}_s^x &\triangleq \Big\{\mathbf{y} : |D(\mathbf{y}) - \mu'(s)| \leq x\sqrt{2\mu''(s)}\Big\} \\ &= \Big\{\mathbf{y} : |D(\mathbf{y}) - \mathbb{E}_{Q_s}\big\{D(\mathbf{y})\big\}| \leq x\sqrt{2\mathrm{Var}_{Q_s}\big\{D(\mathbf{y})\big\}}\Big\} \\ &\geq 1 - \frac{1}{2x^2}. \end{aligned} \tag{5.50}$$

  Clearly, this region particularizes to the region in (5.25) in the particular case where $x = 1$. The lower bound above is meaningful as long as $x > \frac{\sqrt{2}}{2}$, and this generalization is used in the continuation of the derivation of the classical SP67 bound [300] (as outlined in Section 5.3).

- Instead of relying on the simple (though loose) bound on the second derivative of the function μ in (5.34), the authors calculated it exactly in terms of the transition probabilities which specifies the DMC (see [334, Appendix B]). By doing so, they also circumvented the need for the term $\ln \frac{e}{\sqrt{P_{\min}}}$ which originally prevented to use the classical SP67 bound for memoryless continuous-output channels (e.g., the AWGN channel for which the SP59 bound is valid [299], but not the classical SP67 bound) due to the fact that $P_{\min}$ in Theorem 5.5 becomes infinitesimally small for memoryless continuous-output channels.

- Instead of relying on the upper bound (5.46) on the binomial coefficient which behaves polynomially in N, the authors in [334] calculate the exact value of the binomial coefficient $\binom{N+K-1}{K-1}$.

These items lead to the following theorem which suggests an improvement over the classical SP67 bound in terms of the pre-exponent. Note that a lower bound on the ML decoding error probability follows as a particular case of a list decoder with size $L = 1$.

Theorem 5.6 (Improvement on the 1967 Sphere-Packing Bound for Discrete Memoryless Channels)**.** [334, Theorem 7]: Under the assumptions and notation used in Theorem 5.5, the *average decoding error probability* is lower bounded by

$$P_{\mathrm{e}}(N, M, L) \geq \exp\Big\{-N\widetilde{E}_{\mathrm{sp}}(R, N)\Big\} \tag{5.51}$$

where

$$\begin{aligned}\widetilde{E}_{\mathrm{sp}}(R, N) \triangleq \sup_{x > \frac{\sqrt{2}}{2}} \Big\{E_0(\rho_x) - \rho_x\Big(R - O_1\Big(\frac{1}{N}, x\Big)\Big) \\ + O_2\Big(\frac{1}{\sqrt{N}}, x, \rho_x\Big)\Big\}\end{aligned} \tag{5.52}$$

and

$$R \triangleq \frac{\ln\left(\frac{M}{L}\right)}{N} \tag{5.53}$$

$$O_1\Big(\frac{1}{N}, x\Big) \triangleq \frac{\ln 8}{N} + \frac{\ln\binom{N+K-1}{K-1}}{N} - \frac{\ln\left(2-\frac{1}{x^2}\right)}{N} \tag{5.54}$$

$$O_2\Big(\frac{1}{\sqrt{N}}, x, \rho\Big) \triangleq x\sqrt{\frac{8}{N}\sum_{k=1}^{K} q_{k,\rho}\nu_k^{(2)}(\rho)} + \frac{\ln 8}{N} - \frac{\ln\left(2-\frac{1}{x^2}\right)}{N} \tag{5.55}$$

$$\nu_k^{(1)}(\rho) \triangleq \frac{\sum_{j=1}^{J}\beta_{j,k,\rho}\ln\frac{\beta_{j,k,\rho}}{P(j|k)}}{\sum_{j=1}^{J}\beta_{j,k,\rho}} \tag{5.56}$$

$$\nu_k^{(2)}(\rho) \triangleq \frac{\sum_{j=1}^{J}\beta_{j,k,\rho}\ln^2\frac{\beta_{j,k,\rho}}{P(j|k)}}{\sum_{j=1}^{J}\beta_{j,k,\rho}} - \left[\nu_k^{(1)}(\rho)\right]^2 \tag{5.57}$$

where $\mathbf{q}_\rho \triangleq (q_{1,\rho}, \ldots, q_{K,\rho})$ designates the input distribution which maximizes $E_0(\rho, \mathbf{q})$ in (5.43), and the parameter ρ_x is determined by solving the equation

$$R - O_1\Big(\frac{1}{N}, x\Big) = -\frac{1}{\rho_x}\sum_{k} q_{k,\rho_x}\nu_k^{(1)}(\rho_x) + \frac{x}{\rho_x}\sqrt{\frac{2}{N}\sum_{k=1}^{K} q_{k,\rho}\nu_k^{(2)}(\rho)}. \tag{5.58}$$

Proof. See [334, Appendix B]. □

The new version of the SP67 bound in Theorem 5.6 applies to an arbitrary memoryless continuous-output channel where sums over the channel output alphabet are replaced by integrals. This bound is particularized in [334, Section 4] to the special case of a transmission over the AWGN channel with BPSK modulation. Fig. 5.2 depicts a comparison in terms of regions in the two-dimensional space (R, N) where a bound is better than the two others for three different targets (P_t) of block error probability. For example, the improved SP67 bound presented here is tighter than the SP59 bound for block lengths better than 10,000. However, for a rate of $\frac{3}{4}$ bits per channel use, the improved SP67 bound is better than the SP59 bound for $N \geq 900$.

5.5 Concluding Remarks on Sphere-Packing Bounds

The introduction of structured constructions of turbo-like codes which closely approach the Shannon capacity limit with moderate block lengths stirred new interest in studying the limits on the code performance as a function of the block length. The 1959 sphere-packing bound of Shannon [299] serves for the evaluation of the performance limits of block codes over an additive white Gaussian noise (AWGN) channel where the bound is expressed in terms of the block length and the rate. This bound was used in [91] and [182] as a benchmark in order to study the imperfectness of turbo codes with their sub-optimal iterative decoding algorithms; by comparing computer simulations for the performance obtained by turbo codes over a wide range of rates and block sizes, it was exemplified that the gap between their actual performance and the sphere-packing bound was within 0.7 dB. The study of Valembois and Fossorier [334] was also stimulated by the outstanding performance of turbo-like codes with moderate block lengths and feasible decoding complexities; they have improved the tightness of the classical 1967 sphere-packing bound [300] for moderate block lengths, and extended its validity to memoryless continuous-output channels (and not only

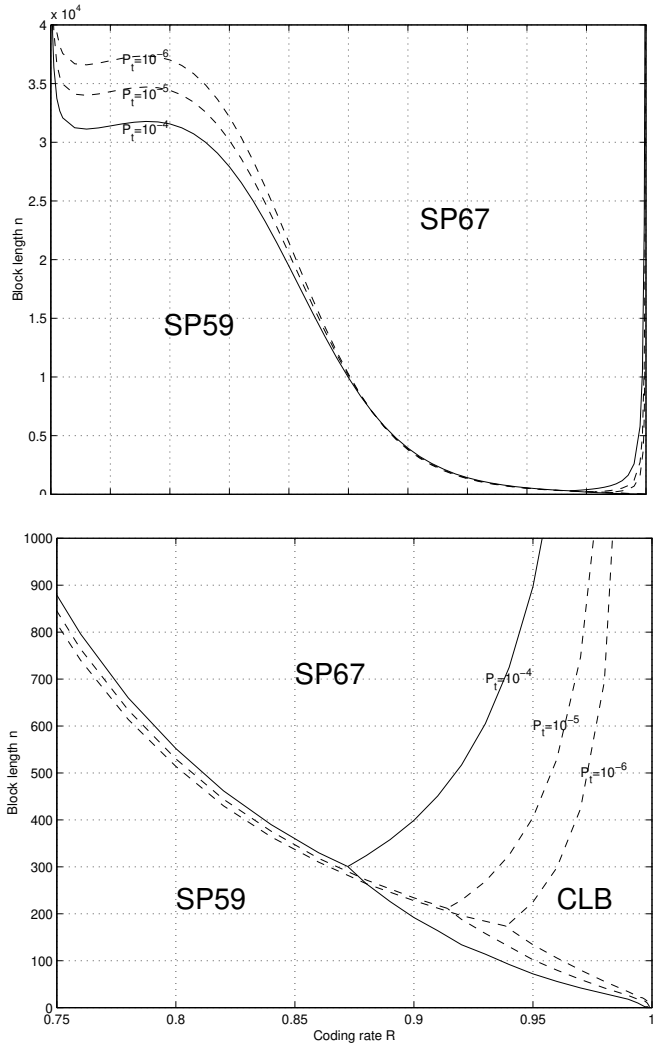


Figure 5.2: Regions in the two-dimensional space (R, N) where a bound is better than the two others for three different targets (P_t) of block error probability, referring to the AWGN channel with BPSK modulation. The figure at the bottom shows a zoom at high rates. The SP59 and the SP67 bounds refers to the sphere-packing bound of Shannon [299] and to the improved version of the SP67 bound by Valembois and Fossorier [334], and the CLB refers to the capacity limit bound. These two figures are reproduced (with permission) from [334].

for discrete memoryless channels as in [300]). The improved version of the 1967 sphere-packing bound in [334] competes well with the 1959 sphere-packing bound which applies to the AWGN channel (see also Fig. 5.2), and their study stimulates further research in the adaptation of sphere-packing bounds for codes of moderate block lengths which are tailored in a more suitable way for specific channel models and certain types of modulations.

In [16], the sphere-packing bound was used to assess the performance limitations of transmission techniques adopting efficient block codes over fading channels. In [117], a sphere-packing bound was derived on the average block error probability of block codes over Rayleigh block fading multiple-input multiple-output (MIMO) channels. The results show that the performance limits, as given by sphere-packing bounds, improve significantly if a code spans a larger number of fading blocks. On the other hand, increasing the block length (i.e., the length of portion of the code within one

block) improves the performance limits only marginally; as we let the block length tend to infinity, the performance limits of space-time codes are determined by the outage probability (see [117]).

Chapter 6

De Caen's Based Lower Bounds and Recent Improvements

Chapter overview: This chapter introduces de Caen's lower bound, and their recent improvements by Cohen and Merhav. It provides lower bounds on the ML decoding error probability of an arbitrary linear block codes which solely depend on the distance spectrum of these codes.

6.1 Introduction

The ubiquitous union bound asserts that the probability of a union of events does not exceed the sum of the probabilities of these events. This upper bound on the probability of a union of events forms a simple upper bound whose obvious drawback is that it may be loose due to the overlap between the individual events. As considered in Chapter 2, for large enough codes, the union bound becomes loose at rates exceeding the channel cut-off rate, thus excluding the portion of the rate region where the outstanding performance of turbo-like codes is most appealing.

In [59], D. de Caen provides a lower bound on the probability of a finite union of events. While an elementary result (essentially, Cauchy-Schwartz inequality), this bound was used to compute lower bounds on the decoding error probability of linear block codes via their distance distribution (see [168] for the binary symmetric channel (BSC), and [294] for the Gaussian channel). In [71], Cohen and Merhav improved de Caen's inequality by introducing a weighting function which is subject to optimization. Their improved bound is presented in the following statement and, like de Caen's inequality, it follows from the Cauchy-Schwartz inequality.

Theorem 6.1. [71, Theorem 2.1] Let $\{A_i\}_{i\in\mathcal{I}}$ be an arbitrary set of events in a probability space $(\Omega, \mathcal{F}, P)$, then the probability of the union of these events is lower bounded by

$$P\left(\bigcup_{i\in\mathcal{I}} A_i\right) \geq \sum_{i\in\mathcal{I}}\left\{\frac{\left(\sum\limits_{x\in A_i} p(x)m_i(x)\right)^2}{\sum\limits_{j\in\mathcal{I}}\sum\limits_{x\in A_i\cap A_j} p(x)m_i(x)^2}\right\} \tag{6.1}$$

where $m_i(x) \geq 0$ is any real function on Ω such that the sums on the RHS of (6.1) converge. Further, equality in (6.1) is achieved when

$$m_i(x) = m^*(x) = \frac{1}{\deg(x)}\,, \quad \forall\, i \in \mathcal{I} \tag{6.2}$$

where for each $x \in \Omega$

$$\deg(x) \triangleq |\{i \in \mathcal{I} \mid x \in A_i\}|. \tag{6.3}$$

Proof. First consider the case where Ω is finite. Using a simple counting argument gives

$$P\left(\bigcup_{i\in\mathcal{I}} A_i\right) = \sum_{i\in\mathcal{I}} \sum_{x\in A_i} \frac{p(x)}{\deg(x)}. \tag{6.4}$$

Let $m_i(x) \geq 0$ be an arbitrary non-negative function on Ω. From the Cauchy-Schwarz inequality, it follows that

$$\begin{aligned}
&\left(\sum_{x\in A_i} \frac{p(x)}{\deg(x)}\right)\left(\sum_{x\in A_i} p(x)m_i^2(x)\deg(x)\right) \\
&\geq \left(\sum_{x\in A_i} \sqrt{\frac{p(x)}{\deg(x)}}\sqrt{p(x)m_i^2(x)\deg(x)}\right)^2 \\
&= \left(\sum_{x\in A_i} p(x)m_i(x)\right)^2
\end{aligned} \tag{6.5}$$

provided that the sums in (6.5) converge. Therefore, (6.4) and (6.5) give

$$\begin{aligned}
P\left(\bigcup_{i\in\mathcal{I}} A_i\right) &\geq \sum_{i\in\mathcal{I}} \frac{\left(\sum_{x\in A_i} p(x)m_i(x)\right)^2}{\sum_{x\in A_i} p(x)m_i^2(x)\deg(x)} \\
&= \sum_{i\in\mathcal{I}} \frac{\left(\sum_{x\in A_i} p(x)m_i(x)\right)^2}{\sum_{j\in\mathcal{I}}\sum_{x\in A_i\cap A_j} p(x)m_i^2(x)}.
\end{aligned} \tag{6.6}$$

Note that $m_i(x)$ may be different for each i in the sum over all $i \in \mathcal{I}$. However, in order to achieve equality in (6.5), we need

$$m_i(x) = \frac{1}{\deg(x)}, \quad \forall\, i \in \mathcal{I}.$$

For a general probability space, as noted in [59] and [175], since there are only finitely many A_i's, the number of Boolean atoms defined by the A_i's unions and intersections is also finite. Thus, the general space can be reduced to a finite probability space. In this case, the sums in (6.1) are replaced by integrals. □

In [71], Cohen and Merhav refer to the choice of $m_i(x) \equiv 1$ as the *trivial* choice of $m_i(x)$. In this particular case, we get the inequality

$$P\left(\bigcup_{i\in\mathcal{I}} A_i\right) \geq \sum_{i\in\mathcal{I}} \left\{ \frac{P(A_i)^2}{\sum_{j\in\mathcal{I}} P(A_i \cap A_j)} \right\}$$

which is de Caen's bound [59]. Thus, de Caen's bound is a special case of the bound suggested by Cohen and Merhav in Theorem 6.1. In this context, it is noted that a recent improvement of de Caen's bound was provided by Kuai, Alajaji and Takahara [175]. However, Dembo has demonstrated in [74] that both de Caen's bound and Kuai, Alajaji and Takahara's bound are

derived by solving the same minimization problem. While the latter bound is obtained by applying a stronger method than the de bound, it improves this bound by a negligible factor which is at most $\frac{9}{8}$.

The essence of the bound in Theorem 6.1 is the ability to choose an appropriate function $m_i(x)$. To define a proper strategy for choosing $m_i(x)$, it was observed by Cohen and Merhav that a multiplication of $m_i(x)$ by a constant factors out in (6.1). Hence, $m_i(x)$ should only define an *essence of behavior*, and not necessarily exact values. When looking for such a behavior, recall that the optimal function m_i is $m_i(x) = \frac{1}{\deg(x)}$. However, in the context of derivation of lower bounds on the ML decoding error probability, the calculation of this function is unfortunately prohibitively complex since it necessitates the characterization of the Voronoi regions of every individual codeword, so Cohen and Merhav suggested in their paper [71] several alternatives to choosing the functions $m_i(x)$ which inherently yield tighter bounds than previously reported lower bounds on the ML decoding error probability which relied on de Caen's bound (e.g., Seguin's bound [294]). By following this approach, Cohen and Merhav relied on (6.1) for the derivation of improved lower bounds on the decoding error probability of linear codes under optimal ML decoding. They exemplified their bounds for BPSK modulated signals which are equally likely to be transmitted among M signals, and the examined communication channels were a BSC and an AWGN channel. In this context, the element x in the RHS of (1.3) is replaced by the received vector $\mathbf{r}$ at the output of the communication channel (whose length K is equal to the dimension of the code), and A_i (where $i = 1, 2, \ldots, M-1$) consists of all the vectors of length K which are closer in the Euclidean sense to the signal $\mathbf{s}_i$ rather than the transmitted signal $\mathbf{s}_0$. Similarly to Seguin's bound [294], the bounds in [71] get (after some loosening in their tightness) final forms which solely depend on the distance spectrum of the code. Recently, two lower bounds on the ML decoding error probability of linear binary block codes were derived by Behnamfar et al. [30] for BPSK-modulated AWGN channels. These bounds are easier for numerical calculation, but are looser than Cohen-Merhav bounds for low to moderate SNRs.

Though not explicitly mentioned in [71], we note that de Caen's based lower bounds on the decoding error probability (see [30], [71], [168] and [294]) are applicable for *specific* codes but not for ensembles; this restriction is due to the fact that Jensen's inequality does not allow to replace the distance spectrum of a linear code in these bounds by the average distance spectrum of ensembles.

6.2 De Caen's Based Lower Bounds and Their Variations

6.2.1 Lower Bounds on the ML Decoding Error Probability for the AWGN Channel

We start our discussion by presenting the lower bound of Cohen and Merhav, as introduced in [71, Proposition 3.1] for the Gaussian channel, and then refer to its two particular cases which are named in [71, Section 3] as the 'norm' and the 'dot-product' lower bounds; these two lower bounds on the ML decoding error probability refer to the situation where the signals are transmitted over the AWGN channel with equal probability. These bounds rely on particular choices of the 'weighting functions' m_i which appear in Theorem 6.1. The following lower bound, which was derived by Cohen and Merhav, is introduced in Theorem 6.2 and it relies on their improvement on the de Caen's lower bound (see Theorem 6.1) with the following parametric weighting function given the transmitted signal is $\mathbf{s}_0$ and the received vector is $\mathbf{r}$

$$m(\mathbf{r}|\mathbf{s}_0) = \exp\left\{-(a||\mathbf{r}||^2 + b\,\langle \mathbf{r}, \mathbf{s}_0\rangle + c||\mathbf{s}_0||^2)\right\} \tag{6.7}$$

with the three free parameters a, b and c, and where $\langle \cdot, \cdot \rangle$ designates the standard inner product. Then, it is easily demonstrated that Seguin's bound [294] forms a particular case of the lower bounds of Cohen and Merhav, and therefore Seguin's bound is in general a looser bound (as exemplified in [71] and in some numerical results in this section). In order to facilitate applying these bounding techniques to get lower bounds on the decoding error probability of binary linear block codes so that the resulting bounds will solely depend on the distance spectrum of the considered code, Seguin has proved a monotonicity property w.r.t. the correlation between two arbitrary codewords of a linear code; this property enables to loosen the lower bounds by replacing the correlation between two arbitrary codewords with a simple upper bound on their correlation which depends on the individual Hamming weights of the codewords and the minimum distance of the code; this upper bound on the correlation of two arbitrary codewords was used by Seguin and by Cohen and Merhav in order to loosen their lower bounds on the decoding error probability in a way which enables them to use the resulting (loosened) bounds to get lower bounds on the ML decoding error probability of linear block codes when only the distance spectrum of the code is available. We finally address Swatzek's bound [317] which was exemplified in [294] to be looser than the bound of Seguin. The discussion concludes by addressing a recently introduced bound of Benhamfar, Alajaji and Linder [30]. Some numerical results finally exemplify the use of these distance-spectrum based bounds for the performance evaluation of some (short) block codes; these bounds are also compared with the 1959 sphere-packing (SP59) bound of Shannon (see Section 5.2) and the TSB upper bound of Poltyrev (see Section 3.2.1).

Theorem 6.2. [71, Proposition 3.1] Let $\mathbf{s}_0, \ldots, \mathbf{s}_{M-1}$ be a set of M signals of dimension K whose transmission takes place over an AWGN channel with two-sided noise spectral density of $\frac{N_0}{2}$. Then, the conditional error probability under ML decoding given the signal $\mathbf{s}_0$ is transmitted is lower bounded by

$$\begin{aligned} P(\text{error}|\mathbf{s}_0) \;\; \geq \;\; & \exp\big((\beta' - 2\beta)||\mathbf{s}_0||^2\big) \left(\frac{N_0'}{\sqrt{N_0 N_0''}} \right)^K \\ & \sum_{i=1}^{M-1} \left\{ \frac{Q^2\big(\kappa(\alpha, \mathbf{s}_i, N_0')\big)}{\sum_{j=1}^{M-1} \Psi\big(\rho_{i,j}, \kappa(\alpha', \mathbf{s}_i, N_0''), \kappa(\alpha', \mathbf{s}_j, N_0'')\big)} \right\} \end{aligned} \tag{6.8}$$

where

$$Q(x) \triangleq \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\big(-\frac{y^2}{2}\big) \, dy \tag{6.9}$$

is the complementary Gaussian cumulative distribution function,

$$\Psi(\rho, x', y') \triangleq \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{x'}^\infty \int_{y'}^\infty \exp\Big\{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\Big\} \, dx \, dy \tag{6.10}$$

is the normalized (unit variance) two-dimensional Gaussian integral with $\Psi(1, x, x) \triangleq Q(x)$,

$$\rho_{i,j} \triangleq \frac{\langle \mathbf{s}_i - \mathbf{s}_0, \mathbf{s}_j - \mathbf{s}_0 \rangle}{||\mathbf{s}_i - \mathbf{s}_0|| \; ||\mathbf{s}_j - \mathbf{s}_0||} \tag{6.11}$$

designates a correlation, the function κ is given by

$$\kappa(\alpha, \mathbf{s}_i, N_0) \triangleq \frac{||\alpha \mathbf{s}_0 - \mathbf{s}_i||^2 - (\alpha - 1)^2 ||\mathbf{s}_0||^2}{\sqrt{2N_0} \; ||\mathbf{s}_i - \mathbf{s}_0||}$$

and the constants $N_0', N_0'', \alpha, \alpha', \beta$ and β' are given by

$$\begin{aligned} N_0' &\triangleq \tfrac{N_0}{1+\alpha N_0}\ , & N_0'' &\triangleq \frac{N_0}{1+2aN_0} \\ \alpha &\triangleq \frac{\frac{1}{N_0}-\frac{b}{2}}{a+\frac{1}{N_0}}\ , & \alpha' &\triangleq \frac{\frac{1}{N_0}-b}{2a+\frac{1}{N_0}} \\ \beta &\triangleq \frac{\left(\frac{1}{N_0}+a\right)\left(\frac{1}{N_0}+c\right)-\left(\frac{1}{N_0}-\frac{b}{2}\right)^2}{\frac{1}{N_0}+a}\ , & \beta' &\triangleq \frac{\left(\frac{1}{N_0}+2a\right)\left(\frac{1}{N_0}+2c\right)-\left(\frac{1}{N_0}-b\right)^2}{\frac{1}{N_0}+2a} \end{aligned} \tag{6.12}$$

and $a > -\frac{1}{2N_0}$, b and c are arbitrary constants.

For the ease of computation of the lower bound in Theorem 6.2, it is useful to rely on the following alternative representations of the functions Q and Ψ from Eqs. (6.9) and (6.10)

$$Q(x) \triangleq \frac{1}{\pi}\int_0^{\frac{\pi}{2}} \exp\left(-\frac{x^2}{2\sin^2\theta}\right)\ d\theta\ , \quad x>0 \tag{6.13}$$

and

$$\begin{aligned} \Psi(\rho,x,y) = {} & \tfrac{1}{2\pi}\int_0^{\frac{\pi}{2}-\tan^{-1}\left(\frac{y}{x}\right)} \frac{\sqrt{1-\rho^2}}{1-\rho\sin 2\theta}\exp\left(-\frac{x^2}{2}\frac{1-\rho\sin 2\theta}{(1-\rho^2)\sin^2\theta}\right)\ d\theta \\ & + \tfrac{1}{2\pi}\int_0^{\tan^{-1}\left(\frac{y}{x}\right)} \frac{\sqrt{1-\rho^2}}{1-\rho\sin 2\theta}\exp\left(-\frac{y^2}{2}\frac{1-\rho\sin 2\theta}{(1-\rho^2)\sin^2\theta}\right) d\theta \end{aligned} \tag{6.14}$$

where Eq. (6.13) was first introduced by Craig [72], and Eq. (6.14) was derived by Simon and Divsalar (see [307, Section 9]).

Let $\mathcal{C}$ be a binary linear block code whose codewords are BPSK modulated, and transmitted over an AWGN channel. Assume the only information we have about the linear code $\mathcal{C}$ is its distance spectrum, and we are looking for a lower bound on the error probability of this code under ML decoding. Based on the linearity of the code, one can assume that the all-zero codeword is transmitted, and apply the lower bound in Theorem 6.2 to obtain a lower bound on the average decoding error probability. However, the only obstacle in applying the bound in (6.8) to linear block codes is the fact that the correlation two signals (i.e., $\rho_{i,j}$ in (6.11)) does not depend on the Hamming weight of the individual codewords, but also on the number of agreements (and disagreements) between these two codewords. A pleasing property of the two-dimensional Gaussian integral $\Psi(\rho,\cdot,\cdot)$ in (6.10) is that this function is a monotonically increasing function of the correlation ρ (this property was formally proved by Seguin in [294, Eqs. (23)–(25)]); hence, as suggested by Seguin, one can replace the correlation $\rho_{i,j}$ in the RHS of (6.8) by an *upper bound on this correlation* which only depends on the Hamming weights of the individual codewords, and in this way obtain a looser lower bound on the decoding error probability which solely depends on the distance spectrum of the code. To this end, Seguin relied on the following upper bound on the correlation:

$$\rho_{i,j} \le \rho(i,j) \triangleq \min\left\{\sqrt{\frac{w_{\mathrm{H}}(\mathbf{c}_i)}{w_{\mathrm{H}}(\mathbf{c}_j)}}, \sqrt{\frac{w_{\mathrm{H}}(\mathbf{c}_j)}{w_{\mathrm{H}}(\mathbf{c}_i)}}, \frac{w_{\mathrm{H}}(\mathbf{c}_i)+w_{\mathrm{H}}(\mathbf{c}_j)-d_{\min}}{2\sqrt{w_{\mathrm{H}}(\mathbf{c}_i)w_{\mathrm{H}}(\mathbf{c}_j)}}\right\} \tag{6.15}$$

where $w_{\mathrm{H}}(\mathbf{c})$ designates the Hamming weight of a codeword $\mathbf{c}$, and $d_{\min}$ designated the minimal distance of the code $\mathcal{C}$. In [71], Cohen and Merhav relied on the same upper bound on the correlation in order to obtain a looser lower bound on the ML decoding error probability with solely depends

on the distance spectrum of a binary linear block code; to this end, the correlation $\rho_{i,j}$ in the RHS of (6.8) with the upper bound on $\rho_{i,j}$, as given in (6.15), and obtain an overall lower bound which solely depends on the distance spectrum of $\mathcal{C}$.

From the statement in Theorem 6.1 and the context of Theorem 6.2 which addresses a hypothesis testing problem for the AWGN channel, the corresponding optimized weighting function m_i which maximizes the lower bound in the RHS of (6.6) is independent of the index i, and it scales like the inverse of the number of signals which are closer to the received vector $\mathbf{r}$ than the transmitted signal $\mathbf{s}_0$, i.e.,

$$m_{\mathrm{opt}}(\mathbf{r}|\mathbf{s}_0) = \frac{1}{\deg(\mathbf{r}|\mathbf{s}_0)} \tag{6.16}$$

where

$$\deg(\mathbf{r}|\mathbf{s}_0) \triangleq \big|\{i : ||\mathbf{r}-\mathbf{s}_i|| < ||\mathbf{r}-\mathbf{s}_0||\}\big| \tag{6.17}$$

Based on the three-parameters weighting function introduced in (6.7), Cohen and Merhav consider in [71] the following three particular cases of the bound in Theorem 6.2:

- *Norm Lower bound*: This lower bound follows by looking at the weighting functions whose behavior is like the exponent of the squared Euclidean distance between the received vector $\mathbf{r}$ and the signal $\mathbf{s}_0$. The motivation for considering this case is because, according to the optimized weighting function given in (6.16), the optimized weighting function is inversely proportional to the number of signals which fall in the interior of a K-dimensional sphere of radius $||\mathbf{r}-\mathbf{s}_0||$ whose center is at the received vector $\mathbf{r}$, so the optimized function is monotonically decreasing in the distance $||\mathbf{r}-\mathbf{s}_0||$. Hence, Cohen and Merhav considered the one-parameter weighting function

  $$m(\mathbf{r}|\mathbf{s}_0) \triangleq \exp(-a||\mathbf{r}-\mathbf{s}_0||^2) \ , \quad a \geq 0 \tag{6.18}$$

  which follows as a particular case of the three-parameters weighting function in (6.7) when $b = -2a$ and $a = c$. The optimized bound which corresponds to this particular case is called the 'norm bound', and it follows as a particular case of the lower bound in Theorem 6.2 in the above setting of the parameters a, b and c (so that the optimization problem is reduced to a single-parameter optimization w.r.t. $a \geq 0$.

- *Dot-Product Lower Bound*: For equal-energy signals, the degree function in (6.17) can be rewritten in the form

  $$\begin{aligned} \deg(\mathbf{r}|\mathbf{s}_0) &= \big|\{i : \langle \mathbf{r}, \mathbf{s}_i\rangle > \langle \mathbf{r}, \mathbf{s}_0\rangle\}\big| \\ &= \big|\{i : \theta_{\mathbf{r},i} < \theta_{\mathbf{r},0}\}\big| \end{aligned} \tag{6.19}$$

  where (f, g) designates the standard inner product of the two functions f and g, and

  $$\theta_{\mathbf{r},i} \triangleq \cos^{-1}\left(\frac{\langle \mathbf{s}_i, \mathbf{r}\rangle}{||\mathbf{s}_i||||\mathbf{r}||}\right), \quad 0 \leq \theta_{\mathbf{r},i} < \pi.$$

  Had the equal-energy signals were uniformly distributed on the surface of a sphere centered at the origin (this assumption holds for the optimal code, and gives the 1959 sphere-packing bound of Shannon [299]), Eqs. (6.16) and (6.19) would imply that the optimized weighting function $m(\mathbf{r}|\mathbf{s}_0)$ is a monotonically decreasing with respect to the absolute value of the angle between $\mathbf{r}$ and $\mathbf{s}_0$. This intuition suggested Cohen and Merhav to consider the particular weighting function

  $$m(\mathbf{r}|\mathbf{s}_0) \triangleq \exp\big(b\ \langle \mathbf{s}_0, \mathbf{r}\rangle\big) \ , \quad b \geq 0 \tag{6.20}$$

  which forms a particular case of the three-parameters function in (6.7) when $a = c = 0$ and b is replaced by $-b$ (so that $b \geq 0$). Hence, the dot-product lower bound is also subject to a single-parameter optimization (w.r.t. the parameter b).

- *Segiun's Lower Bound*: Seguin's lower bound on the ML decoding error probability was introduced in [294], and it follows as a particular case of the Cohen-Merhav bound in Theorem 6.2 in the setting where $a = b = c = 0$. The reason for that is because Sequin's lower bound follows from de Caen's bound, where the latter forms a particular case of the improvement on de Caen's lower bound (see Theorem 6.1) in the setting where all the weighting functions m_i are equal to unity. Hence, by substituting $a = b = c = 0$ in the RHS of (6.7), the weighing function $m(\mathbf{r}|\mathbf{s}_0)$ turns to be equal identically to unity. Note that Seguin's bound also forms a particular case of the norm bound and the dot-product bound.

Based on the particular choice of the weighting function in (6.7) and the upper bound on the correlation $\rho_{i,j}$ as given in (6.15), Theorem 6.2 particularizes to [71, Eqs. (24) and (25)]. However, we will rewrite here these equations in a different form which has two advantages: the first one, is that the lower bound will be expressed explicitly in terms of the signal to noise ratio (as opposed to constants which appear in (6.12) which depends on the noise spectral density N_0 and the signal energy per symbol E_{s} separately; the second advantage is that it shows explicitly that the arbitrary parameter c in (6.7) *does not affect the bound*, and hence, the calculation of the lower bound is reduced to an optimization over two parameters (instead of three parameters); this significantly reduces the computational complexity of the resulting lower bound. Before introducing the final form of this lower bound on the AWGN channel, note that the fact that the effect of c cancels out in this bound is attributed to the fact that the term $c||\mathbf{s}_0||^2$ inside the exponent of the RHS of (6.7) is equivalent to a scaling of the weighting function $m(\mathbf{r}|\mathbf{s}_0)$, and (as indicated in Section 6.1), since a multiplication of the weighting function by a constant does not affect the improved version of de Caen's bound in Theorem 6.1, then clearly the parameter c in (6.7) should not affect the resulting lower bound on the decoding error probability. The following corollary rephrases [71, Eqs. (24) and (25)] to a more convenient form, as explained above.

Corollary 6.1 (Lower Bound on the Decoding Error Probability of Binary Linear Block Codes over an AWGN Channel)**.** Let $\mathcal{C}$ be a binary linear block code of length N and dimension K, let $R \triangleq \frac{K}{N}$ be the code rate of $\mathcal{C}$ in bits per channel use, $d_{\min}$ be the minimal distance of the code $\mathcal{C}$, and let $\{S_i\}_{i=0}^{N}$ designate the distance spectrum of $\mathcal{C}$. Assume that the code is BPSK modulated, and the transmission takes place over an AWGN channel. Then, the block error probability of the code $\mathcal{C}$ under ML decoding is lower bounded by

$$\begin{aligned} P_{\mathrm{e}}(\mathcal{C}) \;\geq\; & \exp\left\{-\frac{NRE_{\mathrm{b}}}{N_0}\left[1+\frac{(1-b)^2}{1+2a}-\frac{(2-b)^2}{2(1+a)}\right]\right\}\left(\frac{\sqrt{1+2a}}{1+a}\right)^K \\ & \cdot \sum_{i\neq 0: S_i>0} \frac{S_i\, Q(\gamma_i)}{Q(\delta_i)+(S_i-1)\Psi\big(\rho(i,i),\delta_i,\delta_i\big)+\sum\limits_{j\neq\{0,i\}} S_j \Psi\big(\rho(i,j),\delta_i,\delta_j\big)} \end{aligned} \tag{6.21}$$

where

$$\gamma_i \triangleq \sqrt{\frac{(2-b)^2}{2(1+a)}\frac{iRE_{\mathrm{b}}}{N_0}} \tag{6.22}$$

$$\delta_i \triangleq \sqrt{\frac{2(1-b)^2}{1+2a}\frac{iRE_{\mathrm{b}}}{N_0}} \tag{6.23}$$

$$\rho(i,j) \triangleq \min\left\{\sqrt{\frac{i}{j}}, \sqrt{\frac{j}{i}}, \frac{i+j-d_{\min}}{2\sqrt{ij}}\right\} \tag{6.24}$$

and $a > -\frac{1}{2}$, $b \in \mathbb{R}$ are two arbitrary constants which are subject to optimization. The functions Q and Ψ in the RHS of (6.21) are given in (6.9) and (6.10), respectively (these functions are also expressed in (6.13) and (6.14) as integrals over finite intervals.)

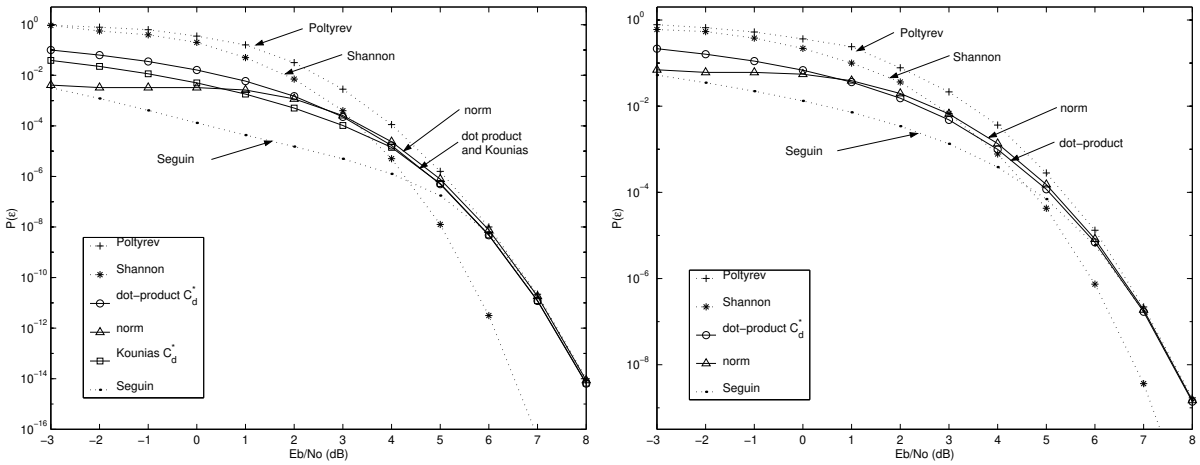


Figure 6.1: Bounds on the decoding error probability of two binary linear block codes under soft-decision ML decoding. The upper plot refers to the (63, 24) BCH codes, and the lower plot refers to the (24, 12) Golay code where the transmission of both codes takes place over the binary-input AWGN channel. Poltyrev's upper bound [259] refers to the tangential-sphere bound (TSB) in Section 3.2.1; the other bounds which are lower bounds on the ML decoding error probability include Seguin's bound [294], the 'dot-product bound' and the 'norm bound' introduced by Cohen and Merhav [71], and the 1959 sphere-packing bound (SP59) of Shannon [299] which is introduced in Section 5.2 (where the latter bound solely depends on the block length and the rate of the code). This figure was reproduced (with the permission of the authors) from [71].

As explained above and since the parameter c in the weighting function (6.7), the following three bounds follow as particular cases of the bound given in Corollary 6.1:

- By setting $b = -2a,\ a \geq 0$ and optimizing over a in (6.21), one obtains the 'norm bound' [71].
- By setting $a = 0$ and optimizing over b in (6.21), one obtains the 'dot-product bound' [71]
- By setting $a = b = 0$ in (6.21) (which makes the weighting function to be equal identically to 1), one obtains Seguin's bound [294].

A comparison of the 'norm bound', the 'dot-product bound' [71] and Seguin's bound [294] which follow as particular cases of Corollary 6.1, and the 1959 sphere-packing lower bound [299] (see Section 5.2) is shown in Fig. 6.1 for two short linear block codes (the upper plot refers to the (63, 24) BCH codes, and the lower plot refers to the (24, 12) Golay code). As expected from the discussion above, Seguin's bound is looser than the 'norm bound' and the 'dot-product' bound. In order to examine the tightness of these lower bounds on the decoding error probability, they are compared with the tangential-sphere bound of Poltyrev [259] (see Section 3.2.1); the TSB is one of the tightest reported upper bounds on the ML decoding error probability for the AWGN channel, especially for short block codes (see [282]).

The sensitivity of the 'norm bound' to the parameter a is exemplified in Fig. 6.2 for the $(23, 12, 7)$ Golay code. This figure (which was reproduced with permission from [71]) shows a three dimensional

mesh of the lower bound on the decoding error probability of the Golay code as a function of the energy per bit to spectral noise density $\left(\frac{E_{\mathrm{b}}}{N_0}\right)$ and the parameter $a > -\frac{1}{2}$ of the lower bound in (6.21) which is subject to optimization.

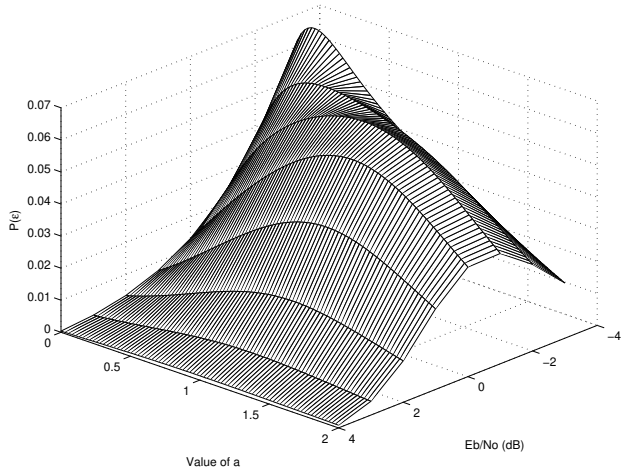


Figure 6.2: A three-dimensional mesh of the 'norm lower bound' on the decoding error probability of the $(23, 12, 7)$ Golay code under ML decoding. The graph of this lower bound is shown as a function of the energy per bit to spectral noise density $\left(\frac{E_{\mathrm{b}}}{N_0}\right)$ and the parameter $a > -\frac{1}{2}$ of the lower bound in (6.21) which is subject to optimization. This figure was reproduced (with the permission of the authors) from [71].

In [30], new lower bounds on the ML decoding error probability of binary linear block codes over the Gaussian channel were recently introduced. They suggest a reduction in the computational complexity as compared to the Cohen-Merhav bound, and are therefore easy for calculation; however, they are less tight than the Cohen and Merhav bounds for low values of signal to noise ratio.

6.2.2 Lower Bounds for the Binary Symmetric Channel

We discuss in this sub-section lower bounds on the ML decoding error probability of binary linear block codes over the BSC, and refer to Keren and Litsyn bound [168] and to a bound of Cohen and Merhav which was derived for the BSC [71].

In this case, the transmitted codeword is one of $M = 2^K$ equiprobable codewords $\mathbf{c}_0, \ldots, \mathbf{c}_{M-1}$ of length N where the transmission takes place over a binary symmetric channel (BSC) with crossover probability p (without any loss of generality, one can assume that $p < \frac{1}{2}$). Let $\mathbf{c}_0$ be the transmitted codeword, then the ML decoder chooses the codeword whose Hamming distance from the received vector is minimized. Let $\mathbf{x} = \mathbf{c}_0 + \mathbf{e}$ be the received vector where $\mathbf{e} \in \mathrm{GF}(2)^N$ is the error vector. Assume the code $\mathcal{C}$ is a binary linear block code, then the symmetry of the channel yields that the average decoding error probability is equal to the conditional error probability given that the all-zero codeword is transmitted. Hence, the error event $\mathcal{E}$ can be expressed as a union of events $\mathcal{E} = \bigcup_{i=1}^{M-1} \mathcal{E}_{0,i}$ (which typically have overlaps, and forms the source of the weakness of the union bound for bad channel conditions) where

$$\mathcal{E}_{0,i} \triangleq \big\{\mathbf{x} \in \mathrm{GF}(2)^N : \ w_{\mathrm{H}}(\mathbf{x} + \mathbf{c}_i) < w_{\mathrm{H}}(\mathbf{x})\big\}. \tag{6.25}$$

In [71, Section 4], Cohen and Merhav derive a lower bound on the decoding error probability for the BSC which solely depends on the distance spectrum of the binary linear block code and the crossover probability p of the BSC. Similarly to the derivation of lower bounds for the AWGN channel (see Section 6.2.1), they rely on the improvement on de Caen's lower bound as stated in Theorem 6.1. In order to simplify the final form of their bound and make it expressible in terms of the distance spectrum, they chose the setting where the non-negative weighting function $m_i(\mathbf{x})$ which appears in the RHS of (6.1) depends on the received vector $\mathbf{x}$ only through its Hamming weight (so, for all vectors $\mathbf{x} \in \mathrm{GF}(2)^N$ of the same composition, the function $m_i(\mathbf{x})$ is assumed to get the same value). For the BSC model, the transition probability is given by

$$\Pr(\mathbf{x}|\mathbf{c}_0 = \mathbf{0}) = p^{w_{\mathrm{H}}(\mathbf{x})}(1-p)^{N-w_{\mathrm{H}}(\mathbf{x})}.$$

From (6.3) (where in the context of this testing hypothesis problem, the event $\mathcal{A}_i$ appearing in Theorem 6.1 is replaced with the pairwise event event $\mathcal{E}_{0,i}$ as given in (6.25)), the degree function can be expressed as follows:

$$\begin{aligned} \deg(\mathbf{x}|\mathbf{c}_0) &= \big|\{\mathbf{c}_i \in \mathcal{C}, i \neq 0 : w_{\mathrm{H}}(\mathbf{x}+\mathbf{c}_i) < w_{\mathrm{H}}(\mathbf{x})\}\big| \\ &= |\mathcal{C}| \ \Pr\{w_{\mathrm{H}}(\mathbf{x}+\mathbf{c}) < w_{\mathrm{H}}(\mathbf{x})\} \end{aligned} \tag{6.26}$$

where a uniform distribution over the codewords of $\mathcal{C}$ is assumed. From the statement in Theorem 6.1, the optimal weighting function $m_i(\mathbf{x})$ is independent of the index i, and it is inversely proportional to the value of $\deg(\mathbf{x}|\mathbf{c}_0)$. In the continuation to their discussion, Cohen and Merhav suggest two alternatives for this optimized weighting function; both of them are amenable to analysis and yield in their final form lower bounds which solely depend on the distance spectrum of the code and the crossover probability of the channel. Since the weighting function m_i is assumed in [71, Section 4] to depend only on the Hamming weight of the received vector $\mathbf{x}$, this function is given in the form

$$m_i(\mathbf{x}) = \mu_i\big(w_{\mathrm{H}}(\mathbf{x})\big), \quad \mu_i : \mathbb{N}^+ \to \mathbb{R}.$$

The interested reader is referred to [71, Section 4] for further details regarding the derivation of the lower bound on the ML decoding error probability for the BSC (where the concept of the derivation is similar to the derivation of the lower bounds for the Gaussian channel in Section 6.2.1; it relies on the lower bound on a union of events, as given in Theorem 6.1). To this end, Cohen and Merhav suggest two functions μ_i which are based on certain approximations of the function $\deg(\mathbf{x}|\mathbf{c}_0)$; the calculation of one of these approximations depends on the knowledge of the distance spectrum of the whole code, and it is obtained from the Chernoff bounding technique; the other approximation depends on more elementary properties of the code (e.g., its size, length and minimum distance). In the final from of these bounds, both of them are subject to a one-parameter optimization in order to obtain the tightest lower bound within the considered family.

The lower bound of Keren and Litsyn [168] also applies to binary linear block codes whose transmission takes place over the BSC; however, since the derivation of their bound follows from de Caen's bound (i.e., the weighting functions in the RHS of (6.1) are all equal to unity), the bound in [168] is looser than the bound of Cohen and Merhav. Both bounds are calculable in terms of the distance spectrum of the code.

Chapter 7

Performance vs. Complexity: An Information-Theoretic Approach

Chapter overview: The chapter is focused on the tradeoff between performance and decoding complexity per iteration for LDPC codes in terms of their gap (in rate) to capacity. The study of this tradeoff is done via information-theoretic bounds which also enable to get an indication on the sub-optimality of message-passing iterative decoding algorithms (as compared to optimal ML decoding). The bounds are generalized for parallel channels, and are applied to ensembles of punctured LDPC codes where both intentional and random puncturing are addressed. The chapter derives information-theoretic lower bounds on the bit error probability of codes defined on graphs, and gives a quantitative measure to the statement that good error correcting codes (even under ML decoding) should have cycles in their representations by standard Tanner graphs.

7.1 Introduction

Error-correcting codes which employ iterative decoding algorithms are widely considered state of the art in channel coding. Their outstanding performance and feasible encoding and decoding complexities motivate a theoretical study of the tradeoff between their performance and complexity. While it would be very interesting to explore this tradeoff for finite block lengths, this central issue is still beyond the scope of rigorous analysis. Via information-theoretic bounds, we study this tradeoff in the asymptotic case where the block length tends to infinity. This study is focused on low-density parity-check (LDPC) codes, but we also address other constructions of codes on graphs which closely approach channel capacity with moderate complexity.

LDPC codes are efficiently encoded and decoded due to the sparseness of their parity-check matrices [272]. An ensemble of LDPC codes is characterized ed by a pair of degree distributions $\lambda(x) = \sum_{i\geq 2} \lambda_i x^{i-1}$ and $\rho(x) = \sum_{i\geq 2} \rho_i x^{i-1}$ where λ_i (ρ_i) is the fraction of edges in the bipartite graph adjacent to variable (parity-check) nodes of degree i. Consider the number of ones in a parity-check matrix which represents a binary linear block code, and normalize it per information bit (i.e., with respect to the dimension of the code). This quantity (defined as the *parity-check density* [290]) is linearly proportional to the number of messages produced in each iteration of the message-passing iterative (MPI) decoder, normalized per information bit. Therefore, the density of the parity-check matrix can be regarded as a measure for the decoding complexity per iteration of LDPC codes normalized per information bit. Our study of the tradeoff between performance and complexity per iteration is initiated by two questions which were addressed by Burshtein et al. [56] and Sason and Urbanke [290]:

Question 1. How sparse can parity-check matrices of binary linear codes be, as a function of the gap between the code rate for which reliable communication can be achieved and the channel

capacity?

Question 2. How good can LDPC codes be (even under maximum-likelihood (ML) decoding), as a function of their degree distributions?

The answer to the first question was given in [290] in the form of lower bounds on the parity-check density (which serve as lower bounds on the decoding complexity per iteration); these lower bounds were later improved by the authors [353], and the improved bounds will be introduced in this paper. Since the density of a parity-check matrix is related to the number of fundamental cycles in the corresponding bipartite graph (see [290]), this also gives a quantitative measure to the statement that bipartite graphs representing good error-correcting codes should have cycles [106]. Referring to the second question, the answer is given in the form of upper bounds on the achievable rates whose derivation is based on performance bounds under ML decoding. Comparing these upper bounds with exact thresholds under iterative decoding, calculated using density evolution (DE) [271], leads to a quantitative measure of the inherent loss in performance due to the sub-optimality of MPI decoding. The derivation of these bounds also allows to derive lower bounds on the bit/block-error probability under ML decoding (see [290, 353]).

In his thesis [122, Theorem 3.3], Gallager provided an upper bound on the rate of right-regular LDPC codes (i.e., LDPC codes with a constant degree (a_{R}) of the parity-check nodes) which is a necessary condition for achieving reliable communications over a binary symmetric channel (BSC), even under optimal ML decoding. This bound shows that right-regular LDPC codes with fixed right degree cannot achieve the channel capacity on a BSC, even under optimal ML decoding. The inherent gap between the achievable rates and the channel capacity is well approximated by an expression which decreases to zero exponentially fast in a_{R}. Burshtein *et al.* have generalized Gallager's bound for general ensembles of LDPC codes transmitted over memoryless binary-input output-symmetric (MBIOS) channels [56]; to this end, they applied a two-level quantization to the log-likelihood ratio (LLR) of these channels which essentially turns them into a BSC.

In [169], Khandekar and McEliece have suggested to study the encoding and decoding complexities of ensembles of iteratively decoded codes on graphs as a function of their gap to capacity. They conjectured that if the achievable rate under MPI decoding is a fraction $1-\varepsilon$ of the channel capacity, then for a wide class of channels, the number of iterations required for decoding with vanishing bit error probability scales like $\frac{1}{\varepsilon}$ and the density of the parity-check matrix scales like $\ln \frac{1}{\varepsilon}$. We note that recent results seem to suggest an analytical tool for proving the conjecture that the average number of iterations scales like the inverse of the multiplicative gap (ε) to capacity [218]. Based on this conjecture, it follows that for irregular repeat-accumulate codes, the encoding complexity scales like $\ln \frac{1}{\varepsilon}$ and the decoding complexity scales like $\frac{1}{\varepsilon} \ln \frac{1}{\varepsilon}$. The only exception is the binary erasure channel (BEC) where the decoding complexity behaves like $\ln \frac{1}{\varepsilon}$ (same as encoding complexity) because of the absolute reliability of the messages passed through the edges of the graph (hence, every edge can be used only once during the iterative decoding process).

Based on the work in [56], Sason and Urbanke derived an information-theoretic lower bound on the asymptotic density of parity-check matrices [290, Theorem 2.1] where this bound applies to every MBIOS channel and every sequence of binary linear block codes achieving a fraction $1-\varepsilon$ of the channel capacity with vanishing bit error probability. It holds for an arbitrary representation of full-rank parity-check matrices for these codes, and is of the form $\frac{K_1+K_2 \ln \frac{1}{\varepsilon}}{1-\varepsilon}$ where K_1 and K_2 are constants which only depend on the channel. For the BEC they used considerations relating to the rank of the parity-check matrix to get an improved bound, which is also logarithmic in $\frac{1}{\varepsilon}$. A natural question arises:

Question 3. Is the logarithmic behavior of the lower bound on the parity-check density true, or is it just an artifact of the bounding technique ?

Sason and Urbanke addressed this question by showing that for any MBIOS channel, the logarithmic behavior of the parity-check density is achievable under ML decoding [290, Theorem 2.2]. For the BEC, they showed that the sequence of right-regular ensembles of Shokrollahi [302] achieves the improved bound (up to a small additive constant) even under (sub-optimal) MPI decoding [290, Theorem 2.3]. A recent analysis in this direction was also presented in [146].

In [217], Measson *et al.* derived an upper bound on the thresholds of ensembles of LDPC codes under maximum a posteriori (MAP) decoding where the transmission of these codes takes place over the BEC. Their general approach relies on extrinsic information transfer (EXIT) charts. Generalized extrinsic information transfer (GEXIT) charts were recently introduced by Measson *et al.* ([216, 218]); they form an extension of EXIT charts to general MBIOS channels, and they fulfill the area theorem. This preservation of the area theorem for general MBIOS channels enables to get upper bounds on the thresholds of turbo-like ensembles under MAP decoding. Under some conditions, the bound in [217] was shown to be tight for the BEC.

A new method for analyzing LDPC codes and low-density generator-matrix (LDGM) codes under bit-MAP decoding is introduced by Montanari in [228]. The method is based on a rigorous approach to spin glasses, and allows a derivation of an upper bound on the achievable rates of these codes under bit-MAP decoding. The computational complexity of this bound grows exponentially with the maximal right and left degrees (see [228, Eqs. (6.2) and (6.3)]). Since the bounds in [217, 228] are derived for ensembles of codes, they are probabilistic in their nature; based on concentration arguments, they hold asymptotically in probability 1 as the block length goes to infinity. It was conjectured that the bounds in [228], which hold for general LDPC and LDGM ensembles over MBIOS channels, are tight.

In [9] and [374], Ardakani *et al.* suggest an approximated one-dimensional analysis which enables to optimize (numerically) the degree distributions for achieving a good tradeoff between performance and complexity.

The derivation of the results presented in this section was motivated by the fact that the information-theoretic bounds in [56] and [290] rely on a two-level quantization of the LLR and do not take into consideration the entire statistics of the communications channel. In this paper we present a new information-theoretic upper bound on the achievable rates of LDPC codes transmitted over MBIOS channels. This bound holds for optimal ML decoding and is therefore valid under any sub-optimal decoding algorithm. We also discuss an improvement of the information-theoretic lower bound on the parity-check density. These bounds hold for every sequence of codes and do not rely on probabilistic arguments. The bounds in this section are generalized for the case where the codes are transmitted over a set of independent parallel MBIOS channels, where each code bit is a-priory assigned to one of the parallel channels. This setup can be used to model different scenarios which include non-uniform error protection [253], punctured LDPC codes [133, 254] and LDPC-coded modulation schemes. The bounds for parallel channels are used to bound the performance and the decoding complexity per iteration of punctured LDPC codes transmitted over a single MBIOS channel.

Due to space limitations, the proofs of the statements in this section are relegated to the full paper versions (see [292] and [353]); further discussion on the relations between these theorems and some previously reported results from [56, 249, 253, 290] are presented there.

7.2 Preliminaries

We introduce here some definitions and theorems from [56, 290] which serve as preliminary material for the rest of the section. Definitions 7.1 and 7.2 are taken from [290, Section 2].

Definition 7.1 (Capacity-Approaching Codes)**.** Let $\{\mathcal{C}_m\}$ be a sequence of codes of rate R_m, and assume that for every m, the codewords of the code $\mathcal{C}_m$ are transmitted with equal probability over a channel whose capacity is C. This sequence is said to *achieve a fraction* $1-\varepsilon$ *of the channel capacity with vanishing bit error probability* if $\lim_{m\to\infty} R_m = (1-\varepsilon)C$, and if there exists a decoding algorithm under which the average bit error probability of the code $\mathcal{C}_m$ tends to zero in the limit where $m \to \infty$.

Definition 7.2 (Parity-Check Density)**.** Let $\mathcal{C}$ be a binary linear code of rate R and block length n, which is represented by a parity-check matrix H. We define the *density* of H, call it $\Delta = \Delta(H)$, as the normalized number of ones in H *per information bit*. The total number of ones in H is therefore equal to $nR\Delta$.

Definition 7.3 (Log-Likelihood Ratio)**.** Let $p_{Y|X}(\cdot|\cdot)$ be the conditional *pdf* of an arbitrary MBIOS channel. The log-likelihood ratio (LLR) at the output of the channel is

$$\mathrm{LLR}(y) \triangleq \ln\left(\frac{p_{Y|X}(y|X=0)}{p_{Y|X}(y|X=1)}\right).$$

Throughout the section, we assume that all the codewords of a binary linear block code are equally likely to be transmitted. Also, we use the notation $h_2(\cdot)$ for the binary entropy function to base 2, i.e., $h_2(\cdot) = -x\log_2(x) - (1-x)\log_2(1-x)$.

Theorem 7.1 (Upper Bound on the Achievable Rates)**.** [56, Theorem 2]: Consider a sequence $\{\mathcal{C}_m\}$ of binary linear block codes of rate R_m, and assume that their block length tends to infinity as $m \to \infty$. Let H_m be a full-rank parity-check matrix of the code $\mathcal{C}_m$, and assume that $\Gamma_{k,m}$ designates the fraction of the parity-check equations involving k variables. Let

$$\Gamma_k \triangleq \lim_{m\to\infty} \Gamma_{k,m}, \quad R \triangleq \lim_{m\to\infty} R_m \tag{7.1}$$

where these limits are assumed to exist. Suppose that the transmission of these codes takes place over an MBIOS channel with capacity C bits per channel use, and let

$$w \triangleq \frac{1}{2}\int_{-\infty}^{\infty} \min\big(f(y), f(-y)\big)\, dy \tag{7.2}$$

where $f(y) \triangleq p_{Y|X}(y|X=1)$ designates the conditional *pdf* of the output of the MBIOS channel. Then, a necessary condition for vanishing block error probability as $m \to \infty$ is

$$R \leq 1 - \frac{1-C}{\displaystyle\sum_k \left\{\Gamma_k\, h_2\left(\frac{1-(1-2w)^k}{2}\right)\right\}}.$$

Theorem 7.2 (Lower Bounds on the Parity-Check Density)**.** [290, Theorem 2.1]: Let $\{\mathcal{C}_m\}$ be a sequence of binary linear block codes achieving a fraction $1-\varepsilon$ of the capacity of an MBIOS channel with vanishing bit error probability. Denote Δ_m as the density of a full-rank parity-check matrix of the code $\mathcal{C}_m$. Then, the asymptotic density satisfies

$$\liminf_{m\to\infty} \Delta_m > \frac{K_1 + K_2 \ln\frac{1}{\varepsilon}}{1-\varepsilon} \tag{7.3}$$

where

$$K_1 = \frac{(1-C)\ \ln\left(\frac{1}{2\ln 2}\ \frac{1-C}{C}\right)}{2C\ \ln\left(\frac{1}{1-2w}\right)}\ , \quad K_2 = \frac{1-C}{2C\ \ln\left(\frac{1}{1-2w}\right)} \tag{7.4}$$

and w is defined in (7.2). For a BEC with erasure probability p, the coefficients K_1 and K_2 in (7.4) are improved to

$$K_1 = \frac{p\ \ln\left(\frac{p}{1-p}\right)}{(1-p)\ \ln\left(\frac{1}{1-p}\right)}\ , \quad K_2 = \frac{p}{(1-p)\ \ln\left(\frac{1}{1-p}\right)}\ . \tag{7.5}$$

Using standard notation, an ensemble of (n, λ, ρ) LDPC codes is characterized by its length n, and the polynomials $\lambda(x) = \sum_{i=2}^{\infty} \lambda_i x^{i-1}$ and $\rho(x) = \sum_{i=2}^{\infty} \rho_i x^{i-1}$, where λ_i (ρ_i) is equal to the fraction of edges which are connected to variable (parity-check) nodes of degree i. The variables (parity-check sets) are represented by the left (right) nodes of a bipartite graph which represents an LDPC code. The design rate of this ensemble is given by

$$R_{\mathrm{d}} = 1 - \frac{\int_0^1 \rho(x)\ dx}{\int_0^1 \lambda(x)\ dx}\ .$$

7.3 Results for Memoryless Binary-Input Output-Symmetric Channels

In this section, we present upper bounds on the achievable rates of LDPC codes, and lower bounds on their bit-error probability. The transmission of the codes is assumed to take place over an MBIOS channel and the bounds refer to ML decoding. We also consider an improved lower bound on the parity-check density of a sequence of LDPC codes which achieve a fraction $1 - \varepsilon$ of the capacity of an MBIOS channel with vanishing bit error probability. These bounds tighten the results in [56, 290]. The upper bound on the achievable rates and the lower bound on the parity-check density in [56, 290] are both derived from a lower bound of the conditional entropy of the transmitted codeword given the received sequence at the output of the communication channel. For the derivation of this bound Burshtein *et al.* considered the entropy of the syndrome of the received sequence, this requires that the received sequence be binary. Hence, a two level quantization which essentially turns the MBIOS channel into the BSC was applied to the LLR at the output of the communication channel. This quantization causes an inherent loss in the tightness of the bounds in [56, 290].

We introduce two approaches to circumvent the two-level quantization of the LLR. The first approach forms a generalization of the approach introduced by Bursthein *et al.* and allows a finer quantization of the LLR to a number of levels which is an arbitrary integer power of 2. Each quantization level is mapped to a member of the Galois field $\mathrm{GF}(2^d)$ (where the number of quantization levels is 2^d and $d \in \mathbb{N}$), and the concept of syndrome is generalized for members of $\mathrm{GF}(2^d)$. These bounds are subject to optimization of the quantization levels, as to achieve the tightest bounds within their form. For optimally chosen quantization levels it can be shown that the bounds become tighter as the number of levels grows. A second approach is to consider the soft value of the LLR which forms a sufficient statistics of ML decoding. In this approach the LLR is split into its sign and its absolute value, due to the symmetry of the channel it is possible to define the syndrome w.r.t the sign of the LLR and use its absolute value as side information. This

second approach leads to bounds which are uniformly tighter than the first approach and require no optimization. However, comparing the results of both approaches gives insight on the effect of quantization of the LLR at the decoder due to implementation restrictions. In the following, we only present bounds which are based on the second approach. Some numerical results, presented at the end of this section, demonstrate the results given by both approaches.

As noted above, the bounds on the achievable rates, the bit error probability and the parity-check density are derived from a lower bound on the entropy of the transmitted codeword conditioned on the received sequence. We present an improved lower bound on this entropy, as compared to the one derived in [56].

Theorem 7.3 (Lower Bound on the Conditional Entropy)**.** [353, Proposition 4.1]: Let $\mathcal{C}$ be a binary linear code of length n and rate R transmitted over an MBIOS channel. Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ be the transmitted codeword and the received sequence, respectively. For an arbitrary representation of the code $\mathcal{C}$ by a full-rank parity-check matrix, let Γ_k be the fraction of the parity-check equations of degree k, and $\Gamma(x) \triangleq \sum_k \Gamma_k x^k$ designate the corresponding right degree distribution of $\mathcal{C}$. Then the conditional entropy of the transmitted codeword given the received sequence satisfies

$$\frac{H(\mathbf{X}|\mathbf{Y})}{n} \geq 1 - C - (1-R)\left(1 - \frac{1}{2\ln(2)} \sum_{p=1}^{\infty} \frac{\Gamma(g_p)}{p(2p-1)}\right)$$

where

$$g_p \triangleq \int_0^{\infty} a(l)(1+e^{-l}) \tanh^{2p}\left(\frac{l}{2}\right) dl \ , \quad p \in \mathbb{N}, \tag{7.6}$$

and a is the conditional pdf of the LLR given the channel input is 1 (where the possible inputs to the channel are $\{+1, -1\}$).

From Fano's inequality and since the code is binary, the conditional entropy of the transmitted codeword given the received sequence satisfies

$$\frac{H(\mathbf{X}|\mathbf{Y})}{n} \leq R h(P_{\mathrm{b}}) \tag{7.7}$$

where R is the rate of the code, P_{b} is the bit-error probability of the ML decoder, and $h_2(x)$ is the binary entropy function to base two. By combining Theorem 7.3 and Eq. (7.7), it is possible to derive an upper bound on the achievable rates of sequences of binary linear block codes and a lower bound on their bit-error probability, these bounds are presented in the next corollaries.

Corollary 7.1 (Upper Bound on Achievable Rates under ML Decoding)**.** [353, Corollary 4.1]: Let $\{C_m\}$ be a sequence of binary linear block codes whose codewords are transmitted with equal probability over an MBIOS channel, and assume that the block lengths of these codes tend to infinity as $m \to \infty$. Let $\Gamma_{k,m}$ be the fraction of the parity-check nodes of degree k for an arbitrary representation of the code $\mathcal{C}_m$ by a bipartite graph which corresponds to a full-rank parity-check matrix, and assume the limit $\Gamma_k \triangleq \lim_{m\to\infty} \Gamma_{k,m}$ exits. Then in the limit where $m \to \infty$, a necessary condition on the asymptotic achievable rate (R) for obtaining vanishing bit error probability is

$$R \leq 1 - \frac{1-C}{1 - \frac{1}{2\ln(2)} \sum_{p=1}^{\infty} \frac{\Gamma(g_p)}{p(2p-1)}}$$

where $\Gamma(x) \triangleq \sum_k \Gamma_k x^k$, and g_p is given in (7.6).

Corollary 7.2 (Lower Bound on the Bit Error Probability under ML Decoding)**.** [353, Corollary 4.2]: Let $\mathcal{C}$ be a binary linear block code of rate R whose transmission takes place over an MBIOS channel with capacity C. For an arbitrary full-rank parity-check matrix H of the code $\mathcal{C}$, let Γ_k designate the fraction of parity-check equations that involve k variables, and $\Gamma(x) \triangleq \sum_k \Gamma_k x^k$ be the right degree distribution from the node perspective which refers to the corresponding bipartite graph of $\mathcal{C}$. Then, under ML decoding (or any other decoding algorithm), the bit error probability (P_{b}) of the code satisfies

$$h_2(P_{\mathrm{b}}) \geq 1 - \frac{C}{R} + \frac{1-R}{2R\ln(2)} \sum_{p=1}^{\infty} \frac{\Gamma(g_p)}{p(2p-1)}$$

where g_p is introduced in (7.6).

The requirement of full-rank parity-check matrices is not natural when dealing with ensembles of codes; from [56], it can be removed by replacing the rate by the design rate of the ensemble.

We now introduce the definition of normalized parity-check density from [290], and derive an improved lower bound on the bit error probability (as compared to [290, Theorem 2.5]) in terms of this quantity.

Definition 7.4 (Normalized parity-check density [290])**.** Let $\mathcal{C}$ be a binary linear code of rate R, which is represented by a parity-check matrix H whose density is Δ. The *normalized density* of H, call it $t = t(H)$, is defined to be $t = \frac{R\Delta}{2-R}$.

In the following, we clarify the motivation for the definition of a normalized parity-check density. Let us assume that $\mathcal{C}$ is a binary linear block code of length n and rate R, and suppose that it can be represented by a bipartite graph which is *cycle-free*. From [290, Lemma 2.1], since this bipartite graph contains $(2-R)n-1$ edges, connecting n variable nodes with $(1-R)n$ parity-check nodes without any cycles, then the parity-check density of such a cycle-free code is $\Delta = \frac{2-R}{R} - \frac{1}{nR}$. Hence, in the limit where we let n tend to infinity, the normalized parity-check density of a cycle-free code tends to 1. For codes which are represented by bipartite graphs with cycles, the normalized parity-check density is above 1. As shown in [290, Corollary 2.5], the number of fundamental cycles in a bipartite graph which represents an arbitrary linear block $\mathcal{C}$ grows linearly with the normalized parity-check density. The normalized parity-check density therefore provides a measure for the number of cycles in bipartite graphs representing linear block codes. It is well known that cycle-free codes are not good in terms of performance, even under optimal ML decoding [106]; hence, good error-correcting codes (e.g., LDPC codes) should be represented by bipartite graphs with cycles. Following [290], providing a lower bound on the asymptotic normalized parity-check density in terms of their rate and gap to capacity gives a quantitative measure for the number of fundamental cycles of bipartite graphs representing good error correcting codes. In the following, we provide such an improved bound as compared to the bound given in [290, Theorem 2.5]. In the continuation to this section, the resulting improvement is exemplified.

First, we note that from Definition 7.4, it follows that the relation between the normalized density of a full-rank parity-check matrix and the corresponding average right degree is $t = \left(\frac{1-R}{2-R}\right) a_{\mathrm{R}}$ so the normalized parity-check density grows linearly with the average right degree (which is directly linked to the decoding complexity per iteration of LDPC codes under MPI decoding) where the scaling factor depends on the code rate R.

Since $\sum_k k\Gamma_k = a_{\mathrm{R}}$, then by applying Jensen's inequality to the RHS of (7.2), we get the

following lower bound on the bit error probability:

$$h_2(P_{\mathrm{b}}) \geq 1 - \frac{C}{R} + \frac{1-R}{2\ln(2)R} \sum_{p=1}^{\infty} \frac{g_p^{\frac{(2-R)t}{1-R}}}{p(2p-1)}. \tag{7.8}$$

This lower bound on the bit error probability is tighter than the bound given in [290, Eq. (23)], as explained in [353].

By combining Theorem 7.3 and Eq. (7.7), one can also derive a lower bound on the asymptotic parity-check density of a sequence of LDPC codes which achieve a fraction $1-\varepsilon$ with vanishing bit-error probability. Similarly to the bound of Sason and Urbanke [290, Theorem 2.1], this bound also grows like the logarithm of the inverse of the multiplicative gap (in rate) to capacity. It is also interesting to note that for the BEC the new bound merges with the bound of Sason and Urbanke which was shown to be tight even under MPI decoding [290, Theorem 2.2].

Theorem 7.4 (Lower Bound on Parity-Check Density)**.** [353, Theorem 4.1] Let $\{C_m\}$ be a sequence of binary linear codes achieving a fraction $1-\varepsilon$ of the capacity C of an MBIOS channel with vanishing bit error probability. Let H_m be an arbitrary *full-rank* parity-check matrix of the code $\mathcal{C}_m$, and denote its density by Δ_m. Then, the asymptotic parity-check density satisfies

$$\liminf_{m\to\infty} \Delta_m \geq \frac{K_1 + K_2 \ln \frac{1}{\varepsilon}}{1-\varepsilon} \tag{7.9}$$

where

$$K_1 = \frac{1-C}{C} \, \frac{\ln\left(\frac{\xi\,(1-C)}{C}\right)}{\ln\left(\frac{1}{g_1}\right)}, \quad K_2 = \frac{1-C}{C} \, \frac{1}{\ln\left(\frac{1}{g_1}\right)},$$

g_1 is introduced in (7.6), and

$$\xi \triangleq \begin{cases} 1 & \text{for a BEC} \\ \frac{1}{2\ln(2)} & \text{otherwise} \end{cases}.$$

For a BEC, one can verify that the lower bound on the asymptotic parity-check density stated in Theorem 7.4 coincides with the bound in [290, Eq. (3)]. This lower bound was demonstrated in [290, Theorem 2.3] to be tight. This is proved by showing that the sequence of right-regular LDPC ensembles of Shokrollahi [302] is optimal in the sense that it achieves (up to a small additive coefficient) the lower bound on the asymptotic parity-check density for the BEC.

The lower bound on the parity-check density in Theorem 7.4 is uniformly tighter than the one in [290, Theorem 2.1] (except for the BSC and BEC where they coincide). For a proof of this claim, the reader is referred to [353].

In the following, we present numerical results for the information-theoretic bounds introduced in this section. As expected, they significantly improve the numerical results presented in [56, Section 4] and [290, Section 4]. This improvement is attributed to the fact that, in contrast to [56, 290], for the derivation of the bounds introduced in this section, a two-level quantization of the LLR is not performed (which in essence would convert the arbitrary MBIOS channel (whose output may be continuous) to a BSC). All the numerical results refer to the case where the transmission of the codes takes place over the binary-input AWGN channel.

We note that the statements in this section refer to the case where the parity-check matrices are full rank. For ensembles of LDPC codes, it follows from the proofs in [353, Sections 3 and

4] that the statements stay valid with the following modifications: the rate R of a code which is randomly picked from the ensemble is replaced by the design rate (R_{d}) of the ensemble, and $\{\Gamma_k\}$ becomes the degree distribution of the parity-check nodes referring to the original bipartite graph which represents a parity-check matrix, possibly not of full rank. By doing this modification, the bound becomes looser when the asymptotic rate of the codes is strictly above the design rate of the ensemble. In light of this modification, we note that the fraction of nodes of degree k, Γ_k, is calculated in terms of the right degree distribution (ρ) by the equation

$$\Gamma_k = \frac{\rho_k}{k} \, \frac{1}{\displaystyle\int_0^1 \rho(x) \, dx}.$$

7.3.1 Thresholds of LDPC Ensembles under ML Decoding

Table 7.1 provides bounds on the thresholds under ML decoding of several ensembles of irregular rate $\frac{1}{2}$ LDPC codes from [270], where the transmission takes place over the binary input AWGN channel. The upper bounds on the achievable rates provide lower bounds on the $\frac{E_{\mathrm{b}}}{N_0}$ thresholds under ML decoding. The bounds shown are the two-level quantization bound of Burshtein *et al.* [56], the four and eight-level quantization bounds based on the first approach discussed here, which are derived in [353, Corollaries 3.1, 3.2], and the un-quantized bound of Theorem 7.3. Comparing these bounds with the exact thresholds under MPI decoding calculated using density evolution (DE) shown in the rightmost column, gives an indication of the loss in performance due to the sub-optimality of the iterative decoder. As mentioned above, comparing the results of the quantized bounds with the un-quantized bound gives an indication on the effect of quantization at the decoder. Table 7.1 shows that the numerical values of the bounds which correspond to the optimized eight quantization levels are very close to the bounds where there is no quantization of the LLR.

$\lambda(x)$	$\rho(x)$	2-Level Bound	4-Level Bound	8-Level Bound	Un-Quantized Lower Bound	DE Threshold
$0.38354x + 0.04237x^2 + 0.57409x^3$	$0.24123x^4 + 0.75877x^5$	0.269 dB	0.370 dB	0.404 dB	0.417 dB	0.809 dB
$0.23802x + 0.20997x^2 + 0.03492x^3 + 0.12015x^4 + 0.01587x^6 + 0.00480x^{13} + 0.37627x^{14}$	$0.98013x^7 + 0.01987x^8$	0.201 dB	0.226 dB	0.236 dB	0.239 dB	0.335 dB
$0.21991x + 0.23328x^2 + 0.02058x^3 + 0.08543x^5 + 0.06540x^6 + 0.04767x^7 + 0.01912x^8 + 0.08064x^{18} + 0.22798x^{19}$	$0.64854x^7 + 0.34747x^8 + 0.00399x^9$	0.198 dB	0.221 dB	0.229 dB	0.232 dB	0.310 dB
$0.19606x + 0.24039x^2 + 0.00228x^5 + 0.05516x^6 + 0.16602x^7 + 0.04088x^8 + 0.01064x^9 + 0.00221x^{27} + 0.28636x^{29}$	$0.00749x^7 + 0.99101x^8 + 0.00150x^9$	0.194 dB	0.208 dB	0.214 dB	0.216 dB	0.274 dB

Table 7.1: Comparison of thresholds for rate one-half ensembles of irregular LDPC codes whose transmission is over the binary-input AWGN channel. The Shannon capacity limit corresponds to $\frac{E_{\mathrm{b}}}{N_0} = 0.187$ dB. The 2-level, 4-level, 8-level and un-quantized lower bounds on the threshold refer to ML decoding, and are based on [56, Theorem 2], [353, Corollaries 3.1, 3.2] and Theorem 7.3, respectively. The degree distributions of the ensembles and their DE thresholds are based on density evolution under sum-product decoding [270], and are taken from [271, Tables 1 and 2].

The plots in Figure 7.1 compare different lower bounds on the $\frac{E_{\mathrm{b}}}{N_0}$-threshold under ML decoding of right-regular LDPC ensembles. The plots refer to a right degree of 6 (left plot) or 10 (right plot).

The following lower bounds are depicted in these plots: the Shannon capacity limit, the 2-level quantization lower bound in [56, Theorem 1], the 4 and 8-level quantization bounds of the LLR in [353, Section 3], and finally, the bound introduced in this section (see [353, Section 4]) where no quantization of the LLR is performed. It can be observed from the two plots in Figure 7.1 that the range of code rates where there exists a visible improvement with the new lower bounds depends on the degree of the parity-check nodes. In principle, the larger the value of the right-degree is, then the improvement obtained by these bounds is more pronounced starting from a higher rate code rate (e.g., for a right degree of 6 or 10, the improvement obtained by the new bounds is observed for code rates starting from 0.35 and 0.55 bits per channel use, respectively).

The comparison between the quantized and un-quantized bounds gives insight on the effect of the number of quantization levels of the LLR (even if they are chosen optimally) on the achievable rates, as compared to the ideal case where no quantization is done. From the numerical results in Table 7.1 (see [353, Section 5]), the improvement in the tightness of the bounds when more than 8 levels of quantization are used is marginal.

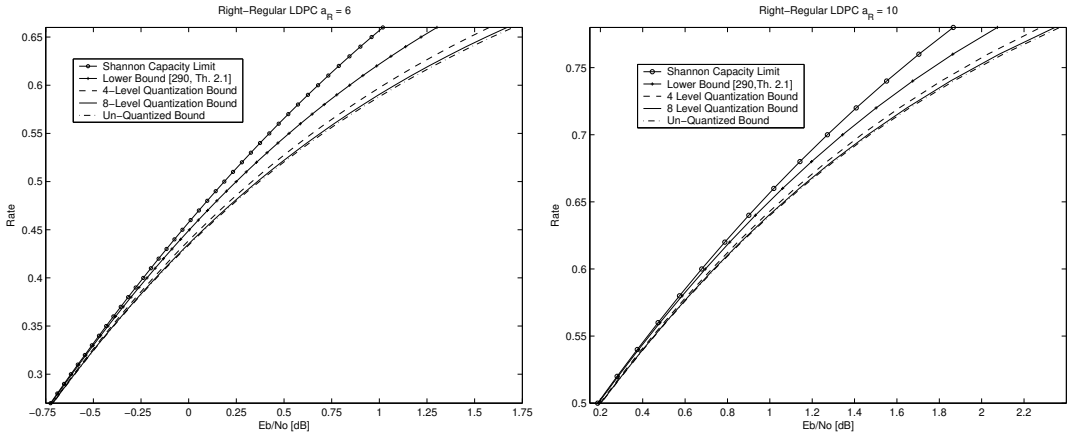


Figure 7.1: Comparison between different lower bounds on the threshold under ML decoding for right-regular LDPC ensembles with $a_{\mathrm{R}} = 6$ (upper plot) and $a_{\mathrm{R}} = 10$ (lower plot). The transmission takes place over the binary-input AWGN channel. The lower bound in [290, Theorem 2.1] relies on a two-level quantization of the LLR, and the bounds which refer to 4-level and 8-level quantization, as well the bound which refers to the case where the LLR is not quantized rely on [353].

7.3.2 Lower Bounds on the Asymptotic Parity-Check Density

The lower bound on the parity-check density presented in Theorem 7.4 enables to assess the tradeoff between asymptotic performance and asymptotic decoding complexity (per iteration) of an MPI decoder. This bound tightens the lower bound on the asymptotic parity-check density derived in [290, Theorem 2.1]. Fig. 7.2 compares these bounds for codes of rate $\frac{1}{2}$ (left plot) and $\frac{3}{4}$ (right plot) where the bounds are plotted as a function of $\frac{E_{\mathrm{b}}}{N_0}$. It can be observed from Fig. 7.2 that as $\frac{E_{\mathrm{b}}}{N_0}$ increases, the advantage of the bound in Theorem 7.4 over the bound in [290, Theorem 2.1] diminishes. This follows from the fact that as the value of $\frac{E_{\mathrm{b}}}{N_0}$ is increased, the two-level quantization of the LLR used in [56] and [290, Theorem 2.1] better captures the true behavior of the MBIOS channel.

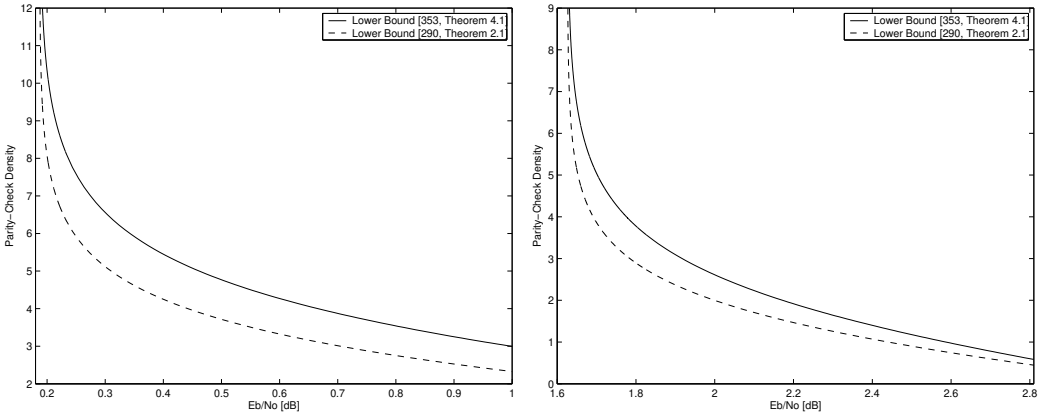


Figure 7.2: Comparison between lower bounds on the asymptotic parity-check density of binary linear block codes transmitted over a binary-input AWGN channel. The dashed line refers to [290, Theorem 2.1], and the solid line refers to Theorem 7.4. The upper and lower plots refer to code rates of $\frac{1}{2}$ and $\frac{3}{4}$, respectively. The Shannon capacity limit for these code rates corresponds to $\frac{E_b}{N_0}$ of 0.187 dB and 1.626 dB, respectively.

7.3.3 Lower Bounds on the Bit Error Probability of LDPC Codes

The statement in Corollary 7.2 (see [353, Corollary 4.2]) introduces an improved lower bound on the bit error probability of binary linear block codes, as compared to the one given in [290, Theorem 2.5]. The plot of Fig. 7.3 presents a comparison of these lower bounds for binary linear block codes where the bounds rely on (7.8) and [290, Theorem 2.5]. They are plotted as a function of the normalized density of an arbitrary parity-check matrix (see Definition 7.4). In our setting, the capacity of the channel is $\frac{1}{2}$ bit per channel use, and the bounds are depicted for binary linear block codes whose rate is a fraction $1-\varepsilon$ of the channel capacity. To demonstrate the advantage of the lower bound on the bit error probability in (7.8) over the lower bound derived in [290, Theorem 2.5], let us assume that one wants to design a binary LDPC code which achieves a bit-error probability of 10^{-6} at a rate which is 99% of the channel capacity. The curve of the lower bound from [290] for $\varepsilon = 0.01$ implies that the normalized density of an arbitrary parity-check matrix which represents the code (see Definition 7.4 in p. 128) should be at least 4.33, while the curve depicting the bound from (7.8) strengthens this requirement to a normalized density (of each parity-check matrix) of at least 5.68. Translating this into terms of parity-check density (which is also the complexity per iteration for MPI decoding) yields minimal parity-check densities of 13.16 and 17.27, respectively (the minimal parity-check density is given by $\Delta_{\min} = \frac{(2-R)t_{\min}}{R}$). It is reflected from Fig. 7.3 that as the gap to capacity ε tends to zero, the lower bound on the normalized density of an arbitrary parity-check matrix (t), which represents a code which achieves low error probability for a rate of $R = (1-\varepsilon)C$ grows significantly.

7.4 Generalization of the Bounds for Parallel Channels

In [292], the authors focus primarily on the information-theoretic limitations of low-density parity-check (LDPC) codes whose transmission takes place over a set of parallel channels. Parallel channels means a communication system or model where the distortion introduced by each channel is independent of the signal and the distortion in all of the other channels. The first main result presented in this section is the derivation of an upper bound on the achievable rates of ensembles of LDPC codes under optimal maximum-likelihood (ML) decoding when we assume that the codes are

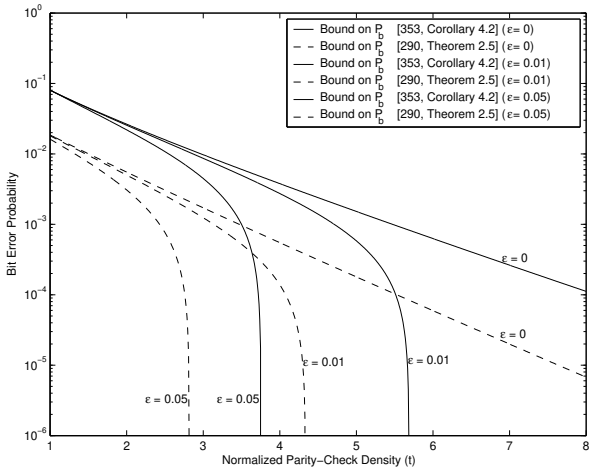


Figure 7.3: Lower bounds on the bit error probability for any binary linear block code transmitted over a binary-input AWGN channel whose capacity is $\frac{1}{2}$ bits per channel use. The bounds are depicted in terms of the normalized density of an arbitrary parity-check matrix which represents the code, and the curves correspond to code rates which are a fraction $1-\varepsilon$ of the channel capacity (for different values of ε). The bounds depicted in dashed lines are based on [290, Theorem 2.5], and the bounds in solid lines are given in Eq. (7.8) (see [353, Corollary 4.2]).

communicated over parallel memoryless binary-input output-symmetric (MBIOS) channels. This result forms a non-trivial generalization of the "un-quantized" bound derived by the authors for a single MBIOS channel [353], where the latter bound improves the tightness of the bound in [56] which was based on a two-level quantization approach. The second main result introduced in this section is a derivation of a lower bound on the decoding complexity of ensembles of LDPC codes for parallel MBIOS channels; this bound assumes an iterative message-passing decoder, and the decoding complexity is normalized per iteration. In this respect, it continues the study in [202] which was focused on upper bounds on the parity-check density of ensembles of LDPC codes for parallel MBIOS channels. The two main issues addressed in this section for parallel channels are applied to punctured LDPC codes. The bounds on the achievable rates and decoding complexity of LDPC codes over statistically independent parallel channels are applied to the case of transmission of ensembles of punctured LDPC codes over a single MBIOS channel.

The interested reader is referred to the full paper version [292] which provides proofs to all of the theorems presented here, and further discusses the relations between these theorems and some other previously reported results from [56, 249, 253, 255, 290, 353].

7.5 An Upper bound on the Achievable Rates of LDPC codes over Parallel Channels

In this section, we present an upper bound on the design rate of a sequence of ensembles of LDPC codes whose transmission takes place over a set of statistically independent parallel MBIOS channels, and achieves vanishing bit error probability under ML decoding. This bound is used in the next section for the derivation of an upper bound on the design rate of an arbitrary sequence of ensembles of punctured LDPC codes.

Let us assume that a binary LDPC code $\mathcal{C}$ of length n is transmitted over a set of J statistically

independent parallel MBIOS channels. Denote the number of code bits of $\mathcal{C}$ which are transmitted over the jth channel by $n^{[j]}$, and the fraction of bits transmitted over the jth channel by

$$p_j \triangleq \frac{n^{[j]}}{n}, \quad j \in \{1, \ldots, J\}. \tag{7.10}$$

Let $\mathcal{G}$ be a bipartite graph which represents the code $\mathcal{C}$, and E be the set of edges in $\mathcal{G}$. Let $E^{[j]}$ designate the set of edges connected to variable nodes which correspond to code bits transmitted over the jth channel, and

$$q_j \triangleq \frac{|E^{[j]}|}{|E|}, \quad j \in \{1, \ldots, J\} \tag{7.11}$$

denote the fraction of edges connected to these variable nodes. Referring to the edges from the subset $E^{[j]}$, let $\lambda_i^{[j]}$ designate the fraction of these edges which are connected to variable nodes of degree i, and define the following J degree distributions from the edge perspective:

$$\lambda^{[j]}(x) \triangleq \sum_{i=2}^{\infty} \lambda_i^{[j]} x^{i-1}, \qquad j \in \{1, \ldots, J\}$$

which correspond to each of the J parallel channels. For the simplicity of the notation, let us define a vector of degree distributions for the variable nodes from the edge perspective to be $\underline{\lambda}(x) = \big(\lambda^{[1]}(x), \ldots, \lambda^{[J]}(x)\big)$. Following the notation in [253], the ensemble $(n, \underline{\lambda}, \rho)$ is defined to be the set of LDPC codes of length n, which according to their representation by bipartite graphs and the assignment of their code bits to the parallel channels, imply left and right degree distributions of $\underline{\lambda}$ and ρ, respectively.

In the following, we introduce a sequence of ensembles of LDPC codes, say $\{(n_r, \underline{\lambda}_r, \rho)\}_{r=1}^{\infty}$ where the fraction of code bits assigned to each of the J parallel channels is uniform over all the codes of each ensemble, and ρ is fixed for all the ensembles of this sequence (i.e., it is independent of r). Since λ which corresponds to the overall left degree distribution of the edges is independent of r (and so is the right degree distribution ρ), one can consider here the common design rate of the sequence of ensembles $\{(n_r, \underline{\lambda}_r, \rho)\}_{r=1}^{\infty}$ which does not depend on r.

This setting is general enough for applying the following theorem to various applications which form particular cases of communication over parallel channels. In this setting, the fraction of code bits assigned to the jth channel, $p_{j,r}$, depends on $j \in \{1, \ldots, J\}$ and $r \in \mathbb{N}$, but not on the particular code chosen from each ensemble. It follows from [292, Lemma 4.2] that the same property also holds for $q_{j,r}$ which designates the fraction of edges which are connected to variable nodes whose code bits are assigned to the jth channel. In the following, we assume that the limits

$$p_j \triangleq \lim_{r \to \infty} p_{j,r}, \quad q_j \triangleq \lim_{r \to \infty} q_{j,r} \tag{7.12}$$

exist and are positive for all $j \in \{1, \ldots, J\}$.

Theorem 7.5 (Upper Bound on the Achievable Rates of Ensembles of LDPC Codes over Parallel MBIOS Channels)**.** [292, Theorem 4.1]: Let a sequence of LDPC ensembles $\{(n_r, \underline{\lambda}_r, \rho)\}_{r=1}^{\infty}$ be transmitted over a set of J statistically independent parallel MBIOS channels, and assume that the block length (n_r) goes to infinity as we let r tend to infinity. Let C_j denote the capacity of the jth channel, and $a(\cdot; j)$ designate the *pdf* of the LLR at the output of the jth channel given its input is 1. If in the limit where r tends to infinity, the bit error probability of this sequence under ML

decoding vanishes, then the common design rate R_{d} of these ensembles satisfies

$$R_{\mathrm{d}} \leq 1 - \frac{1 - \sum_{j=1}^{J} p_j C_j}{1 - \frac{1}{2\ln 2} \sum_{p=1}^{\infty} \left\{ \frac{1}{p(2p-1)} \, \Gamma\left(\sum_{j=1}^{J} q_j \; g_{j,p} \right) \right\}} \tag{7.13}$$

where Γ denotes the right degree distribution from the node perspective, and for all $j \in \{1, \ldots, J\}$ and $p \in \mathbb{N}$, $g_{j,p}$ is defined as follows:

$$g_{j,p} \triangleq \int_0^{\infty} a(l; j) \; (1 + e^{-l}) \tanh^{2p}\left(\frac{l}{2}\right) dl. \tag{7.14}$$

Example 7.1. For the particular case where the J parallel MBIOS channels are binary erasure channels where the erasure probability of the jth channel is ε_j, we get from (7.14)

$$g_{j,p} = 1 - \varepsilon_j \; , \quad \forall \; j \in \{1, \ldots, J\}, \;\; p \in \mathbb{N}. \tag{7.15}$$

Since $g_{j,p}$ is independent of p for a BEC, and based on the equality $\sum_{p=1}^{\infty} \frac{1}{2p(2p-1)} = \ln 2$, we obtain from Theorem 7.5 that the common design rate of the sequence of LDPC ensembles is upper bounded by

$$R_{\mathrm{d}} \leq 1 - \frac{\sum_{j=1}^{J} p_j \varepsilon_j}{1 - \Gamma\left(1 - \sum_{j=1}^{J} q_j \; \varepsilon_j\right)} \; . \tag{7.16}$$

This particular result coincides with [253, Theorem 2].

7.6 Upper Bounds on the Achievable Rates of Punctured LDPC Codes under ML Decoding

In this section, we present information-theoretic upper bounds on the achievable rates under ML decoding of ensembles of punctured LDPC codes whose transmission takes place over an MBIOS channel. The transmission of punctured codes can be interpreted as a special case of transmitting the original (un-punctured) codes over a set of parallel channels where these component channels are formed by a mixture of the communication channel and BECs whose erasure probabilities are the puncturing rates of different subsets of code bits. Hence, the analysis in this section relies on the bound presented in Section 7.5.

Randomly Punctured LDPC Codes In this section, we consider the achievable rates of randomly punctured LDPC (RP-LDPC) codes. We assume that the transmission of these codes takes place over an MBIOS channel, and refer to their achievable rates under optimal ML decoding. The upper bound on the achievable rates of ensembles of RP-LDPC codes relies on Section 7.5 where we presented an upper bound on the achievable rates of LDPC codes for parallel channels.

In the following, we assume that the communication takes place over an MBIOS channel with capacity C, and g_p is defined in (7.6).

Theorem 7.6 (Achievable Rates of Randomly Punctured LDPC Codes)**.** [292, Theorem 5.1]: Let $\{(n_r, \lambda, \rho)\}_{r=1}^{\infty}$ be a sequence of ensembles of LDPC codes whose block length (n_r) tends to infinity

as $r \to \infty$. Assume that a sequence of ensembles of RP-LDPC codes is constructed in the following way: for each code from an ensemble of the original sequence, a subset of αn_r code bits is a-priori selected, and these bits are randomly punctured at a fixed rate ($P_{\rm pct}$). Assume that the punctured codes are transmitted over an MBIOS channel with capacity C, and that in the limit where r approaches infinity, the sequence of ensembles of RP-LDPC codes achieves vanishing bit error probability under some decoding algorithm. Then in probability 1 w.r.t. the random puncturing patterns, the asymptotic design rate ($R_{\rm d}$) of the new sequence satisfies

$$R_{\rm d} \leq \frac{1}{1-\alpha P_{\rm pct}} \left(1 - \frac{1-(1-\alpha P_{\rm pct})C}{1 - \frac{1}{2\ln 2} \sum_{p=1}^{\infty} \left\{ \frac{1}{p(2p-1)} \, \Gamma\Big((1 - P_{\rm pct} + \xi) g_p \Big) \right\}} \right) \tag{7.17}$$

where Γ denotes the right degree distribution (from the node perspective) of the original sequence, g_p is introduced in (7.6), and ξ is the following positive number:

$$\xi \triangleq 2(1-\alpha) P_{\rm pct} \int_0^1 \lambda(x) \, dx. \tag{7.18}$$

Intentionally Punctured LDPC Codes In [133], Ha and McLaughlin show that good codes can be constructed by puncturing good ensembles of LDPC codes using a technique called "intentional puncturing". In this approach, the code bits are partitioned into disjoint sets so that each set contains all the code bits whose corresponding variable nodes have the same degree. The code bits in each one of these sets are randomly punctured at a fixed puncturing rate.

We briefly present the notation used in [133] for the characterization of ensembles of intentionally punctured LDPC (IP-LDPC) codes. Consider an ensemble of LDPC codes with left and right edge degree distributions λ and ρ, respectively. For each degree j such that $\lambda_j > 0$, a puncturing rate $\pi_j \in [0,1]$ is determined for randomly puncturing the set of code bits which correspond to variable nodes of degree j. The polynomial associated with this puncturing pattern is

$$\pi^0(x) \triangleq \sum_{j=1}^{\infty} \pi_j x^{j-1}. \tag{7.19}$$

An ensemble of IP-LDPC codes can be therefore represented by the quadruplet $(n, \lambda, \rho, \pi^0)$ where n designates the block length of these codes, λ and ρ are the left and right degree distributions from the edge perspective, respectively, and π^0 is the polynomial which corresponds to the puncturing pattern, as given in (7.19). The average fraction of punctured bits is given by $p^{(0)} = \sum_{j=1}^{\infty} \Lambda_j \pi_j$ where Λ is the left node degree distribution of the original LDPC ensemble. The following statement, which relies on Theorem 7.5, provides an upper bound on the common design rate of a sequence of ensembles of IP-LDPC codes. This bound refers to ML decoding (and hence, to any sub-optimal decoding algorithm).

Theorem 7.7 (Achievable Rates of Intentionally Punctured LDPC Codes)**.** [292, Theorem 5.2]: Let $\{(n_r, \lambda, \rho, \pi^0)\}_{r=1}^{\infty}$ be a sequence of ensembles of IP-LDPC codes transmitted over an MBIOS channel, and assume that n_r tends to infinity as $r \to \infty$. Let C be the channel capacity, and a be the *pdf* of the LLR at the output of the channel given its input is 1. If the asymptotic bit error probability of this sequence vanishes under ML decoding (or any sub-optimal decoding algorithm) as $r \to \infty$, then in probability 1 w.r.t. the puncturing patterns, the common design rate $R_{\rm d}$ of

these ensembles satisfies

$$R_{\mathrm{d}} \leq \frac{1}{1-p^{(0)}} \left[1 - \frac{1-(1-p^{(0)})C}{1 - \frac{1}{2\ln 2} \sum_{p=1}^{\infty} \left\{ \frac{1}{p(2p-1)} \, \Gamma\left(\left(1 - \sum_{j=1}^{\infty} \lambda_j \pi_j \right) g_p \right) \right\}} \right] \tag{7.20}$$

where Γ denotes the right degree distribution from the node perspective,

$$p^{(0)} \triangleq \sum_{j=1}^{\infty} \Lambda_j \pi_j \tag{7.21}$$

designates the average puncturing rate of the code bits, and g_p is the functional of the MBIOS channel introduced in (7.6).

Numerical Results for Intentionally Punctured LDPC Codes In this section, we compare between thresholds under message-passing iterative (MPI) decoding and bounds on thresholds under ML decoding for ensembles of IP-LDPC codes. It is assumed that the transmission of the punctured LDPC codes takes place over a binary-input AWGN channel. The pairs of degree distributions and the corresponding puncturing patterns were originally presented in [133]. We use these ensembles in order to study their inherent gap to capacity, and also study how close to optimal is iterative decoding for these ensembles (in the asymptotic case where the block length goes to infinity). Table 7.2 refers to an ensemble of rate$-\frac{1}{2}$ LDPC codes which by puncturing, the rate varies between 0.50 and 0.91.

$\pi^0(x)$ (puncturing pattern)	Design rate	Capacity limit	Lower bound (ML decoding)	Iterative (IT) Decoding	Fractional gap to capacity (ML vs. IT)
0	0.500	0.187 dB	0.270 dB	0.393 dB	$\geq 40.3\%$
$0.07886x + 0.01405x^2 + 0.06081x^3 + 0.07206x^9$	0.528	0.318 dB	0.397 dB	0.526 dB	$\geq 37.9\%$
$0.20276x + 0.09305x^2 + 0.03356x^3 + 0.16504x^9$	0.592	0.635 dB	0.716 dB	0.857 dB	$\geq 36.4\%$
$0.25381x + 0.15000x^2 + 0.34406x^3 + 0.019149x^9$	0.629	0.836 dB	0.923 dB	1.068 dB	$\geq 37.3\%$
$0.31767x + 0.18079x^2 + 0.05265x^3 + 0.24692x^9$	0.671	1.083 dB	1.171 dB	1.330 dB	$\geq 35.6\%$
$0.36624x + 0.24119x^2 + 0.49649x^3 + 0.27318x^9$	0.719	1.398 dB	1.496 dB	1.664 dB	$\geq 36.9\%$
$0.41838x + 0.29462x^2 + 0.05265x^3 + 0.30975x^9$	0.774	1.814 dB	1.927 dB	2.115 dB	$\geq 37.2\%$
$0.47074x + 0.34447x^2 + 0.02227x^3 + 0.34997x^9$	0.838	2.409 dB	2.547 dB	2.781 dB	$\geq 37.1\%$
$0.52325x + 0.39074x^2 + 0.01324x^3 + 0.39436x^9$	0.912	3.399 dB	3.607 dB	3.992 dB	$\geq 35.1\%$

Table 7.2: Comparison of thresholds for ensembles of IP-LDPC codes where the original ensemble before puncturing has the degree distributions $\lambda(x) = 0.25105x + 0.30938x^2 + 0.00104x^3 + 0.43853x^9$ and $\rho(x) = 0.63676x^6 + 0.36324x^7$ (so its design rate is equal to $\frac{1}{2}$). The transmission of these codes takes place over a binary-input AWGN channel. The table compares values of $\frac{E_{\mathrm{b}}}{N_0}$ referring to the capacity limit, the bound given in Theorem 7.7 (which provides a lower bound on $\frac{E_{\mathrm{b}}}{N_0}$ under ML decoding), and thresholds under iterative message-passing decoding. The fractional gap to capacity (see the rightmost column) measures the ratio of the gap to capacity under optimal ML decoding and the achievable gap to capacity under (sub-optimal) iterative message-passing decoding. The pair of degree distributions for the ensemble of LDPC codes, and the polynomials which correspond to its puncturing patterns are taken from [133, Table 2].

Table 7.2 provides lower bounds on the inherent gap to capacity under optimal ML decoding (based on Theorem 7.7); these values are compared to the corresponding gaps to capacity under

iterative message-passing decoding (whose calculation is based on the density evolution analysis). On one hand, Table 7.2 provides a quantitative assessment of the loss in the asymptotic performance which is attributed to the sub-optimality of iterative decoding (as compared to optimal ML decoding), and on the other hand, they provide an assessment of the inherent loss in performance which is attributed to the structure of the ensembles, even if optimal ML decoding could be applied to decode these codes. The loss in performance in both cases is measured in terms of $\frac{E_b}{N_0}$ in decibels. It is demonstrated in Table 7.2 that the asymptotic loss in performance due to the code structure is still non-negligible as compared to the corresponding loss due to the sub-optimality of iterative decoding. For all the ensembles of IP-LDPC codes considered in Table 7.2 (which were originally introduced in [133, Table 2]), the gap to capacity under the sum-product iterative decoding algorithm does not exceed 0.6 dB; however, under ML decoding, the gap to capacity is always greater than $\frac{1}{3}$ of the corresponding gap to capacity under this iterative decoding algorithm; therefore, the results in Table 7.2 regarding the thresholds under ML decoding further emphasize the efficiency of the sum-product decoding algorithm for these ensembles, especially in light of its moderate complexity. Table 7.2 also shows that the performance of the punctured LDPC codes is degraded at high rates, where one needs to pay a considerable penalty for using punctured codes. This phenomenon was explained in [254, Theorem 1] by the threshold effect for ensembles of IP-LDPC codes.

7.7 On the Decoding Complexity of LDPC Codes for Parallel Channels

We present in this section a lower bound on the decoding complexity of LDPC codes for parallel MBIOS channels and lower bounds on the decoding complexity of punctured LDPC codes, and later (see Section 7.8) relate the new bounds to previously reported results.

A Lower Bound on the Decoding Complexity for Parallel MBIOS Channels Consider a binary linear block code which is represented by a bipartite graph, and assume that the graph serves for the decoding with an iterative algorithm. Following [202] and [249], the decoding complexity under MPI decoding is defined as the number of edges in the graph normalized per information bit. This quantity measures the number of messages which are delivered through the edges of the graph (from left to right and vice versa) during a single iteration. Equivalently, since there is a one-to-one correspondence between a bipartite graph and the parity-check matrix H which represents the code, the decoding complexity is also equal to the number of non-zero elements in H normalized per information bit (i.e., the density of the parity-check matrix [290, Definition 2.2]). Hence, the decoding complexity (as well as the performance) of iteratively decoded binary linear block codes depends on the specific representation of the code by a parity-check matrix. Since the average right degree (a_{R}) of a bipartite graph is equal to the number of edges normalized per parity-check equation, then the average right degree and the decoding complexity are related quantities. Consider an ensemble of LDPC codes whose design rate is R_{d}. It is natural to relate the decoding complexity of the ensemble, say χ_D, to its average right degree and design rate, as follows:

$$\chi_{\mathrm{D}} = \frac{1 - R_{\mathrm{d}}}{R_{\mathrm{d}}} \, a_{\mathrm{R}} \, .$$

We note that a_{R} is fixed for all the codes from an ensemble of LDPC codes with a given pair of degree distributions.

Consider a sequence of ensembles of LDPC codes, $\{(n_r, \underline{\lambda}_r, \rho)\}_{r=1}^{\infty}$, whose transmission takes place over a set of J statistically independent parallel MBIOS channels. Let C_j and p_j be the capacity and the fraction of code bits which are assigned to the jth channel, respectively (where

$j \in \{1, \ldots, J\}$). We now present a lower bound on the decoding complexity per iteration under MPI decoding for this sequence.

Theorem 7.8 (Lower Bound on the Decoding Complexity Per Iteration for Ensembles of LDPC Codes over Parallel MBIOS Channels)**.** [292, Theorem 6.1]: Let a sequence of ensembles of LDPC codes, $\{(n_r, \underline{\lambda}_r, \rho)\}_{r=1}^{\infty}$, be transmitted over a set of J statistically independent parallel MBIOS channels. Assume that the capacities C_j of these channels are all positive, and denote the average capacity by $\overline{C} \triangleq \sum_{j=1}^{J} p_j C_j$. If this sequence achieves a fraction $1-\varepsilon$ of $\overline{C}$ with vanishing bit error probability, then the asymptotic decoding complexity under MPI decoding satisfies

$$\chi_{\mathrm{D}}(\varepsilon) \geq K_1 + K_2 \ln\left(\frac{1}{\varepsilon}\right). \tag{7.22}$$

The coefficients $K_{1,2}$ in this lower bound are as follows:

$$K_1 = -\frac{(1-\overline{C})\ \ln\left(\frac{1}{2\ln 2}\frac{1-\overline{C}}{\overline{C}}\right)}{\overline{C}\ \ln\left(\sum_{j=1}^{J} q_j g_{j,1}\right)}, \quad K_2 = -\frac{1-\overline{C}}{\overline{C}\ \ln\left(\sum_{j=1}^{J} q_j g_{j,1}\right)} \tag{7.23}$$

where $g_{j,1}$ is introduced in (7.14), and q_j is introduced in (7.12) and is assumed to be positive for all $j \in \{1, \ldots, J\}$. For parallel BECs, the term $\frac{1}{2\ln 2}$ can be removed from the numerator in the expression of K_1.

Lower Bounds on the Decoding Complexity for Punctured LDPC Codes Consider an ensemble of LDPC codes of length n and design rate R'_{d}, and let the code bits be partitioned into J disjoint sets where the jth set contains a fraction p_j of these bits ($j \in \{1, \ldots, J\}$). Assume that the bits in the jth set are randomly punctured at rate π_j, and let the punctured codes be transmitted over an MBIOS channel whose capacity is C. As shown in the previous section, this is equivalent to transmitting the original (un-punctured) codes over a set of J parallel channels, where the jth set of code bits is transmitted over a channel whose capacity is $C_j = (1-\pi_j)C$. The average capacity of this set of J parallel channels is given by $\overline{C} = \sum_{j=1}^{J} p_j\ (1-\pi_j)C = (1-\gamma)C$ where $\gamma \triangleq \sum_{j=1}^{J} p_j \pi_j$ is the overall puncturing rate. Denote the design rate of the punctured codes by $R_{\mathrm{d}} \triangleq \frac{R'_{\mathrm{d}}}{1-\gamma}$, then it follows that the multiplicative gap to capacity of the punctured codes is given by $\varepsilon = 1 - \frac{R_{\mathrm{d}}}{C} = 1 - \frac{R'_{\mathrm{d}}}{\overline{C}}$. In the following, we present a lower bound on the decoding complexity of RP-LDPC codes.

Theorem 7.9 (Lower Bound on the Decoding Complexity Per Iteration for Ensembles of Randomly Punctured LDPC Codes)**.** [292, Theorem 6.2]: Let $\{(n_r, \lambda, \rho)\}_{r=1}^{\infty}$ be a sequence of ensembles of LDPC codes whose block length (n_r) tends to infinity as $r \to \infty$. Assume that a sequence of ensembles of RP-LDPC codes is constructed in the following way: for each code from an ensemble of the original sequence, a subset of αn_r code bits is a-priori selected, and these bits are randomly punctured at a fixed rate (P_{pct}). Assume that the punctured codes are transmitted over an MBIOS channel with capacity C, and that as r tends to infinity, the sequence of ensembles of punctured codes achieves a fraction $1-\varepsilon$ of the capacity with vanishing bit error probability. Then in probability 1 w.r.t. the random puncturing patterns, the decoding complexity of this sequence under MPI decoding satisfies

$$\chi_{\mathrm{D}}(\varepsilon) \geq K_1 + K_2 \ln\left(\frac{1}{\varepsilon}\right). \tag{7.24}$$

The coefficients $K_{1,2}$ in this lower bound are are given explicitly in [292, Eq. (72)].

The upper bound on the decoding complexity for sequences of ensembles of IP-LDPC codes is also given in terms of the gap between the rate of the punctured rate and the channel capacity.

Theorem 7.10 (Lower Bound on the Decoding Complexity Per Iteration for Ensembles of Intentionally Punctured LDPC Codes)**.** [292, Theorem 6.3]: Let $\{(n_r, \lambda, \rho, \pi^0)\}_{r=1}^{\infty}$ be a sequence of ensembles of IP-LDPC codes transmitted over an MBIOS channel whose capacity is C. If this sequence achieves a fraction $1-\varepsilon$ of the capacity with vanishing bit error probability, then in probability 1 w.r.t. the random puncturing patterns, the decoding complexity of this sequence under MPI decoding satisfies

$$\chi_{\mathrm{D}}(\varepsilon) \geq K_1 + K_2 \ln\left(\frac{1}{\varepsilon}\right). \tag{7.25}$$

The coefficients $K_{1,2}$ in this lower bound are given explicitly in [292, Eq. (75)].

7.8 Summary and Outlook

One of the main results in this section, Theorem 7.5, provides an upper bound on the asymptotic rate of a sequence of ensembles of LDPC codes which achieves vanishing bit error probability. We assume that the communication takes place over a set of parallel memoryless binary-input output-symmetric (MBIOS) channels. The derivation of Theorem 7.5 relies on upper and lower bounds on the conditional entropy of the transmitted codeword given the received sequence at the output of the parallel channels (see [292, Section 3]), and it is valid under optimal ML decoding (or any sub-optimal decoding algorithm). This theorem enables the derivation of a lower bound on the decoding complexity (per iteration) of ensembles of LDPC codes under message-passing iterative decoding when the transmission of the codes takes place over parallel MBIOS channels. The latter bound is given in terms of the gap between the rate of these codes for which reliable communication is achievable and the channel capacity. Similarly to a lower bound on the decoding complexity of ensembles of LDPC codes for a single MBIOS channel [290], the lower bound on the decoding complexity which is presented for parallel channels also grows like the log of the inverse of the gap to capacity.

Theorem 7.5 can be used for various applications which form particular cases of communication over parallel channels, e.g., intentionally punctured LDPC codes [133], non-uniformly error protected LDPC codes [253], and LDPC-coded modulation schemes. In [292, Section 5], we use Theorem 7.5 for the derivation of upper bounds on the achievable rates under ML decoding of (randomly and intentionally) punctured LDPC codes whose transmission takes place over an MBIOS channel. It is exemplified numerically that for various good ensembles of IP-LDPC codes, the asymptotic loss in performance due to the code structure is still non-negligible as compared to the corresponding loss due to the sub-optimality of iterative decoding (as compared to optimal ML decoding). Looser versions of the bounds presented here for punctured LDPC codes suggest a simplified re-derivation of previously reported bounds on the decoding complexity of randomly punctured LDPC codes (see [292]). Interconnections between the theorems introduced in this section and some other previously reported results are shown in Fig. 7.4. The proofs of all the theorems in this section and further discussion on the theorems are provided in [292].

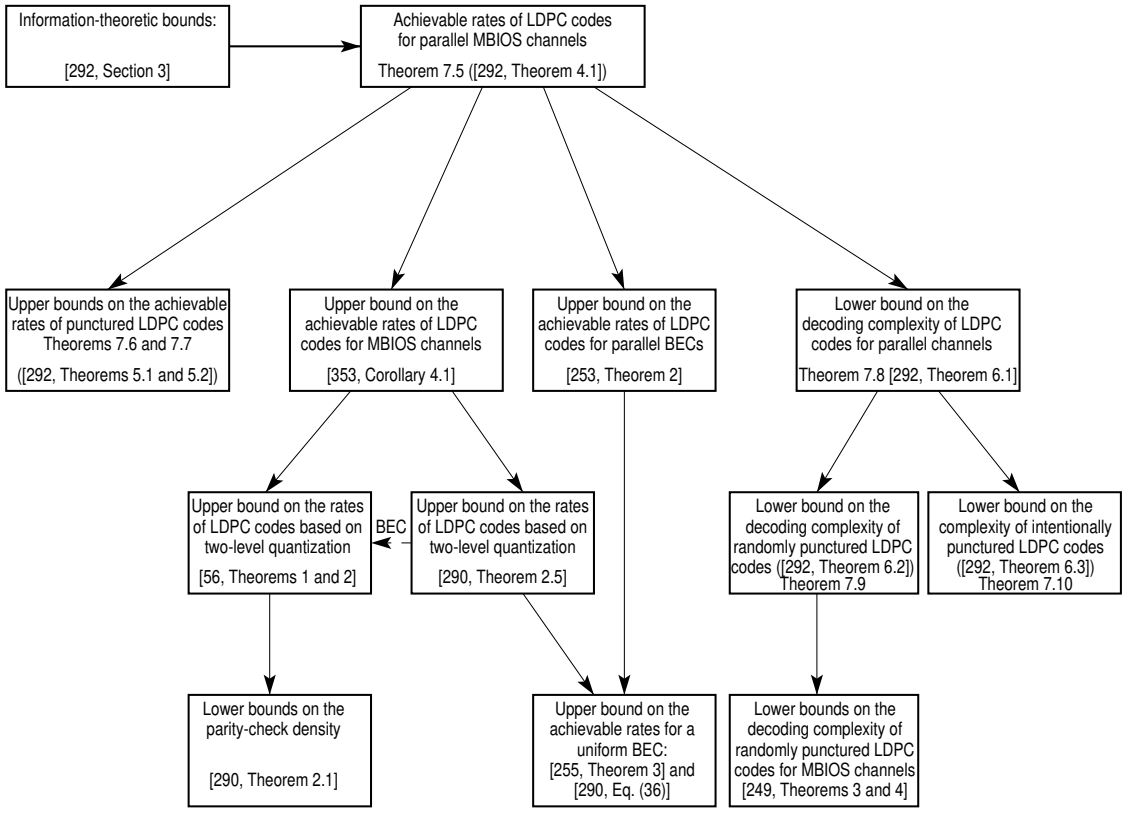


Figure 7.4: An interconnections diagram among the bounds in this section and previously reported bounds.

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